

Statistics for Insurance

Theory and methods for life and non-life insurance data

Martin Bladt

University of Copenhagen

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Preface

The practice of actuarial science has undergone a profound transformation in recent decades. Where the field once primarily relied on more deterministic and aggregate methods, it now commonly deals with the complexity of stochastic processes and the sheer volume of granular data that describes them. The fundamental challenge, however, remains the same: to manage the risk of processes arising in insurance and financial operations. The discipline of statistics for insurance is our primary scientific tool for this task. It offers the required methods to describe the behaviour of the underlying risks, which is the first and arguably the most fundamental step in designing any effective risk mitigation strategy.

Modern insurance data comes in a great variety of shapes and sizes, and the statistical methods required to analyse it must be equally varied and sophisticated. In this monograph, we study this complexity from different angles. We are interested in data that can give rise to a claim, which may possess both discrete and continuous components, such as in the interplay between claim frequency and severity. We consider processes that evolve through time, requiring pathwise analysis, and we address the practical challenges of varying exposure periods and reporting delays. We also face rare or catastrophic events that manifest as heavy-tails, the influence of different sources of randomness, and the common issues of truncated or censored data. The main aim of these notes is to provide a representative and central exposition of statistical tools that can tackle these characteristics. While the classical distinction between life and non-life insurance has its place, a further goal of this work is to close the gap between these two perspectives, presenting a more unified approach, striving to use the same set of mathematical tools to treat both cases.

In selecting the material, a deliberate choice was made to emphasise not primarily the application of statistical methods, but rather their mathematical foundations. This aligns with my conviction that a deep understanding of the mathematical structure is what drives intuition the most, even when practical application is the ultimate goal. The development of novel methods to solve the emerging problems in insurance often builds upon established theories. For this reason, we strive to document and understand how mathematical statistics quantifies risk, and in particular embarking on the understanding of the detailed proofs of precise statements. Practical examples, however, are also amply provided in both the main text and the exercises.

The order of the presented chapters creates a path through the core of modern statistics for insurance, though clearly only one of many. The material has

grown from a shorter set of lecture notes used to teach a semester course at the masters level; in its current length, a selection of about half of the material is recommended. We begin with preliminaries on stochastic convergence, before moving to exploratory tools, the study of heavy-tailed data, and parametric models. We then explore selected topics of machine learning, with an emphasis on model selection and explainability, before turning to specialised topics like stochastic claims reserving, survival analysis, and nonparametric inference for Markov processes. The later chapters cover more advanced topics on multi-state modelling, including both parametric and nonparametric approaches. A Bayesian chapter finalises our treatment. A notable omission is a systematic treatment of parametric dependence modelling and time-series analysis. However, the reader may find that the mathematical tools presented can often be adapted to cover such cases. Numerous references have been pivotal to the development of this monograph, and they are purposefully collected in an organised manner at the end of each chapter, albeit merely representative and far from being comprehensive.

This work would not have been possible without the support and inspiration of many people, but I prominently want to thank my masters and PhD students at the University of Copenhagen, as well as my collaborators. They have been a constant source of motivation, and their intellectual curiosity, dedication to learning and teaching the material, and their help with earlier versions of the manuscript have been essential. Above all, their enthusiasm for the subject has sharpened my own understanding in many ways.

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Chapter 1

Probabilistic Foundations

The mathematical foundations of statistics rest on results in probability theory, such as laws of large numbers, central limit theorems, and general principles of stochastic convergence. These results provide the framework for analysing the behaviour of statistical procedures and for establishing their validity in large samples. A broad range of methods in insurance statistics, developed in later chapters, build directly on these tools. We begin with finite-dimensional convergence and limit theorems that are used repeatedly later on, then move to empirical summaries and process-level convergence.

1.1 Stochastic convergence in finite dimensions

We now recall some of the main results used to analyse independent and identically distributed (iid) random vectors. Here, we interpret such vectors as real-valued risks, e.g. claim sizes from a non-life insurance portfolio. We start by fixing notation for deterministic and stochastic orders, and then list standard convergence tools. From real analysis we recall the following notation, which makes precise the notion of speed of convergence.

Definition 1.1 (*o*-notation) Let a_n, b_n be sequences of real numbers. We write $a_n = o(b_n)$ if $\lim_{n \rightarrow \infty} a_n/b_n = 0$. We write $a_n = O(b_n)$ if there exists $K < \infty$ such that $|a_n/b_n| \leq K$ for all $n \geq 1$.

When working with stochastic convergence, this notation can be ambiguous because several modes of convergence exist. Nevertheless, it can be adapted as a convenient shorthand that proves useful in practice.

Definition 1.2 (Types of convergence) Let (Z_n) and Z be $\mathbb{R}^p$ -valued random variables defined on some probability space. We say that Z_n converges almost surely (a.s.) to Z , denoted by $Z_n \xrightarrow{a.s.} Z$, if for the metric d of interest

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} d(Z_n, Z) = 0\right) = 1.$$

We say that Z_n converges in probability to Z , denoted by $Z_n \xrightarrow{\mathbb{P}} Z$, if for all

$\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(d(Z_n, Z) > \varepsilon) = 0.$$

Finally, we say that Z_n converges in distribution to Z , denoted by $Z_n \xrightarrow{d} Z$, if for all bounded continuous (equivalently, bounded Lipschitz) functions $f : \mathbb{R}^p \rightarrow \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \mathbb{E}[f(Z_n)] = \mathbb{E}[f(Z)].$$

Convergence in distribution is also characterised by convergence of cumulative distribution functions at continuity points, or by convergence of characteristic functions.

Almost sure statements are often proved by controlling whether a sequence of exceptional events happens infinitely many times. The next definition and lemma formalise this idea and are used repeatedly in this chapter.

Definition 1.3 (Infinitely often) For events (A_n) , define

$$\{A_n \text{ i.o.}\} := \bigcap_{n=1}^{\infty} \bigcup_{k \geq n} A_k.$$

This is the event that infinitely many of the events A_n occur.

Lemma 1.1 (Borel–Cantelli) *Let $(A_n)_{n \geq 1}$ be events.*

1. *If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then*

$$\mathbb{P}(A_n \text{ i.o.}) = 0.$$

2. *If, instead, the events A_n are independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then*

$$\mathbb{P}(A_n \text{ i.o.}) = 1.$$

In particular, under independence,

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty \quad \Leftrightarrow \quad \mathbb{P}(A_n \text{ i.o.}) = 0.$$

Proof. If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then by monotone convergence,

$$\mathbb{E}\left[\sum_{n=1}^{\infty} 1_{A_n}\right] = \sum_{n=1}^{\infty} \mathbb{E}[1_{A_n}] = \sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty.$$

Hence $\sum_{n=1}^{\infty} 1_{A_n} < \infty$ almost surely, which means only finitely many A_n occur, i.e. $\mathbb{P}(A_n \text{ i.o.}) = 0$. Now assume independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$. For $m \geq n$,

$$\mathbb{P}\left(\bigcup_{k=n}^m A_k\right) = 1 - \prod_{k=n}^m \mathbb{P}(A_k^c) = 1 - \prod_{k=n}^m (1 - \mathbb{P}(A_k)) \geq 1 - \exp\left(-\sum_{k=n}^m \mathbb{P}(A_k)\right),$$

where we used $1 - x \leq e^{-x}$. Letting $m \rightarrow \infty$ and by continuity from below, $\mathbb{P}(\cup_{k \geq n} A_k) = 1$ for every n . Since these events decrease in n , we get by continuity from above $\mathbb{P}(A_n \text{ i.o.}) = 1$. $\square$

The above theorem is often specialised to a criterion of almost sure convergence.

Corollary 1.1 (Borel–Cantelli criterion for almost sure convergence) *Let (Z_n) and Z be $\mathbb{R}^p$ -valued random variables. If for every $\varepsilon > 0$,*

$$\sum_{n=1}^{\infty} \mathbb{P}(\|Z_n - Z\| > \varepsilon) < \infty,$$

then $Z_n \xrightarrow{a.s.} Z$.

Proof. For $\varepsilon > 0$, define $A_n^\varepsilon = \{\|Z_n - Z\| > \varepsilon\}$. By Lemma 1.1, $\mathbb{P}(A_n^\varepsilon \text{ i.o.}) = 0$ for every $\varepsilon > 0$. Hence, for each $m \geq 1$, with probability one, only finitely many of the events $A_n^{1/m}$ occur. Intersecting over $m \geq 1$, we obtain, with probability one, that for every $m \geq 1$ there exists $N_m < \infty$ such that $\|Z_n - Z\| \leq 1/m$ for all $n \geq N_m$. $\square$

Example 1.1 (Laws of large numbers under fourth moment condition) Let (Z_n) be iid real-valued random variables with $\mathbb{E}[Z_1] = 0$, $\mathbb{E}[Z_1^4] < \infty$, and define

$$S_n = \frac{1}{n} \sum_{i=1}^n Z_i, \quad n \geq 1.$$

For any $\varepsilon > 0$, Markov's inequality gives $\mathbb{P}(|S_n| > \varepsilon) \leq \mathbb{E}[S_n^4]/\varepsilon^4$. Using

$$\mathbb{E}\left[\left(\sum_{i=1}^n Z_i\right)^4\right] = n\mathbb{E}[Z_1^4] + 3n(n-1)(\mathbb{E}[Z_1^2])^2,$$

it follows that there exists $C < \infty$ such that $\mathbb{P}(|S_n| > \varepsilon) \leq C/n^2\varepsilon^4$, which immediately yields $S_n \xrightarrow{\mathbb{P}} 0$. Moreover, since $\sum_{n=1}^{\infty} n^{-2} = \pi^2/6 < \infty$, Corollary 1.1 strengthens the statement to $S_n \xrightarrow{a.s.} 0$. $\diamond$

A complementary tool for weak convergence is a set-wise characterisation. One can test convergence in distribution through probabilities of open or closed sets. This is also a main tool used below.

Theorem 1.1 (Portmanteau theorem) *Let Z and (Z_n) be a random element, and sequence of random elements, respectively, in a common metric space (S, d) . The following are equivalent:*

1. $Z_n \xrightarrow{d} Z$.
2. $\liminf_{n \rightarrow \infty} \mathbb{P}(Z_n \in G) \geq \mathbb{P}(Z \in G)$ for every open set $G \subset S$.
3. $\limsup_{n \rightarrow \infty} \mathbb{P}(Z_n \in F) \leq \mathbb{P}(Z \in F)$ for every closed set $F \subset S$.

Proof. Assume convergence in distribution, and let $G \subset S$ be open. Define

$$f_m(s) = \min\{1, m d(s, G^c)\}, \quad s \in S, \quad m \geq 1.$$

Then each f_m is continuous, $0 \leq f_m \leq 1_G$, and $f_m \uparrow 1_G$ pointwise. Hence

$$\mathbb{E}[f_m(Z_n)] \leq \mathbb{P}(Z_n \in G).$$

Taking $\liminf_{n \rightarrow \infty}$ and using the assumption, $\mathbb{E}[f_m(Z)] \leq \liminf_{n \rightarrow \infty} \mathbb{P}(Z_n \in G)$. Now let $m \rightarrow \infty$ and apply monotone convergence to obtain the second condition. The equivalence between the second and third conditions follows by complements. Finally, assume the second condition, and let $f \geq 0$ be continuous on S . Since $\{f > t\}$ is open for every $t > 0$, the assumption gives

$$\mathbb{P}(f(Z) > t) \leq \liminf_{n \rightarrow \infty} \mathbb{P}(f(Z_n) > t).$$

Using Fatou's lemma,

$$\begin{aligned} \mathbb{E}[f(Z)] &= \int_0^\infty \mathbb{P}(f(Z) > t) \, dt \leq \int_0^\infty \liminf_{n \rightarrow \infty} \mathbb{P}(f(Z_n) > t) \, dt \\ &\leq \liminf_{n \rightarrow \infty} \int_0^\infty \mathbb{P}(f(Z_n) > t) \, dt = \liminf_{n \rightarrow \infty} \mathbb{E}[f(Z_n)]. \end{aligned}$$

Now let f be bounded continuous, with $|f| \leq c < \infty$. Applying the previous inequality to $c \pm f$ yields

$$\mathbb{E}[f(Z)] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[f(Z_n)] \quad \text{and} \quad \limsup_{n \rightarrow \infty} \mathbb{E}[f(Z_n)] \leq \mathbb{E}[f(Z)].$$

Hence $\mathbb{E}[f(Z_n)] \rightarrow \mathbb{E}[f(Z)]$ for every bounded continuous f , so convergence in distribution holds. $\square$

The above types of convergence are the three most commonly encountered in statistics, and they are intrinsically related in several ways. The next lemma collects the standard implications:

Lemma 1.2 (Relation between types of convergence) *Let (Z_n) and Z be $\mathbb{R}^p$ -valued random variables. Then*

1. $Z_n \xrightarrow{a.s.} Z$ implies $Z_n \xrightarrow{\mathbb{P}} Z$.
2. $Z_n \xrightarrow{\mathbb{P}} Z$ implies $Z_n \xrightarrow{d} Z$.
3. Let c be a constant. Then $Z_n \xrightarrow{\mathbb{P}} c$ if and only if $Z_n \xrightarrow{d} c$.

Proof. Assume first that $Z_n \xrightarrow{a.s.} Z$. Fix $\varepsilon > 0$ and set

$$A_n = \{\|Z_n - Z\| > \varepsilon\}.$$

Then $1_{A_n} \rightarrow 0$ almost surely and $0 \leq 1_{A_n} \leq 1$, so dominated convergence yields

$$\mathbb{P}(A_n) = \mathbb{E}[1_{A_n}] \rightarrow 0,$$

which is exactly $Z_n \xrightarrow{\mathbb{P}} Z$. Next assume $Z_n \xrightarrow{\mathbb{P}} Z$, and let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ be bounded and continuous. Take an arbitrary subsequence (Z_{n_k}) . Since $Z_{n_k} \xrightarrow{\mathbb{P}} Z$, we can choose a further subsequence $(Z_{n_{k_j}})$ such that for all $j \geq 1$

$$\mathbb{P}(\|Z_{n_{k_j}} - Z\| > 2^{-j}) < 2^{-j}$$

By Lemma 1.1 (first part), $\|Z_{n_{k_j}} - Z\| \xrightarrow{a.s.} 0$, hence $f(Z_{n_{k_j}}) \xrightarrow{a.s.} f(Z)$. Because f is bounded, the dominated convergence theorem gives

$$\mathbb{E}[f(Z_{n_{k_j}})] \rightarrow \mathbb{E}[f(Z)].$$

Thus every subsequence of $\mathbb{E}[f(Z_n)]$ has a further subsequence converging to $\mathbb{E}[f(Z)]$, so $\mathbb{E}[f(Z_n)] \rightarrow \mathbb{E}[f(Z)]$, and this proves $Z_n \xrightarrow{d} Z$. Finally, assume $Z_n \xrightarrow{d} c$ and fix $\varepsilon > 0$. The set

$$F_\varepsilon = \{x \in \mathbb{R}^p : \|x - c\| \geq \varepsilon\}$$

is closed, so Theorem 1.1 yields

$$\limsup_{n \rightarrow \infty} \mathbb{P}(\|Z_n - c\| \geq \varepsilon) = \limsup_{n \rightarrow \infty} \mathbb{P}(Z_n \in F_\varepsilon) \leq \mathbb{P}(c \in F_\varepsilon) = 0.$$

Hence $\mathbb{P}(\|Z_n - c\| > \varepsilon) \rightarrow 0$, i.e. $Z_n \xrightarrow{\mathbb{P}} c$. $\square$

It is interesting to note that the converse implications in general fail.

Example 1.2 (Converse implications fail) The implications in Lemma 1.2 are, in general, not reversible.

1. *Convergence in distribution does not imply convergence in probability.* Let Y satisfy $\mathbb{P}(Y = 1) = \mathbb{P}(Y = -1) = 1/2$, and define

$$Z_n = (-1)^n Y, \quad n \geq 1.$$

Since Y and $-Y$ have the same law, each Z_n has the same distribution as Y , hence $Z_n \xrightarrow{d} Y$. However,

$$\|Z_{n+1} - Z_n\| = 2|Y|,$$

so $\mathbb{P}(\|Z_{n+1} - Z_n\| > 1) = 1$ for all n . If (Z_n) converged in probability to some random variable Z , then

$$\mathbb{P}(\|Z_{n+1} - Z_n\| > 2\varepsilon) \leq \mathbb{P}(\|Z_{n+1} - Z\| > \varepsilon) + \mathbb{P}(\|Z_n - Z\| > \varepsilon) \rightarrow 0$$

for each $\varepsilon > 0$, a contradiction.

2. *Convergence in probability does not imply almost sure convergence.* Let $U \sim \text{Unif}(0, 1)$. For $n = 2^m + j$ with $m \geq 0$ and $j = 0, \dots, 2^m - 1$, set

$$A_n = [j2^{-m}, (j+1)2^{-m}), \quad Z_n = 1_{A_n}(U).$$

Equivalently, (A_n) is the typewriter sequence

$$[0, 1), [0, 1/2), [1/2, 1), [0, 1/4), [1/4, 1/2), [1/2, 3/4), [3/4, 1), \dots$$

Then $\mathbb{P}(Z_n = 1) = |A_n| = 2^{-m} \rightarrow 0$, hence $Z_n \xrightarrow{\mathbb{P}} 0$. On the other hand, fix $u \in (0, 1)$ and define the m -th block of indices

$$B_m = \{2^m, \dots, 2^{m+1} - 1\}.$$

In each block B_m , exactly one interval among

$$[0, 2^{-m}), [2^{-m}, 2 \cdot 2^{-m}), \dots, [(2^m - 1)2^{-m}, 1)$$

contains u . Hence, for exactly one $n \in B_m$ we have $Z_n(u) = 1$, and for all other $n \in B_m$ we have $Z_n(u) = 0$. Therefore

$$\limsup_{n \rightarrow \infty} Z_n(u) = 1, \quad \liminf_{n \rightarrow \infty} Z_n(u) = 0,$$

so $(Z_n(u))$ has no limit for any $u \in (0, 1)$, and in particular Z_n does not converge almost surely.

◇

Remark 1.1 (Verifying convergences) There are several useful ways of verifying convergence in probability and in distribution apart from the definition. For instance *Chebyshev's inequality* states that for any increasing function $\varphi : [0, \infty) \rightarrow [0, \infty)$ one has, for any $\varepsilon > 0$,

$$\mathbb{P}(\|Z_n - Z\| > \varepsilon) \leq \frac{\mathbb{E}[\varphi(\|Z_n - Z\|)]}{\varphi(\varepsilon)}, \quad (1.1)$$

so if the numerator on the right-hand side tends to zero, convergence in probability follows. Likewise, the *Cramér–Wold device* states that

$$Z_n \xrightarrow{d} Z \Leftrightarrow a^\top Z_n \xrightarrow{d} a^\top Z, \quad \forall a \in \mathbb{R}^p,$$

linking one-dimensional convergence in distribution, which can be verified using standard results such as convergence of distribution functions at continuity points, and multidimensional convergence. Characteristic functions provide yet another standard route. ○

Theorem 1.2 (Strong law of large numbers) *Let (Z_n) be iid random variables with $\mathbb{E}[|Z_1|] < \infty$. Then*

$$\frac{1}{n} \sum_{i=1}^n Z_i \xrightarrow{a.s.} \mathbb{E}[Z_1].$$

Proof. It is enough to treat $\mathbb{E}[Z_1] = 0$. Write $S_n = Z_1 + \dots + Z_n$. For any $\varepsilon > 0$ define

$$A_\varepsilon = \left\{ \limsup_{n \rightarrow \infty} S_n/n > \varepsilon \right\}, \quad Z_i^\varepsilon = (Z_i - \varepsilon)1_{A_\varepsilon}, \quad S_n^\varepsilon = \sum_{i=1}^n Z_i^\varepsilon.$$

Let $M_n = S_1 \vee \dots \vee S_n$ and $M'_n = \max_{1 \leq i \leq n} (Z_2 + \dots + Z_{i+1})$, we have for $i \leq n$, $S_i = Z_1 + (Z_2 + \dots + Z_i) \leq Z_1 + M'_n$, hence $M_n \leq Z_1 + M'_n$. By stationarity, M'_n has the same distribution as M_n , so

$$\mathbb{E}[Z_1; M_n > 0] \geq \mathbb{E}[M_n - (M'_n)_+; M_n > 0] \geq \mathbb{E}[(M_n)_+ - (M'_n)_+] = 0,$$

and letting $n \rightarrow \infty$ gives $\mathbb{E}[Z_1; \sup_n S_n > 0] \geq 0$, which holds for any stationary sequence.

Now, A_ε is measurable with respect to $\mathcal{T} := \cap_n \sigma(Z_i : i > n)$, and since $Z_1, \dots, Z_n$ are independent of $\mathcal{T}$, a generating-class argument yields $\sigma(Z_n : n \geq 1) \perp \mathcal{T}$, so $\mathcal{T} \perp \mathcal{T}$, and hence $\mathbb{P}(A_\varepsilon) = 1$ or 0 . Moreover $\{\sup_n S_n^\varepsilon > 0\} = \{\sup_n S_n/n > \varepsilon\} \cap A_\varepsilon = A_\varepsilon$. Further, since A_ε is shift-invariant, (Z_n^ε) is stationary. Then, by the above inequality,

$$0 \leq \mathbb{E}[Z_1^\varepsilon; \sup_n S_n^\varepsilon > 0] = \mathbb{E}[Z_1 - \varepsilon; A_\varepsilon] = -\varepsilon \mathbb{P}(A_\varepsilon),$$

and hence $\mathbb{P}(A_\varepsilon) = 0$. Thus $\limsup_n S_n/n \leq 0$ a.s. Applying the same argument to $-S_n$ yields $\liminf_n S_n/n \geq 0$ a.s., so $S_n/n \xrightarrow{a.s.} 0$. $\square$

A further useful inequality, which helps bound the speed of weak convergence from above, is the following. It can also be paired with the Borel–Cantelli Lemma 1.1 when proving almost sure convergence.

Theorem 1.3 (Bernstein’s inequality) *Let (Z_n) be independent random variables with values in a measurable space $\mathcal{Z}$ and let γ be a real-valued function on $\mathcal{Z}$ such that $\mathbb{E}[\gamma(Z_n)] = 0$ for all $n \geq 1$. Assume that for some constant $\kappa > 0$,*

$$\frac{1}{n} \sum_{i=1}^n \mathbb{E}[|\gamma(Z_i)|^m] \leq \frac{m!}{2} \kappa^{m-2}, \quad m = 2, 3, \dots$$

Then for any $t > 0$,

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n \gamma(Z_i) \geq \kappa t + \sqrt{2t}\right) \leq e^{-nt}.$$

Proof. For any centred real-valued random variable X with $\mathbb{E}[X] = 0$ and $c > 0$,

$$\exp[X - c] - 1 \leq \frac{\exp[X]}{1 + c} - 1 = \frac{e^X - 1 - X + X - c}{1 + c} \leq \frac{e^{|X|} - 1 - |X| + X - c}{1 + c}.$$

Choosing $c = \mathbb{E}[e^{|X|}] - 1 - \mathbb{E}[|X|]$ and taking expectations gives $\mathbb{E}[e^{X-c}] \leq 1$, or

$$\log \mathbb{E}[e^X] \leq c = \mathbb{E}[e^{|X|}] - 1 - \mathbb{E}[|X|].$$

Applying this to $X_i = \gamma(Z_i)/L$, expanding the exponential, and using the moment condition we obtain

$$\begin{aligned} \log \mathbb{E}\left[\exp\left(\frac{S_n}{L}\right)\right] &\leq \sum_{i=1}^n (\mathbb{E}[e^{\gamma(Z_i)/L}] - 1 - \mathbb{E}[\gamma(Z_i)/L]) \\ &\leq \sum_{m=2}^{\infty} \sum_{i=1}^n \frac{\mathbb{E}[|\gamma(Z_i)|^m]}{L^m m!} \\ &\leq \sum_{m=2}^{\infty} \frac{1}{L^m m!} \left(\frac{nm!}{2} \kappa^{m-2}\right) = \frac{n}{2L^2(1 - \kappa/L)}. \end{aligned}$$

Let $S_n = \sum_{i=1}^n \gamma(Z_i)$. Fix $a > 0$ and let $L > \kappa$. By Markov’s inequality

$$\mathbb{P}(S_n \geq a) = \mathbb{P}(e^{S_n/L} \geq e^{a/L}) \leq \exp\left(-\frac{a}{L} + \frac{n}{2(L^2 - L\kappa)}\right).$$

Optimising over L by taking $\kappa/L = 1 - (1 + 2a\kappa/n)^{-1/2}$ and then reparametrising by taking $a/n = \kappa t + \sqrt{2t}$ yields the result. $\square$

Remark 1.2 In Monte Carlo estimation, $n^{-1} \sum_{i=1}^n \gamma(Z_i)$ is an average of iid simulation outputs. Bernstein's inequality provides a non-asymptotic error bound for

$$\mathbb{P}\left(\left|n^{-1} \sum_{i=1}^n \gamma(Z_i) - \mathbb{E}[\gamma(Z_1)]\right| \geq \varepsilon\right),$$

under moment/boundedness conditions, which can be used to pick a simulation size n for a desired accuracy and confidence. $\circ$

Numeric example 1.1 (SLLN paths and Bernstein concentration for bounded risks) Let (Z_n) be iid with $Z_i \sim \text{Unif}(-1, 1)$, and define $\bar{Z}_n = n^{-1} \sum_{i=1}^n Z_i$. Then $\mathbb{E}[Z_i] = 0$ and $\text{Var}(Z_i) = 1/3$. Taking $\gamma(z) = z$ in Theorem 1.3, the moment condition is satisfied (for instance) with $\kappa = 1/3$, so that for $t > 0$,

$$\mathbb{P}(\bar{Z}_n \geq \kappa t + \sqrt{2t}) \leq e^{-nt}.$$

By symmetry, for any fixed $\varepsilon > 0$ and t_ε defined by $\varepsilon = \kappa t_\varepsilon + \sqrt{2t_\varepsilon}$, we obtain

$$\mathbb{P}(|\bar{Z}_n| \geq \varepsilon) \leq 2e^{-nt_\varepsilon}.$$

For comparison, Chebyshev gives

$$\mathbb{P}(|\bar{Z}_n| \geq \varepsilon) \leq \frac{\text{Var}(Z_1)}{n\varepsilon^2} = \frac{1}{3n\varepsilon^2}.$$

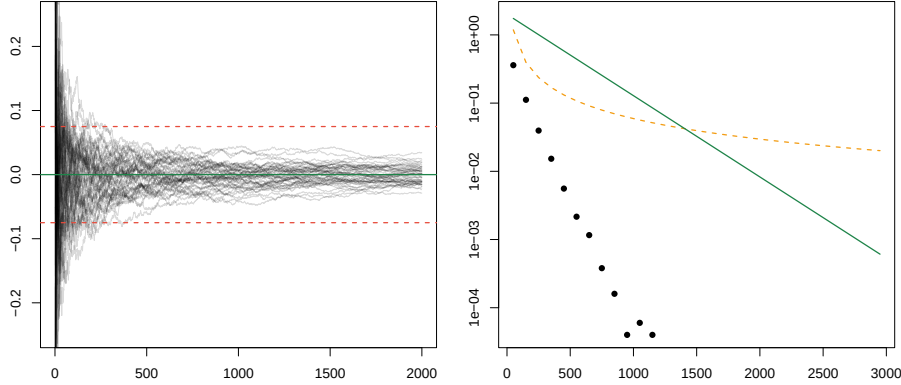


Figure 1.1: Left: 50 simulated trajectories of $\bar{Z}_n$ (black) with horizontal reference lines at 0 (green) and $\pm\varepsilon$ (red dashed), with $\varepsilon = 0.075$. Right (log scale): Monte Carlo empirical probabilities $\mathbb{P}(|\bar{Z}_n| \geq \varepsilon)$ (black points), Chebyshev bound $1/(3n\varepsilon^2)$ (orange dashed), and Bernstein bound $2e^{-nt_\varepsilon}$ (green solid), where t_ε solves $\varepsilon = \kappa t + \sqrt{2t}$ with $\kappa = 1/3$.

The left panel of Figure 1.1 visualises the pathwise statement behind Theorem 1.2, with visible early fluctuations, and trajectories stabilising and eventually remaining inside the $\pm\varepsilon$ region. The right panel highlights the rate contrast on a logarithmic scale, and although both inequalities are not tight, Chebyshev's inequality decays polynomially in n , while Bernstein's inequality decays exponentially in n . $\diamond$

Example 1.3 (Law of large numbers for non-iid risks) Let (Z_n) be independent (not necessarily identically distributed) random variables. Write $\mu_i = \mathbb{E}[Z_i]$ and $Y_i = Z_i - \mu_i$. Assume there exist constants $v > 0$ and $\kappa > 0$ such that for all i and all $m \geq 2$,

$$\mathbb{E}[|Y_i|^m] \leq \frac{m!}{2} v^2 \kappa^{m-2}.$$

Applying Theorem 1.3 to the centred variables $Y_1, \dots, Y_n$ with the scaling function $\gamma(y) = y/v$, the moment condition holds with parameter κ/v . Hence,

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n Y_i \geq \kappa t + v\sqrt{2t}\right) \leq e^{-nt}, \quad t > 0,$$

and the same bound applies to $-Y_i$. For any $\varepsilon > 0$, set $t = \kappa^{-2}(\sqrt{v^2 + 2\kappa\varepsilon} - v)^2/2$, so that $\kappa t + v\sqrt{2t} = \varepsilon$. Therefore

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n (Z_i - \mu_i)\right| > \varepsilon\right) \leq 2e^{-nt},$$

which is clearly summable in n . Therefore, Lemma 1.1 yields

$$\frac{1}{n} \sum_{i=1}^n (Z_i - \mu_i) \xrightarrow{\text{a.s.}} 0.$$

◇

Two results that are often combined to analyse the asymptotic behaviour of complex expressions by reducing to simpler components are the following.

Theorem 1.4 (Continuous mapping theorem) *Let Z and (Z_n) be an $\mathbb{R}^p$ -valued random variables and sequence, respectively, and let $g : \mathbb{R}^p \rightarrow \mathbb{R}^q$ be continuous. Then*

1. $Z_n \xrightarrow{\text{a.s.}} Z$ implies $g(Z_n) \xrightarrow{\text{a.s.}} g(Z)$.
2. $Z_n \xrightarrow{\mathbb{P}} Z$ implies $g(Z_n) \xrightarrow{\mathbb{P}} g(Z)$.
3. $Z_n \xrightarrow{d} Z$ implies $g(Z_n) \xrightarrow{d} g(Z)$.

Proof. For the first statement, let $\Omega_0 = \{\omega : Z_n(\omega) \rightarrow Z(\omega)\}$. Then $\mathbb{P}(\Omega_0) = 1$. For $\omega \in \Omega_0$, continuity of g gives $g(Z_n(\omega)) \rightarrow g(Z(\omega))$. Hence $g(Z_n) \xrightarrow{\text{a.s.}} g(Z)$. For the second statement, fix $\varepsilon, \eta > 0$. Choose M such that $\mathbb{P}(\|Z\| > M) \leq \eta$ and define

$$K = \{x \in \mathbb{R}^p : \|x\| \leq M + 1\}.$$

Since g is continuous on the compact set K , it is uniformly continuous on K , so there exists $\delta \in (0, 1)$ such that

$$\|x - y\| \leq \delta, \quad x, y \in K \quad \Rightarrow \quad \|g(x) - g(y)\| \leq \varepsilon.$$

On the event $\{\|Z\| \leq M, \|Z_n - Z\| \leq \delta\}$ we have $\|Z_n\| \leq M + 1$, so $Z, Z_n \in K$, hence $\|g(Z_n) - g(Z)\| \leq \varepsilon$. Therefore

$$\mathbb{P}(\|g(Z_n) - g(Z)\| > \varepsilon) \leq \mathbb{P}(\|Z_n - Z\| > \delta) + \mathbb{P}(\|Z\| > M).$$

Using $Z_n \xrightarrow{\mathbb{P}} Z$ now gives

$$\limsup_{n \rightarrow \infty} \mathbb{P}(\|g(Z_n) - g(Z)\| > \varepsilon) \leq \eta.$$

Since $\eta > 0$ is arbitrary, $g(Z_n) \xrightarrow{\mathbb{P}} g(Z)$. Finally, for the third statement, let $F \subset \mathbb{R}^q$ be closed. Then $g^{-1}(F) \subset \mathbb{R}^p$ is closed, and by Theorem 1.1 (iii),

$$\limsup_{n \rightarrow \infty} \mathbb{P}(g(Z_n) \in F) = \limsup_{n \rightarrow \infty} \mathbb{P}(Z_n \in g^{-1}(F)) \leq \mathbb{P}(Z \in g^{-1}(F)) = \mathbb{P}(g(Z) \in F).$$

Applying Theorem 1.1 again, with the reverse implication, yields $g(Z_n) \xrightarrow{d} g(Z)$. $\square$

Lemma 1.3 (Slutsky's lemma for random variables) *Let (Z_n) and Z be $\mathbb{R}^p$ -valued random variables. If $Z_n \xrightarrow{d} Z$ and $W_n \xrightarrow{d} c$ with c a constant, then*

$$(Z_n, W_n) \xrightarrow{d} (Z, c).$$

Proof. We first compare (Z_n, W_n) to (Z_n, c) in probability, and then use convergence in distribution for the latter. We have that $\|(Z_n, W_n) - (Z_n, c)\| = \|W_n - c\| \xrightarrow{\mathbb{P}} 0$ by the third point of Lemma 1.2. We also have $(Z_n, c) \xrightarrow{d} (Z, c)$ by the definition of convergence in distribution. Indeed, for any bounded continuous function f , the map $g(x) = f(x, c)$ is again bounded and continuous, so

$$\mathbb{E}[f(Z_n, c)] = \mathbb{E}[g(Z_n)] \rightarrow \mathbb{E}[g(Z)] = \mathbb{E}[f(Z, c)].$$

The two pieces together imply the result, as per Exercise 6. $\square$

We now state a useful approximation theorem which helps to prove $Z_n \xrightarrow{d} Z$ by choosing approximations $Y_n^{(k)}$ of Z_n and $Y^{(k)}$ of Z such that $Y_n^{(k)} \xrightarrow{d} Y^{(k)}$ for fixed k , when the approximation errors are uniformly small.

Theorem 1.5 (Weak convergence by approximation) *Let (S, d) be a metric space and let $Z, Z_n, Y^{(k)}, Y_n^{(k)}$ be S -valued random elements. Assume that for each fixed k , $Y_n^{(k)} \xrightarrow{d} Y^{(k)}$ as $n \rightarrow \infty$, and that $Y^{(k)} \xrightarrow{d} Z$ as $k \rightarrow \infty$. If in addition, for any given $\varepsilon > 0$,*

$$\lim_{k \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(d(Y_n^{(k)}, Z_n) > \varepsilon) = 0, \quad (1.2)$$

then $Z_n \xrightarrow{d} Z$.

Proof. Let $F \subset S$ be closed and fix $\varepsilon > 0$. Write $F^\varepsilon = \{s \in S : d(s, F) \leq \varepsilon\}$. Then

$$\mathbb{P}(Z_n \in F) \leq \mathbb{P}(Y_n^{(k)} \in F^\varepsilon) + \mathbb{P}(d(Y_n^{(k)}, Z_n) > \varepsilon).$$

For fixed k , Theorem 1.1 (iii) yields

$$\limsup_{n \rightarrow \infty} \mathbb{P}(Z_n \in F) \leq \mathbb{P}(Y^{(k)} \in F^\varepsilon) + \limsup_{n \rightarrow \infty} \mathbb{P}(d(Y_n^{(k)}, Z_n) > \varepsilon).$$

By (1.2) the second term vanishes as $k \rightarrow \infty$. Since F^ε is closed and $Y^{(k)} \xrightarrow{d} Z$, another application of Theorem 1.1 gives

$$\limsup_{k \rightarrow \infty} \mathbb{P}(Y^{(k)} \in F^\varepsilon) \leq \mathbb{P}(Z \in F^\varepsilon).$$

Hence

$$\limsup_{n \rightarrow \infty} \mathbb{P}(Z_n \in F) \leq \mathbb{P}(Z \in F^\varepsilon).$$

Letting $\varepsilon \downarrow 0$ and using continuity from above for the closed sets $F^\varepsilon \downarrow F$ yields $\limsup_{n \rightarrow \infty} \mathbb{P}(Z_n \in F) \leq \mathbb{P}(Z \in F)$. Since F was arbitrary, Theorem 1.1 implies $Z_n \xrightarrow{d} Z$. $\square$

Then we may make the following definitions, which are the stochastic analogues of Definition 1.1.

Definition 1.4 (Stochastic o -notation) Let (Z_n) be $\mathbb{R}^p$ -valued random variables and r_n be positive random variables. We say that Z_n is bounded in probability, denoted by $Z_n = O_{\mathbb{P}}(1)$ if

$$\lim_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(\|Z_n\| > M) = 0.$$

Likewise, we say that Z_n converges to zero in probability, denoted by $Z_n = o_{\mathbb{P}}(1)$ if $Z_n \xrightarrow{\mathbb{P}} 0$. We also define $Z_n = O_{\mathbb{P}}(r_n)$ if $Z_n/r_n = O_{\mathbb{P}}(1)$ and similarly $Z_n = o_{\mathbb{P}}(r_n)$ if $Z_n/r_n = o_{\mathbb{P}}(1)$.

An immediate consequence is that bounded in probability follows from distributional convergence, a useful fact which is repeatedly used in the remaining chapters.

Example 1.4 (Convergence in distribution implies bounded in probability) Let (Z_n) be a real-valued random sequence converging in distribution to Z with distribution function F . Then at continuity points $M, -M$ we have

$$\mathbb{P}(Z_n > M) \rightarrow 1 - F(M), \quad \mathbb{P}(Z_n \leq -M) \rightarrow F(-M).$$

Since both of the expressions tend to zero as $M \rightarrow \infty$ it follows that

$$Z_n \xrightarrow{d} Z \quad \Rightarrow \quad Z_n = O_{\mathbb{P}}(1),$$

which is the desired implication. $\diamond$

A companion transfer rule lets us propagate stochastic bounds across approximations, simplifying notation.

Example 1.5 (Convergence in distribution of an approximating sequence) We show that if $Z_n = O_{\mathbb{P}}(a_n)$ and $W_n = o_{\mathbb{P}}(\|Z_n\|)$, then $W_n = o_{\mathbb{P}}(a_n)$.

The assumptions that $Z_n = O_{\mathbb{P}}(a_n)$ and $W_n = o_{\mathbb{P}}(\|Z_n\|)$ mean that

$$\lim_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(\|Z_n\|/a_n > M) = 0, \quad \lim_{\varepsilon \rightarrow 0} \mathbb{P}(\|W_n\|/\|Z_n\| > \varepsilon) = 0$$

for all $\varepsilon > 0$. We want to show that for all $\varepsilon, \delta > 0$ there exists L such that for $n \geq L$, we have $\mathbb{P}(\|W_n/a_n\| \leq \varepsilon) > 1 - \delta$. For $\varepsilon_1, \varepsilon > 0$ we get

$$\begin{aligned} \mathbb{P}(\|W_n\|/a_n \leq \varepsilon) &\geq \mathbb{P}(\{\|W_n\|/\|Z_n\| < \varepsilon_1\} \cap \{\|Z_n\|/a_n < \varepsilon/\varepsilon_1\}) \\ &\geq \mathbb{P}(\|W_n\|/\|Z_n\| < \varepsilon_1) + \mathbb{P}(\|Z_n\|/a_n < \varepsilon/\varepsilon_1) - 1. \end{aligned}$$

By assumption we can find L such that for $n \geq L$ we have $\mathbb{P}(\|Z_n\|/a_n < \varepsilon/\varepsilon_1) > 1 - \delta/2$ and $\mathbb{P}(\|W_n\|/\|Z_n\| < \varepsilon_1) > 1 - \delta/2$. The result follows. $\diamond$

1.2 Fundamental theorems in finite dimensions

In insurance, we need not only point estimates of risks but also uncertainty quantification. The main results we encounter in asymptotic statistics are limit theorems describing the distribution of suitably normalised estimators. These results fall under the umbrella term “central limit theorems”, or CLTs, and play a central role in describing the *large sample behaviour* of estimators.

Theorem 1.6 (Laplace’s multivariate CLT) *Let (Z_n) be a sequence of iid random vectors in $\mathbb{R}^p$. Assume that $\mathbb{E}[\|Z_1\|^2] < \infty$. It holds that*

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n Z_i - \mathbb{E}[Z_1] \right) \xrightarrow{d} N(0, \text{Var}(Z_1)),$$

where N denotes the p -dimensional normal distribution.

Proof. Let $\mu = \mathbb{E}[Z_1]$ and fix $a \in \mathbb{R}^p$. Put $\xi_k = a^\top (Z_k - \mu)$ so that $\mathbb{E}[\xi_1] = 0$ and $\mathbb{E}[\xi_1^2] = a^\top \text{Var}(Z_1)a$. Let $\varphi(t) = \mathbb{E}[e^{it\xi_1}]$ be the characteristic function of ξ_1 . Since $\mathbb{E}[\xi_1^2] < \infty$ and $|e^{it} - 1| \leq |t|$, dominated convergence yields

$$\varphi'(t) = \mathbb{E}[(i\xi_1)e^{it\xi_1}], \quad \varphi''(t) = \mathbb{E}[(i\xi_1)^2 e^{it\xi_1}], \quad t \in \mathbb{R}.$$

Hence $\varphi'(0) = i\mathbb{E}[\xi_1] = 0$ and $\varphi''(0) = -\mathbb{E}[\xi_1^2]$, and Taylor’s formula gives

$$\varphi(t) = 1 - \frac{t^2}{2} \mathbb{E}[\xi_1^2] + o(t^2), \quad t \rightarrow 0.$$

Hence the characteristic function of $n^{-1/2} \sum_{k=1}^n \xi_k$ is

$$\varphi\left(\frac{t}{\sqrt{n}}\right)^n = \left(1 - \frac{t^2}{2n} \mathbb{E}[\xi_1^2] + o(n^{-1})\right)^n \rightarrow \exp\left(-\frac{t^2}{2} \mathbb{E}[\xi_1^2]\right),$$

which is the characteristic function of $N(0, a^\top \text{Var}(Z_1)a)$. By the Cramér–Wold device this implies

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n Z_i - \mu \right) \xrightarrow{d} N(0, \text{Var}(Z_1)),$$

as claimed. $\square$

Sometimes the distribution of the random variables may depend on the sample size, or the risks may be transformed using an n -dependent function. In that case, it results convenient to work with the following notion.

Definition 1.5 (Triangular array) A triangular array of random variables (or in general of stochastic elements) is a sequence with a representation on the form

$$Z_{nm}, \quad n = 1, 2, \dots, \quad m = 1, \dots, n.$$

We require mutual independence for elements within the same row but do not make any assumptions on the distribution, nor on the dependence between rows. A triangular array can be visualised as

$$\begin{array}{cccc} Z_{11} & & & \\ Z_{21} & Z_{22} & & \\ Z_{31} & Z_{32} & Z_{33} & \\ \vdots & \vdots & \vdots & \ddots \end{array}$$

Theorem 1.7 (Lindeberg's multivariate CLT) *Let (Z_{nm}) be a triangular array of random variables with values in $\mathbb{R}^p$ with $\mathbb{E}[Z_{nm}] = 0$ and $\mathbb{E}[\|Z_{nm}\|^2] < \infty$ for all $m \leq n$. Assume that*

$$\sum_{m=1}^n \text{Var}(Z_{nm}) \rightarrow \Sigma,$$

for a fixed $p \times p$ positive semi-definite matrix Σ , and that the associated real-valued array $(\|Z_{nm}\|)$ satisfies Lindeberg's condition, i.e. for all $c > 0$:

$$\sum_{m=1}^n \mathbb{E}[\|Z_{nm}\|^2; \|Z_{nm}\| > c] \rightarrow 0.$$

Then

$$\sum_{m=1}^n Z_{nm} \xrightarrow{d} N(0, \Sigma),$$

where N denotes the p -dimensional normal distribution.

Proof. Fix $a \in \mathbb{R}^p$ and set $X_{nm} = a^\top Z_{nm}$. If $a = 0$ there is nothing to prove, so assume $a \neq 0$. Then $\mathbb{E}[X_{nm}] = 0$ and

$$c_{nm} = \mathbb{E}[X_{nm}^2] = a^\top \text{Var}(Z_{nm})a, \quad c_n = \sum_{m=1}^n c_{nm} \rightarrow a^\top \Sigma a.$$

Moreover, for any $\varepsilon > 0$,

$$\sum_{m=1}^n \mathbb{E}[X_{nm}^2; |X_{nm}| > \varepsilon] \leq \|a\|^2 \sum_{m=1}^n \mathbb{E}[\|Z_{nm}\|^2; \|Z_{nm}\| > \varepsilon/\|a\|] \rightarrow 0,$$

so the (one-dimensional) Lindeberg condition holds for (X_{nm}) . It therefore suffices, by the Cramér–Wold device, to prove that $\sum_{m=1}^n X_{nm} \xrightarrow{d} N(0, a^\top \Sigma a)$.

We first note that for any $\varepsilon > 0$,

$$\sup_m c_{nm} \leq \varepsilon^2 + \sup_m \mathbb{E}[X_{nm}^2; |X_{nm}| > \varepsilon] \leq \varepsilon^2 + \sum_{m=1}^n \mathbb{E}[X_{nm}^2; |X_{nm}| > \varepsilon],$$

so $\sup_m c_{nm} \rightarrow 0$ by the Lindeberg condition. Let $\varphi_{nm}(t) = \mathbb{E}[e^{itX_{nm}}]$. Introduce independent $\zeta_{nm} \sim N(0, c_{nm})$ with characteristic functions $\psi_{nm}(t) = \exp(-t^2 c_{nm}/2)$, so $\sum_{m=1}^n \zeta_{nm} \sim N(0, c_n)$. Using the elementary bound (proved by induction)

$$\left| \prod_{m=1}^n z_m - \prod_{m=1}^n z'_m \right| \leq \sum_{m=1}^n |z_m - z'_m|, \quad |z_m| \leq 1, \quad |z'_m| \leq 1,$$

To control the remainders, set $h_m(x) = e^{ix} - \sum_{k=0}^m (ix)^k/k!$. Then $h_m(x) = i \int_0^x h_{m-1}(s) ds$, $|h_{-1}| \equiv 1$, and $|h_0| \leq 2$, so by induction

$$\left| e^{ix} - \sum_{k=0}^m \frac{(ix)^k}{k!} \right| \leq \frac{2|x|^m}{m!} \wedge \frac{|x|^{m+1}}{(m+1)!}.$$

With $m = 2$ this gives $|e^{itx} - 1 - itx + \frac{1}{2}t^2x^2| \leq |tx|^2(1 \wedge |tx|)$, so

$$\begin{aligned} |\varphi_{nm}(t) - 1 + \tfrac{1}{2}t^2c_{nm}| &\leq t^2 \mathbb{E}[X_{nm}^2(1 \wedge |tX_{nm}|)], \\ |\psi_{nm}(t) - 1 + \tfrac{1}{2}t^2c_{nm}| &\leq t^2 \mathbb{E}[\zeta_{nm}^2(1 \wedge |t\zeta_{nm}|)]. \end{aligned}$$

Combining this with the product bound and the triangle inequality yields

$$\begin{aligned} &\left| \prod_{m=1}^n \varphi_{nm}(t) - \prod_{m=1}^n \psi_{nm}(t) \right| \\ &\leq \sum_{m=1}^n |\varphi_{nm}(t) - 1 + \tfrac{1}{2}t^2c_{nm}| + \sum_{m=1}^n |\psi_{nm}(t) - 1 + \tfrac{1}{2}t^2c_{nm}| \\ &\leq |t|^2 \sum_{m=1}^n \mathbb{E}[X_{nm}^2(1 \wedge |tX_{nm}|)] + |t|^2 \sum_{m=1}^n \mathbb{E}[\zeta_{nm}^2(1 \wedge |t\zeta_{nm}|)]. \end{aligned}$$

It remains to note that

$$\sum_{m=1}^n \mathbb{E}[X_{nm}^2(1 \wedge |tX_{nm}|)] \leq \varepsilon |t| \sum_{m=1}^n c_{nm} + \sum_{m=1}^n \mathbb{E}[X_{nm}^2; |X_{nm}| > \varepsilon],$$

vanishes as $n \rightarrow \infty$ and $\varepsilon \downarrow 0$, by the Lindeberg condition and $c_n \rightarrow a^\top \Sigma a$. Likewise,

$$\sum_{m=1}^n \mathbb{E}[\zeta_{nm}^2(1 \wedge |t\zeta_{nm}|)] \leq |t|^3 \mathbb{E}[|\xi|^3] \sum_{m=1}^n c_{nm}^{3/2} \leq |t|^3 \mathbb{E}[|\xi|^3] c_n \sup_m c_{nm}^{1/2} \rightarrow 0,$$

where $\xi \sim N(0, 1)$. □

A simple and effective way of checking Lindeberg's condition is through a moment condition, as given by the following result.

Lemma 1.4 (Lyapounov's condition) *If a triangular array (Z_{nm}) satisfies Lyapounov's condition of order $\alpha > 2$, i.e.*

$$\sum_{m=1}^n \mathbb{E}[\|Z_{nm}\|^\alpha] \rightarrow 0,$$

then Lindeberg's condition is fulfilled.

Proof. Fix $c > 0$. On the event $\{\|Z_{nm}\| > c\}$ we have $\|Z_{nm}\|^2 \leq c^{2-\alpha}\|Z_{nm}\|^\alpha$ since $\alpha > 2$. Hence

$$\sum_{m=1}^n \mathbb{E}[\|Z_{nm}\|^2; \|Z_{nm}\| > c] \leq c^{2-\alpha} \sum_{m=1}^n \mathbb{E}[\|Z_{nm}\|^\alpha] \rightarrow 0,$$

which is exactly Lindeberg's condition. $\square$

To see the Lindeberg condition in action, we consider a portfolio with non-identical risks.

Example 1.6 (Portfolio with heterogeneous risks) Let (Z_n) represent $\mathbb{R}$ -valued independent but not identically distributed risks. By subtracting their means we can assume w.l.o.g. that they have mean zero. Let

$$S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i, \quad \sigma_i = \text{Var}(Z_i)/n, \quad s_n = \left(\sum_{i=1}^n \sigma_i \right)^{1/2}.$$

We are interested in conditions under which $S_n/s_n \xrightarrow{d} N(0, 1)$. One can achieve this by applying Lindeberg's condition to the array defined by $Z_{nm} = n^{-1/2}Z_m/s_n$, assuming s_n has a positive finite limit $\bar{\sigma}$. For instance, if there exists a constant M such that $|Z_m| \leq M$ for all m , then since $n^{1/2}s_n \rightarrow \infty$, there exists N such that for $n \geq N$ we have $n^{1/2}s_n > M/c$, and so

$$\frac{1}{ns_n^2} \sum_{m=1}^n \mathbb{E}[|Z_m|^2 1_{(|Z_m| > cn^{1/2}s_n)}] = 0.$$

for every $n \geq N$, i.e. Lindeberg's condition is fulfilled. $\diamond$

We turn our attention to uniform consistency. By the law of large numbers, we know that the empirical distribution function

$$\mathbb{F}^{(n)}(t) = \frac{1}{n} \sum_{i=1}^n 1_{(Z_i \leq t)}$$

converges almost surely to $F(t) = \mathbb{P}(Z \leq t)$ for any fixed t . Sometimes, we may be interested in providing a pathwise result of the convergence, for instance guaranteeing that the latter convergence is uniform in t . That is exactly what the following celebrated theorem states.

Theorem 1.8 (Glivenko–Cantelli) *Let (Z_n) be a sequence of iid real-valued random variables with distribution function F . It holds that*

$$\sup_{t \in \mathbb{R}} |\mathbb{F}^{(n)}(t) - F(t)| = \sup_{t \in \mathbb{R}} \left| \frac{1}{n} \sum_{i=1}^n 1_{(Z_i \leq t)} - F(t) \right| \xrightarrow{\text{a.s.}} 0.$$

Using the same proof technique, we prove a slightly more general result which is useful when generalising to other estimators that are not bounded but still non-decreasing and càdlàg (right-continuous with left limits); then the above result follows as a corollary. Crucially, the limit must be bounded, but the requirement does not extend to the approximating sequence.

Lemma 1.5 (Abstract Glivenko–Cantelli) *Assume that $T: \mathbb{R} \rightarrow \mathbb{R}$ is a non-decreasing, bounded and càdlàg function. Assume that $(T^{(n)})$ is a sequence of random non-decreasing càdlàg maps $T^{(n)}: \mathbb{R} \rightarrow \mathbb{R}$ such that $T^{(n)}(x) \xrightarrow{\text{a.s.}} T(x)$ and $T^{(n)}(x-) \xrightarrow{\text{a.s.}} T(x-)$ for all $x \in \mathbb{R}$. It then holds that*

$$\sup_{x \in \mathbb{R}} |T^{(n)}(x) - T(x)| \xrightarrow{\text{a.s.}} 0.$$

Proof. The idea is to control oscillations between jumps by a finite partition and use pointwise convergence on the grid. Given $\varepsilon > 0$ choose some $k \in \mathbb{N}$ and a sequence $-\infty = x_0 < \dots < x_k = \infty$ satisfying $T(x_j-) - T(x_{j-1}) < \varepsilon$. We can do this by including possible jumps in the sequence. By assumption it holds for each finite point in the partition x_j that

$$|T^{(n)}(x_j) - T(x_j)| \xrightarrow{\text{a.s.}} 0, \quad |T^{(n)}(x_j-) - T(x_j-)| \xrightarrow{\text{a.s.}} 0.$$

Let

$$\Delta^{(n)} = \max_{j=1, \dots, k-1} \{|T^{(n)}(x_j) - T(x_j)|, |T^{(n)}(x_j-) - T(x_j-)|\}$$

Let $x \in \mathbb{R}$ be arbitrary. By construction there is a j such that $x \in [x_{j-1}, x_j)$ and we have

$$T^{(n)}(x_{j-1}) - T(x_{j-1}) - \varepsilon \leq T^{(n)}(x_{j-1}) - T(x_j-) \leq T^{(n)}(x) - T(x)$$

and also

$$T^{(n)}(x) - T(x) \leq T^{(n)}(x_j-) - T(x_{j-1}) \leq T^{(n)}(x_j-) - T(x_j-) + \varepsilon.$$

Then

$$\begin{aligned} T^{(n)}(x_{j-1}) - T(x_{j-1}) - \varepsilon &\leq T^{(n)}(x) - T(x) \leq T^{(n)}(x_j-) - T(x_j-) + \varepsilon \\ \Rightarrow |T^{(n)}(x) - T(x)| &\leq \Delta^{(n)} + \varepsilon. \end{aligned}$$

Since x was arbitrary, it holds that

$$\sup_{x \in \mathbb{R}} |T^{(n)}(x) - T(x)| \leq \Delta^{(n)} + \varepsilon.$$

Note that $\Delta^{(n)}$ is the maximum of finitely many terms converging to zero almost surely, so $\Delta^{(n)} \xrightarrow{\text{a.s.}} 0$, and in turn

$$\lim_{n \rightarrow \infty} \sup_{x \in \mathbb{R}} |T^{(n)}(x) - T(x)| \leq \varepsilon, \quad \text{a.s.}$$

Since $\varepsilon > 0$ was arbitrary, the result follows. $\square$

A similar result can be established for arrays, though the proof is not available through standard techniques, see Theorem 3.2.1 in [Shorack and Wellner \[2009\]](#).

Theorem 1.9 (Extended Glivenko–Cantelli) *Let (Z_{nm}) be a triangular array of real-valued random variables with arbitrary distribution functions F_{nm} and empirical distribution function*

$$\mathbb{F}^{(n)}(t) = \frac{1}{n} \sum_{m=1}^n 1_{(Z_{nm} \leq t)}$$

for every n . Let $F_n = \frac{1}{n} \sum_{m=1}^n F_{nm}$. Then it holds that

$$\sup_{t \in \mathbb{R}} |\mathbb{F}^{(n)}(t) - F_n(t)| \xrightarrow{a.s.} 0.$$

We now introduce an important tool for transferring asymptotic distributions through transformations of estimators. It is to be distinguished from the continuous mapping theorem in that the transformation happens before normalisation.

Theorem 1.10 (δ -method) *Let (T_n) and Z be $\mathbb{R}^p$ -valued random variables and (a_n) be a positive deterministic sequence. Let $h : \mathbb{R}^p \rightarrow \mathbb{R}$ be a differentiable function at c , with derivative $\nabla h(c)$. If $a_n(T_n - c) \xrightarrow{d} Z$ then*

$$a_n(h(T_n) - h(c)) \xrightarrow{d} \nabla h(c)^\top Z.$$

Proof. We linearise h around c and then show the remainder is negligible at the scale a_n . By Slutsky’s Lemma [1.3](#) (with the constant sequence) and the continuous mapping Theorem [1.4](#) (for the inner product operation) we have that

$$\nabla h(c)^\top a_n(T_n - c) \xrightarrow{d} \nabla h(c)^\top Z.$$

Now, from Example [1.4](#) we know that $a_n(T_n - c) = O_{\mathbb{P}}(1)$, or in other words $(T_n - c) = O_{\mathbb{P}}(a_n^{-1})$. By Taylor expansion and Example [1.5](#) we get

$$h(T_n) - h(c) = \nabla h(c)^\top (T_n - c) + o(\|T_n - c\|) = \nabla h(c)^\top (T_n - c) + o_{\mathbb{P}}(a_n^{-1}),$$

so that by multiplication with a_n we may piece the two results together to obtain

$$a_n(h(T_n) - h(c)) = \nabla h(c)^\top a_n(T_n - c) + o_{\mathbb{P}}(1) \xrightarrow{d} \nabla h(c)^\top Z$$

by Exercise [6](#). □

1.3 Basic exploratory tools

In anticipation of specialised tools in the subsequent chapters, we now introduce some basic functionals for exploring the distribution of a dataset and indicate how the above results can be used to quantify their uncertainty. The focus is on functionals that later appear in tail modelling and dependence analysis, and we emphasise their large-sample behaviour.

1.3.1 Some concrete estimators

We start with variance as a baseline measure of dispersion and then turn to tail-oriented diagnostics. For measuring the spread of a distribution, it is common to use the empirical variance, which is defined as

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (Z_i - \bar{Z}_n)^2, \quad \bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i.$$

Proposition 1.1 (Asymptotics of the empirical variance) *Let (Z_n) be an iid sequence with finite fourth moment, and central moments $\mu^{(k)} = \mathbb{E}[(Z_1 - 1_{(k>1)}\mathbb{E}[Z_1])^k]$, $k = 1, \dots, 4$. Then we have*

$$\begin{aligned} \hat{\sigma}_n^2 &\xrightarrow{a.s.} \mu^{(2)}, \\ \sqrt{n}(\hat{\sigma}_n^2 - \mu^{(2)}) &\xrightarrow{d} N(0, \mu^{(4)} - (\mu^{(2)})^2). \end{aligned}$$

Proof. First note that

$$\hat{\sigma}_n^2 = h(T_n), \quad T_n = \left(\frac{1}{n} \sum_{i=1}^n Z_i, \frac{1}{n} \sum_{i=1}^n Z_i^2 \right)^\top, \quad h(x, y) = y - x^2.$$

From this decomposition, the first part follows from the law of large numbers and the continuous mapping Theorem 1.4. Let the raw moments be given by μ_k , $k = 1, \dots, 4$. By Laplace's CLT, Theorem 1.6, $\sqrt{n}(T_n - (\mu_1, \mu_2)^\top) \xrightarrow{d} N(0, \Sigma)$ where

$$\Sigma = \begin{pmatrix} \mu_2 - \mu_1^2 & \mu_3 - \mu_1\mu_2 \\ \mu_3 - \mu_1\mu_2 & \mu_4 - \mu_2^2 \end{pmatrix}$$

We then apply the δ -method, Theorem 1.10, to the function h , with gradient $\nabla h(x, y) = (-2x, 1)^\top$. The resulting variance is calculated by

$$\nabla h(\mu_1, \mu_2)^\top \Sigma \nabla h(\mu_1, \mu_2) = \mu^{(4)} - (\mu^{(2)})^2.$$

□

As an illustration of the multivariate delta method, consider the sample correlation.

Example 1.7 (Sample correlation and Fisher's z -transform) Let (X_n, Y_n) be iid bivariate random variables with $\mathbb{E}[X_i] = \mathbb{E}[Y_i] = 0$, $\mathbb{E}[X_i^2] = \mathbb{E}[Y_i^2] = 1$, and $\text{Corr}(X_i, Y_i) = \rho \in (-1, 1)$. Define

$$S_{XX} = \frac{1}{n} \sum_{i=1}^n X_i^2, \quad S_{YY} = \frac{1}{n} \sum_{i=1}^n Y_i^2, \quad S_{XY} = \frac{1}{n} \sum_{i=1}^n X_i Y_i,$$

and the sample correlation

$$r_n = \frac{S_{XY}}{\sqrt{S_{XX}S_{YY}}}.$$

By Laplace's CLT, Theorem 1.6,

$$\sqrt{n}((S_{XY}, S_{XX}, S_{YY}) - (\rho, 1, 1)) \xrightarrow{d} N(0, \Sigma),$$

where direct moment calculations for the bivariate normal yield

$$\Sigma = \begin{pmatrix} 1 + \rho^2 & 2\rho & 2\rho \\ 2\rho & 2 & 2\rho^2 \\ 2\rho & 2\rho^2 & 2 \end{pmatrix}.$$

Let $g(a, b, c) = a/(bc)^{1/2}$. Then $\nabla g(\rho, 1, 1) = (1, -\rho/2, -\rho/2)^\top$, and the δ -method, Theorem 1.10, gives

$$\sqrt{n}(r_n - \rho) \xrightarrow{d} N(0, (1 - \rho^2)^2).$$

Define the Fisher z -transform $z_n = \frac{1}{2} \log((1 + r_n)/(1 - r_n))$. Since $h(r) = \frac{1}{2} \log((1 + r)/(1 - r))$ has derivative $h'(\rho) = 1/(1 - \rho^2)$, another application of the δ -method yields

$$\sqrt{n}(z_n - \frac{1}{2} \log(\frac{1+\rho}{1-\rho})) \xrightarrow{d} N(0, 1).$$

◇

The tail of a distribution refers to its survival behaviour at very large values. Such values are sometimes deemed “catastrophic” from a non-life insurance perspective, and their accurate estimation is crucial for risk management in both insurance and reinsurance. For determining or classifying tail behaviour, one can look at the following function.

Definition 1.6 (Empirical mean excess function) Let u be a threshold. Then

$$\mathfrak{e}_n(u) = \frac{\sum_{i=1}^n Z_i 1(Z_i > u)}{\sum_{i=1}^n 1(Z_i > u)} - u,$$

is called the empirical mean excess function.

Clearly, the analysis of $\mathfrak{e}_n(u)$ is equivalent to that of $\mathfrak{e}_n(u) + u$. The following result establishes convergence to the true mean excess function, defined as $e_F(u) = \mathbb{E}[Z_1 - u | Z_1 > u]$. Define also $e_F^{(2)}(u) = \mathbb{E}[(Z_1 - u)^2 | Z_1 > u]$. A list of mean excess functions of several parametric families may be found in [Embrechts et al., 1997, p. 161].

Theorem 1.11 (Asymptotics of the empirical mean excess function) Let (Z_n) be an iid sequence with finite second moment and with distribution function F . For any u such that $F(u) < 1$ we have

$$\begin{aligned} \mathfrak{e}_n(u) &\xrightarrow{a.s.} e_F(u), \\ \sqrt{n}(\mathfrak{e}_n(u) - e_F(u)) &\xrightarrow{d} N\left(0, \frac{e_F^{(2)}(u) - e_F(u)^2}{1 - F(u)}\right). \end{aligned}$$

Proof. We express $e_n(u)$ as a smooth function of two sample averages and apply a bivariate CLT. The first part follows directly from the law of large numbers and the first part of the continuous mapping Theorem 1.4. By Laplace's CLT, Theorem 1.6,

$$T_n = \left(\frac{1}{n} \sum_{i=1}^n Z_i 1_{(Z_i > u)}, \frac{1}{n} \sum_{i=1}^n 1_{(Z_i > u)} \right)^\top$$

converges, when normalised, to a centred Gaussian vector with covariance matrix Σ with entries

$$\begin{aligned} \sigma_1^2 &= \mathbb{E}[Z_1^2 1_{(Z_1 > u)}] - \mathbb{E}[Z_1 1_{(Z_1 > u)}]^2 \\ \sigma_2^2 &= \mathbb{P}(Z_1 > u) \mathbb{P}(Z_1 \leq u) \\ \sigma_{12} &= \mathbb{E}[Z_1 1_{(Z_1 > u)}] \mathbb{P}(Z_1 \leq u). \end{aligned}$$

We then apply the δ -method, Theorem 1.10, to the function $h(x, y) = x/y$, with gradient $\nabla h(x, y) = (1/y, -x/y^2)^\top$. The resulting variance is calculated straightforwardly through

$$\nabla h(\mathbb{E}[Z_1 1_{(Z_1 > u)}], \mathbb{P}(Z_1 > u))^\top \Sigma \nabla h(\mathbb{E}[Z_1 1_{(Z_1 > u)}], \mathbb{P}(Z_1 > u)).$$

□

Notice, however, that the above result is pointwise in u and does not describe the behaviour of the process $u \mapsto e_n(u)$ across thresholds.

Definition 1.7 (Empirical cumulative hazard without product integral) Let (Z_n) be an iid sequence. We define its empirical cumulative hazard by

$$\Lambda^{(n)}(t) = \log(n) - \log \left(\sum_{i=1}^n 1_{(Z_i > t)} \right).$$

The following result establishes convergence to the cumulative hazard function

$$\Lambda(t) = -\log(1 - F(t)),$$

which for absolutely continuous F coincides with the usual integral representation

$$\int_{-\infty}^t \frac{dF(s)}{ds} \frac{1}{1 - F(s)} ds = \int_{-\infty}^t \frac{1}{1 - F(s)} dF(s).$$

Theorem 1.12 (Asymptotics of the cumulative hazard) Let (Z_n) be an iid sequence with distribution function F . For any t such that $F(t) < 1$ we have

$$\begin{aligned} \Lambda^{(n)}(t) &\xrightarrow{a.s.} \Lambda(t), \\ \sqrt{n}(\Lambda^{(n)}(t) - \Lambda(t)) &\xrightarrow{d} N\left(0, \frac{F(t)}{1 - F(t)}\right). \end{aligned}$$

Proof. This is another direct δ -method application, now for the log transform of the empirical CDF. The first part follows directly from the law of large numbers

and the first part of the continuous mapping Theorem 1.4. By Laplace's CLT, Theorem 1.6,

$$\sqrt{n}(\mathbb{F}^{(n)}(t) - F(t)) \xrightarrow{d} N(0, F(t)(1 - F(t))).$$

Notice that

$$\Lambda^{(n)}(t) = -\log(1 - \mathbb{F}^{(n)}(t)).$$

Thus, we apply the δ -method, Theorem 1.10, to the function $h(x) = -\log(1-x)$, with gradient $h'(x) = 1/(1-x)$. The resulting variance is given by

$$h'(F(t))^2 F(t)(1 - F(t)) = \frac{F(t)}{1 - F(t)}.$$

□

1.3.2 Finite sample behaviour

Recall that given any estimator T_n with $\sqrt{n}(T_n - \eta) \xrightarrow{d} N(0, \sigma^2)$, one can approximate its distribution by considering

$$T_n \overset{d}{\approx} N(\eta, \sigma^2/n) \quad (1.3)$$

and obtain quantiles on the right-hand side expression to match the ones on the left-hand side. Of course, the variance is in general unknown, so that a common strategy is to approximate it by a suitable estimator $\hat{\sigma}_n^2$ and plug it into the relevant quantile calculation.

It should be noted that the quality of the approximation (1.3) can vary substantially. For instance, Figure 1.2 shows that for the given sample sizes, the cumulative hazard performs better visually than the mean excess function. The behaviour of an estimator before its limit is well approximated is often referred to as *finite sample behaviour*.

We see from the above theorems that we are able to capture the behaviour of several of the exploratory tools for a fixed time (or threshold). Similar results hold when considering finitely many time points. For instance,

$$\sqrt{n}(\mathbb{F}^{(n)}(t) - F(t), \mathbb{F}^{(n)}(s) - F(s)) \xrightarrow{d} N(0, \Sigma(s, t)). \quad (1.4)$$

where $\sigma_{12}(s, t) = \mathbb{E}[1_{(Z_1 \leq t)} 1_{(Z_1 \leq s)}] - \mathbb{E}[1_{(Z_1 \leq t)}] \mathbb{E}[1_{(Z_1 \leq s)}] = F(s \wedge t) - F(s)F(t)$, and the variances are unchanged from the unidimensional case. This raises the question of whether one-dimensional central limit theorems suffice to describe the full behaviour of $\mathbb{F}^{(n)}$, especially for hypotheses that depend on the entire curve. The answer is negative. For instance, in Figure 1.3 we plot empirical versus true cumulative distribution functions for two sample sizes, where we wish to test whether F belongs to one of three given distributions. The probability that an empirical curve jumps at any fixed time $t_0 < \dots < t_m$ is zero, so the maximal distance between curves requires a genuinely continuous inspection of the paths.

In fact, although finite-dimensional distributions are enough to characterise the distribution of a stochastic process, their convergence is not enough to establish *pathwise* (see next section) convergence between e.g. $t \mapsto \mathbb{F}^{(n)}(t)$ and a

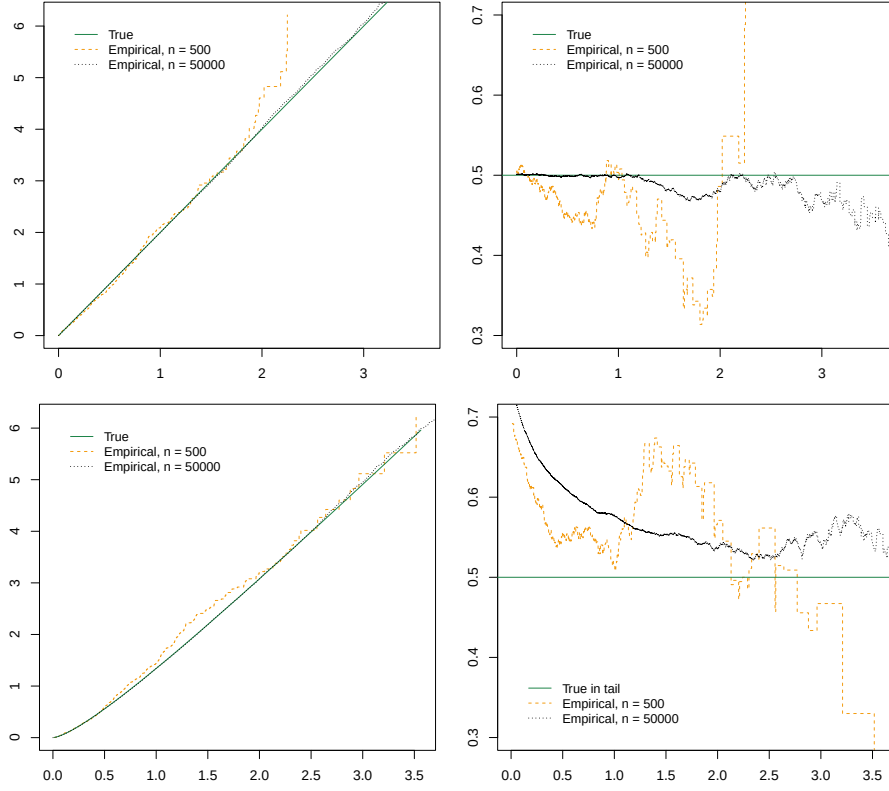


Figure 1.2: Empirical versus theoretical cumulative hazard (left, as a function of t) and mean excess functions (right, as a function of threshold u) for Exponential (top) and Gamma (bottom) distributed random variables.

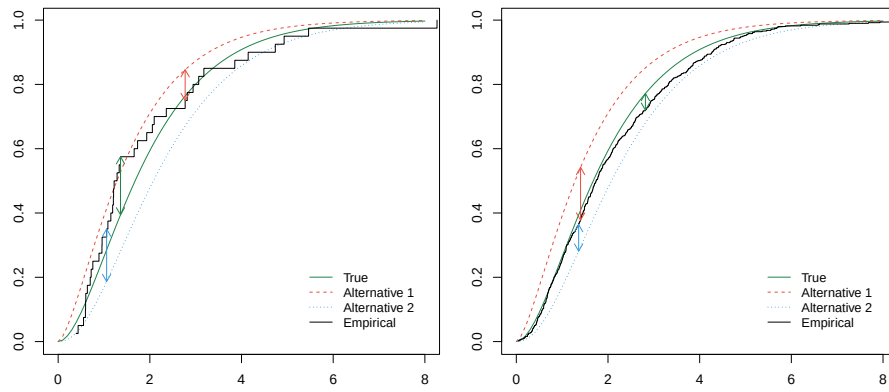


Figure 1.3: Empirical versus theoretical cumulative distribution functions $F(t)$ for Gamma distributed samples of size $n = 40$ (left) and $n = 500$ (right). Vertical arrows provide the maximal distance between curves.

suitable limit process. Therefore, our next task is to define pathwise convergence and provide additional conditions under which we may formulate *functional central limit theorems*.

1.4 Convergence of stochastic processes

In this section, we provide an overview of the relevant theory on stochastic processes and stochastic convergence, along with a couple of preliminary examples. The main application of these methods lies in nonparametric statistical procedures. The notation introduced here is used extensively in the sequel, particularly when establishing limit results for models in parametric models, extremes, survival analysis, and multi-state models. Additional results on the convergence of stochastic processes and the functional δ -method may be found in references such as [van der Vaart and Wellner \[1996\]](#) and [van der Vaart \[1998\]](#).

1.4.1 Introduction to stochastic processes

A stochastic process in continuous time is a collection of random variables indexed by some real subset, that is $X = (X_t)$, where typically $t \in I \subseteq \mathbb{R}$. Equivalently, from a pathwise point of view, given an underlying probability space $(\Omega, \mathcal{A}, \mathbb{P})$, the maps

$$\omega \mapsto X(\omega) = (X_t(\omega))_{t \in I}$$

define a random element in a suitable metric space of functions $(\mathbb{D}, d)$, where d is a suitable metric. Let $\mathcal{D}$ be the Borel σ -field generated by d . We usually refer to the space as $(\mathbb{D}, \mathcal{D})$ when no ambiguity arises. A schematic representation is provided in Figure 1.4.

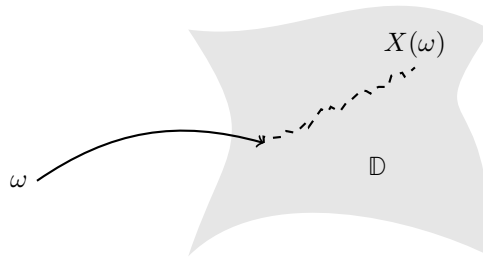


Figure 1.4: A schematic representation of a trajectory of a stochastic process X in the space $\mathbb{D}$.

Common choices for this metric space of functions are:

1. A finite-dimensional Euclidean space $\mathbb{D} = \mathbb{R}^p$, equipped with any norm.
2. $\mathbb{D} = C[a, b]$, the set of all continuous functions $z: [a, b] \rightarrow \mathbb{R}$ for $a, b \in \bar{\mathbb{R}}$, equipped with the uniform norm $\|z\|_{[a, b]} = \sup_{s \in [a, b]} |z(s)|$.

3. $\mathbb{D} = D[a, b]$, the set of all càdlàg functions $z: [a, b] \rightarrow \mathbb{R}$ for $a, b \in \bar{\mathbb{R}}$, equipped with the uniform norm $\|z\|_{[a, b]} = \sup_{s \in [a, b]} |z(s)|$. Another common choice is the Skorokhod norm.
4. $\mathbb{D} = \ell^\infty(\mathcal{F})$, the set of all uniformly bounded real functions on an arbitrary set $\mathcal{F}$: all functions $z: \mathcal{F} \rightarrow \mathbb{R}$ such that $\|z\|_{\mathcal{F}} = \sup_{f \in \mathcal{F}} |z(f)| < \infty$, equipped with the uniform norm $\|z\|_{\mathcal{F}}$. For instance, $\mathcal{F}$ could be a Euclidean subset.

Note that the convergence theory sketched below reduces to the classical one in finite dimensions for the first case above. The key observation in that case is that all norms in finite-dimensional spaces are equivalent.

In any of the four cases, to speak about the distribution of X , and hence about convergence of distributions, we define $C_b(\mathbb{D})$, the set of all bounded continuous functions on $\mathbb{D}$, and consider expectations of the kind

$$\mathbb{E}[f(X)], \quad f \in C_b(\mathbb{D}). \quad (1.5)$$

The spaces in items 1 and 2 are usually measurable and thus the above expectation is well-defined. Unfortunately, when dealing with processes in 3 or 4, subtle but important measurability issues arise when using the supremum norm. For instance, for $\mathbb{F}^{(n)}(t) = n^{-1} \sum_{i=1}^n 1_{(Z_i \leq t)}$ the process

$$t \mapsto X_n(t) = \sqrt{n}(\mathbb{F}^{(n)}(t) - F(t)), \quad t \in [a, b],$$

has paths in $\mathbb{D} = D[a, b]$ but the Borel σ -field $\mathcal{D}$ is too large for the pre-image to satisfy $X_n^{-1}(\mathcal{D}) \subseteq \mathcal{B}^n$, where $\mathcal{A} = \mathcal{B}^n$ is the underlying product space supporting $Z_1, \dots, Z_n$.

Equipping the target space with the sup-norm is natural, since it allows us to study uniform convergence of paths. However, as mentioned, it also causes measurability issues. We address these by working with outer expectations, which coincide with the usual expectation when measurability holds, and allow us to extend standard convergence results.

1.4.2 Convergence of processes

We introduce the concept of *weak convergence* for stochastic processes. Note, however, that thanks to the wealth of theorems available for this extended notion, we seldom have to deal with the foundational definitions directly. First, we introduce the so-called *outer integrals* and *outer probabilities*. They play a fundamental role in the rigorous construction of the theory, though their practical role is minor in terms of applications to statistics for insurance.

Definition 1.8 (Inner and outer integral and probability) Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let $T: \Omega \rightarrow \bar{\mathbb{R}}$ be an arbitrary map. The outer integral of T with respect to $\mathbb{P}$ is defined as

$$\mathbb{E}^*[T] = \inf \{ \mathbb{E}[U] \text{ such that } U \geq T, U: \Omega \rightarrow \bar{\mathbb{R}} \text{ measurable with } \mathbb{E}[U] < \infty \}.$$

The outer probability of a subset $B \subseteq \Omega$ is

$$\mathbb{P}^*(B) = \inf \{ \mathbb{P}(A) : B \subseteq A, A \in \mathcal{A} \}.$$

The inner integral and inner probability are respectively defined as

$$\mathbb{E}_*[T] = -\mathbb{E}^*[-T], \quad \mathbb{P}_*(B) = 1 - \mathbb{P}^*(\Omega \setminus B).$$

Outer probabilities enter the picture because sup-norm functionals over uncountable index sets need not be measurable. In most applications we work with separable or càdlàg versions, in which case the outer and ordinary notions coincide for sets of interest (more precisely on sets generated by balls in $\mathbb{D}$; see [van der Vaart and Wellner, 1996, Section 1.7]), so the machinery is primarily technical rather than conceptual. Nonetheless, it is important to remember that there is a subtle distinction.

Convergence in distribution of a stochastic process evaluated at fixed indices is referred to as convergence in finite dimensions. The latter notion is implied by the following stronger condition.

Definition 1.9 (Weak convergence) Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. Let $(\mathbb{D}, d)$ be a metric space and let $\mathcal{D}$ be the Borel σ -field on $\mathbb{D}$. Let (X_n) be a sequence of arbitrary maps $X_n: \Omega \rightarrow \mathbb{D}$. The sequence (X_n) converges weakly to a Borel probability measure¹ L if

$$\mathbb{E}^*[f(X_n)] \rightarrow \int f \, dL \quad \text{for every } f \in C_b(\mathbb{D}).$$

This is denoted $X_n \xrightarrow{d} L$ or $X_n \xrightarrow{d} X$ if $X \sim L$.

Notice that the definition subsumes the distributional convergence in finite dimensions, and therefore the use of the same symbol is justified.

Remark 1.3 The limit X is usually a continuous stochastic process, which is measurable with the uniform metric, and so we can write $X_n \xrightarrow{d} X$ whenever, equivalently,

$$\mathbb{E}^*[f(X_n)] \rightarrow \mathbb{E}[f(X)] \quad \text{for every } f \in C_b(\mathbb{D}).$$

Also note that the metric on $\mathbb{D}$ does not play a role in the existence within this definition, since the outer expectation always exists (we have $\inf \emptyset = \infty$). $\circ$

The following lemma shows that establishing weak convergence in a larger space than required is not a problem, as it is equivalent to showing it in the possibly smaller space where the processes take values.

Lemma 1.6 (Subspaces) Let $\mathbb{D}_0 \subseteq \mathbb{D}$ be arbitrary metric spaces equipped with the same metric. If X and every X_n take their values in $\mathbb{D}_0$, then $X_n \xrightarrow{d} X$ as maps in $\mathbb{D}_0$ if and only if $X_n \xrightarrow{d} X$ as maps in $\mathbb{D}$.

For stochastic processes we may also define convergence in probability and almost surely using the outer probability notion. Again, the definitions coincide in finite dimensions and therefore the same notation is justified.

Definition 1.10 (Convergence in outer probability and outer almost surely) Let (X_n) be a sequence of arbitrary maps $X_n: \Omega \rightarrow \mathbb{D}$ and let $X: \Omega \rightarrow \mathbb{D}$.

¹A Borel probability measure L is a probability measure defined on the Borel σ -field $\mathcal{D}$.

- (i) (X_n) converges in *outer probability* to X if $\mathbb{P}^*(d(X_n, X) > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$. This is denoted $X_n \xrightarrow{\mathbb{P}} X$.
- (ii) (X_n) converges *outer almost surely* to X if $\mathbb{P}^*(\lim_{n \rightarrow \infty} d(X_n, X) = 0) = 1$ for d the metric of interest. This is denoted $X_n \xrightarrow{a.s.} X$.

The following result generalises Lemma 1.2.

Lemma 1.7 (Relation between types of convergence) *Let (X_n) be a sequence of arbitrary maps and X Borel measurable. Then*

1. $X_n \xrightarrow{a.s.} X$ implies $X_n \xrightarrow{\mathbb{P}} X$.
2. $X_n \xrightarrow{\mathbb{P}} X$ implies $X_n \xrightarrow{d} X$.
3. Let $c \in \mathbb{D}$. $X_n \xrightarrow{\mathbb{P}} c$ if and only if $X_n \xrightarrow{d} c$.

Sometimes it can be convenient to deal with stochastic processes and their distributional representations using results from real analysis directly; in that connection the following result allows for such treatment in an appropriate probability space, sometimes referred to as a Skorokhod construction.

Theorem 1.13 (A.s. representation) *Let $X_n : \Omega \rightarrow \mathbb{D}$ be an arbitrary sequence of maps, and let X be Borel measurable and separable. If $X_n \xrightarrow{d} X$, then there exists a probability space $(\tilde{\Omega}, \tilde{\mathcal{A}}, \tilde{\mathbb{P}})$ and maps $\tilde{X}_n : \tilde{\Omega} \rightarrow \mathbb{D}$ with*

1. $\tilde{X}_n \xrightarrow{a.s.} \tilde{X}$;
2. $\mathbb{E}^*[f(\tilde{X}_n)] = \mathbb{E}^*[f(X_n)]$, for every bounded $f : \mathbb{D} \rightarrow \mathbb{R}$ and every n .

The following definition is quite technical, but required for full understanding of weak convergence of stochastic processes. All limit processes considered in the sequel are tight, as a consequence of some general-purpose central limit theorems given below.

Definition 1.11 (Tightness) A Borel probability measure L is *tight* if for every $\varepsilon > 0$ there exists a compact set K with $L(K) \geq 1 - \varepsilon$. Likewise we say the process $X \sim L$ is tight if L is tight.

The following lemma allows us to establish the joint weak convergence of two stochastic processes from that of the individual processes and the cross-finite-dimensional distributions. This is useful since the latter condition usually follows easily by elementary means.

Lemma 1.8 (From marginal to joint weak convergence) *Let $X_n, X : \Omega \rightarrow \ell^\infty(T)$ and $Y_n, Y : \Omega \rightarrow \ell^\infty(S)$, satisfy $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ with tight X and Y . If additionally*

$$(X_n(t_1), \dots, X_n(t_k), Y_n(s_1), \dots, Y_n(s_l)) \xrightarrow{d} (X(t_1), \dots, X(t_k), Y(s_1), \dots, Y(s_l)),$$

for any $t_1, \dots, t_k \in T$ and $s_1, \dots, s_l \in S$, then there exist versions of X and Y with bounded sample paths and $(X_n, Y_n) \xrightarrow{d} (X, Y)$ in $\ell^\infty(T) \times \ell^\infty(S)$.

We state a general version of the continuous mapping theorem. Slutsky's lemma in combination with the continuous mapping theorem is a strong tool used to conclude weak convergence for different kinds of processes in what follows.

Theorem 1.14 (Continuous mapping) *Let $\mathbb{D}_n \subseteq \mathbb{D}$ and $g_n: \mathbb{D}_n \rightarrow \mathbb{E}$ satisfy the following: if $x_n \rightarrow x$ with $x_n \in \mathbb{D}_n$ for every n and $x \in \mathbb{D}_0$, then $g_n(x_n) \rightarrow g(x)$, where $\mathbb{D}_0 \subseteq \mathbb{D}$ and $g: \mathbb{D}_0 \rightarrow \mathbb{E}$. Let (X_n) be a sequence of maps with values in $\mathbb{D}_n$, let X be Borel measurable and tight, and take values in $\mathbb{D}_0$. Then*

1. $X_n \xrightarrow{a.s.} X$ implies $g_n(X_n) \xrightarrow{a.s.} g(X)$.
2. $X_n \xrightarrow{\mathbb{P}} X$ implies $g_n(X_n) \xrightarrow{\mathbb{P}} g(X)$.
3. $X_n \xrightarrow{d} X$ implies $g_n(X_n) \xrightarrow{d} g(X)$.

Lemma 1.9 (Slutsky's lemma for general random elements) *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$ with X tight and c a constant then $(X_n, Y_n) \xrightarrow{d} (X, c)$.*

Notice that the following theorem is akin to the Glivenko–Cantelli result, and the proof is also very similar.

Theorem 1.15 (Dini's theorem, for monotone processes) *Let (X_n) be a sequence of monotone stochastic processes indexed by (some subset of) $\mathbb{R}$ converging marginally in probability to a continuous deterministic function f . Then the convergence is uniform in probability on compact subsets of the index set.*

Proof. Monotonicity lets us reduce uniform deviations to a finite grid. Assume w.l.o.g. that the processes are non-decreasing. Let $\eta > 0$ and consider a partition $a = s_0 < s_1 < \dots < s_K = b$ such that $f(s_i) - f(s_{i-1}) \leq \eta/2$. For $s \in [s_i, s_{i+1})$ we have

$$\begin{aligned} X_n(s) - f(s) &\leq X_n(s_{i+1}) - f(s_{i+1}) + f(s_{i+1}) - f(s) \\ &\leq X_n(s_{i+1}) - f(s_{i+1}) + \eta/2 \leq \max_{0 \leq k \leq K} |X_n(s_k) - f(s_k)| + \eta/2. \end{aligned}$$

Similarly $X_n(s) - f(s) \geq -\max_{0 \leq k \leq K} |X_n(s_k) - f(s_k)| - \eta/2$, which gives

$$\sup_{s \in [a, b]} |X_n(s) - f(s)| \leq \max_{0 \leq k \leq K} |X_n(s_k) - f(s_k)| + \eta/2.$$

Using the convergence in probability of the K finite terms on the right-hand side, we obtain

$$\begin{aligned} &\limsup_{n \rightarrow \infty} \mathbb{P} \left(\sup_{s \in [a, b]} |X_n(s) - f(s)| > \eta \right) \\ &\leq \limsup_{n \rightarrow \infty} \mathbb{P} \left(\max_{0 \leq k \leq K} |X_n(s_k) - f(s_k)| > \eta/2 \right) = 0. \end{aligned}$$

This finishes the proof. □

1.4.3 The functional δ -method

When dealing with stochastic processes, the nature of transformations that we can apply to their paths can be rather complex, since they may vary in a pathwise manner. The *functional δ -method* is a tool that preserves weak convergence of stochastic processes when transformed under pathwise maps that satisfy the natural but important requirement of *Hadamard differentiability*.

Definition 1.12 (Hadamard differentiability) Let $\mathbb{E}$ and $\mathbb{D}$ be two normed spaces. A map $\phi: \mathbb{D}_\phi \subseteq \mathbb{D} \rightarrow \mathbb{E}$ is called Hadamard differentiable at $\theta \in \mathbb{D}_\phi$ if there is a continuous linear map $\phi'_\theta: \mathbb{D}_\phi \rightarrow \mathbb{E}$ such that

$$\frac{\phi(\theta + t_n h_n) - \phi(\theta)}{t_n} \rightarrow \phi'_\theta(h)$$

for all converging scalar sequences $t_n \rightarrow 0$ and functions $h_n \rightarrow h$ such that $\theta + t_n h_n \in \mathbb{D}_\phi$ for every n . We say that ϕ is *Hadamard differentiable tangentially* to a set $\mathbb{D}_0 \subseteq \mathbb{D}$ by requiring that every $h_n \rightarrow h$ in the definition has $h \in \mathbb{D}_0$ and the derivative need only be defined on $\mathbb{D}_0$. For a specific h , we speak of Hadamard differentiability in the direction of h .

Remark 1.4 (Hadamard differentiability implies continuity) Hadamard differentiability implies continuity at θ in the norm of $\mathbb{E}$ (e.g. the supremum norm). Indeed, let $x_n \rightarrow \theta$ in $\mathbb{D}$ and set $r_n = x_n - \theta$. Define $t_n = \|r_n\|^{1/2}$ and $h_n = r_n/t_n$, so that $t_n \rightarrow 0$, $h_n \rightarrow 0$, and $\theta + t_n h_n = x_n$. The definition of Hadamard differentiability yields

$$\frac{\phi(\theta + r_n) - \phi(\theta)}{t_n} \rightarrow \phi'_\theta(0),$$

hence $\|\phi(x_n) - \phi(\theta)\| = o(t_n) \rightarrow 0$. Thus ϕ is continuous at θ for whichever norm is placed on $\mathbb{E}$, as long as Hadamard differentiability holds in that norm. $\circ$

Notice that the above defines a pathwise derivative as a map of functions, so one needs to know how it transforms functions to be able to evaluate it. An effective tool is the following.

Theorem 1.16 (Chain rule) *Let the following maps be Hadamard differentiable:*

1. $\phi: \mathbb{D}_\phi \subseteq \mathbb{D} \rightarrow \mathbb{E}_\phi$ at $\theta \in \mathbb{D}_\phi$ tangentially to $\mathbb{D}_0$, and
2. $\psi: \mathbb{E}_\psi \supseteq \mathbb{E}_\phi \rightarrow \mathbb{F}$ at $\phi(\theta)$ tangentially to $\phi'_\theta(\mathbb{D}_0)$.

Then $\psi \circ \phi$ is Hadamard differentiable at θ tangentially to $\mathbb{D}_0$ with derivative $\psi'_{\phi(\theta)} \circ \phi'_\theta$.

Proof. We may simply rewrite

$$\frac{\psi \circ \phi(\theta + t_n h_n) - \psi \circ \phi(\theta)}{t_n} = \frac{\psi(\phi(\theta) + t_n \{ \frac{\phi(\theta + t_n h_n) - \phi(\theta)}{t_n} \}) - \psi(\phi(\theta))}{t_n},$$

and obtain the limit by the composition rule for limits in normed spaces, using the Hadamard differentiability of ϕ with derivative ϕ'_θ for the expression in brackets, and then the Hadamard differentiability of ψ . $\square$

Before providing more specialised Hadamard differentiability results of intricate functions, we provide a few elementary examples, all with respect to the uniform norm.

Proposition 1.2 (Elementary Hadamard differentiability) *Fix $\tau > 0$ and equip $\mathbb{D} = \ell^\infty([0, \tau])$ with the sup norm $\|\cdot\|_\infty$.*

1. *Weighted integral functional. Let w be bounded and measurable on $[0, \tau]$, and define*

$$\phi(F) = \int_0^\tau F(t)w(t) \, dt, \quad F \in \mathbb{D}.$$

Then $\phi : \mathbb{D} \rightarrow \mathbb{R}$ is Hadamard differentiable at every F tangentially to $\mathbb{D}$ with derivative $\phi'_F(h) = \int_0^\tau h(t)w(t) \, dt$.

2. *Cumulative integral map. Define*

$$\Phi(F)(t) = \int_0^t F(s) \, ds, \quad 0 \leq t \leq \tau.$$

Then $\Phi : \mathbb{D} \rightarrow \mathbb{D}$ is Hadamard differentiable at every F tangentially to $\mathbb{D}$ with derivative $\Phi'_F(h)(t) = \int_0^t h(s) \, ds$.

3. *Ratio of integrals. Let w be bounded and measurable on $[0, \tau]$, and set*

$$\psi(F) = \frac{\int_0^\tau F(t) \, dt}{\int_0^\tau F(t)w(t) \, dt}, \quad F \in \mathbb{D}_\psi := \left\{ F \in \mathbb{D} : \int_0^\tau F(t)w(t) \, dt \neq 0 \right\}.$$

Then ψ is Hadamard differentiable at any $F \in \mathbb{D}_\psi$, with

$$\psi'_F(h) = \frac{\int_0^\tau h(t) \, dt}{\int_0^\tau F(t)w(t) \, dt} - \frac{\int_0^\tau F(t) \, dt}{\left(\int_0^\tau F(t)w(t) \, dt\right)^2} \int_0^\tau h(t)w(t) \, dt.$$

Proof. For the first statement, let $t_n \rightarrow 0$ and $h_n \rightarrow h$ in $\|\cdot\|_\infty$, and simply note that

$$\frac{\phi(F + t_n h_n) - \phi(F)}{t_n} - \phi'_F(h) = \int_0^\tau (h_n(t) - h(t))w(t) \, dt.$$

Hence

$$\left| \int_0^\tau (h_n - h)w \, dt \right| \leq \|h_n - h\|_\infty \int_0^\tau |w(t)| \, dt \leq \tau \|w\|_\infty \|h_n - h\|_\infty \rightarrow 0.$$

For the second statement, for each t , $\{\Phi(F + t_n h_n) - \Phi(F)\}(t)/t_n = \int_0^t h_n(s) \, ds$. Thus

$$\sup_{0 \leq t \leq \tau} \left| \int_0^t (h_n - h)(s) \, ds \right| \leq \tau \|h_n - h\|_\infty \rightarrow 0,$$

so the convergence holds in $\|\cdot\|_\infty$. For the third statement, let $L_1(F) = \int_0^\tau F(t) \, dt$ and $L_2(F) = \int_0^\tau F(t)w(t) \, dt$. By the previous statements and Remark 1.4 both

are continuous linear functionals on $\mathbb{D}$. For $F \in \mathbb{D}_\psi$, we have $L_2(F) \neq 0$. Hence, by continuity, there exists N such that for $n \geq N$,

$$|L_2(F + t_n h_n) - L_2(F)| < \frac{1}{2} |L_2(F)|,$$

so $|L_2(F + t_n h_n)| \geq \frac{1}{2} |L_2(F)|$ and $F + t_n h_n \in \mathbb{D}_\psi$. Using linearity,

$$\frac{\psi(F + t_n h_n) - \psi(F)}{t_n} = \frac{L_1(h_n)L_2(F) - L_1(F)L_2(h_n)}{L_2(F + t_n h_n)L_2(F)}.$$

Since $L_1(h_n) \rightarrow L_1(h)$ and $L_2(h_n) \rightarrow L_2(h)$, and $L_2(F + t_n h_n) \rightarrow L_2(F)$, the limit is the stated derivative. $\square$

We now provide some more complicated maps, which are used sporadically in the remaining chapters.

Lemma 1.10 (Hadamard differentiability of inverses and compositions) *The following hold.*

1. *We consider the inverse map with different support assumptions.*

(i) *Let $0 < u_1 < u_2 < 1$ and let F be continuously differentiable on $[a, b] = [F^{-1}(u_1) - \varepsilon, F^{-1}(u_2) + \varepsilon]$ with strictly positive derivative f . Then the inverse $\phi(h) = h^{-1}$ as a map from the set of restrictions of distribution functions to $[a, b]$, to $\ell^\infty([u_1, u_2])$ is Hadamard differentiable at F tangentially to $C[a, b]$. The derivative is*

$$\phi'_F(h)(\cdot) = -(h/f) \circ F^{-1}(\cdot).$$

(ii) *Let F have compact support $[a, b]$ and be continuously differentiable on $[a, b]$ with strictly positive derivative f . Then the inverse $\phi(h) = h^{-1}$ as a map from the set of distribution functions supported on $[a, b]$, to $\ell^\infty(0, 1)$ is Hadamard differentiable at F tangentially to $C[a, b]$. The derivative is again*

$$\phi'_F(h)(\cdot) = -(h/f) \circ F^{-1}(\cdot).$$

2. *Let $\varphi : (a, b) \rightarrow \mathbb{R}$ be a function with uniformly continuous and bounded derivative. Let $\phi(F) = \varphi(F)$ be a map from the subset of functions in $\ell^\infty(\mathcal{F})$ with image in (a, b) , to $\ell^\infty(\mathcal{F})$. The derivative of ϕ for F in such subset is*

$$\phi'_F(h)(\cdot) = \varphi'(F(\cdot))h(\cdot).$$

3. *Suppose $F_1 : \mathcal{X} \rightarrow \mathcal{Y}$ is a map, and $F_2 : \mathcal{Y} \rightarrow \mathcal{Z}$ is Fréchet differentiable² uniformly in y in the range of F_1 , with derivatives $F'_{2,y}$ such that $y \mapsto F'_{2,y}$ is uniformly norm-bounded. Then the composition $\phi(F_1, F_2)(x) = F_2 \circ F_1(x)$ is Hadamard differentiable at (F_1, F_2) tangentially to the set $\ell^\infty(\mathcal{X}, \mathcal{Y}) \times UC(\mathcal{Y}, \mathcal{Z})$ ³ with derivative*

$$\phi'_{(F_1, F_2)}(h_1, h_2)(x) = h_2 \circ F_1(x) + F'_{2, F_1(x)}(h_1(x)).$$

²A technical strengthening of Hadamard differentiability, where the limit should be continuous and linear, and the convergence uniform in h on bounded subsets of $\mathbb{D}_\phi$.

³The space of uniformly continuous functions from $\mathcal{Y} \rightarrow \mathcal{Z}$.

Proof. We treat the three items in turn.

1. Let $t_n \rightarrow 0$ and let $h_n \rightarrow h$ uniformly on $[a, b]$, with $h \in C[a, b]$, such that $F + t_n h_n$ lies in the stated domain for all n . For $u \in (0, 1)$ set $q_u = F^{-1}(u)$ and $q_{u,n} = (F + t_n h_n)^{-1}(u)$.

We first prove a generic pointwise expansion for fixed $u \in (0, 1)$. Choose $\varepsilon_n > 0$ with $\varepsilon_n = o(t_n)$. By the definition of the inverse,

$$(F + t_n h_n)(q_{u,n} - \varepsilon_n) \leq u \leq (F + t_n h_n)(q_{u,n}).$$

Since (h_n) is uniformly bounded, this yields

$$F(q_{u,n} - \varepsilon_n) + O(t_n) \leq u \leq F(q_{u,n}) + O(t_n).$$

Since F is monotone and continuous with $F(q_u) = u$ and strictly positive derivative at q_u , the function F is bounded away from u outside any neighbourhood of q_u . Hence the preceding inequalities imply $q_{u,n} \rightarrow q_u$. Moreover, uniform convergence of h_n and continuity of h give $h_n(q_{u,n} - \varepsilon_n) \rightarrow h(q_u)$. By Taylor's theorem at q_u we obtain

$$\begin{aligned} u + (q_{u,n} - q_u)f(q_u) - o(|q_{u,n} - q_u|) + O(\varepsilon_n) + t_n h(q_u) - o(t_n) &\leq u \\ &\leq u + (q_{u,n} - q_u)f(q_u) + o(|q_{u,n} - q_u|) + O(\varepsilon_n) + t_n h(q_u) + o(t_n), \end{aligned}$$

and hence $q_{u,n} - q_u = O(t_n)$ and

$$\frac{q_{u,n} - q_u}{t_n} \rightarrow -\frac{h(q_u)}{f(q_u)}.$$

- (i) The continuity and strict positivity of f on $[a, b]$ imply uniform continuity and a positive lower bound. The Taylor remainder above is therefore uniform in $u \in [u_1, u_2]$, and the same argument yields

$$\sup_{u \in [u_1, u_2]} \left| \frac{q_{u,n} - q_u}{t_n} + \frac{h(q_u)}{f(q_u)} \right| \rightarrow 0,$$

which proves Hadamard differentiability as a map into $\ell^\infty([u_1, u_2])$ with the stated derivative.

- (ii) Set $\varepsilon_{u,n} = t_n^2 \wedge (q_{u,n} - a) > 0$ for each $u \in (0, 1)$. Because $F + t_n h_n$ lies in the stated domain, it is supported on $[a, b]$, so $a < q_{u,n} \leq b$ and thus

$$(F + t_n h_n)(q_{u,n} - \varepsilon_{u,n}) \leq u \leq (F + t_n h_n)(q_{u,n}).$$

The smoothness of F on $[a, b]$ implies that

$$F(q_{u,n} - \varepsilon_{u,n}) = F(q_{u,n}) + O(\varepsilon_{u,n}),$$

uniformly in $u \in (0, 1)$. Since $h_n \rightarrow h$ uniformly, we may replace h_n by h at the cost of an $o(t_n)$ term. Hence

$$-t_n h(q_{u,n}) + o(t_n) \leq F(q_{u,n}) - F(q_u) \leq -t_n h(q_{u,n} - \varepsilon_{u,n}) + o(t_n),$$

with $o(t_n)$ uniform in u . Since f is continuous and strictly positive on $[a, b]$, the middle term is bounded above and below by a constant times $|q_{u,n} - q_u|$. This yields $|q_{u,n} - q_u| = O(t_n)$ uniformly in u . Using uniform convergence of h_n and uniform continuity of h , the preceding Taylor expansion is uniform in $u \in (0, 1)$, and we obtain

$$\sup_{u \in (0,1)} \left| \frac{q_{u,n} - q_u}{t_n} + \frac{h(q_u)}{f(q_u)} \right| \rightarrow 0.$$

This proves Hadamard differentiability as a map into $\ell^\infty(0, 1)$ with the stated derivative.

2. Let $t_n \rightarrow 0$ and let $h_n \rightarrow h$ in $\ell^\infty(\mathcal{F})$ such that $F + t_n h_n$ remains in the stated domain for all n . For each $x \in \mathcal{F}$, the mean value theorem yields a $\xi_{n,x} \in (0, 1)$ with

$$\varphi(F(x) + t_n h_n(x)) = \varphi(F(x)) + \varphi'(F(x) + \xi_{n,x} t_n h_n(x)) t_n h_n(x).$$

Therefore

$$\begin{aligned} & \left\| \frac{\phi(F + t_n h_n) - \phi(F)}{t_n} - \phi'(F)h \right\|_\infty \\ & \leq \|(\phi'(F + \xi_n t_n h_n) - \phi'(F))h_n\|_\infty + \|\phi'(F)(h_n - h)\|_\infty, \end{aligned}$$

where ξ_n denotes the pointwise selection $x \mapsto \xi_{n,x}$. Since $h_n \rightarrow h$ in the sup norm, the sequence (h_n) is bounded and $t_n \|h_n\|_\infty \rightarrow 0$. By uniform continuity of ϕ' on (a, b) the first term vanishes, and the second term tends to 0 because ϕ' is bounded. Hence ϕ is Hadamard differentiable at F with derivative $\phi'_F(h)(\cdot) = \phi'(F(\cdot))h(\cdot)$.

3. Let $t_n \rightarrow 0$ and let $h_{1n} \rightarrow h_1$ and $h_{2n} \rightarrow h_2$ in $\ell^\infty(\mathcal{X}, \mathcal{Y})$ and $\ell^\infty(\mathcal{Y}, \mathcal{Z})$, respectively, with $h_2 \in UC(\mathcal{Y}, \mathcal{Z})$, and such that $F_1 + t_n h_{1n}$ and $F_2 + t_n h_{2n}$ remain in the stated domain for all n . Write

$$F'_{2,F_1}(h_1)(x) = F'_{2,F_1(x)}(h_1(x)), \quad x \in \mathcal{X},$$

and similarly for $F'_{2,F_1}(h_{1n})$. Then

$$\begin{aligned} & \frac{(F_2 + t_n h_{2n}) \circ (F_1 + t_n h_{1n}) - F_2 \circ F_1}{t_n} - h_2 \circ F_1 - F'_{2,F_1}(h_1) \\ & = (h_{2n} - h_2) \circ (F_1 + t_n h_{1n}) + h_2(F_1 + t_n h_{1n}) - h_2(F_1) \\ & \quad + \frac{F_2(F_1 + t_n h_{1n}) - F_2(F_1)}{t_n} - F'_{2,F_1}(h_{1n}) + F'_{2,F_1}(h_{1n} - h_1). \end{aligned}$$

The first term converges to zero in $\ell^\infty(\mathcal{X}, \mathcal{Z})$ because

$$\|(h_{2n} - h_2) \circ (F_1 + t_n h_{1n})\|_\infty \leq \|h_{2n} - h_2\|_\infty \rightarrow 0.$$

The second term converges to zero since h_2 is uniformly continuous and $\|t_n h_{1n}\|_\infty \rightarrow 0$. For the third term, uniform Fréchet differentiability of F_2 in the range of F_1 yields

$$\sup_{x \in \mathcal{X}} \left\| \frac{F_2(F_1(x) + t_n h_{1n}(x)) - F_2(F_1(x))}{t_n} - F'_{2,F_1(x)}(h_{1n}(x)) \right\|_\infty \rightarrow 0.$$

Finally, by uniform boundedness of $y \mapsto F'_{2,y}$,

$$\|F'_{2,F_1}(h_{1n} - h_1)\|_\infty \leq \sup_y \|F'_{2,y}\| \|h_{1n} - h_1\|_\infty \rightarrow 0.$$

Thus the difference quotient converges to $h_2 \circ F_1 + F'_{2,F_1}(h_1)$, and the stated derivative follows. $\square$

The two maps presented below are useful because of their link to the cumulative hazard function and to the distribution function and because they are Hadamard differentiable. They are used extensively in the final three chapters, where censoring plays an important role.

Lemma 1.11 (Hadamard differentiability of integrals) *Let $\psi: [0, M] \rightarrow \mathbb{R}$ be a C^2 -function. Then the function*

$$g(F_1, F_2)(\cdot) = \int_{(-\infty, \cdot]} \psi(F_1(s)) \, dF_2(s)$$

is Hadamard differentiable as a map from the domain⁴

$$\mathbb{D}_g = \text{BV}_M[-\infty, \infty] \times \text{BV}_M[-\infty, \infty] \subseteq D[-\infty, \infty] \times D[-\infty, \infty] = \mathbb{D}$$

to $\mathbb{E} = D[-\infty, \infty]$ where we equip the spaces with the supremum norm. The derivative is given by

$$\begin{aligned} g'_{(F_1, F_2)}(h_1, h_2)(\cdot) &= h_2(s)\psi(F_1(s)) \Big|_{s=-\infty}^{s=\cdot} - \int_{(-\infty, \cdot]} h_2(s-) \, d\psi(F_1(s)) \\ &\quad + \int_{(-\infty, \cdot]} \psi'(F_1(s))h_1(s) \, dF_2(s). \end{aligned}$$

Proof. Let $t_n \downarrow 0$ and let $h_{1n} \rightarrow h_1$, $h_{2n} \rightarrow h_2$ in $D[-\infty, \infty]$ be such that $F_{2n} = F_2 + t_n h_{2n} \in \text{BV}_M$ for all n . Then each h_{2n} has bounded variation. Set $F_{1n} = F_1 + t_n h_{1n}$ and fix $x \in [-\infty, \infty]$. By adding and subtracting $\psi'(F_1)h_1$ we obtain

$$\begin{aligned} &\frac{g(F_{1n}, F_{2n})(x) - g(F_1, F_2)(x)}{t_n} \\ &= \int_{(-\infty, x]} \left(\frac{\psi(F_{1n}) - \psi(F_1)}{t_n} - \psi'(F_1)h_1 \right) dF_{2n} \\ &\quad + \int_{(-\infty, x]} \psi(F_1) \, dh_{2n} + \int_{(-\infty, x]} \psi'(F_1)h_1 \, dF_{2n}. \end{aligned}$$

For the first term, Taylor's theorem and boundedness of (h_{1n}) in the supremum norm yield

$$\|(\psi(F_{1n}) - \psi(F_1))/t_n - \psi'(F_1)h_1\|_\infty \leq \|\psi'\|_\infty \|h_{1n} - h_1\|_\infty + \frac{1}{2} \|\psi''\|_\infty t_n \|h_{1n}\|_\infty^2,$$

⁴Here, we use BV_M to denote the space of càdlàg functions of bounded variation with total variation at most M .

and therefore

$$\begin{aligned} \sup_x \left| \int_{(-\infty, x]} \left(\frac{\psi(F_{1n}) - \psi(F_1)}{t_n} - \psi'(F_1)h_1 \right) dF_{2n} \right| \\ \leq (\|\psi'\|_\infty \|h_{1n} - h_1\|_\infty + \tfrac{1}{2}\|\psi''\|_\infty t_n \|h_{1n}\|_\infty^2) M, \end{aligned}$$

since $F_{2n} \in \text{BV}_M$ and as $t_n \rightarrow 0$, this term converges to zero uniformly in x .

By integration by parts,

$$\int_{(-\infty, x]} \psi(F_1) dh_{2n} = (\psi(F_1(s)) h_{2n}(s)) \Big|_{s=-\infty}^{s=x} - \int_{(-\infty, x]} h_{2n}(s-) d\psi(F_1(s)),$$

and the right-hand side converges uniformly in x to the corresponding expression with h_2 , because $h_{2n} \rightarrow h_2$ in supremum norm and $\psi(F_1)$ is of bounded variation.

Finally, write

$$\int_{(-\infty, x]} \psi'(F_1)h_1 dF_{2n} = \int_{(-\infty, x]} \psi'(F_1)h_1 dF_2 + \int_{(-\infty, x]} \psi'(F_1)h_1 d(F_{2n} - F_2).$$

Set $r = \psi'(F_1)h_1$, which is bounded and càdlàg on $[-\infty, \infty]$. Given $\varepsilon > 0$, choose a finite partition $-\infty = u_0 < \dots < u_m = \infty$ such that the oscillation of r on each $[u_{i-1}, u_i]$ is at most ε . Then, uniformly in x ,

$$\left| \int_{(-\infty, x]} r d(F_{2n} - F_2) \right| \leq 2M\varepsilon + \sum_{i=1}^m |r(u_{i-1})| |(F_{2n} - F_2)[u_{i-1}, u_i]|.$$

For fixed partition the sum tends to zero because $F_{2n} \rightarrow F_2$ uniformly, and the first term is bounded by $2M\varepsilon$. Hence the second integral vanishes uniformly in x as $n \rightarrow \infty$.

Collecting terms shows that the difference quotient converges uniformly in x to $g'_{(F_1, F_2)}(h_1, h_2)$, proving the required Hadamard differentiability. $\square$

Definition 1.13 (Product integral) Let $F \in D(0, b]$ and let $\Delta F(t) = F(t) - F(t-)$ and $F^c(t) = F(t) - \sum_{0 < s \leq t} \Delta F(s)$. The product integral is defined as

$$\phi(F)(t) = \prod_{0 < s \leq t} (1 + dF(s)) = \exp(F^c(t)) \prod_{0 < s \leq t} (1 + \Delta F(s)).$$

As additional notation we have

$$\phi(F)(s, t] = \prod_{s < u \leq t} (1 + dF(u)) = \frac{\phi(F)(t)}{\phi(F)(s)}, \quad \text{for } s < t.$$

Let X be a finite signed $(k \times k)$ matrix-valued measure defined on the Borel sets $((0, b], \mathcal{B}(0, b])$. By this we mean that each entry $X(i, j)$ is a finite signed measure on $((0, b], \mathcal{B}(0, b])$. The following definition does not require that X is commutative:

$$\prod_{0 < s \leq t} (\text{Id} + dX(s)) = \lim_{\max |t_j - t_{j-1}| \rightarrow 0} \prod_j (\text{Id} + X(t_{j-1}, t_j]),$$

for partitions $0 = t_0 < \dots < t_n = t$ of $(0, t]$.

Remark 1.5 The product integral can be defined in several ways, and more generally than Definition 1.13. For some equivalent definitions, see Gill and Johansen [1990]. $\circ$

Lemma 1.12 (Hadamard differentiability of the product integral) *For fixed, finite, positive constants b and M , the product integral map $\phi: \text{BV}_M(0, b] \subseteq D(0, b] \rightarrow D(0, b]$ is Hadamard differentiable with derivative*

$$\phi'_F(h)(\cdot) = \int_{(0, \cdot]} \phi(F)(0, u-) \phi(F)(u, \cdot] \, dh(u).$$

The scalar result above is the template for product-integral arguments throughout the book. In multi-state settings we need the matrix analogue, so we provide it next before invoking the general functional δ -method.

Lemma 1.13 (Hadamard differentiability of the product integral - matrix version) *Let $\mathbb{D}$ be a space of matrices with function entries each of bounded variation, indexed by $[0, \theta]$, and equip it with the uniform norm. For $F \in \mathbb{D}$, define*

$$\phi(F)(t, s) = \prod_{(t, s]} (\text{Id} + dF), \quad 0 \leq t < s \leq \theta.$$

The map ϕ is Hadamard differentiable at each $F \in \mathbb{D}$ w.r.t. the uniform norm on $[0, \theta]$ and has derivative

$$\phi'_F(h)(t, s) = \int_{(t, s]} \prod_t^{u-} (\text{Id} + dF) \, dh(u) \prod_u^s (\text{Id} + dF), \quad 0 \leq t < s \leq \theta,$$

where the integral w.r.t. h is defined by the integration by parts formula.

Theorem 1.17 (Functional δ -method) *Let $\mathbb{D}$ and $\mathbb{E}$ be normed spaces. Let $\phi: \mathbb{D}_\phi \subseteq \mathbb{D} \rightarrow \mathbb{E}$ be Hadamard differentiable at θ tangentially to $\mathbb{D}_0$. Let $X_n: \Omega \rightarrow \mathbb{D}_\phi$ be maps with $r_n(X_n - \theta) \xrightarrow{d} X$ for some sequence of constants $r_n \rightarrow \infty$, where X is tight and takes its values in $\mathbb{D}_0$. Then $r_n(\phi(X_n) - \phi(\theta)) \xrightarrow{d} \phi'_\theta(X)$.*

Proof. We simply rescale via a clever function g_n and then apply the continuous mapping theorem to the weak limit. Indeed, define the function $g_n(h) = r_n(\phi(\theta + r_n^{-1}h) - \phi(\theta))$ on $\mathbb{D}_n = \{h: \theta + r_n^{-1}h \in \mathbb{D}_\phi\}$. By definition of Hadamard differentiability $g_n(h_n) \rightarrow \phi'_\theta(h)$ for any $h_n \rightarrow h \in \mathbb{D}_0$. Then the continuous mapping Theorem 1.14 gives

$$r_n(\phi(X_n) - \phi(\theta)) = g_n(r_n(X_n - \theta)) \xrightarrow{d} \phi'_\theta(X).$$

$\square$

Remark 1.6 The phrase *tangentially to $\mathbb{D}_0$* in Theorem 1.17 highlights that ϕ is defined on the full domain $\mathbb{D}_\phi$, but differentiability is only required along directions in $\mathbb{D}_0$. In the theorem, the perturbations are precisely $h = r_n(X_n - \theta)$, and the assumption $X \in \mathbb{D}_0$ means these directions live in $\mathbb{D}_0$. For example, if X_n are càdlàg and $\sqrt{n}(X_n - \theta) \xrightarrow{d} X$ with X a continuous Gaussian process on $[0, \tau]$, then $\theta \in \mathbb{D}[0, \tau]$ while $h = X \in C[0, \tau]$, so it suffices that ϕ be differentiable along continuous directions even if the derivative may or may not be defined on the whole of $\mathbb{D}[0, \tau]$. $\circ$

1.4.4 Entropy theorems

There exist similar central limit theorems for stochastic processes that turn out very useful in the sequel. Many times, the estimators we deal with are sums of iid terms. Therefore, before introducing these results, it is useful to define the empirical distribution function indexed by a function class. Thus, we introduce the so-called *empirical process*.

The empirical measure $\mathbb{P}^{(n)}$ of a sample of iid random variables $Z_1, \dots, Z_n \sim L$ on a measurable space $(\mathcal{Z}, \mathcal{B})$ is the discrete measure given by

$$\mathbb{P}^{(n)}(C) = \frac{1}{n} \sum_{i=1}^n \delta_{Z_i}(C),$$

for a measurable set $C \in \mathcal{B}$. Let $\mathcal{F}$ be a collection of measurable functions $f: \mathcal{Z} \rightarrow \mathbb{R}$. Then the empirical measure induces the following map from $\mathcal{F}$ to $\mathbb{R}$

$$f \mapsto \mathbb{P}^{(n)} f = \int f \, d\mathbb{P}^{(n)} = \frac{1}{n} \sum_{i=1}^n f(Z_i).$$

We scale and centre the above map to obtain the $\mathcal{F}$ -indexed empirical process $\mathbb{G}^{(n)} = \sqrt{n}(\mathbb{P}^{(n)} - L)$ given by

$$f \mapsto \mathbb{G}^{(n)} f = \int f \, d\mathbb{G}^{(n)} = \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n f(Z_i) - \int f \, dL \right).$$

If $\sup_{f \in \mathcal{F}} |f(z) - \int f \, dL| < \infty$, for almost every $z \in \mathcal{Z}$, then we view the empirical process $(\mathbb{G}^{(n)} f)_{f \in \mathcal{F}}$ as a well-defined stochastic process with paths in $\ell^\infty(\mathcal{F})$.

The complexity of $\mathcal{F}$ is later quantified via covering or bracketing numbers. For instance, the class of indicator half-lines $\{1_{(-\infty, t]} : t \in \mathbb{R}\}$ has small bracketing numbers because each function can be bracketed by neighbouring indicators with close thresholds, which is why this class has favourable asymptotic properties, as shown below.

Definition 1.14 (Glivenko–Cantelli class) Given an empirical measure $\mathbb{P}^{(n)}$ of a random sample (Z_n) defined on a measurable space $(\mathcal{Z}, \mathcal{B})$ where every Z_i has law L , a class $\mathcal{F}$ of measurable functions $f: \mathcal{Z} \rightarrow \mathbb{R}$, is called a $\mathbb{P}$ -Glivenko–Cantelli class if

$$\|\mathbb{P}^{(n)} - L\|_{\mathcal{F}} \rightarrow 0,$$

where the convergence is in outer probability or outer almost surely.

Definition 1.15 (Donsker class) A collection $\mathcal{F}$ of measurable functions $f: \mathcal{Z} \rightarrow \mathbb{R}$ is a $\mathbb{P}$ -Donsker class if

$$\mathbb{G}^{(n)} = \sqrt{n}(\mathbb{P}^{(n)} - L) \xrightarrow{d} \mathbb{G} \quad \text{in } \ell^\infty(\mathcal{F}),$$

for a tight Gaussian limit process $\mathbb{G} \in \ell^\infty(\mathcal{F})$.

Remark 1.7 Sometimes we may treat empirical processes $\mathbb{G}^{(n)}$ as stochastic processes indexed exchangeably by either: a) classes of functions $\mathcal{F}$; or b) intervals. For instance, if we consider sets on the form

$$\mathcal{F} = \{f: [0, \infty) \rightarrow \{0, 1\} \text{ s.t. } f(z) = 1_{[0, t]}(z) \text{ for } t \in [0, T]\},$$

implying that $\mathbb{G}^{(n)}f = \sqrt{n}(\frac{1}{n} \sum_{i=1}^n 1_{(Z_i \leq t)} - F(t))$, for $f = 1_{[0, t]}$, and we may rewrite weak convergence as

$$\mathbb{G}^{(n)} = \sqrt{n}(\mathbb{P}^{(n)} - L) \xrightarrow{d} \mathbb{G} \quad \text{in } \ell^\infty([0, T]),$$

meaning that the t -indexed stochastic process $(\mathbb{G}^{(n)}(t))_{t \in [0, T]}$ converges weakly to a pathwise bounded process. $\circ$

Definition 1.16 (Bracketing number) Given two functions l and u the bracket $[l, u]$ is the set of all functions f with $l \leq f \leq u$. An ε -bracket is a bracket with $\|u - l\| < \varepsilon$ where $\|\cdot\|$ is some norm. The bracketing number $N_{[]}(\varepsilon, \mathcal{F}, \|\cdot\|)$ is the minimum number of ε -brackets needed to cover $\mathcal{F}$. The entropy (with bracketing) is defined as $\log N_{[]}(\varepsilon, \mathcal{F}, \|\cdot\|)$.

The bracketing number increases as ε decreases. Intuitively speaking, the larger the entropy, the more the class of functions wiggles, and the more difficult it becomes to establish limit theorems. The precise size is quantified in the theorems below, collectively referred to as entropy theorems.

The next three results form a progression. Finite L^1 -bracketing controls uniform laws, finite L^2 entropy integral yields Donsker limits, and Theorem 1.20 extends the same logic to changing classes, where n -dependent functions allow us to extend from the iid case to arrays.

Theorem 1.18 (Glivenko–Cantelli) *A class $\mathcal{F}$ is $\mathbb{P}$ -Glivenko–Cantelli if*

$$N_{[]}(\varepsilon, \mathcal{F}, L^1(\mathbb{P})) < \infty$$

for every $\varepsilon > 0$.

Theorem 1.19 (Bracketing CLT, Donsker’s theorem) *A class $\mathcal{F}$ is $\mathbb{P}$ -Donsker if*

$$J_{[]} (1, \mathcal{F}, L^2(\mathbb{P})) = \int_{(0,1)} \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L^2(\mathbb{P}))} \, d\varepsilon < \infty.$$

The theorem below is stated for wide classes of separable processes. For practical purposes, we only apply this theorem in the following chapters for càdlàg processes indexed by euclidean subsets, which are separable. It is also worth remarking that even separability can be relaxed.

Theorem 1.20 (Bracketing CLT for changing functions) *Let (X_n) be a sequence of iid separable stochastic processes. Define the classes $\mathcal{F}_n = \{f_{n,t} : t \in I\}$ indexed by a Euclidean subspace I . We require a specific norm, so redefine the bracketing number $N_{[]}(\varepsilon \|F_n(X_1)\|_{L^2}, \mathcal{F}_n, L^2(\mathbb{P}))$ as the minimal number of*

sets N_ε in a partition $\mathcal{F}_n = \bigcup_{j=1}^{N_\varepsilon} \mathcal{F}_{\varepsilon_j}^n$ of the index set into sets $\mathcal{F}_{\varepsilon_j}^n$ such that for every one of these sets it holds that

$$\mathbb{E} \left[\sup_{f_{n,t}, f_{n,s} \in \mathcal{F}_{\varepsilon_j}^n} |f_{n,t}(X_1) - f_{n,s}(X_1)|^2 \right] \leq \varepsilon^2 \|F_n(X_1)\|_{L^2}^2,$$

where $F_n = \sup_t |f_{n,t}|$ is an envelope function of $\mathcal{F}^n$. Suppose

$$\mathbb{E} [F_n(X_1)^2] = O(1) \quad (1.6)$$

$$\mathbb{E} [F_n(X_1)^2 1_{(F_n(X_1) > \eta\sqrt{n})}] \rightarrow 0 \quad \text{for every } \eta > 0 \quad (1.7)$$

$$\sup_{\|s-t\| \leq \delta_n} \mathbb{E} [(f_{n,t}(X_1) - f_{n,s}(X_1))^2] \rightarrow 0 \quad \text{for every } \delta_n \rightarrow 0 \quad (1.8)$$

$$\int_{[0, \delta_n]} \sqrt{\log(N_{[\cdot]}(\varepsilon \|F_n(X_1)\|_{L^2}, \mathcal{F}_n, L^2(\mathbb{P})))} d\varepsilon \rightarrow 0 \quad \text{for every } \delta_n \rightarrow 0. \quad (1.9)$$

Then the sequence $(\mathbb{G}^{(n)} f_{n,t})_{f_{n,t} \in \mathcal{F}_n}$ converges in distribution in $\ell^\infty(I)$ to a tight Gaussian process (with a continuous version) provided the sequence of covariance functions $\mathbb{E} [f_{n,t}(X_1) f_{n,s}(X_1)] - \mathbb{E} [f_{n,s}(X_1)] \mathbb{E} [f_{n,t}(X_1)]$ converge pointwise for each $(s, t) \in I^2$. If for each n the partition points (in t) can be chosen independently of n , then the condition in (1.8) is unnecessary.

The following examples illustrate the basic idea behind the entropy theorems. For low-dimensional parametric classes, a simple grid in the parameter space often yields a polynomial bound on the bracketing number.

Example 1.8 (A one-parameter linear class) Let (Z_n) be iid real-valued random variables with law L and assume $\mathbb{E}[Z_1^2] < \infty$. Consider

$$\mathcal{F}_1 = \{f_\theta : \mathbb{R} \rightarrow \mathbb{R} : f_\theta(z) = \theta z, \theta \in [-1, 1]\}.$$

Fix $\varepsilon > 0$ and choose a grid $-1 = \theta_0 < \theta_1 < \dots < \theta_m = 1$ with mesh size at most $\varepsilon / \|Z_1\|_{L^2}$. For $j = 0, \dots, m-1$, define

$$l_j(z) = \min\{f_{\theta_j}(z), f_{\theta_{j+1}}(z)\}, \quad u_j(z) = \max\{f_{\theta_j}(z), f_{\theta_{j+1}}(z)\}.$$

Then every f_θ with $\theta \in [\theta_j, \theta_{j+1}]$ belongs to $[l_j, u_j]$, and

$$\|u_j - l_j\|_{L^2} = (\theta_{j+1} - \theta_j) \|Z_1\|_{L^2} \leq \varepsilon.$$

Hence

$$N_{[\cdot]}(\varepsilon, \mathcal{F}_1, L^2(\mathbb{P})) \leq \left\lceil \frac{2\|Z_1\|_{L^2}}{\varepsilon} \right\rceil.$$

Since $\|g\|_{L^1} \leq \|g\|_{L^2}$, we also have $N_{[\cdot]}(\varepsilon, \mathcal{F}_1, L^1(\mathbb{P})) < \infty$ for every $\varepsilon > 0$, so $\mathcal{F}_1$ is $\mathbb{P}$ -Glivenko–Cantelli by Theorem 1.18. Moreover, a simple calculation shows $J_{[\cdot]}(1, \mathcal{F}_1, L^2(\mathbb{P})) < \infty$ and thus $\mathcal{F}_1$ is even $\mathbb{P}$ -Donsker by Theorem 1.19. $\diamond$

Example 1.9 (A two-parameter affine class) Under the same moment assumption as the previous example, consider

$$\mathcal{F}_2 = \{f_{a,b} : \mathbb{R} \rightarrow \mathbb{R} : f_{a,b}(z) = a + bz, (a, b) \in [-1, 1]^2\}.$$

Fix $\varepsilon > 0$ and choose grids in $[-1, 1]$ with mesh sizes $\Delta_a = \varepsilon/2$ and $\Delta_b = \varepsilon/(2\|Z_1\|_{L^2})$. This partitions $[-1, 1]^2$ into rectangles $R_{jk} = [a_j, a_{j+1}] \times [b_k, b_{k+1}]$. For each rectangle, define

$$l_{jk}(z) = \inf_{(a,b) \in R_{jk}} (a + bz), \quad u_{jk}(z) = \sup_{(a,b) \in R_{jk}} (a + bz).$$

Then every $f_{a,b}$ with $(a, b) \in R_{jk}$ belongs to $[l_{jk}, u_{jk}]$. Since $a + bz$ is linear in (a, b) , the extrema occur at the corners, and therefore

$$u_{jk}(z) - l_{jk}(z) \leq \Delta_a + \Delta_b |z|.$$

Consequently,

$$\|u_{jk} - l_{jk}\|_{L^2} \leq \Delta_a + \Delta_b \|Z_1\|_{L^2} \leq \varepsilon.$$

The number of rectangles is at most $\lceil 2/\Delta_a \rceil \lceil 2/\Delta_b \rceil$, and hence

$$N_{[]}(\varepsilon, \mathcal{F}_2, L^2(\mathbb{P})) \leq \frac{C}{\varepsilon^2}$$

for some constant $C < \infty$. As in the previous example, this implies finite $L^1(\mathbb{P})$ -bracketing numbers for every $\varepsilon > 0$, so $\mathcal{F}_2$ is $\mathbb{P}$ -Glivenko–Cantelli by Theorem 1.18, and the entropy integral in Theorem 1.19 is again finite, hence $\mathcal{F}_2$ is $\mathbb{P}$ -Donsker. $\diamond$

1.4.5 k -nearest neighbours empirical processes

The tools above allow the study of fixed-bandwidth localisation, as is conveniently exploited in the following chapter. However, one can sometimes instead use a k -nearest-neighbour (k -NN) approach, where the neighbourhood is selected by spatial search and therefore adapts automatically to the covariate density. The corresponding local empirical-process CLT can be viewed as a corollary of the bracketing CLT for changing classes, Theorem 1.20, but the proof is technically involved; we therefore refer to the detailed arguments in Portier [2025]. Equivalently, the k -NN rule may be interpreted as a kernel method with a data-dependent bandwidth, when choosing a uniform kernel on the random ball centred at x whose radius is the k -NN distance.

Definition 1.17 (Local k -nearest-neighbour empirical measure) Let $(S, \mathcal{S})$ be a measurable space and let (X_n, Y_n) be iid random elements in $(\mathbb{R}^d \times S, \mathcal{B}(\mathbb{R}^d) \otimes \mathcal{S})$, with $x \in \mathbb{R}^d$ fixed. Let $\mathcal{K}_{n,\kappa}(x)$ be the index set of the κ nearest covariates X_i to x (ties are broken lexicographically). For $A \in \mathcal{S}$, define

$$\hat{\mu}_{n,\kappa,x}(A) = \frac{1}{\kappa} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} 1_{(Y_i \in A)}.$$

For measurable $g: S \rightarrow \mathbb{R}$, write $\hat{\mu}_{n,\kappa,x}(g) = \int g \, d\hat{\mu}_{n,\kappa,x}$.

The next theorem is the local k -NN counterpart of Theorem 1.20, adapted from Portier [2025].

Theorem 1.21 (Bracketing CLT for local k -NN empirical processes) *Let $(S, \mathcal{S})$ be a measurable space. Let (X_n, Y_n) be iid random elements in $(\mathbb{R}^d \times S, \mathcal{B}(\mathbb{R}^d) \otimes \mathcal{S})$, let μ_z denote the regular conditional law of Y given $X = z$, and fix $x \in \mathbb{R}^d$. Assume X has density f_X with $f_X(x) > 0$ and f_X continuous at x . Let $\mathcal{F}$ be a class of measurable functions on S with envelope F , and for $u > 0$ define*

$$\mu_{x,u}(A) = \frac{\mathbb{E}[\mu_X(A)1_{(X \in B(x, u^{1/d}))}]}{\mathbb{E}[1_{(X \in B(x, u^{1/d}))}]}, \quad A \in \mathcal{S}.$$

Here $B(x, r) = \{z \in \mathbb{R}^d : \|z - x\| \leq r\}$. Suppose there exists $\delta > 0$ such that with

$$\rho_\delta(f_1, f_2) = \sup_{|u| \leq \delta} \|f_1 - f_2\|_{L^2(\mu_{x,u})}$$

the following holds:

$$\sup_{|u| \leq \delta} \int_{[0, \delta_n]} \sqrt{\log N_{[]}(\varepsilon \|F\|_{L^2(\mu_{x,u})}, \mathcal{F}, L^2(\mu_{x,u}))} \, d\varepsilon \rightarrow 0 \quad \text{for every } \delta_n \rightarrow 0, \quad (1.10)$$

$$\sup_{|u| \leq \delta} \mu_{x,u}(F^2 1_{(F > M)}) \rightarrow 0 \quad \text{as } M \rightarrow \infty, \quad (1.11)$$

$$\sup_{f \in \mathcal{F}} \min_{1 \leq j \leq m_\eta} \rho_\delta(f, f_j^\eta) \leq \eta \quad \text{for every } \eta > 0 \text{ and some } f_1^\eta, \dots, f_{m_\eta}^\eta \in \mathcal{F}, \quad (1.12)$$

$$z \mapsto (\mu_z(f_1 f_2), \mu_z(f_1), \mu_z(F^2)) \text{ is continuous at } x, \text{ for every } f_1, f_2 \in \mathcal{F}. \quad (1.13)$$

If $\kappa = \kappa_n$ satisfies $\kappa \rightarrow \infty$ and $\kappa/n \rightarrow 0$, then

$$\mathbb{U}_{\kappa,x}^{(n)}(f) := \kappa^{-1/2} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} (f(Y_i) - \mu_{X_i}(f)), \quad f \in \mathcal{F},$$

converges weakly in $\ell^\infty(\mathcal{F})$ to a tight centred Gaussian process $\mathbb{U}_x$ with covariance

$$\text{Cov}(\mathbb{U}_x(f_1), \mathbb{U}_x(f_2)) = \mu_x(f_1 f_2) - \mu_x(f_1)\mu_x(f_2).$$

The next corollary adds a deterministic localisation-bias bound and an explicit rate condition to pass from the centred process to the full local empirical measure.

Corollary 1.2 (From centred to full k -NN empirical-process CLT) *Assume the conditions of Theorem 1.21. Suppose additionally that there are constants $\tau_x, L_x > 0$ such that*

$$\sup_{f \in \mathcal{F}} |(\mu_z - \mu_x)(f)| \leq L_x \|z - x\|, \quad z \in B(x, \tau_x), \quad (1.14)$$

$$\kappa^{(d+2)/2}/n \rightarrow 0. \quad (1.15)$$

Then

$$\{\sqrt{\kappa}(\hat{\mu}_{n,\kappa,x}(f) - \mu_x(f))\}_{f \in \mathcal{F}} \xrightarrow{d} \mathbb{U}_x \quad \text{in } \ell^\infty(\mathcal{F}).$$

As a first elementary example we show how the previous results recover asymptotics for a k -NN regression from a singleton class. It also highlights that, when the response is bounded, the local Lindeberg requirement is automatic.

Example 1.10 (k -NN local regression at a fixed covariate) Let (X_n, Y_n) be iid with values in $\mathbb{R}^d \times \mathbb{R}$, and fix $x \in \mathbb{R}^d$. Write

$$m(z) = \mathbb{E}[Y_1 | X_1 = z], \quad \hat{m}_{n,\kappa}(x) = \frac{1}{\kappa} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} Y_i.$$

Assume first that $|Y_1| \leq B$ a.s. for some $B < \infty$, that $f_X(x) > 0$ with f_X continuous at x , and that $z \mapsto \mathbb{E}[Y_1 | X_1 = z]$ and $z \mapsto \mathbb{E}[Y_1^2 | X_1 = z]$ are continuous at x . Assume also that there exist $\tau_x, L_x > 0$ such that

$$|m(z) - m(x)| \leq L_x \|z - x\|, \quad z \in B(x, \tau_x),$$

and let $\kappa = \kappa_n$ satisfy $\kappa \rightarrow \infty$, and $\kappa^{(d+2)/2}/n \rightarrow 0$. Set $\mathcal{F} = \{g\}$ with $g(y) = y$. Then $\hat{m}_{n,\kappa}(x) = \hat{\mu}_{n,\kappa,x}(g)$, and:

$$\begin{aligned} N_{[]}(\varepsilon \|F\|_{L^2(\mu_{x,u})}, \mathcal{F}, L^2(\mu_{x,u})) &= 1, \quad \varepsilon > 0, \\ \sup_{|u| \leq \delta} \mu_{x,u}(F^2 1_{(F>M)}) &= 0 \quad \text{for } M > B, \end{aligned}$$

with envelope $F \equiv B$. Hence (1.10)–(1.12) are immediate; condition (1.13) is exactly continuity of the conditional first and second moments at x . The boundedness assumption can be weakened; it is enough that for some $\delta_0 > 0$ and $\tau'_x > 0$,

$$\sup_{\|z-x\| \leq \tau'_x} \mathbb{E}[|Y_1|^{2+\delta_0} | X_1 = z] < \infty.$$

Indeed, taking $F(y) = |y|$ and $\delta < (\tau'_x)^d$ gives

$$\sup_{|u| \leq \delta} \mu_{x,u}(F^2 1_{(F>M)}) \leq M^{-\delta_0} \sup_{|u| \leq \delta} \mu_{x,u}(F^{2+\delta_0}) \rightarrow 0,$$

so (1.11) still holds. Therefore Theorem 1.21 gives

$$\kappa^{-1/2} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} (Y_i - m(X_i)) \xrightarrow{d} N(0, \sigma^2(x)),$$

where $\sigma^2(x) = \text{Var}(Y_1 | X_1 = x)$. By (1.14) and (1.15), Corollary 1.2 further yields

$$\sqrt{\kappa}(\hat{m}_{n,\kappa}(x) - m(x)) \xrightarrow{d} N(0, \sigma^2(x)).$$

This is the k -NN analogue of the fixed-point Nadaraya–Watson asymptotic normality, to be treated in the following chapter, with normalisation $\sqrt{\kappa}$ and without explicit bandwidth. $\diamond$

The next example treats discrete responses through a finite class of indicator functions. This gives, in one step, binary, categorical and multinomial asymptotics together with the familiar conditional multinomial covariance structure.

Example 1.11 (Binary, categorical and multinomial k -NN local probabilities) Let (X_n, Y_n) be iid and fix $x \in \mathbb{R}^d$. Assume $f_X(x) > 0$, that f_X is continuous at x , and let $\kappa = \kappa_n$ satisfy $\kappa \rightarrow \infty$ and $\kappa/n \rightarrow 0$.

For the categorical case, suppose $Y_1 \in \{1, \dots, J\}$. Define

$$p_j(z) = \mathbb{P}(Y_1 = j | X_1 = z), \quad \hat{p}_{j,\kappa}(x) = \frac{1}{\kappa} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} 1_{(Y_i=j)}, \quad j = 1, \dots, J.$$

Take $\mathcal{F} = \{g_1, \dots, g_J\}$ with $g_j(y) = 1_{(y=j)}$ and envelope $F \equiv 1$. Then, for every $\varepsilon \in (0, 1)$ and every probability measure Q ,

$$N_{[\cdot]}(\varepsilon \|F\|_{L^2(Q)}, \mathcal{F}, L^2(Q)) \leq J,$$

so (1.10) holds; condition (1.11) is immediate since $F \equiv 1$, and condition (1.12) holds because $\mathcal{F}$ is finite. If $z \mapsto p_j(z)$ is continuous at x for each j , then (1.13) holds as well, since

$$\mu_z(g_j) = p_j(z), \quad \mu_z(g_j g_r) = \mu_z(1_{(Y=j)} 1_{(Y=r)}).$$

Hence, if $\kappa \rightarrow \infty$ and $\kappa/n \rightarrow 0$,

$$\left(\kappa^{-1/2} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} (g_j(Y_i) - \mu_{X_i}(g_j)) \right)_{j=1, \dots, J} \xrightarrow{d} N_J(0, \Sigma_x),$$

with

$$[\Sigma_x]_{jr} = p_j(x) 1_{(j=r)} - p_j(x) p_r(x), \quad 1 \leq j, r \leq J.$$

If, in addition, there exist $\tau_x, L_x > 0$ such that

$$\max_{1 \leq j \leq J} |p_j(z) - p_j(x)| \leq L_x \|z - x\|, \quad z \in B(x, \tau_x),$$

and $\kappa^{(d+2)/2}/n \rightarrow 0$, then Corollary 1.2 gives

$$\sqrt{\kappa}(\hat{p}_{\cdot,\kappa}(x) - p(x)) \xrightarrow{d} N_J(0, \Sigma_x),$$

where $\hat{p}_{\cdot,\kappa}(x) = (\hat{p}_{1,\kappa}(x), \dots, \hat{p}_{J,\kappa}(x))^\top$ and $p(x) = (p_1(x), \dots, p_J(x))^\top$.

The binary case is $J = 2$, yielding

$$\sqrt{\kappa}(\hat{p}_{1,\kappa}(x) - p_1(x)) \xrightarrow{d} N(0, p_1(x)(1 - p_1(x))).$$

For multinomial responses with a fixed number m of trials, write $Y = (Y_1, \dots, Y_J)$ with $\sum_{j=1}^J Y_j = m$, and apply the same argument to $g_j(y) = y_j/m \in [0, 1]$, which is valid because $g_j(Y_1)$ is the within-observation category- j proportion and satisfies $\mathbb{E}[g_j(Y_1) | X_1 = x] = p_j(x)$. The statistical similarity with the categorical case means they estimate the same conditional probability vector, with multinomial observations carrying within-observation replication so the covariance is divided by m (the categorical case corresponding to $m = 1$). One obtains the same limit covariance form, scaled by $1/m$:

$$[\Sigma_x]_{jr} = \frac{1}{m} (p_j(x) 1_{(j=r)} - p_j(x) p_r(x)), \quad 1 \leq j, r \leq J.$$

◇

1.5 Empirical distributions, quantiles and copulas

With the functional theorems in place, we can now transfer them to quantile- and dependence-based summaries that are central to risk measurement. We consider once again the process $t \mapsto \mathbb{F}^{(n)}(t) = n^{-1} \sum_{i=1}^n 1_{(Z_i \leq t)}$. We may extend the proof of Theorem 1.12 to recognise that its finite-dimensional distributions converge to a centred Gaussian process $\mathbb{G}$ with covariance function

$$\text{Cov}(\mathbb{G}(t), \mathbb{G}(s)) = F(s \wedge t) - F(s)F(t),$$

or in other words, $\mathbb{G} = \mathbb{B} \circ F$ a.s. where $\mathbb{B}$ is a standard Brownian bridge⁵. We can now assert the full pathwise convergence.

Theorem 1.22 (Pathwise convergence of the empirical distribution function) *Let (Z_n) be an iid sequence of positive random variables with distribution function F . Then*

$$\sqrt{n}(\mathbb{F}^{(n)} - F) \xrightarrow{d} \mathbb{B} \circ F \quad \text{in } \ell^\infty([0, \infty)).$$

Moreover, consider another sample (W_n) and denote the joint empirical and true distribution functions for the pair (Z_n, W_n) respectively by $\mathbb{H}^{(n)}$ and H . Then

$$\sqrt{n}(\mathbb{H}^{(n)} - H) \xrightarrow{d} \mathbb{G}_H \quad \text{in } \ell^\infty([0, \infty)^2)$$

with $\mathbb{G}_H$ a tight Gaussian process satisfying

$$\text{Cov}(\mathbb{G}_H(s, t), \mathbb{G}_H(u, v)) = H(s \wedge u, t \wedge v) - H(s, t)H(u, v).$$

Proof. Given that we already have recognised the limit process, we simply apply the Bracketing CLT, Theorem 1.19. Define $\mathcal{F} = \{f: [0, \infty) \rightarrow \{0, 1\} \text{ s.t. } f(z) = 1_{[0, t]}(z) \text{ for } t \in [0, \infty)\}$. We calculate the bracketing number. Consider for $j = 1, \dots, k$ brackets on the form $l_j(x) = 1_{[0, t_{j-1})}(x)$, $u_j(x) = 1_{[0, t_j]}(x)$ for sequences on the form $0 = t_0 < t_1 < \dots < t_k = \infty$. Let $\mathcal{F}_j = \{f \in \mathcal{F} : l_j \leq f \leq u_j\}$. Then clearly we have the partition $\mathcal{F} = \bigcup_{j=1}^k \mathcal{F}_j$. Let $\varepsilon > 0$ be given, and pick a sequence for $k \leq \lceil 2/\varepsilon^2 \rceil$ such that

$$F(t_j -) - F(t_{j-1}) < \varepsilon^2 \quad \text{for } 1 \leq j \leq k,$$

which is possible because F is a distribution function, in particular, it is càdlàg, non-decreasing and bounded by 1. Then this sequence covers all of $\mathcal{F}$ with brackets satisfying

$$\|u_j(Z_1) - l_j(Z_1)\|_{L^2} = (F(t_j -) - F(t_{j-1}))^{1/2} < \varepsilon.$$

Consequently $N_{[]}(\varepsilon, \mathcal{F}, L^2(\mathbb{P})) \leq k \leq \lceil 2/\varepsilon^2 \rceil$. We then obtain

$$\int_{(0,1)} \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L^2(\mathbb{P}))} \, d\varepsilon \leq \int_{(0,1)} \sqrt{\log(2/\varepsilon^2 + 1)} \, d\varepsilon < \infty,$$

which shows that $\mathcal{F}$ is $\mathbb{P}$ -Donsker. The second part is similar and left for the reader. $\square$

⁵A Brownian motion and Brownian bridge are centred stochastic processes on $\mathbb{R}$ and $[0, 1]$, respectively. Their covariance functions are given by $(s, t) \mapsto t \wedge s$ and $(s, t) \mapsto t \wedge s - ts$, respectively.

Quantiles are crucial in insurance for assessing risk, and they are related to the concept of Value at Risk (VaR). Indeed the VaR at a confidence level $\alpha \in (0, 1)$ is defined simply as the α -quantile of the loss distribution. Formally, for a loss random variable Z , the VaR at level α is given by:

$$\text{VaR}_\alpha(Z) = \inf\{z \in \mathbb{R} \mid \mathbb{P}(Z \leq z) \geq \alpha\} = F^{-1}(\alpha).$$

In other words, $\text{VaR}_\alpha(Z)$ represents the maximum loss not exceeded with a confidence level of α . As a risk measure VaR possesses several important properties:

1. *Monotonicity*: if $Z_1 \leq Z_2$ almost surely, then $\text{VaR}_\alpha(Z_1) \leq \text{VaR}_\alpha(Z_2)$.
2. *Translation invariance*: for any constant $c \in \mathbb{R}$, $\text{VaR}_\alpha(Z+c) = \text{VaR}_\alpha(Z) + c$.
3. *Homogeneity*: for any positive scalar $\lambda > 0$, $\text{VaR}_\alpha(\lambda Z) = \lambda \text{VaR}_\alpha(Z)$.
4. *Subadditivity for elliptical distributions*⁶: for loss variables Z_1 and Z_2 , and $\alpha \geq 1/2$ we have $\text{VaR}_\alpha(Z_1 + Z_2) \leq \text{VaR}_\alpha(Z_1) + \text{VaR}_\alpha(Z_2)$.

For instance, Solvency II requires that pensions fulfil the Solvency Capital Requirement that enough capital is held to withstand a “200-year event”. The loss of such 200-year event can be estimated as the 99.5% upper quantile of the total loss.

Given the widespread use of VaR in the insurance industry, in particular as one of the gold standard risk measures, finding limit results for nonparametric estimators for the quantile process, which considers all quantiles simultaneously, is of considerable interest. Generally speaking, nonparametric methods provide a flexible and robust approach to estimating risk without relying on specific distributional assumptions. This is true for medium and large quantiles using the method below. However, it should also be noted that for estimating *very high quantiles*, other nonparametric tools from Extreme Value Theory are better suited.

Theorem 1.23 (Pathwise convergence of the quantile process) *Let $0 < p < q < 1$. Let (Z_n) be an iid sequence of real-valued random variables such that their distribution function F is continuously differentiable on $[a, b] = [F^{-1}(p) - \varepsilon, F^{-1}(q) + \varepsilon]$ for some $\varepsilon > 0$ (which may span the support of F if compact) with strictly positive derivative f . Then*

$$\sqrt{n}([F^{(n)}]^{-1} - F^{-1}) \xrightarrow{d} -\frac{\mathbb{B}}{f \circ F^{-1}} \quad \text{in } \ell^\infty([p, q]).$$

Proof. By the first part of Lemma 1.10 we know that the inverse map is Hadamard differentiable tangentially to the space of functions that are continuous on $[a, b]$,

⁶Distributions with $\mathbb{E}[e^{i\langle t, (Z_1, \dots, Z_r) - \mu \rangle}] = \psi(t\Sigma t^\top)$ for every $t \in \mathbb{R}^r$, for some location parameter μ , some nonnegative-definite matrix Σ and some scalar function ψ . For instance, Gaussian random vectors.

and hence we may apply the functional δ -method, Theorem 1.17, to the already established convergence from Theorem 1.22. This yields that

$$\sqrt{n}([\mathbb{F}^{(n)}]^{-1} - F^{-1}) \xrightarrow{d} -\frac{(\mathbb{B} \circ F)}{f} \circ F^{-1} \quad \text{in } \ell^\infty([p, q]),$$

as desired. $\square$

Quantile limits immediately deliver distribution theory for robust scale; the interquartile range is the prototypical case.

Example 1.12 (Asymptotics of the interquartile range) Let $q_p = F^{-1}(p)$ and $\hat{q}_p = [\mathbb{F}^{(n)}]^{-1}(p)$. Under the assumptions of Theorem 1.23 with $p = 1/4$ and $q = 3/4$, we have

$$\sqrt{n}(\hat{q}_p - q_p) \xrightarrow{d} -\frac{\mathbb{B}(p)}{f(q_p)},$$

so the vector $\sqrt{n}(\hat{q}_{1/4} - q_{1/4}, \hat{q}_{3/4} - q_{3/4})$ is asymptotically normal with covariance matrix

$$\Sigma = \begin{pmatrix} \frac{(1/4)(3/4)}{f(q_{1/4})^2} & \frac{(1/4)(1-3/4)}{f(q_{1/4})f(q_{3/4})} \\ \frac{(1/4)(1-3/4)}{f(q_{1/4})f(q_{3/4})} & \frac{(1/4)(3/4)}{f(q_{3/4})^2} \end{pmatrix},$$

since $\text{Cov}(\mathbb{B}(p_1), \mathbb{B}(p_2)) = p_1 \wedge p_2 - p_1 p_2$. The interquartile range is $\text{IQR} = q_{3/4} - q_{1/4}$ and $\widehat{\text{IQR}} = \hat{q}_{3/4} - \hat{q}_{1/4}$, so by the continuous mapping theorem,

$$\sqrt{n}(\widehat{\text{IQR}} - \text{IQR}) \xrightarrow{d} N(0, \sigma_{\text{IQR}}^2),$$

with

$$\sigma_{\text{IQR}}^2 = \frac{3}{16} \left(\frac{1}{f(q_{3/4})^2} + \frac{1}{f(q_{1/4})^2} \right) - \frac{1}{8} \frac{1}{f(q_{1/4})f(q_{3/4})}.$$

$\diamond$

We now examine the Lorenz curve, Gini coefficient, and expected shortfall, which are defined as integral functionals of the quantile process.

Example 1.13 (Lorenz curve and Gini coefficient) Let $Z \geq 0$ with mean $\mu = \mathbb{E}[Z] \in (0, \infty)$ and distribution function F with strictly positive density f . Define the quantile function $q_p = F^{-1}(p)$. Fix $0 < \delta < 1/2$ and assume that f is continuous and strictly positive on $[q_\delta, q_{1-\delta}]$. The *trimmed* Lorenz curve is

$$L_\delta(p) = \frac{1}{\mu_\delta} \int_\delta^p q_u \, du, \quad p \in [\delta, 1-\delta], \quad \mu_\delta = \int_\delta^{1-\delta} q_u \, du.$$

Let $\hat{q}_p$ be the empirical quantile and $\hat{\mu}_\delta = \int_\delta^{1-\delta} \hat{q}_u \, du$. By Theorem 1.23,

$$\sqrt{n}(\hat{q} - q) \xrightarrow{d} G \quad \text{in } \ell^\infty([\delta, 1-\delta]), \quad G(p) = -\frac{\mathbb{B}(p)}{f(q_p)}.$$

The map $q \mapsto \int_{\delta}^{(\cdot)} q_u \, du$ is Hadamard differentiable by Lemma 1.11, and the quotient map is Hadamard differentiable on $(0, \infty)^2$. Hence, by the functional δ -method,

$$\sqrt{n}(\widehat{L}_{\delta}(p) - L_{\delta}(p)) \xrightarrow{d} \frac{1}{\mu_{\delta}} \int_{\delta}^p G(u) \, du - \frac{L_{\delta}(p)}{\mu_{\delta}} \int_{\delta}^{1-\delta} G(u) \, du, \quad p \in [\delta, 1 - \delta].$$

Define the trimmed Gini coefficient by

$$G_{\delta} = 1 - \frac{2}{1 - 2\delta} \int_{\delta}^{1-\delta} L_{\delta}(p) \, dp, \quad \widehat{G}_{\delta} = 1 - \frac{2}{1 - 2\delta} \int_{\delta}^{1-\delta} \widehat{L}_{\delta}(p) \, dp.$$

Another application of Lemma 1.11 yields

$$\sqrt{n}(\widehat{G}_{\delta} - G_{\delta}) \xrightarrow{d} -\frac{2}{1 - 2\delta} \int_{\delta}^{1-\delta} \left(\frac{1}{\mu_{\delta}} \int_{\delta}^p G(u) \, du - \frac{L_{\delta}(p)}{\mu_{\delta}} \int_{\delta}^{1-\delta} G(u) \, du \right) dp,$$

which is mean-zero normal. If additional regularity holds near $p = 0$ and $p = 1$ (for instance f is bounded away from 0 on a compact support), one may let $\delta \downarrow 0$ and recover limit results for the untruncated Lorenz and Gini definitions. $\diamond$

Remark 1.8 The quantile process limit in Theorem 1.23 is only guaranteed on compact subintervals of $(0, 1)$. Near the endpoints, the factor $1/f(q_p)$ may blow up when $f(q_p)$ is small or vanishes (as often happens for distributions on $[0, \infty)$). The trimmed versions avoid this problem; as mentioned above, extending to $\delta = 0$ requires extra endpoint regularity. $\circ$

Example 1.14 (Expected shortfall) Let $0 < \alpha < 1$ and define the *trimmed* expected shortfall

$$\text{ES}_{\alpha, \delta}(Z) = \frac{1}{1 - \alpha - \delta} \int_{\alpha}^{1-\delta} q_u \, du, \quad 0 < \delta < 1 - \alpha,$$

with estimator $\widehat{\text{ES}}_{\alpha, \delta} = (1 - \alpha - \delta)^{-1} \int_{\alpha}^{1-\delta} \widehat{q}_u \, du$. Under the assumptions of Theorem 1.23 on $[\alpha, 1 - \delta]$, the map $q \mapsto \int_{\alpha}^{(\cdot)} q_u \, du$ is Hadamard differentiable by Lemma 1.11, and thus

$$\sqrt{n}(\widehat{\text{ES}}_{\alpha, \delta} - \text{ES}_{\alpha, \delta}) \xrightarrow{d} \frac{1}{1 - \alpha - \delta} \int_{\alpha}^{1-\delta} G(u) \, du, \quad G(u) = -\frac{\mathbb{B}(u)}{f(q_u)}.$$

If additional endpoint regularity holds so that the quantile process extends to $[\alpha, 1)$ and $f(q_u)^{-1}$ is integrable near $u = 1$, one may let $\delta \downarrow 0$ and recover the usual expected shortfall ES_{α} . $\diamond$

A *copula* is a mathematical function that allows us to study and model the dependence structure between multiple random variables independently of their marginal distributions. Formally, a copula is a multivariate distribution function with uniform marginals on the interval $[0, 1]$. According to Sklar's Theorem, for any multivariate distribution function H with continuous marginals $F_1, F_2, \dots, F_r$, there exists a copula $C : [0, 1]^r \rightarrow [0, 1]$ such that

$$H(x_1, x_2, \dots, x_r) = C(F_1(x_1), F_2(x_2), \dots, F_r(x_r)).$$

Conversely, if C is a copula and $F_1, F_2, \dots, F_r$ are distribution functions, then H defined as above is a multivariate distribution function with marginals $F_1, \dots, F_r$.

In risk management and insurance, copulas are commonly used for their ability to model dependencies between different risk factors. Some fundamental properties of copulas include:

1. *Uniform marginals:* for a copula $C(u_1, u_2, \dots, u_r)$, the marginal distribution of each variable is uniform on $[0, 1]$.
2. *Grounded and monotone:* a copula is grounded, meaning $C(u_1, \dots, u_r) = 0$ if any $u_i = 0$. It is also r -increasing, that is C is non-decreasing in each argument.
3. *Invariant under strictly increasing transformations:* the copula remains unchanged under strictly increasing transformations of the marginal distributions.
4. *Bounds:* copulas are bounded by the Fréchet–Hoeffding bounds, where the lower bound is $\max(\sum_{i=1}^r u_i - r + 1, 0)$ and the upper bound is given by $\min(u_1, u_2, \dots, u_r)$.

Many parametric copula functions exist, and we refer the reader to [Nelsen \[2006\]](#) for general methods on how to construct parametric copulas. We now define a nonparametric estimator of a bivariate copula, i.e. for $r = 2$ and with marginals F_1 and F_2 , referred to as the *empirical copula* function:

$$\mathbb{C}^{(n)}(u, v) = \mathbb{H}^{(n)}([\mathbb{F}_1^{(n)}]^{-1}(u), [\mathbb{F}_2^{(n)}]^{-1}(v)).$$

Theorem 1.24 (Convergence of the empirical copula process) *Let $0 < p < q < 1$. Let (Z_n, W_n) be iid vectors with joint distribution function H and marginal distribution functions F_1 and F_2 . Assume the latter are continuously differentiable on $[a, b] = [F_1^{-1}(p) - \varepsilon, F_1^{-1}(q) + \varepsilon]$ and $[c, d] = [F_2^{-1}(p) - \varepsilon, F_2^{-1}(q) + \varepsilon]$ with positive derivatives f_1 and f_2 , respectively, for some $\varepsilon > 0$. Additionally, assume that $\partial H / \partial x, \partial H / \partial y$ are continuous on the product of the intervals. Then*

$$\sqrt{n}(\mathbb{C}^{(n)} - C) \xrightarrow{d} \mathbb{G}_C \quad \text{in } \ell^\infty([a, b] \times [c, d]).$$

where with $\mathbb{G}_H$ as in Theorem 1.22 we have

$$\begin{aligned} \mathbb{G}_C &= \mathbb{G}_H(F_1^{-1}, F_2^{-1}) \\ &- \frac{\partial H}{\partial x}(F_1^{-1}, F_2^{-1}) \frac{\mathbb{G}_H(F_1^{-1}, \infty)}{f_1(F_1^{-1})} - \frac{\partial H}{\partial y}(F_1^{-1}, F_2^{-1}) \frac{\mathbb{G}_H(F_2^{-1}, \infty)}{f_2(F_2^{-1})}. \end{aligned}$$

Proof. Notice that we may construct the copula map as a composition of Hadamard differentiable maps. Indeed, define the maps

$$\begin{aligned} \phi(H)(x, y) &= (H(x, y), H(x, \infty), H(\infty, y)) \\ \psi(H, F_1, F_2)(x, y) &= (H(x, y), F_1^{-1}(x), F_2^{-1}(y)) \\ \nu(H, F_1^{-1}, F_2^{-1})(x, y) &= H \circ (F_1^{-1}(x), F_2^{-1}(y)), \end{aligned}$$

with Hadamard derivatives given by

$$\begin{aligned}\phi'(a)(x, y) &= (a(x, y), a(x, \infty), a(\infty, y)) \\ \psi'(a, b, c)(x, y) &= (a(x, y), -(b/f_1) \circ F_1^{-1}(x), -(c/f_2) \circ F_2^{-1}(y)) \\ \nu'(a, b, c)(x, y) &= a \circ (F_1^{-1}(x), F_2^{-1}(y)) \\ &\quad + (\partial H/\partial x, \partial H/\partial y) \circ (F_1^{-1}, F_2^{-1})(x, y)(b, c)^\top,\end{aligned}$$

where we have used the second, first, third, and second statements in Lemma 1.10 for the first, second, and third maps, respectively (one can also apply the definition directly to see that these are indeed the derivatives). Thus, by the chain rule, Theorem 1.16, we obtain that $\nu \circ \psi \circ \phi$ is Hadamard differentiable at H with derivative

$$\begin{aligned}\nu'_{\psi \circ \phi(H)} \circ \psi'_{\phi(H)} \circ \phi'_H(h)(u, v) &= h(F_1^{-1}(u), F_2^{-1}(v)) \\ &\quad + \left\langle (\partial H/\partial x, \partial H/\partial y) \circ (F_1^{-1}, F_2^{-1})(u, v), \left(-\frac{h(F_1^{-1}(u), \infty)}{f_1(F_1^{-1}(u))}, -\frac{h(F_2^{-1}(v), \infty)}{f_2(F_2^{-1}(v))} \right) \right\rangle,\end{aligned}$$

so that we may apply the functional δ -method, Theorem 1.17, to Theorem 1.22 to obtain the desired result. $\square$

Numeric example 1.2 (Empirical copula approximation) In Figure 1.5 we illustrate how sample size improves both the empirical copula and the associated empirical copula process. The copula considered is the Gaussian copula, constructed as per $C(u, v) = H(F_1^{-1}(u), F_2^{-1}(v))$. More precisely, the right panels display the centred and scaled field

$$(u, v) \mapsto \sqrt{n}(\mathbb{C}^{(n)}(u, v) - C(u, v)),$$

that is, the process appearing in Theorem 1.24. For $n = 100$, the empirical copula surface still exhibits visible sampling roughness, and the empirical-process surface does not yet resemble a typical realisation of the Gaussian limit particularly well. For $n = 10,000$, both visual approximations are substantially more stable. The true copula is not depicted, since it is virtually indistinguishable from the $n = 10,000$ approximation to the eye. $\diamond$

1.6 Stochastic simulation

We have already encountered some simulation studies, without providing details on *how* to simulate. Although stochastic simulation is not, strictly speaking, a topic of statistics for insurance, it is an indispensable tool throughout modern actuarial work, and throughout this book. It is the foundation for parametric bootstrap methods, enables scenario generation for both random variables and stochastic processes, and makes it possible to approximate integrals and probabilities when closed-form expressions are unavailable or intractable. Below, we give a very basic treatment, and leave more sophisticated methods to the chapters which require them.

At the root of essentially every simulation method lies the ability to generate values that behave like independent draws from the standard uniform distribution. In practice these numbers are produced by deterministic pseudorandom

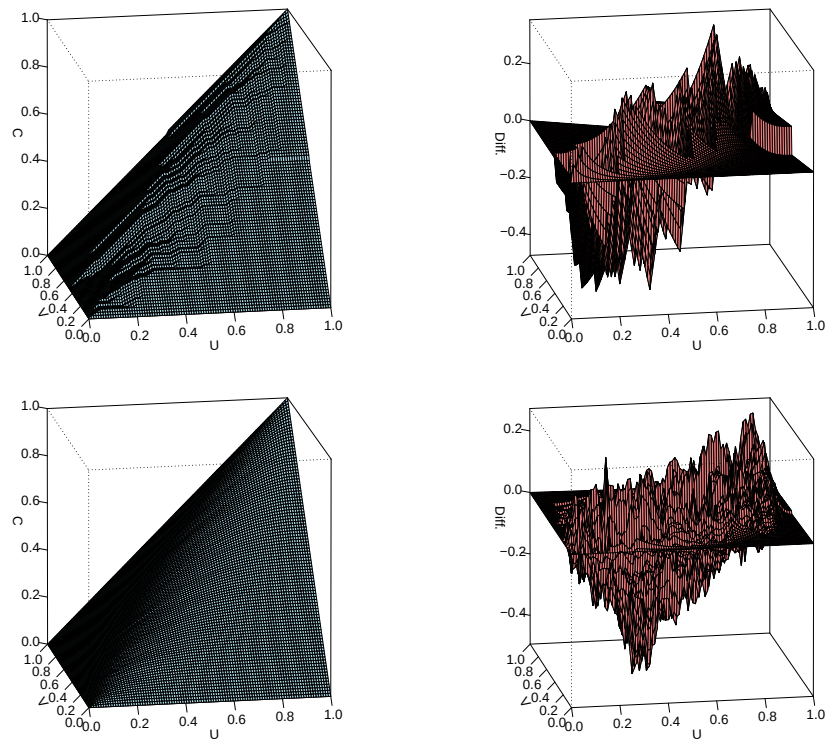


Figure 1.5: Empirical copulas (left) and their associated empirical processes (right) for simulated Gaussian vectors (correlation $\rho = 0.8$) of sample size $n = 100$ (top) and $n = 10,000$ (bottom).

number generators, designed so that their output has good statistical properties and appears random for computational purposes. Here we adopt the standard working assumption that the underlying random number generator provides independent and identically distributed samples from $\text{Unif}(0, 1)$, and we build subsequent simulation algorithms from that starting point.

Discrete random variables are then immediate to simulate. If Z is discrete and takes values $x_1, \dots, x_N$ with probabilities $p_1, \dots, p_N$, define cumulative sums $c_j = \sum_{i=1}^j p_i$ with $c_0 = 0$ and $c_N = 1$. Draw $U \sim \text{Unif}(0, 1)$ and return $Z = x_j$ whenever $c_{j-1} < U \leq c_j$. In other words, we split $(0, 1)$ into intervals of lengths $p_1, \dots, p_N$ and map each interval to its corresponding value.

For continuous random variables Z with density f , this direct interval partition is no longer available, and three general-purpose methods are common in practice: inversion, acceptance-rejection, and Fourier inversion. Beyond these, many specialised simulation routines exploit specific structural properties of the distribution of Z . We present theory for univariate laws. For multivariate laws, the most general direct construction is to factor the joint law into successive conditional laws and simulate componentwise (known as the Rosenblatt construction); when these conditional laws are not tractable, one typically turns to more specialised case-by-case techniques.

1.6.1 Inversion method

The inversion method is the most simple and robust and general-purpose simulation technique. Its caveat is that computing the inverse of a function might be numerically costly.

Definition 1.18 (Generalised inverse) Let F be a distribution function on $\mathbb{R}$. Its generalised inverse is

$$F^{-1}(u) = \inf\{y \in \mathbb{R} : F(y) \geq u\}, \quad u \in (0, 1).$$

Proposition 1.3 (Generalised inverse and inversion) Let F be a distribution function and let F^{-1} be its generalised inverse.

1. For $u \in (0, 1)$ and $z \in \mathbb{R}$,

$$u \leq F(z) \quad \Leftrightarrow \quad F^{-1}(u) \leq z.$$

2. If $U \sim \text{Unif}(0, 1)$, then $F^{-1}(U)$ has distribution function F .
3. If Z has distribution function F and F is continuous, then $F(Z) \sim \text{Unif}(0, 1)$.

Proof. The equivalence in the first statement follows directly from the definition

$$F^{-1}(u) = \inf\{y \in \mathbb{R} : F(y) \geq u\},$$

since $\{y : F(y) \geq u\} = [F^{-1}(u), \infty)$. Therefore, for any $z \in \mathbb{R}$,

$$\mathbb{P}(F^{-1}(U) \leq z) = \mathbb{P}(U \leq F(z)) = F(z),$$

which proves that $F^{-1}(U)$ has cdf F . Now assume $Z \sim F$ and that F is continuous. For $u \in (0, 1)$,

$$\mathbb{P}(F(Z) \geq u) = \mathbb{P}(Z \geq F^{-1}(u)).$$

By continuity of F , $\mathbb{P}(Z \geq a) = \mathbb{P}(Z > a) = 1 - F(a)$ for every a , hence

$$\mathbb{P}(F(Z) \geq u) = 1 - F(F^{-1}(u)).$$

Now $F(F^{-1}(u)) \geq u$ by definition of F^{-1} . If $y_n < F^{-1}(u)$ and $y_n \uparrow F^{-1}(u)$, then $F(y_n) < u$ for all n , and continuity gives $F(F^{-1}(u)) \leq u$. Therefore $F(F^{-1}(u)) = u$, so

$$\mathbb{P}(F(Z) \geq u) = 1 - u, \quad u \in (0, 1),$$

which is equivalent to $F(Z) \sim \text{Unif}(0, 1)$. $\square$

In simulation, the key statement is the second one: a random variable with cdf F is obtained by generating $U \sim \text{Unif}(0, 1)$ and setting $Z = F^{-1}(U)$.

Example 1.15 (Exponential building blocks and inversion) Let $F(z) = 1 - \exp(-\lambda z)$, $z \geq 0$, be the exponential cdf with rate $\lambda > 0$. Its inverse is

$$F^{-1}(u) = -\frac{\log(1-u)}{\lambda}, \quad u \in (0, 1),$$

so if $U \sim \text{Unif}(0, 1)$ then

$$Z = -\frac{\log(1-U)}{\lambda} \sim \text{Exp}(\lambda).$$

Since $1 - U \stackrel{d}{=} U$, one may equivalently use $Z = -\log(U)/\lambda$. This basic construction is often exploited to generate richer laws. For instance, if $E_1, \dots, E_p$ are independent $\text{Exp}(\lambda)$ random variables and $p \in \mathbb{N}$, then

$$G_p = \sum_{i=1}^p E_i \sim \text{Gamma}(p, \lambda).$$

Likewise, if (E_n) are iid $\text{Exp}(\lambda)$ and

$$N = \max \left\{ n \geq 0 : \sum_{i=1}^n E_i \leq 1 \right\},$$

then $N \sim \text{Poisson}(\lambda)$. Indeed, this N is the unit-time count of the process with iid exponential inter-arrival times, hence of a Poisson process with rate λ . Writing $E_i = -\log(U_i)/\lambda$ with iid $U_i \sim \text{Unif}(0, 1)$ gives the equivalent product representation

$$N = \max \left\{ n \geq 0 : \prod_{i=1}^n U_i > e^{-\lambda} \right\}.$$

At the same time, direct inversion is available for both families. One may generate

$$G_p = F_\Gamma^{-1}(U), \quad N = F_{\text{Pois}}^{-1}(U),$$

where F_Γ and F_{Pois} denote the Gamma and Poisson cdf, respectively. For Gamma this means numerical inversion of the Gamma cdf, while for Poisson it is the smallest $m \in \mathbb{N}_0$ such that

$$\sum_{k=0}^m e^{-\lambda} \frac{\lambda^k}{k!} \geq U.$$

The broader message is that simulation can proceed either by direct inversion of the target cdf or by exploiting probabilistic identities that transform simpler random inputs into the target law. $\diamond$

1.6.2 Acceptance-rejection

We now turn to the second general-purpose method. Let f be a target density and let g be a proposal density that is easy to simulate, with

$$f(z) \leq Cg(z), \quad z \in \mathbb{R},$$

for some finite constant $C \geq 1$. The method repeatedly draws $Y \sim g$ and $U \sim \text{Unif}(0, 1)$, independently, and accepts the proposal whenever

$$U \leq \frac{f(Y)}{Cg(Y)}.$$

If the proposal is rejected, one draws a new pair and repeats until acceptance. The next result guarantees that such a method correctly samples from f .

Proposition 1.4 (Acceptance-rejection) *Let (Y_n, U_n) be iid with $Y_n \sim g$, $U_n \sim \text{Unif}(0, 1)$, and $Y_n \perp U_n$. Define*

$$A_n = \left\{ U_n \leq \frac{f(Y_n)}{Cg(Y_n)} \right\}, \quad T = \inf\{n \geq 1 : A_n \text{ occurs}\}, \quad Z = Y_T.$$

Then Z has density f .

Proof. Set $p = \mathbb{P}(A_1)$. Since U_1 is independent of Y_1 ,

$$p = \mathbb{E}\left[\mathbb{P}\left(U_1 \leq \frac{f(Y_1)}{Cg(Y_1)} \mid Y_1\right)\right] = \mathbb{E}\left[\frac{f(Y_1)}{Cg(Y_1)}\right] = \frac{1}{C} \int f(z) \, dz = \frac{1}{C}.$$

Now let $B \subseteq \mathbb{R}$ be measurable. Then

$$\begin{aligned} \mathbb{P}(Z \in B) &= \sum_{n=1}^{\infty} \mathbb{P}(T = n, Y_n \in B) = \sum_{n=1}^{\infty} \mathbb{P}(A_1^c, \dots, A_{n-1}^c, A_n, Y_n \in B) \\ &= \sum_{n=1}^{\infty} (1-p)^{n-1} \mathbb{P}(A_1, Y_1 \in B), \end{aligned}$$

by independence of the iid trials. Furthermore,

$$\mathbb{P}(A_1, Y_1 \in B) = \int_B g(z) \frac{f(z)}{Cg(z)} dz = \frac{1}{C} \int_B f(z) dz = p \int_B f(z) dz.$$

Hence

$$\mathbb{P}(Z \in B) = \sum_{n=1}^{\infty} (1-p)^{n-1} p \int_B f(z) dz = \int_B f(z) dz,$$

which proves that Z has density f . $\square$

Remark 1.9 (Efficiency) The acceptance probability is

$$\mathbb{P}(A_1) = \mathbb{E} \left[\mathbb{P} \left(U_1 \leq \frac{f(Y_1)}{Cg(Y_1)} \mid Y_1 \right) \right] = \frac{1}{C},$$

so the expected number of proposals per accepted draw equals C . Therefore the method is efficient when C is close to 1, which in turn requires a proposal g that resembles f closely on both body and tail. $\circ$

This following simple construction illustrates the bounded-support case.

Example 1.16 (Beta simulation) Suppose the target is $\text{Beta}(\alpha, \beta)$ on $(0, 1)$ with $\alpha, \beta \geq 1$, so its density

$$f(z) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} z^{\alpha-1} (1-z)^{\beta-1}, \quad 0 < z < 1,$$

is bounded. Let

$$M = \sup_{0 < z < 1} f(z) < \infty, \quad g(z) = 1_{(0,1)}(z),$$

so g is the uniform density on $(0, 1)$ and $f(z) \leq Mg(z)$ on $\mathbb{R}$. Taking $C = M$ gives a valid acceptance-rejection scheme with acceptance probability $1/M$. $\diamond$

On unbounded domains, or when densities are sharply peaked, good proposals are typically chosen by matching concavity and tail behaviour so that C remains moderate. When f itself is also costly to evaluate, the following technique is useful.

Remark 1.10 (Squeezing) When f is expensive to evaluate, one can reduce calls to f by adding simple bounds. Suppose there are simple functions l and u such that

$$0 \leq l(z) \leq \frac{f(z)}{Cg(z)} \leq u(z) \leq 1.$$

Given a proposal Y and a uniform U , accept immediately if $U \leq l(Y)$ and reject immediately if $U > u(Y)$. Only on the intermediate event $l(Y) < U \leq u(Y)$ is the exact ratio $f(Y)/(Cg(Y))$ evaluated. This leaves the target law unchanged while reducing expensive evaluations. $\circ$

1.6.3 Fourier inversion

In some models the distribution function F or density f is not available in explicit form, while a transform is. Let $Z \sim F$ and define the characteristic function

$$\psi(s) = \widehat{F}[is] = \mathbb{E}[e^{isZ}], \quad s \in \mathbb{R}.$$

This permits simulation constructions based directly on Fourier inversion. The first result gives the density representation directly from ψ .

Proposition 1.5 (Fourier inversion) *If $\psi \in L^1(\mathbb{R})$, then Z has a bounded continuous density*

$$f(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \psi(s) e^{-isz} \, ds, \quad z \in \mathbb{R}.$$

Proof. Define f by the displayed integral. Since $\psi \in L^1(\mathbb{R})$, dominated convergence implies that f is bounded and continuous. Moreover, for $a < b$,

$$\int_a^b f(z) \, dz = \lim_{M \rightarrow \infty} \frac{1}{2\pi} \int_{-M}^M \psi(s) \frac{e^{-isa} - e^{-isb}}{is} \, ds.$$

By the Lévy inversion formula, the right-hand side equals $F(b) - F(a)$ at continuity points a, b of F . Hence the finite measure with density f agrees with F on intervals $(a, b]$ and therefore coincides with F . $\square$

The next example gives the practical approximation route for simulation, and shows how the equidistant-grid case leads to an FFT implementation.

Example 1.17 (Grid approximation and FFT implementation) Assume the density f of Z is continuous and supported on $[a, b]$. For any fixed integer $n \geq 2$, we define an equidistant grid of step size $\epsilon = (b - a)/(n - 1)$ with points

$$z_m = a + (m - 1)\epsilon, \quad m = 1, \dots, n.$$

We construct a discrete random variable $\tilde{Z}_n$ taking values in $\{z_1, \dots, z_n\}$ with probabilities proportional to the density:

$$p_m = \mathbb{P}(\tilde{Z}_n = z_m) = \frac{f(z_m)}{\sum_{j=1}^n f(z_j)}.$$

This discrete variable $\tilde{Z}_n$ approximates Z as $n \rightarrow \infty$. Indeed, for every bounded continuous function h on $[a, b]$, Riemann-sum convergence yields

$$\mathbb{E}[h(\tilde{Z}_n)] = \frac{\sum_{m=1}^n h(z_m) f(z_m) \epsilon}{\sum_{m=1}^n f(z_m) \epsilon} \rightarrow \int_a^b h(z) f(z) \, dz = \mathbb{E}[h(Z)],$$

hence $\tilde{Z}_n \xrightarrow{d} Z$. Typically, such an approximation requires the point values $f(z_m)$ to be evaluated via numerical quadrature (e.g. using Proposition 1.5).

However, because the grid is explicitly equidistant, we can significantly accelerate this computation using the Fast Fourier Transform (FFT) applied directly

to the characteristic function $\psi(s) = \mathbb{E}[e^{isZ}]$. Let $w = e^{2\pi i/n}$ and define the discrete frequencies $s_r = 2\pi(r-1)/(n\epsilon)$ for $r = 1, \dots, n$. The Discrete Fourier Transform (DFT) of the probability vector (p_m) is given by

$$\hat{p}_r = \frac{1}{n} \sum_{m=1}^n p_m w^{(r-1)(m-1)} = \frac{e^{-is_r z_1}}{n} \sum_{m=1}^n p_m e^{is_r z_m}.$$

Recognising the sum as a discrete approximation to the characteristic function $\psi(s_r)$, we can bypass the direct computation of $f(z_m)$ by defining proxy Fourier coefficients:

$$\hat{q}_r = \frac{e^{-is_r z_1}}{n} \psi(s_r).$$

Applying the inverse transform to $(\hat{q}_r)$ yields candidate probability masses $\tilde{p}_m$:

$$\tilde{p}_m = \sum_{r=1}^n \hat{q}_r w^{-(r-1)(m-1)}.$$

To correct for minor numerical artifacts and ensure a valid distribution, we take the absolute values and re-normalise:

$$p_m^* = |\tilde{p}_m|, \quad p_m = \frac{p_m^*}{\sum_{j=1}^n p_j^*}.$$

The resulting probability vector (p_m) over the grid (z_m) allows the approximate simulation of Z to be reduced to standard discrete sampling. $\diamond$

The final result gives an exact transform-based acceptance-rejection route, avoiding grid discretisation.

Proposition 1.6 (Transform bounds for acceptance-rejection) *Assume $\psi, \psi', \psi'' \in L^1(\mathbb{R})$. Let f be the density of Z , and define*

$$c = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\psi(s)| \, ds, \quad k = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\psi''(s)| \, ds.$$

Then

$$f(z) \leq c \wedge \frac{k}{z^2}, \quad z \neq 0.$$

Consequently, with $b = \sqrt{k/c}$ and $A = 4\sqrt{kc}$, the density

$$g(z) = \frac{1}{A} (c 1_{(|z| \leq b)} + k z^{-2} 1_{(|z| > b)})$$

yields the valid acceptance-rejection envelope $f(z) \leq Ag(z)$.

Proof. From Proposition 1.5,

$$|f(z)| \leq \frac{1}{2\pi} \int_{-\infty}^{\infty} |\psi(s)| \, ds = c.$$

Because $\psi, \psi', \psi'' \in L^1$ we have $\psi(s) \rightarrow 0$ and $\psi'(s) \rightarrow 0$ as $|s| \rightarrow \infty$. Hence for $z \neq 0$, integration by parts gives

$$f(z) = \frac{1}{2\pi iz} \int_{-\infty}^{\infty} \psi'(s) e^{-isz} \, ds,$$

and integrating by parts again gives

$$f(z) = -\frac{1}{2\pi z^2} \int_{-\infty}^{\infty} \psi''(s) e^{-isz} \, ds.$$

Hence

$$|f(z)| \leq \frac{1}{2\pi z^2} \int_{-\infty}^{\infty} |\psi''(s)| \, ds = \frac{k}{z^2}.$$

Therefore $f(z) \leq c \wedge kz^{-2}$ for $z \neq 0$, while $f(0) \leq c$. Now,

$$A = \int_{-\infty}^{\infty} (c \wedge kz^{-2}) \, dz = 2 \int_0^b c \, dz + 2 \int_b^{\infty} kz^{-2} \, dz = 2cb + \frac{2k}{b} = 4\sqrt{kc}.$$

So g is a density. Also, on $|z| \leq b$, one has $Ag(z) = c \geq f(z)$; on $|z| > b$, one has $Ag(z) = kz^{-2} \geq f(z)$. Hence $f \leq Ag$. $\square$

Simulation from g is straightforward. With $b = \sqrt{k/c}$, the density g is a mixture of a uniform law on $(-b, b)$ and a symmetric shifted Pareto component with density $(b/2)z^{-2}1_{(|z|>b)}$. A convenient representation for the tail component is Sb/U , where $S \in \{-1, 1\}$ is symmetric and $U \sim \text{Unif}(0, 1)$. The acceptance-rejection step is then to accept Z with probability

$$\frac{f(Z)}{Ag(Z)},$$

where $A = 4\sqrt{kc}$ and $f(Y)$ is evaluated from Proposition 1.5.

1.7 Notes and Comments

The classical limit results and tools on stochastic convergence have long histories that can be traced back to the development of measure-theoretic probability in the early twentieth century. In [Kolmogoroff \[1933\]](#) the modern axiomatic framework was laid out, within which the principal convergence modes are formulated. The Law of Large Numbers and the Central Limit Theorem both have eighteenth-century origins; for the CLT under broad moment conditions, key proofs are due to [Lyapunov \[1901\]](#) and [Lindeberg \[1922\]](#). The technical tools of asymptotic statistics were developed in parallel, with Slutsky's lemma [[Slutsky, 1925](#)], the Continuous Mapping Theorem [[Mann and Wald, 1943](#)], and the Cramér–Wold device [[Cramér and Wold, 1936](#)] all entering the literature in this period, together with the Glivenko–Cantelli theorem on uniform convergence of the empirical distribution function [[Glivenko, 1933](#), [Cantelli, 1933](#)]. Stochastic order notation was formalised in [Mann and Wald \[1943\]](#), and [Cramér \[1946\]](#) gave a systematic asymptotic treatment that helped make this notation and delta-method arguments standard tools. The classical monograph [Feller \[1971\]](#) offers a thorough probabilistic treatment of these results, while modern references that present these results systematically include [van der Vaart \[1998\]](#), [Billingsley \[1999\]](#), [Kallenberg \[2002\]](#). At the finite-dimensional level, standard references for estimator examples and finite-sample behaviour include [Serfling \[1980\]](#), together with sharp nonasymptotic inequalities for empirical distributions such as [Dvoretzky et al. \[1956\]](#), [Massart \[1990\]](#).

Convergence in law for stochastic processes required further structural tools. Tightness and relative compactness criteria were given by [Prokhorov \[1956\]](#), and almost-sure representations were introduced by [Skorokhod \[1956\]](#), forming the early foundations of weak convergence theory in metric spaces. A complementary development was martingale theory, as provided in [Doob \[1953\]](#), where maximal inequalities, optional-sampling arguments, and martingale convergence theorems became standard tools in stochastic-process limit proofs. The invariance principle of [Chung and Fuchs \[1951\]](#), establishing convergence of rescaled partial-sum processes to Brownian motion, marks the early beginning of functional central limit theorems. Various of these results were unified by [Billingsley \[1999\]](#), whose work largely systematised weak convergence of probability measures on function spaces.

Building on the foundational work, the development of empirical process theory is treated systematically in [Dudley \[1984\]](#), [Pollard \[1984\]](#), [Shorack and Wellner \[2009\]](#), [van der Vaart and Wellner \[1996\]](#). The functional delta method, formulated in terms of Hadamard differentiability, connects these weak convergence results to asymptotic expansions for estimators and test statistics [[van der Vaart, 1998](#), [van der Vaart and Wellner, 1996](#)]. The formulation of weak convergence using outer probability and outer expectation, which elegantly deals with measurability issues, follows Hoffmann–Jørgensen's work [[Hoffmann-Jørgensen, 1974](#), [Hoffmann-Jørgensen and Pisier, 1976](#)] and has become standard in the empirical-process literature. The k -nearest neighbours empirical process treatment is taken from [Portier \[2025\]](#) and builds on the same empirical-process framework as [van der Vaart and Wellner \[1996\]](#). The entropy-theorem block, based on bracketing arguments, fits directly into this same empirical-process line of work, as does the section on empirical distributions and quantiles fol-

lows standard weak-convergence treatments from [Shorack and Wellner \[2009\]](#), [van der Vaart and Wellner \[1996\]](#).

Finally, the simulation subsections are classically covered in random-variates monographs such as [Devroye \[1986\]](#), [Hörmann et al. \[2004\]](#), [Ripley \[1987\]](#), [Robert and Casella \[2004\]](#), with the material being inspired by the treatment of [Asmussen and Glynn \[2007\]](#). Inversion and acceptance-rejection are treated there in depth, while acceptance-rejection traces back to [von Neumann \[1951\]](#), and transform-based inversion dates back to the inversion identity of [Gil-Pelaez \[1951\]](#) and practical numerical inversion methods such as [Abate and Whitt \[1995\]](#).

Beyond the provided material, these theoretical results have found systematic application in statistical methodology, as expanded in subsequent chapters; here we only mention a few directions. Survival analysis and event-history modelling rely heavily on martingale limit theorems and counting-process techniques, as developed in [Andersen et al. \[1993\]](#), [Fleming and Harrington \[2005\]](#). Extreme value theory rests on weak convergence arguments for maxima, with standard references in [Resnick \[1987\]](#), [de Haan and Ferreira \[2006\]](#). Dependence modelling via copulas has led to the study of empirical copula processes; foundational copula references include [Sklar \[1959\]](#), [Nelsen \[2006\]](#), [Joe \[2014\]](#), [Durante and Sempi \[2015\]](#). Other specialized developments include treatment of stochastic-process limits in [Whitt \[2002\]](#), on Markov processes in [Ethier and Kurtz \[1986\]](#), and a semimartingale framework in [Jacod and Shiryaev \[2003\]](#). The rigorous treatment of stochastic simulation methods of [Asmussen and Glynn \[2007\]](#) is used throughout the book.

1.8 Exercises

1. By applying Theorem 1.3 to $-\gamma$, show that we also obtain

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^n\gamma(Z_i)\right|\geq\kappa t+\sqrt{2t}\right)\leq 2e^{-nt}.$$

Equivalently, show that there exists a constant $C > 0$ such that for all $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^n\gamma(Z_i)\right|\geq\varepsilon\right)\leq 2\exp\left(-\frac{Cn\varepsilon^2}{1+\kappa\varepsilon}\right).$$

2. Let (Z_n) be iid with $\mathbb{P}(Z_1 > z) = z^{-\alpha}$ for $z \geq 1$ and $\alpha \in (0, 1)$.

(a) Show that $\mathbb{E}[|Z_1|] = \infty$.

(b) Show that $\limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n Z_i = \infty$, and hence Theorem 1.2 fails without the moment assumption.

Hint: Use Borel–Cantelli with the events $\{Z_n > Mn\}$ for any $M > 0$.

3. Let (Z_n) be iid symmetric with $\mathbb{P}(|Z_1| > z) = z^{-\alpha}$ for $z \geq 1$ and $\alpha \in (1, 2)$. Show that the Lindeberg condition for the normalisation $\sqrt{n}$ fails, by proving that for any fixed $\varepsilon > 0$,

$$n \mathbb{E}[Z_1^2; |Z_1| > \varepsilon\sqrt{n}] \rightarrow \infty.$$

Show that $n^{-1/2} \sum_{i=1}^n Z_i$ cannot converge in distribution to a non-degenerate normal law. Conclude that Theorem 1.6 fails when $\mathbb{E}[Z_1^2] = \infty$.

4. Let $Z_n = Z_1$ almost surely for all n , where Z_1 has a continuous distribution function F . Show that $\mathbb{F}^{(n)}(t) = 1_{(Z_1 \leq t)}$ for all n and hence

$$\sup_{t \in \mathbb{R}} |\mathbb{F}^{(n)}(t) - F(t)| = \max\{F(Z_1), 1 - F(Z_1)\} \geq \frac{1}{2} \quad \text{a.s.}$$

Conclude that the Glivenko–Cantelli theorem fails without independence.

5. Explain why the extended Glivenko–Cantelli Theorem 1.9 cannot be proven using a combination of Theorem 1.7 and Lemma 1.5.

6. Let (Z_n) and (W_n) be $\mathbb{R}^p$ -valued random variables. Show that if $\|Z_n - W_n\| = o_{\mathbb{P}}(1)$ and $Z_n \xrightarrow{d} Z$, then also $W_n \xrightarrow{d} Z$.

Hint: Use the definition of convergence in distribution.

7. Go through the details of Example 1.5, and discuss the practical implications of this result.

8. Prove a multidimensional version of (1.4). That is, for $-\infty < t_1 < t_2 < \dots < t_M < \infty$, $M \in \mathbb{N}$, show that

$$\sqrt{n}(\mathbb{F}^{(n)}(t_1) - F(t_1), \dots, \mathbb{F}^{(n)}(t_M) - F(t_M)) \xrightarrow{d} N(0, \Sigma(t_1, \dots, t_M)),$$

where

$$(\Sigma(t_1, \dots, t_M))_{i,j} = \sigma(t_i, t_j) = F(t_i \wedge t_j) - F(t_i)F(t_j).$$

9. Derive the mean excess function for each of the following distributions:

- (a) If Z is a Pareto-distributed random variable with scale parameter $\kappa > 0$ and shape parameter $\alpha > 0$, then $\bar{F}_Z(z) = (\kappa/z)^\alpha$, $z \geq \kappa$. Show that if $\alpha > 1$ then $e_F(u) = u/(\alpha - 1)$, $u \geq \kappa$, where $e_F(u) = \mathbb{E}[Z - u | Z > u]$ is the mean excess function.
- (b) If Z is log-normally distributed with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, then the distribution function of Z is given by

$$G_Z(z) = \Phi\left(\frac{\log z - \mu}{\sigma}\right), \quad z > 0,$$

where Φ denotes the distribution function for a standard normal random variable. Straightforward calculations yield that

$$e_G(u) = e^{\mu + \frac{\sigma^2}{2}} \frac{\Phi\left(\frac{(\mu + \sigma^2) - \log u}{\sigma}\right)}{\Phi\left(\frac{\mu - \log u}{\sigma}\right)} - u.$$

Show that as $u \rightarrow \infty$,

$$e_G(u) = \frac{\sigma^2 u}{\log u - (\mu + \sigma^2)} (1 + o(1)),$$

where $o(1)$ is understood as $u \rightarrow \infty$.

Hint: Recall Mills' ratio, which states that for $z \rightarrow -\infty$ we have that $\Phi(z) \sim -\phi(z)/z$, where ϕ denotes the density of a standard normal distribution.

- 10. Generalise Proposition 1.1 to independent but not identically distributed risks, using a uniformly bounded condition as in Example 1.6.
- 11. In Proposition 1.1 we saw that the empirical variance is asymptotically Gaussian. While empirical variance is commonly used for measuring the spread of a distribution, it is also common to use the empirical skewness to measure the asymmetry of a distribution.

$$\hat{\xi}_n = \frac{\frac{1}{n} \sum_{i=1}^n (Z_i - \bar{Z}_n)^3}{\left(\frac{1}{n} \sum_{i=1}^n (Z_i - \bar{Z}_n)^2\right)^{\frac{3}{2}}}, \quad \bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i,$$

which targets the standardised central third moment

$$\xi := \frac{\mathbb{E}[(Z - \mathbb{E}[Z])^3]}{(\mathbb{E}[(Z - \mathbb{E}[Z])^2])^{\frac{3}{2}}}.$$

Let (Z_n) be iid $\mathbb{R}$ -valued random variables with finite sixth moment.

- (a) Show that $\hat{\xi}_n \xrightarrow{\mathbb{P}} \xi$.
- (b) State and prove a central limit theorem for $\sqrt{n}(\hat{\xi}_n - \xi)$.

12. Show that $\int_{(0,1]} \sqrt{\log(2/\varepsilon^2 + 1)} \, d\varepsilon < \infty$.

13. Prove Lemma 1.9 by applying Lemma 1.8.

14. Go through the details of the proof of Theorem 1.22 in the univariate case, and provide the details in the bivariate case.
15. In Chapter 1 we proved that the empirical process (scaled difference) of the cumulative hazard is asymptotically Gaussian for each t with $F(t) < 1$. The goal of this exercise is to prove a functional (pathwise) CLT for the empirical process of the cumulative hazard. A useful route is to first identify the functional map and its derivative, and then apply the functional δ -method.

(a) Consider the map ϕ , given by

$$\phi(h)(t) = -\log(1 - h(t)),$$

such that $h(t) < 1$ for all t . Show that the Hadamard derivative of ϕ at a function f in the direction of h is given by

$$\phi'_f(h)(t) = \frac{h(t)}{1 - f(t)}.$$

Hint: For small x , $-\log(1 - x)$ is approximated by the identity function.

- (b) State and prove a functional limit theorem of $\sqrt{n}(\mathbb{A}^{(n)} - \Lambda)$ and compute the corresponding covariance function.
16. State and prove a functional CLT for the mean excess function.
17. Go through the details for obtaining the derivatives of Theorem 1.24.
18. Prove the following:

Lemma 1.14 (Duhamel equation) *Let $B_1 = \phi(A_1)$ and $B_2 = \phi(A_2)$ be two product integrals. They satisfy the Duhamel equation*

$$B_2(s, t] - B_1(s, t] = \int_{(s, t]} B_1(s, u) B_2(u, t] (A_2 - A_1) \, du.$$

19. Prove the following:

Lemma 1.15 (Integration by parts for the Duhamel equation) *Suppose that $A, B \in D(0, \infty)$ and that $h \in D(0, \infty)$ is of bounded variation. Then it holds that*

$$\begin{aligned} & \int_{(0, t]} \prod (1 + dB)(0, u) \prod (1 + dA)(u, t] \, dh(u) \\ &= h(t) + \int_{(0, t]} \prod (1 + dB)(0, s) [h(t) - h(s)] \, dB(s) \\ & \quad + \int_{(0, t]} h(r-) \prod (1 + dA)(r, t] \, dA(r) \\ & \quad + \int_{(0, t]} \int_{(s, t]} \prod (1 + dB)(0, s) [h(r-) - h(s)] \prod (1 + dA)(r, t] \, dA(r) \, dB(s). \end{aligned}$$

20. Let (Z_n) be iid with characteristic function $\varphi(t) = \mathbb{E}[e^{itZ_1}]$ and define the empirical characteristic function

$$\varphi_n(t) = \frac{1}{n} \sum_{i=1}^n e^{itZ_i}, \quad t \in [-T, T].$$

- (a) Show that the class $\{e^{it\cdot} : t \in [-T, T]\}$ is Donsker and conclude that $\sqrt{n}(\varphi_n - \varphi) \xrightarrow{d} \mathbb{G}$ in $\ell^\infty([-T, T])$ for a centred Gaussian process with covariance

$$\text{Cov}(\mathbb{G}(s), \mathbb{G}(t)) = \varphi(s - t) - \varphi(s)\varphi(-t).$$

- (b) For fixed t_0 , use the δ -method (viewing $\varphi_n(t_0)$ as a complex-valued random element) to derive the asymptotic distribution of $|\varphi_n(t_0)|$.

21. Let $w: [0, 1] \rightarrow \mathbb{R}$ be Lipschitz and define the L -statistic functional

$$T(F) = \int_0^1 w(u) F^{\leftarrow}(u) \, du.$$

Assume F is continuous with strictly positive density on an interval containing the support of w . Show that T is Hadamard differentiable at F and derive the asymptotic distribution of $T(\mathbb{F}^{(n)})$ by the functional δ -method.

22. Let H be a bivariate distribution function with continuous marginals F and G , and copula C .

- (a) Show that if a and b are strictly increasing, then the copula of $(a(X), b(Y))$ equals C .
 (b) Assume that H has a density h with marginal densities f and g . Show that the copula density exists and is given by

$$c(u, v) = \frac{h(F^{\leftarrow}(u), G^{\leftarrow}(v))}{f(F^{\leftarrow}(u))g(G^{\leftarrow}(v))}, \quad (u, v) \in (0, 1)^2.$$

23. Establish consistency of the Lorenz curve and Gini coefficient. Comment on the truncated versus untruncated case.

24. Let Z be a discrete random variable with values $x_1, \dots, x_N$ and probabilities $p_1, \dots, p_N$, and define $c_j = \sum_{i=1}^j p_i$ with $c_0 = 0$ and $c_N = 1$. Let F be the distribution function of Z and F^{-1} its generalised inverse.

Use Proposition 1.3 to show that if $U \sim \text{Unif}(0, 1)$ and $\tilde{Z} = F^{-1}(U)$, then

$$\tilde{Z} = x_j \quad \Leftrightarrow \quad c_{j-1} < U \leq c_j,$$

and therefore

$$\mathbb{P}(\tilde{Z} = x_j) = c_j - c_{j-1} = p_j, \quad j = 1, \dots, N.$$

Conclude that the standard discrete simulation algorithm (partitioning $(0, 1)$ into intervals of lengths p_j and mapping each interval to x_j) is exactly the inversion method applied to a discrete target law.

25. Prove that the squeezing variant of acceptance-rejection method of Remark 1.10 produces a random variable with density f .

Practical exercises

1. Verify through simulation the variance of the approximations in Theorems 1.1 and 1.11.
 - (a) Let (Z_n) be a sequence of iid random variables where $Z_n \sim N(0, 1)$ for all n . Verify through simulation the asymptotic distribution $\sqrt{n}(\hat{\sigma}_n^2 - \mu^{(2)})$, where $\mu^{(2)} = \mathbb{E}[(Z_1 - \mathbb{E}[Z_1])^2]$.
 - (b) Let (Z_n) be a sequence of iid random variables where Z_n is Pareto distributed with shape parameter $\alpha = 3$ and scale parameter $\kappa = 100$ for all n . Verify through simulation the asymptotic distribution for $u = 200$ of $\sqrt{n}(\hat{e}_n(u) - e_F(u))$.
Hint: In R you can simulate Pareto distributed variables using the function `rPareto()` from the R package `Pareto`.
2. In the R package `CASdatasets` Dutang and Charpentier [2024] one may find the dataset `frecomfire`, within which there is a column called `ClaimCost2007`, a vector of claim costs in thousand euros from commercial French fire insurance, normalised to 2007. In this exercise we assume that the data in `ClaimCost2007` are iid and examine the variance, mean excess, and cumulative hazard.
 - (a) Plot a histogram of the column `ClaimCost2007`,
 - (b) Construct a 95% confidence interval for the variance using Theorem 1.1,
 - (c) Construct a 95% confidence interval for the mean excess function in $u = 5000$, using Theorem 1.11,
 - (d) Construct a 95% confidence interval for the cumulative hazard in $t = 2000$, using Theorem 1.12.
3. Consider the following simulation studies:
 - (a) For $Z \sim \text{Exp}(\lambda)$, $\lambda > 0$ we have $F_Z(z) = 1 - e^{-\lambda z}$. In R simulate $n \in \{10, 100, 1000\}$ iid $\text{Exp}(1/2)$ -variables, construct and plot $\mathbb{F}^{(n)}(z)$ for $z \in (0, 10]$ with 95% confidence bounds using Theorem 1.22.
 - (b) Denote $D_n = \sup_x |\mathbb{F}^{(n)}(x) - F(x)|$. State the limit distribution for $\sqrt{n}D_n$. With the same setup as in (a), simulate 100 realisations of $\sqrt{n}D_n$ for each n . Plot the histograms and compare with the limit distribution of $\sqrt{n}D_n$ either through the exact limit density or through simulations; for the latter, the function `rbridge()` from the R package `e1071` can be useful.
Hint: The supremum of a continuously time-changed Brownian bridge is the same as a non-time-changed one.
4. If $Y \sim N(\mu, \Sigma)$ is an r -dimensional multivariate random vector then its corresponding copula is the so-called *Gaussian copula*. The marginal distributions can be standardised through strictly increasing transformations to obtain a new vector $X \sim N(0, \rho_\Sigma)$, where $\rho_\Sigma = \rho$ is the correlation

matrix of Y . Since copulas are invariant under strictly increasing transformations it follows that Y and X have the same copula. It is given by

$$\begin{aligned} C_\rho^{\text{Ga}} &= \mathbb{P}(\Phi(X_1) \leq u_1, \dots, \Phi(X_r) \leq u_r) \\ &= \Phi_\rho(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_r)), \end{aligned}$$

where Φ_ρ is the cdf of an r -dimensional zero-mean Gaussian vector with covariance matrix ρ . We consider $r = 2$ and X a centred Gaussian vector with $\sigma_1^2 = \sigma_2^2 = 1$ and $\sigma_{12} = 0.7$.

- (a) Simulate $n = 1000$ observations from $N(0, \rho)$ and set $U_i = \Phi(X_{i1})$, $V_i = \Phi(X_{i2})$. Construct the empirical copula

$$\mathbb{C}^{(n)}(u, v) = \frac{1}{n} \sum_{i=1}^n 1_{(U_i \leq u, V_i \leq v)}$$

on a grid in $(0, 1)^2$ and compare it visually with the true Gaussian copula $C(u, v) = \Phi_\rho(\Phi^{-1}(u), \Phi^{-1}(v))$. (Either numerically evaluate or approximate C by Monte Carlo simulation using a very large sample).

- (b) To investigate the scaled differences $\sqrt{n}(\mathbb{C}^{(n)} - C)$, simulate 1000 realisations of sample size $n = 1000$ and then compute $\sup_{(u,v) \in [0,1]^2} |\sqrt{n}(\mathbb{C}^{(n)}(u, v) - C(u, v))|$ for each simulation. Plot a histogram of these values. Compare with $n = 100, 000$.
5. Implement simulation of $\text{Gamma}(p, \lambda)$ (with integer p) and $\text{Poisson}(\lambda)$ by two conceptually different methods. Use the distributional identities from the chapter to construct $\text{Gamma}(p, \lambda)$ as a sum of p iid exponentials with rate λ , and to construct $\text{Poisson}(\lambda)$ from exponential inter-arrival times (or equivalently from the product representation involving uniforms). Then implement direct inversion, i.e. $F_\Gamma^{-1}(U)$ and $F_{\text{Pois}}^{-1}(U)$ with $U \sim \text{Unif}(0, 1)$. Using a large sample, compare the two routes through empirical means and variances against the theoretical values, compare computation times, and discuss briefly which method is preferable in your implementation and why.
6. Consider the three distribution functions

$$F_P(z) = 1 - z^{-1/\gamma}, \quad z \geq 1, \quad F_{Fr}(z) = \exp(-z^{-1/\gamma}), \quad z > 0,$$

and

$$F_B(z) = 1 - (1 + (z/\theta)^\eta)^{-\beta}, \quad z \geq 0, \quad \theta, \eta, \beta > 0,$$

which are the Pareto, Fréchet and Burr parameterisations used in Chapter 3. Derive their inverse distribution functions and implement inversion-based simulation for all three laws from a uniform sample.

7. Consider the standard normal target law, for which $\psi(s) = \exp(-s^2/2)$. Use Example 1.17 and Proposition 1.6 to compare Fourier simulation by FFT approximation and by exact acceptance-rejection.

- (a) Compute explicitly

$$c = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\psi(s)| \, ds, \quad k = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\psi''(s)| \, ds,$$

and deduce $b = \sqrt{k/c}$ and $A = 4\sqrt{kc}$ for the acceptance-rejection envelope.

- (b) Implement the FFT-based method of Example 1.17 on a fixed equidistant grid.
- (c) Implement the exact transform-based acceptance-rejection method from Proposition 1.6.
- (d) Using large samples, compare approximation quality (moments, selected quantiles, cdf values), computational cost, and, for acceptance-rejection, empirical acceptance probability.
- (e) For the FFT method, study sensitivity with respect to truncation window and mesh size.

Conclude which method is preferable in your implementation for a given accuracy target.

Chapter 2

Smoothing covariates

In this chapter, we introduce nonparametric regression estimators. They are a cornerstone of statistical methods for investigating the distribution of a risk variable Z and its dependence on covariates X . However, kernel methods work best when the data, including the covariates, are not sparse. Other nonparametric methods from machine learning are better suited for high-dimensional data, but their statistical properties are generally less transparent than those of kernel methods.

Kernel nonparametric methods are valuable in insurance, as they are well suited for estimating skewed, multimodal, or heteroscedastic claim distributions and regression functions, which can be crucial for accurate risk assessment and premium pricing, particularly for medium (or attritional) claims. One important limitation of the methods in this chapter is the accurate modelling of large claims, i.e. heavy-tailed distributions, which is treated in the next chapter. Other actuarial applications of kernel regression include smoothing irregular development patterns or risk classification. We first develop the basic kernel estimators and their asymptotics, then turn to distributional regression, discuss practical bandwidth selection, and finally study k-nearest neighbour distributional regression.

2.1 Kernel estimators

Despite their drawbacks in higher dimensions, we first define the estimators for covariates in $\mathbb{R}^k$; for simplicity we provide proofs for the $k = 1$ case, which is also where kernel methods are most applicable. We refer the interested reader to Section 1.4 in [Ghosh \[2018\]](#) and [Stute \[1986\]](#) for general treatments. We begin by defining the kernel and the two core estimators that are analysed throughout the chapter.

Consider a kernel, defined as a bounded function $K: \mathbb{R}^k \rightarrow \mathbb{R}_+ \cup \{0\}$ with

bounded support¹ and satisfying

$$\int K(u) \, du = 1, \quad \int u_i K(u) \, du = 0,$$

for all $i \in \{1, \dots, k\}$. The first condition guarantees that the kernel is a density function, and the second is a balance property. We assume that

$$\int u_i^2 K(u) \, du > 0,$$

which guarantees that there are no dimensions where the kernel is degenerate. Since the kernel is bounded and of bounded support, we also have

$$\int |u_i|^m K(u) \, du < \infty, \quad m \in \mathbb{N}.$$

Henceforth we assume the kernel is symmetric around the origin, so the first-moment conditions are automatically satisfied.

Definition 2.1 (Parzen–Rosenblatt density estimator) Let (X_n) be a sequence of iid random vectors in $\mathbb{R}^k$ with density f . Let $K: \mathbb{R}^k \rightarrow \mathbb{R}$ be a k -dimensional kernel. Let b_n be a sequence of $k \times k$ non-singular matrices with non-negative entries. The *Parzen–Rosenblatt density estimator* of f is given by

$$\mathbb{f}^{(n)}(x) = \frac{1}{n|b_n|} \sum_{i=1}^n K(b_n^{-1}(x - X_i)),$$

where $|b_n|$ is the determinant of b_n .

The next definition builds directly on this construction. The estimator $\mathfrak{m}^{(n)}$ targets the product $m(x) = e(x)f(x)$, while $\mathfrak{e}^{(n)}$ is obtained by normalising with $\mathbb{f}^{(n)}$.

Definition 2.2 (Priestley–Chao and Nadaraya–Watson estimators) Let (Y_n, X_n) be a sequence of iid random variables in $\mathbb{R} \times \mathbb{R}^k$ with density f for X_1 and conditional expectation $e(x) = \mathbb{E}[Y_1 | X_1 = x]$. Let $K: \mathbb{R}^k \rightarrow \mathbb{R}$ be a k -dimensional kernel. Let b_n be a sequence of $k \times k$ non-singular matrices with non-negative entries. The *Priestley–Chao estimator* of $m(x) = e(x)f(x)$ is given by

$$\mathfrak{m}^{(n)}(x) = \frac{1}{n|b_n|} \sum_{i=1}^n Y_i K(b_n^{-1}(x - X_i)).$$

Similarly, the *Nadaraya–Watson regression estimator* of $e(x)$ is given by

$$\mathfrak{e}^{(n)}(x) = \frac{\mathfrak{m}^{(n)}(x)}{\mathbb{f}^{(n)}(x)} = \sum_{i=1}^n \frac{Y_i K(b_n^{-1}(x - X_i))}{\sum_{i=1}^n K(b_n^{-1}(x - X_i))}.$$

Diagonal bandwidth matrices correspond to variable-wise smoothing (often with a common scalar bandwidth), whereas non-diagonal b_n allow for rotations

¹Kernels of unbounded support are common in the literature; here we stick with the bounded support assumption for mathematical convenience.

and rescalings of the covariate space but are less common in practice. As k grows, the amount of data falling in a local neighbourhood shrinks rapidly, which is why we focus on the univariate case for proofs and applications.

For simplicity, we assume for the remainder of the chapter that the conditional distribution of Y_1 has the first three moments, that is $m^{(j)}(x) = \mathbb{E}[|Y_1|^j | X_1 = x]f(x) < \infty$ for $j = 1, 2, 3$, and we let $x \mapsto m^{(j)}(x)$ and $f(x)$ be C^2 . When necessary, we also assume existence of the joint density. A common choice sets b_n to be diagonal with identical entries h_n , and uses product kernels. In this case, the Parzen–Rosenblatt estimator reduces to

$$\mathbb{f}^{(n)}(x) = \frac{1}{nh_n^k} \sum_{i=1}^n \prod_{j=1}^k K_0\left(\frac{X_{ij} - x_j}{h_n}\right)$$

where X_{ij} is the j –th entry of the vector X_i . The Priestley–Chao and Nadaraya–Watson estimators collapse similarly. We now concentrate on the case $k = 1$.

In practice, Gaussian kernels are a common choice (despite not satisfying the bounded support assumption). Uniform kernels are another simple option, given by

$$K_0(u) = \begin{cases} \frac{1}{2} & u \in [-1, 1] \\ 0 & \text{otherwise.} \end{cases}$$

2.1.1 Density estimator

For coherence, we redefine the estimator for univariate covariates. We write b_n for matrix bandwidths and h_n (or simply h) for scalar bandwidths. Let K be a univariate kernel function. For univariate X , the Parzen–Rosenblatt density estimator is given as

$$\mathbb{f}^{(n)}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right). \quad (2.1)$$

Theorem 2.1 (Weak consistency of the Parzen–Rosenblatt density estimator)
Let $h = h_n$ be a sequence of bandwidths with $h_n \rightarrow 0$ and $nh_n \rightarrow \infty$. Then

$$\mathbb{f}^{(n)}(x) \xrightarrow{\mathbb{P}} f(x).$$

Proof. We expand the bias by Taylor’s theorem and show the variance vanishes under the bandwidth conditions. Notice that by the change of variables $u = (x - y)/h$ and Taylor expansion of f around x , the bias of the estimator is given by

$$\begin{aligned} \mathbb{E}[\mathbb{f}^{(n)}(x)] - f(x) &= \int \frac{1}{h} K((x - y)/h) \{f(y) - f(x)\} dy \\ &= \int K(u) \{f(x - hu) - f(x)\} du \\ &= \int K(u) \{f'(x)(-hu) + \frac{1}{2}f''(x)(hu)^2 + o(h^2)\} du \\ &= \frac{h^2}{2} f''(x) \int u^2 K(u) du + o(h^2), \end{aligned}$$

which vanishes when $h \rightarrow 0$. Notice that under the integral sign the remainder is $o(u^2 h^2)$, and by boundedness of the kernel, $\int u^2 K(u) du < \infty$. A similar calculation for the variance yields

$$\begin{aligned} \text{Var}(\mathbb{f}^{(n)}(x)) &= \frac{1}{nh} \int K^2(u) f(x - hu) du - \frac{1}{n} \left(\int K(u) f(x - hu) du \right)^2 \\ &= \frac{f(x)}{nh} \int K^2(u) du + O(1/n) - O(1/n), \end{aligned} \quad (2.2)$$

which vanishes as $nh \rightarrow \infty$ and $h \rightarrow 0$. We now apply Chebyshev's inequality (1.1) with the second moment to conclude that $\mathbb{f}^{(n)}(x) - f(x) \xrightarrow{\mathbb{P}} 0$. $\square$

Notice the above theorem holds even for $f(x) = 0$. Keeping the bandwidth fixed, however, removes consistency; the next example makes this explicit.

Example 2.1 (Inconsistency if the bandwidth does not shrink) Assume $h_n \equiv h_0 > 0$ is fixed. Then

$$\mathbb{E}[\mathbb{f}^{(n)}(x)] = \int \frac{1}{h_0} K\left(\frac{x-y}{h_0}\right) f(y) dy = (K_{h_0} * f)(x), \quad K_{h_0}(u) = h_0^{-1} K(u/h_0).$$

Moreover $\text{Var}(\mathbb{f}^{(n)}(x)) = O(n^{-1})$, so $\mathbb{f}^{(n)}(x) \xrightarrow{\mathbb{P}} (K_{h_0} * f)(x)$. Unless f is a fixed point of this convolution, the estimator is inconsistent. For instance, if f is the standard normal density and K is the uniform kernel on $[-1, 1]$, then

$$(K_{h_0} * f)(0) = \frac{1}{2h_0} \int_{-h_0}^{h_0} f(u) du < f(0)$$

since f is strictly concave at 0. $\diamond$

Theorem 2.2 (Weak convergence of the kernel density estimator) *Let $h = h_n$ be a sequence of bandwidths with $nh_n^5 \rightarrow 0$ and $nh_n \rightarrow \infty$. Then*

$$\sqrt{nh}(\mathbb{f}^{(n)}(x) - f(x)) \xrightarrow{d} N\left(0, f(x) \int K^2(u) du\right).$$

Proof. We split the error into a stochastic term and a bias term, and apply a CLT to the former. Consider the decomposition

$$\begin{aligned} \sqrt{nh}(\mathbb{f}^{(n)}(x) - f(x)) &= \sqrt{nh}(\mathbb{f}^{(n)}(x) - \mathbb{E}[\mathbb{f}^{(n)}(x)]) + \sqrt{nh}(\mathbb{E}[\mathbb{f}^{(n)}(x)] - f(x)) \\ &=: A_n + B_n. \end{aligned}$$

Recall that $\mathbb{E}[\mathbb{f}^{(n)}(x)] - f(x) = O(h^2)$. Thus B_n is a bias term of order $O(\sqrt{nh^5})$, which vanishes by assumption.

The remaining term A_n can be handled using Lindeberg's CLT for arrays, Theorem 1.7 via Lyapounov's condition as per Lemma 1.4. Namely, we need to verify that the row sums of the variances of the array

$$Z_{ni} = (nh)^{-1/2} \{K(h^{-1}(x - X_i)) - \mathbb{E}[K(h^{-1}(x - X_i))]\}$$

converge, and that the row sums of the absolute central moments of order $\alpha > 2$ vanish. We now make a computation similar to (2.2) to obtain

$$\begin{aligned} \sum_{i=1}^n \sigma_{n,i}^2 &= \sum_{i=1}^n \text{Var}(Z_{ni}) = \int K^2(u) f(x - hu) \, du - h \left(\int K(u) f(x - hu) \, du \right)^2 \\ &= f(x) \int K^2(u) \, du + O(h) - O(h), \end{aligned}$$

which converges to the desired variance. Now note that for the nonnegative random variable $\tilde{Z}_{ni} = (nh)^{-1/2} K(h^{-1}(x - X_i))$, by Minkowski's inequality, positivity, and the fact that L^p norms are increasing in p , we have

$$\mathbb{E}[|Z_{ni}|^3]^{1/3} \leq \mathbb{E}[\tilde{Z}_{ni}^3]^{1/3} + |\mathbb{E}[\tilde{Z}_{ni}]| = \mathbb{E}[\tilde{Z}_{ni}^3]^{1/3} + \mathbb{E}[|\tilde{Z}_{ni}|] \leq 2\mathbb{E}[\tilde{Z}_{ni}^3]^{1/3}.$$

Thus, the sum of third absolute central moments gives

$$\sum_{i=1}^n \mathbb{E}[|Z_{ni}|^3] \leq 2^3 \sum_{i=1}^n \mathbb{E}[\tilde{Z}_{ni}^3] = 8f(x) \frac{1}{\sqrt{nh}} \int K^3(u) \, du + O(h/\sqrt{nh}),$$

which vanishes as $nh \rightarrow \infty$ and $h \rightarrow 0$. This concludes the proof. $\square$

Since we have calculated the bias in the above proof, we may actually examine an asymptotic mean integrated square error (AMISE) calculation, which makes the bias–variance tradeoff explicit and suggests a concrete bandwidth selection rule.

Example 2.2 (AMISE and optimal bandwidth) Assume f is twice continuously differentiable and K is symmetric with $\mu_2(K) = \int u^2 K(u) \, du$ and $R(K) = \int K^2(u) \, du$. A Taylor expansion as above gives the pointwise bias

$$\mathbb{E}[\mathbb{f}^{(n)}(x)] - f(x) = \frac{h^2}{2} \mu_2(K) f''(x) + o(h^2),$$

while Theorem 2.2 yields the variance term $\text{Var}(\mathbb{f}^{(n)}(x)) = \frac{1}{nh} f(x) R(K) + o((nh)^{-1})$. Hence the mean squared error satisfies

$$\text{MSE}(x) = \frac{h^4}{4} \mu_2(K)^2 f''(x)^2 + \frac{1}{nh} f(x) R(K) + o(h^4) + o((nh)^{-1}).$$

Integrating over x gives the asymptotic mean integrated squared error

$$\text{AMISE}(h) = \frac{h^4}{4} \mu_2(K)^2 \int (f''(x))^2 \, dx + \frac{1}{nh} R(K).$$

Minimising yields the optimal bandwidth

$$h_{\text{AMISE}} = \left(\frac{R(K)}{\mu_2(K)^2 \int (f''(x))^2 \, dx} \right)^{1/5} n^{-1/5}.$$

A natural issue with this procedure is that we require an estimate of f'' to compute the optimal bandwidth, which itself is aimed at estimating f . This circular difficulty is in practice solved by inputting some parametric estimate for f'' . $\diamond$

Remark 2.1 Kernel density smoothing on the positive line encounters finite-sample boundary issues, since symmetric kernels place mass outside the support. When estimating near zero, such kernels extend into the negative region, where no data exist, leading to underestimation. Similar issues arise near the edges of compact supports. These effects are negligible asymptotically, but important in finite samples. $\circ$

Example 2.3 (Boundary correction by reflection) Suppose X has support on $[0, \infty)$ and f is C^2 in a neighbourhood of 0. For a symmetric kernel K with bounded support, define the reflected estimator for $x \geq 0$ by

$$\mathbb{F}_R^{(n)}(x) = \frac{1}{nh} \sum_{i=1}^n \left\{ K\left(\frac{x - X_i}{h}\right) + K\left(\frac{x + X_i}{h}\right) \right\}.$$

Let $f_{\text{ref}}(y) = f(|y|)$ denote the even reflection of f . Then a change of variables shows

$$\mathbb{E}[\mathbb{F}_R^{(n)}(x)] = \int K(u) f_{\text{ref}}(x - hu) \, du,$$

while the uncorrected estimator $\mathbb{F}^{(n)}(x)$ satisfies, at $x = 0$,

$$\mathbb{E}[\mathbb{F}^{(n)}(0)] = \int_{-\infty}^0 K(u) f(-hu) \, du = \frac{1}{2}f(0) + hf'(0) \int_0^\infty uK(u) \, du + o(h),$$

where $f'(0)$ denotes the right derivative. Thus the boundary bias of the uncorrected estimator equals $-\frac{1}{2}f(0) + O(h)$. For $x = 0$, using the symmetry of K , we have $\mathbb{E}[\mathbb{F}_R^{(n)}(0)] = 2 \int_0^\infty K(u) f(hu) \, du$. A Taylor expansion yields

$$\mathbb{E}[\mathbb{F}_R^{(n)}(0)] = f(0) + 2hf'(0) \int_0^\infty uK(u) \, du + o(h).$$

Thus the reflected estimator removes the leading boundary loss of order $O(1)$. Generally, a bias of order $O(h)$ remains, involving the first moment of the kernel on the half-line. However, if $f'(0) = 0$ (e.g. if the density has a shoulder at the boundary), the linear term vanishes and the bias becomes of order h^2 . $\diamond$

Numeric example 2.1 (Boundary reflection on a compact support) To illustrate the boundary effect and its correction, consider iid observations (X_n) with density f from the Beta(1, 0.8) density. We take $n = 1000$ and estimate f with the uniform kernel. The reflection idea from the previous example is applied at both boundaries. This yields a reflected estimator of the same form as $\mathbb{F}_R^{(n)}$, now correcting, to some degree, the boundary problems at both 0 and 1.

The resulting comparison is shown in Figure 2.1, where the correction is clearly visible. This specific density has the feature that it diverges to infinity at the right endpoint of the support, which unsurprisingly is not fully corrected by the reflected estimator. $\diamond$

2.1.2 Nonparametric regression

The Nadaraya–Watson estimator is a fundamental tool in nonparametric statistics for estimating the conditional expectation of a dependent risk variable Y

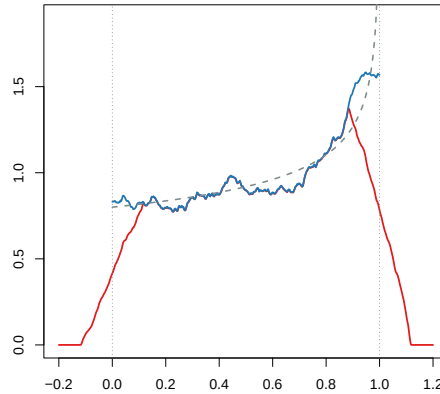


Figure 2.1: Boundary correction by reflection for Beta(1, 0.8) data on $[0, 1]$, using a uniform kernel. Red solid: uncorrected estimator $\mathbb{f}^{(n)}$; blue solid: reflected estimator; grey dashed: true density f . Vertical grey dashed lines delimit the support, and the vertical range is truncated since the true density is unbounded at 1.

given a covariate X . It can be viewed as a regression model that makes no parametric assumptions on the conditional distribution, which offers flexibility for modelling nonlinear relationships. It is therefore often used as a visual or exploratory tool to assess whether parametric models fit adequately. We now study its asymptotics in detail, and notice that for $k = 1$ the estimator is given by

$$\mathfrak{e}^{(n)}(x) = \sum_{i=1}^n \frac{Y_i K(h_n^{-1}(x - X_i))}{\sum_{i=1}^n K(h_n^{-1}(x - X_i))}.$$

Theorem 2.3 (Consistency of the Nadaraya–Watson estimator) *Let $h = h_n$ be a sequence of bandwidths with $h_n \rightarrow 0$ and $nh_n \rightarrow \infty$. Let x be such that the density of X satisfies $f(x) > 0$. Then*

$$\mathfrak{e}^{(n)}(x) \xrightarrow{\mathbb{P}} \mathbb{E}[Y | X = x].$$

If in addition, $nh_n/\log n \rightarrow \infty$ and for some $\delta \in (0, 1]$

$$\sum_{n=1}^{\infty} \left(\frac{\log n}{nh_n} \right)^{\frac{1+\delta}{2}} < \infty,$$

then the convergence strengthens to

$$\mathfrak{e}^{(n)}(x) \xrightarrow{a.s.} \mathbb{E}[Y | X = x].$$

Proof. Recall that $\mathfrak{e}^{(n)}(x) = \mathfrak{m}^{(n)}(x)/\mathbb{f}^{(n)}(x)$ where $\mathfrak{m}^{(n)}(x)$ is the Priestley–Chao estimator. The calculations below only use $m(x) = \mathbb{E}[Y | X = x]f(x)$ and $m^{(2)}(x) = \mathbb{E}[Y^2 | X = x]f(x)$, so the argument applies equally to discrete responses; we keep the density notation for brevity. The latter estimator has

bias given by

$$\begin{aligned}
& \mathbb{E}[\mathfrak{m}^{(n)}(x)] - m(x) \\
&= \int \int K(u) \{yf(y, x - hu) - yf(y, x)\} \, du \, dy \\
&= \int \int K(u) y \left\{ \frac{\partial}{\partial x} f(y, x)(-hu) + \frac{1}{2} \frac{\partial^2}{\partial x^2} f(y, x)(hu)^2 + o(h^2) \right\} \, du \, dy \\
&= \frac{h^2}{2} m''(x) \int u^2 K(u) \, du + o(h^2),
\end{aligned}$$

which goes to zero as $h \rightarrow 0$. The variance is given by

$$\begin{aligned}
\text{Var}(\mathfrak{m}^{(n)}(x)) &= \frac{1}{nh} \int \int y^2 K^2(u) f(y, x - hu) \, du \, dy \\
&\quad - \frac{1}{n} \left(\int \int y K(u) f(y, x - hu) \, du \, dy \right)^2 \\
&= \frac{\mathbb{E}[Y^2 | X = x] f(x)}{nh} \int K^2(u) \, du + O(1/n) - O(1/n),
\end{aligned}$$

which vanishes as $nh \rightarrow \infty$. Thus we may apply Chebyshev's inequality with the second moment to conclude that $\mathfrak{m}^{(n)}(x) \xrightarrow{P} m(x) = e(x)f(x)$. Division with the consistent estimator $\mathfrak{f}^{(n)}(x)$ of $f(x) > 0$ now finishes the proof of consistency by Theorem 1.4.

For almost sure convergence, let $M_n := \sqrt{nh/\log n}$ and decompose the estimator into a truncated part and a remainder:

$$Y_i = Y_i^{(n)} + R_i^{(n)}, \quad Y_i^{(n)} := Y_i 1_{(|Y_i| \leq M_n)}, \quad R_i^{(n)} := Y_i 1_{(|Y_i| > M_n)}.$$

Accordingly, let $\mathfrak{m}^{(n)} = \mathfrak{m}_T^{(n)} + \mathfrak{m}_R^{(n)}$. Since $|Y_i^{(n)}| \leq M_n$ and K is bounded with compact support, the terms in $\mathfrak{m}_T^{(n)}$ are bounded. Bernstein's inequality, Theorem 1.3, implies that for each fixed $\varepsilon > 0$, there exists a constant $C > 0$ such that

$$\mathbb{P}\left(|\mathfrak{m}_T^{(n)}(x) - \mathbb{E}[\mathfrak{m}_T^{(n)}(x)]| > \varepsilon\right) \leq 2 \exp \left\{ -\frac{C\varepsilon^2 nh}{1 + \varepsilon M_n} \right\}.$$

Since $M_n = \sqrt{nh/\log n} \rightarrow \infty$, the exponent is dominated by $-C'\sqrt{nh \log n}$, which ensures that the series $\sum_{n=1}^{\infty} \mathbb{P}(|\mathfrak{m}_T^{(n)}(x) - \mathbb{E}[\mathfrak{m}_T^{(n)}(x)]| > \varepsilon)$ is summable. By the Borel–Cantelli Lemma, $\mathfrak{m}_T^{(n)}(x) - \mathbb{E}[\mathfrak{m}_T^{(n)}(x)] \xrightarrow{a.s.} 0$.

Now, for the remainder term, we use the bound $|y| 1_{(|y| > b)} \leq |y|^{2+\delta} b^{-(1+\delta)}$. Under the assumed $(2 + \delta)$ -moment bounds, we have

$$\mathbb{E}[|\mathfrak{m}_R^{(n)}(x)|] \leq C M_n^{-(1+\delta)} = C \left(\frac{\log n}{nh} \right)^{\frac{1+\delta}{2}}.$$

By Markov's inequality and the assumed growth rate of nh relative to $\log n$, we also obtain $\mathfrak{m}_R^{(n)}(x) \xrightarrow{a.s.} 0$ by summability. Finally, since $\mathbb{E}[\mathfrak{m}^{(n)}(x)] \rightarrow m(x)$ and $\mathbb{E}[\mathfrak{m}_R^{(n)}(x)] \rightarrow 0$, it follows that $\mathbb{E}[\mathfrak{m}_T^{(n)}(x)] \rightarrow m(x)$. Combining these results, we obtain $\mathfrak{m}^{(n)}(x) \xrightarrow{a.s.} m(x)$. The same argument for $\mathfrak{f}^{(n)}(x)$ yields $\mathfrak{f}^{(n)}(x) \xrightarrow{a.s.} f(x)$, and the continuous mapping Theorem 1.4 completes the proof. $\square$

Theorem 2.4 (Weak convergence of the Nadaraya–Watson estimator) *Let $h = h_n$ be a sequence of bandwidths with $nh_n^5 \rightarrow 0$ and $nh_n \rightarrow \infty$. Let x be such that the density of X satisfies $f(x) > 0$. Then*

$$\sqrt{nh}(\mathfrak{e}^{(n)}(x) - \mathbb{E}[Y | X = x]) \xrightarrow{d} N\left(0, \frac{\sigma(x)^2}{f(x)} \int K^2(u) \, du\right),$$

where $\sigma(x)^2 = \mathbb{E}[Y^2 | X = x] - \mathbb{E}[Y | X = x]^2$.

Proof. First consider the following decomposition:

$$\begin{aligned} \sqrt{nh_n}(\mathfrak{e}^{(n)}(x) - \mathbb{E}[Y | X = x]) &= \sqrt{nh_n}(\mathfrak{e}^{(n)}(x) - \mathbb{E}[\mathfrak{m}^{(n)}(x)]/\mathbb{E}[\mathfrak{f}^{(n)}(x)]) \\ &\quad + \sqrt{nh_n}(\mathbb{E}[\mathfrak{m}^{(n)}(x)]/\mathbb{E}[\mathfrak{f}^{(n)}(x)] - m(x)/f(x)) \\ &=: A_n + B_n. \end{aligned}$$

Recall that both $\mathbb{E}[\mathfrak{f}^{(n)}(x)] - f(x) = O(h^2)$ and $\mathbb{E}[\mathfrak{m}^{(n)}(x)] - m(x) = O(h^2)$. Thus the B_n term is $O(\sqrt{nh^5})$, which vanishes by assumption.

We can decompose the A_n term further as

$$\begin{aligned} A_n &= \frac{1}{\mathbb{E}[\mathfrak{f}^{(n)}(x)]\mathfrak{f}^{(n)}(x)} \sqrt{nh_n} \{ \mathfrak{m}^{(n)}(x) \mathbb{E}[\mathfrak{f}^{(n)}(x)] - \mathbb{E}[\mathfrak{m}^{(n)}(x)] \mathfrak{f}^{(n)}(x) \} \\ &= \frac{1}{\mathbb{E}[\mathfrak{f}^{(n)}(x)]\mathfrak{f}^{(n)}(x)} \sqrt{nh_n} \{ \mathfrak{m}^{(n)}(x) (\mathbb{E}[\mathfrak{f}^{(n)}(x)] - \mathfrak{f}^{(n)}(x)) \} \\ &\quad + \frac{1}{\mathbb{E}[\mathfrak{f}^{(n)}(x)]\mathfrak{f}^{(n)}(x)} \sqrt{nh_n} \{ (\mathfrak{m}^{(n)}(x) - \mathbb{E}[\mathfrak{m}^{(n)}(x)]) \mathfrak{f}^{(n)}(x) \} \\ &= \frac{m(x)}{f^2(x)} \sqrt{nh_n} \{ \mathbb{E}[\mathfrak{f}^{(n)}(x)] - \mathfrak{f}^{(n)}(x) \} + \frac{1}{f(x)} \sqrt{nh_n} \{ \mathfrak{m}^{(n)}(x) - \mathbb{E}[\mathfrak{m}^{(n)}(x)] \} \\ &\quad + o_{\mathbb{P}}(\sqrt{nh_n} \{ \mathbb{E}[\mathfrak{f}^{(n)}(x)] - \mathfrak{f}^{(n)}(x) \}) + o_{\mathbb{P}}(\sqrt{nh_n} \{ \mathfrak{m}^{(n)}(x) - \mathbb{E}[\mathfrak{m}^{(n)}(x)] \}). \end{aligned}$$

Thus it suffices to verify that $\mathfrak{f}^{(n)}(x)$ and $\mathfrak{m}^{(n)}(x)$ are asymptotically *jointly* normal, so we can evaluate the sum of the two leading terms of A_n , which by Exercise 6 of Chapter 1 establishes the weak convergence of A_n and hence the theorem.

We handle this joint convergence using Lindeberg's CLT for arrays, Theorem 1.7, via Lyapounov's condition in Lemma 1.4. We show that the array

$$Z_{ni} = (nh)^{-1/2} \{ K(h^{-1}(x - X_i)) \{ \lambda_1 + \lambda_2 Y_i \} - \mathbb{E}[K(h^{-1}(x - X_i)) \{ \lambda_1 + \lambda_2 Y_i \}] \}$$

for arbitrary constants λ_1, λ_2 satisfies Lyapounov's condition. We first check the variance convergence in Theorem 1.7 by computing

$$\begin{aligned} \sum_{i=1}^n \sigma_{n,i}^2 &= \sum_{i=1}^n \text{Var}(Z_{ni}) = \int \int (\lambda_1 + \lambda_2 y)^2 K^2(u) f(y, x - hu) \, du \, dy \\ &\quad - h \left(\int \int (\lambda_1 + \lambda_2 y) K(u) f(y, x - hu) \, du \, dy \right)^2 \\ &= f(x) \mathbb{E}[(\lambda_1 + \lambda_2 Y)^2 | X = x] \int K^2(u) \, du + O(h^2) - O(h). \end{aligned}$$

Similarly, Lyapounov's condition with $\alpha = 3$ follows by defining $\tilde{Z}_{ni} = (nh)^{-1/2}K(h^{-1}(x - X_i))\{\lambda_1 + \lambda_2 Y\}$ and noting that

$$\begin{aligned} \sum_{i=1}^n \mathbb{E}[|Z_{ni}|^3] &\leq 2^3 \sum_{i=1}^n \mathbb{E}[|\tilde{Z}_{ni}|^3] \\ &= 8 \frac{f(x) \mathbb{E}[|\lambda_1 + \lambda_2 Y|^3 | X = x]}{\sqrt{nh}} \int K^3(u) \, du + O(h/\sqrt{nh}), \end{aligned}$$

which vanishes as $nh \rightarrow \infty$ and $h \rightarrow 0$. Thus, by the Cramér–Wold device, joint asymptotic normality holds.

Returning to A_n , choose $\lambda_1 = -\mathbb{E}[Y | X = x]/f(x)$ and $\lambda_2 = 1/f(x)$ to obtain convergence to a normal random variable with mean zero and variance given by

$$\begin{aligned} &f(x) \mathbb{E}[(\lambda_1 + \lambda_2 Y)^2 | X = x] \int K^2(u) \, du \\ &= \frac{\mathbb{E}[Y^2 | X = x] - \mathbb{E}[Y | X = x]^2}{f(x)} \int K^2(u) \, du, \end{aligned}$$

concluding the proof. $\square$

As a corollary, we consider local linear regression, a popular alternative for estimating the regression function $e(x) = \mathbb{E}[Y | X = x]$. The approach improves upon kernel smoothing by fitting a straight line, rather than just a constant, to the data within a local neighbourhood of a target point x .

This is accomplished by solving a weighted least squares problem for coefficients $(\hat{\beta}_0, \hat{\beta}_1)$ that minimise the sum of squared residuals, with weights determined by a kernel function K and bandwidth h :

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1(X_i - x))^2 h^{-1} K(h^{-1}(X_i - x))$$

The resulting local linear estimator, $\mathfrak{r}^{(n)}(x)$, is the intercept of this locally fitted line, $\hat{\beta}_0$. (Actually, generalised versions exist in the literature where higher order terms are subtracted). The following theorem describes its asymptotic properties.

Theorem 2.5 (Asymptotic distribution of the local linear regression estimator) *Let $h = h_n$ be a sequence of bandwidths with $h_n \rightarrow 0$, $nh_n \rightarrow \infty$, and $nh_n^5 \rightarrow 0$. Then for fixed x such that $f(x) > 0$, the Local Linear Regression estimator $\mathfrak{r}^{(n)}(x)$ satisfies:*

$$\sqrt{nh}(\mathfrak{r}^{(n)}(x) - \mathbb{E}[Y | X = x]) \xrightarrow{d} N\left(0, \frac{\sigma(x)^2}{f(x)} \int K^2(u) \, du\right),$$

where $\sigma(x)^2 = \mathbb{E}[Y^2 | X = x] - \mathbb{E}[Y | X = x]^2$.

Proof. The estimator $\mathfrak{r}^{(n)}(x)$ admits the explicit form

$$\mathfrak{r}^{(n)}(x) = \frac{T_{n,0}S_{n,2} - T_{n,1}S_{n,1}}{S_{n,0}S_{n,2} - S_{n,1}^2},$$

where $S_{n,j} := \sum_{i=1}^n h^{-1} K(h^{-1}(X_i - x))(X_i - x)^j$ and $T_{n,j} := \sum_{i=1}^n h^{-1} K(h^{-1}(X_i - x))(X_i - x)^j Y_i$. Notice that $\mathfrak{e}^{(n)}(x) = T_{n,0}/S_{n,0}$. The difference is

$$\mathfrak{r}^{(n)}(x) - \mathfrak{e}^{(n)}(x) = \frac{S_{n,1}(T_{n,0}S_{n,1} - T_{n,1}S_{n,0})}{S_{n,0}(S_{n,0}S_{n,2} - S_{n,1}^2)}.$$

Each individual term in both numerator and denominator can be expanded into bias and stochastic terms. Lengthy but straightforward calculations yield a total rate of $O(h^2) + O_{\mathbb{P}}(\sqrt{h/n})$. Thus, $\sqrt{nh}(\mathfrak{r}^{(n)}(x) - \mathfrak{e}^{(n)}(x)) \xrightarrow{\mathbb{P}} 0$, and the estimators are asymptotically equivalent. By Theorem 2.4, the result follows. $\square$

The consistency of several kernel regression estimators follows directly from the univariate case. For asymptotic normality, a bit more work is required. In this case we have two sequences (Y_n) and (Z_n) of dependent variables and (X_n) of covariates, and we specialise notation to

$$m_Y(x) = \mathbb{E}[Y | X = x]f(x), \quad m_Z(x) = \mathbb{E}[Z | X = x]f(x)$$

which should satisfy the same assumptions as $m(x)$. We may also denote the associated Nadaraya–Watson estimators by $\mathfrak{e}_Y^{(n)}$ and $\mathfrak{e}_Z^{(n)}$. It is also clear how to generalise the following theorem beyond two estimators.

Theorem 2.6 (Weak convergence of two Nadaraya–Watson estimators) *Let $h = h_n$ be a sequence of bandwidths with $nh_n^5 \rightarrow 0$ and $nh_n \rightarrow \infty$. Let Z satisfy the same conditions as Y . Let x be such that the density of X satisfies $f(x) > 0$. Then*

$$\sqrt{nh}(\mathfrak{e}_Y^{(n)}(x) - \mathbb{E}[Y | X = x], \mathfrak{e}_Z^{(n)}(x) - \mathbb{E}[Z | X = x]) \xrightarrow{d} N\left(0, \frac{\int K^2(u) du}{f(x)} \Sigma(x)\right),$$

where $\Sigma(x)$ is the conditional covariance matrix:

$$\begin{aligned} \sigma_1^2(x) &= \mathbb{E}[Y^2 | X = x] - \mathbb{E}[Y | X = x]^2 \\ \sigma_2^2(x) &= \mathbb{E}[Z^2 | X = x] - \mathbb{E}[Z | X = x]^2 \\ \sigma_{12}(x) &= \mathbb{E}[ZY | X = x] - \mathbb{E}[Y | X = x]\mathbb{E}[Z | X = x]. \end{aligned}$$

Proof. Joint normality can be established as in Theorem 2.4, now considering the array

$$\begin{aligned} Z_{ni} &= (nh)^{-1/2} K(h^{-1}(x - X_i))\{\lambda_1 + \lambda_2 Y_i + \lambda_3 Z_i\} \\ &\quad - (nh)^{-1/2} \mathbb{E}[K(h^{-1}(x - X_i))\{\lambda_1 + \lambda_2 Y_i + \lambda_3 Z_i\}] \end{aligned}$$

and invoking the Cramér–Wold device once again. It remains to compute the covariance. Using the same method as for the individual variances, for $A_i = (nh)^{-1/2} K(h^{-1}(x - X_i))\{\alpha_1 + \alpha_2 Y_i\}$ and $B_i = (nh)^{-1/2} K(h^{-1}(x - X_i))\{\beta_1 + \beta_2 Z_i\}$ we obtain

$$\sum_{i=1}^n \text{Cov}(A_i, B_i) = f(x) \mathbb{E}[\{\alpha_1 + \alpha_2 Y\}\{\beta_1 + \beta_2 Z\} | X = x] \int K^2(u) du + O(h^2).$$

It then remains to plug in $\alpha_1 = -\mathbb{E}[Y | X = x]/f(x)$, $\beta_1 = -\mathbb{E}[Z | X = x]/f(x)$ and $\alpha_2 = \beta_2 = 1/f(x)$ and the result follows. $\square$

For finite-dimensional weak convergence in the covariate argument, it is also natural to compare the same Nadaraya–Watson estimator at different target covariate points. This is an easy task, as when the target points are distinct, the kernel windows eventually do not overlap, and the asymptotic fluctuations become independent. The next theorem makes this precise for two fixed covariate values.

Theorem 2.7 (Weak convergence at two distinct covariate points) *Let $h = h_n$ be a sequence of bandwidths with $nh_n^5 \rightarrow 0$ and $nh_n \rightarrow \infty$. Let $x_1 \neq x_2$ be such that the density of X satisfies $f(x_1), f(x_2) > 0$. Then*

$$(\sqrt{nh}(\mathfrak{e}^{(n)}(x_1) - \mathbb{E}[Y | X = x_1]), \sqrt{nh}(\mathfrak{e}^{(n)}(x_2) - \mathbb{E}[Y | X = x_2])) \xrightarrow{d} (G_1, G_2),$$

where G_1 and G_2 are independent and

$$G_j \sim N\left(0, \frac{\sigma(x_j)^2}{f(x_j)} \int K^2(u) \, du\right), \quad j = 1, 2,$$

with $\sigma(x)^2 = \mathbb{E}[Y^2 | X = x] - \mathbb{E}[Y | X = x]^2$.

Proof. By the same decomposition as in the proof of Theorem 2.4, for each $j \in \{1, 2\}$ we have

$$\sqrt{nh}(\mathfrak{e}^{(n)}(x_j) - \mathbb{E}[Y | X = x_j]) = W_n(x_j) + o_{\mathbb{P}}(1),$$

where

$$W_n(x_j) = \frac{1}{f(x_j)\sqrt{nh}} \sum_{i=1}^n \left(K(h^{-1}(x_j - X_i))(Y_i - \mathbb{E}[Y | X = x_j]) \right. \\ \left. - \mathbb{E}[K(h^{-1}(x_j - X_i))(Y_i - \mathbb{E}[Y | X = x_j])] \right).$$

Hence, by Slutsky's theorem, it suffices to establish joint weak convergence of $(W_n(x_1), W_n(x_2))$. The Cramér–Wold and Lindeberg arguments are the same as in Theorem 2.6, so it only remains to identify the limiting covariance structure.

The marginal variances are exactly those in Theorem 2.4. Thus it suffices to show that the cross-covariance tends to zero. Let

$$\xi_{n,i}^{(j)} = (nh)^{-1/2} \frac{1}{f(x_j)} \left(K(h^{-1}(x_j - X_i))(Y_i - \mathbb{E}[Y | X = x_j]) \right. \\ \left. - \mathbb{E}[K(h^{-1}(x_j - X_i))(Y_i - \mathbb{E}[Y | X = x_j])] \right), \quad j = 1, 2.$$

Then $\text{Cov}(W_n(x_1), W_n(x_2)) = \sum_{i=1}^n \text{Cov}(\xi_{n,i}^{(1)}, \xi_{n,i}^{(2)})$. Let $M > 0$ be such that $K(u) = 0$ for $|u| > M$. Since $x_1 \neq x_2$ and $h \rightarrow 0$, for all sufficiently large n we have $2Mh < |x_1 - x_2|$, and therefore

$$K(h^{-1}(x_1 - z))K(h^{-1}(x_2 - z)) = 0, \quad z \in \mathbb{R}.$$

Hence the product term in $\mathbb{E}[\xi_{n,i}^{(1)}\xi_{n,i}^{(2)}]$ is zero for large n . The remaining centring term is negligible because, as in the proof of Theorem 2.4,

$$\mathbb{E}[K(h^{-1}(x_j - X))(Y - \mathbb{E}[Y | X = x_j])] \\ = h(\mathbb{E}[\mathfrak{m}^{(n)}(x_j)] - \mathbb{E}[Y | X = x_j]\mathbb{E}[\mathfrak{f}^{(n)}(x_j)]) = O(h^3), \quad j = 1, 2.$$

Therefore $\text{Cov}(W_n(x_1), W_n(x_2)) = O(h^5) \rightarrow 0$, and consequently we get that $(W_n(x_1), W_n(x_2)) \xrightarrow{d} (G_1, G_2)$. $\square$

2.2 Nonparametric distributional regression

Distributional regression, unlike traditional regression which predicts the mean of the dependent variable given the covariates, models the entire conditional distribution. This provides a fuller picture of the distribution and serves not only as a predictive model but also as a goodness-of-fit tool for checking whether conditional distributions satisfy a prescribed parametric structure. The starting point is the conditional empirical distribution function, whose asymptotics we establish first.

Definition 2.3 (Conditional empirical distribution function) The conditional empirical distribution function (cdf) is given for $t \in [0, \infty)$, $x \in \mathbb{R}$ by

$$\mathbb{F}^{(n)}(t | x) = \sum_{i=1}^n \frac{1_{(Y_i \leq t)} K(h^{-1}(x - X_i))}{\sum_{i=1}^n K(h^{-1}(x - X_i))}.$$

Theorem 2.8 (Pathwise weak convergence of the cdf) *Let $h = h_n$ be a sequence of bandwidths with $nh_n^5 \rightarrow 0$ and $nh_n \rightarrow \infty$. Let x be such that the density of X satisfies $f(x) > 0$. Then*

$$\sqrt{nh}(\mathbb{F}^{(n)}(\cdot | x) - F(\cdot | x)) \xrightarrow{d} \left(\frac{\int K^2(u) du}{f(x)} \right)^{1/2} \mathbb{B} \circ F(\cdot | x)$$

in $\ell^\infty([0, \infty))$, where $\mathbb{B}$ is a standard Brownian bridge.

Proof. We establish finite-dimensional convergence to identify the covariance structure, and then apply a bracketing CLT to control the process in t . For a fixed t , we have a Nadaraya–Watson estimator. Thus, finite-dimensional convergence follows as Theorem 2.6 (for binary responses). Next, we apply the Bracketing CLT for changing functions, Theorem 1.20. However, we know from the proof of Theorem 2.4 that $(nh)^{-1} \sum_{i=1}^n 1_{(Y_i \leq t)} K(h^{-1}(x - X_i))$ and $(nh)^{-1} \sum_{i=1}^n K(h^{-1}(x - X_i))$ are jointly weakly convergent, and this can be extended to finite-dimensional distributions too. Consequently, since the latter term is not a process in t , by Theorem 1.8 it suffices to establish pathwise weak convergence of the first process $\tilde{X}(t) = (nh)^{-1} \sum_{i=1}^n 1_{(Y_i \leq t)} K(h^{-1}(x - X_i))$.

Define

$$\begin{aligned} \mathcal{F}_n &= \{f_{n,t}: [0, \infty) \times \mathbb{R} \rightarrow [0, \infty), \\ &\text{s.t. } f_{n,t}(y, z) = h^{-1/2} 1_{[0,t)}(y) K(h^{-1}(z - x)) \text{ for } t \in [0, \infty)\}. \end{aligned}$$

Clearly the following random variable is an envelope

$$F_n(Y_1, X_1) = h^{-1/2} K(h^{-1}(X_1 - x)).$$

It satisfies by Taylor expansion (as in the proof of Theorem 2.2)

$$\mathbb{E}[F_n(Y_1, X_1)^2] = f(x) \int K^2(u) du + O(h) = O(1),$$

showing condition (1.6). We also have

$$\begin{aligned} \mathbb{E}[F_n(Y_1, X_1)^2 1_{(F_n(Y_1, X_1) > c\sqrt{nh})}] &\leq \frac{1}{c\sqrt{nh}} \mathbb{E}[F_n(Y_1, X_1)^3] \\ &= f(x) \frac{1}{c\sqrt{nh}} \int K^3(u) du + O(h/\sqrt{nh}), \end{aligned}$$

which vanishes as $nh \rightarrow \infty$, showing (1.7). The bracketing number is calculated as in Theorem 1.22, but with slight adaptation. We provide the details.

Consider for $j = 1, \dots, k$ brackets on the form

$$l_j(y, z) = h^{-1/2} 1_{[0, t_{j-1}]}(y) K(h^{-1}(z-x)), u_j(y, z) = h^{-1/2} 1_{[0, t_j)}(y) K(h^{-1}(z-x))$$

for sequences on the form $0 = t_0 < t_1 < \dots < t_k = \infty$. Let $\mathcal{F}_j^n = \{f \in \mathcal{F}_n : l_j \leq f \leq u_j\}$. Then clearly we have the partition $\mathcal{F}_n = \bigcup_{j=1}^k \mathcal{F}_j^n$. Let $\varepsilon > 0$ be given, and pick a sequence for $k \leq \lceil 2/\varepsilon^2 \rceil$ such that

$$F(t_j - |x) - F(t_{j-1} |x) \leq \varepsilon^2/2 \quad \text{for } 1 \leq j \leq k,$$

which is possible because F is a cumulative distribution function, in particular, it is càdlàg, non-decreasing and bounded by 1. Then this sequence covers all of $\mathcal{F}_n$ with brackets satisfying

$$\begin{aligned} & \|u_j(Y_1, X_1) - l_j(Y_1, X_1)\|_{L^2} \\ &= (F(t_j - |x) - F(t_{j-1} |x))^{1/2} \left\{ f(x) \int K^2(u) \, du + O(h) \right\}^{1/2} \\ &= (F(t_j - |x) - F(t_{j-1} |x))^{1/2} \|F_n(Y_1, X_1)\|_{L^2} (1 + o(1)) \\ &\leq \varepsilon \|F_n(Y_1, X_1)\|_{L^2}, \end{aligned}$$

Consequently $N_{[\cdot]}(\varepsilon \|F_n(Y_1, X_1)\|_{L^2}, \mathcal{F}_n, L^2(\mathbb{P})) \leq k \leq \lceil 2/\varepsilon^2 \rceil$. We then obtain

$$\int_{(0,1)} \sqrt{\log N_{[\cdot]}(\varepsilon \|F_n(Y_1, X_1)\|_{L^2}, \mathcal{F}_n, L^2(\mathbb{P}))} \, d\varepsilon \leq \int_{(0,1)} \sqrt{\log(2/\varepsilon^2 + 1)} \, d\varepsilon < \infty,$$

which shows (1.9). Finally, since the partitions were chosen independently of n , (1.8) does not need to be verified. The bias term is handled as in finite dimensions and the proof is done. $\square$

2.3 Bandwidth selection

All the above models crucially depend on an adequate choice of bandwidth. There exist many approaches in the literature for selecting an appropriate bandwidth, most of them minimising certain criterion functions, as in the AMISE example at the beginning of the chapter, or in a data-driven way. This section anticipates one of the fundamental concepts used in machine learning to provide a bandwidth selection strategy based on predictive performance. A broader and more detailed discussion pertains to later chapters. We first illustrate the approach for density estimation, and then note how it extends to conditional mean and distribution estimation.

The bandwidth selection strategy that we present is the *leave-one-out cross-validation* technique and is applicable in similar ways for density, conditional mean (local regression), and conditional distribution estimation.

2.3.1 Density estimator

For density estimators we evaluate the quality of the estimator $\mathbb{F}^{(n)}$ in terms of a specific risk measure called the *mean integrated squared error*, defined as

$$R(h) = \mathbb{E}[L(h)] = \mathbb{E} \int (\mathbb{F}^{(n)}(x) - f(x))^2 dx.$$

However, the true f is not available, which appears not only inside the square term, but also through integration. Note that we may rewrite

$$L(h) = \int \mathbb{F}^{(n)}(x)^2 dx - 2 \int \mathbb{F}^{(n)}(x)f(x) dx + \int f^2(x) dx,$$

and the last term does not depend on the bandwidth. Thus, we are interested in minimising the expected value of

$$\begin{aligned} J(h) &= \int \mathbb{F}^{(n)}(x)^2 dx - 2 \int \mathbb{F}^{(n)}(x)f(x) dx \\ &= \int \mathbb{F}^{(n)}(x)^2 dx - 2\mathbb{E}\mathbb{F}^{(n)}(X). \end{aligned}$$

The expectation in the second term could in principle be approximated by an empirical average of $\mathbb{F}^{(n)}$ at each observed X_i :

$$\frac{1}{n} \sum_{i=1}^n \mathbb{F}^{(n)}(X_i).$$

The problem, however, is that the maximisation of such quantity would tend to drastically overfit the data, since it is optimal to take a very spiky $\mathbb{F}^{(n)}$ at each datapoint. The fundamental technique to overcome this issue is to fit the estimator with all the data except X_i , call it $\mathbb{F}_{-i}^{(n)}$, so that it does not use the value X_i , and then the empirical average can reflect a true loss.

Definition 2.4 (Cross-validation risk estimator and bandwidth, for the density estimator) We define the cross-validation risk estimator as

$$\hat{J}(h) = \int \mathbb{F}^{(n)}(x)^2 dx - \frac{2}{n} \sum_{i=1}^n \mathbb{F}_{-i}^{(n)}(X_i).$$

The minimiser of the above function provides the cross-validation bandwidth, that is

$$\hat{h} = \operatorname{argmin}_h \hat{J}(h).$$

The latter bandwidth selection strategy, along with several other alternatives and refinements are usually implemented in standard software packages².

²For instance via the R function `density()`.

2.3.2 Local regression

The risk in the local regression case is the *mean squared error*, given by

$$R(h) = \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (\mathfrak{e}^{(n)}(X_i) - e(X_i))^2 \right],$$

where we once again encounter a problem of a term being unknown, in this case, the $e(\cdot)$ function. Similarly, the natural guess for estimating the mean squared error is simply

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \mathfrak{e}^{(n)}(X_i))^2,$$

which turns out to be a poor estimator and drastically overfits (or under-smooths) the data for the same reason as in the density. The correct approach is to use cross-validation now with the estimator fitted without the i -th pair (Y_i, X_i) , denoted by $\mathfrak{e}_{-i}^{(n)}$, as the function to test Y_i against.

Definition 2.5 (Cross-validation risk estimator and bandwidth, for the regression estimator) We define the cross-validation risk estimator as

$$\hat{R}(h) = \frac{1}{n} \sum_{i=1}^n (Y_i - \mathfrak{e}_{-i}^{(n)}(X_i))^2.$$

The minimiser of the above function provides the cross-validation bandwidth, that is

$$\hat{h} = \operatorname{argmin}_h \hat{R}(h).$$

A good rule of thumb is to *always* plot $\hat{R}(h)$ to verify that the minimum is well-behaved. Other approaches may be found for instance in Chapter 5 of [Wasserman \[2006\]](#).

2.3.3 Distributional regression

In the case of distributional regression, things can be a bit trickier to define. Any bandwidth criterion must aggregate over t , for instance by integrating over t or by focusing on a set of quantiles of interest. A naive choice can over-emphasise the central part of the distribution, so tail-sensitive goals may require reweighting. We would like to minimise the following risk function

$$\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (\mathbb{F}^{(n)}(t | X_i) - F(t | X_i))^2 \right]$$

for all t at the same time. Several approaches come directly to mind:

$$\begin{aligned} R_1(h) &= \frac{1}{n} \sum_{i=1}^n \int (\mathbb{F}^{(n)}(t | X_i) - F(t | X_i))^2 dt \\ R_2(h) &= \frac{1}{n} \sum_{i=1}^n \int (\mathbb{F}^{(n)}(t | X_i) - F(t | X_i))^2 dF(t | X_i). \end{aligned}$$

However – and quite surprisingly – this *does not work*. Several workarounds are possible, though not entirely straightforward. The issue is that the asymptotic behaviour of the leave-one-out cross-validation in this case is equivalent to not leaving any observation out (cf. Altman and Léger [1995]), in stark contrast to density estimation and even local regression where this asymptotic equivalence does not hold. A simple practical solution which works reasonably well is to find the optimal $\hat{h}$ for density estimation of X as per Definition 2.4 and plug it into $\mathbb{F}^{(n)}(t | X_i)$.

Numeric example 2.2 (Conditional distributional regression) One convenient aspect of distributional regression is that, apart from a global bandwidth selection, no model choice is required. For instance, consider risks Y with conditional Weibull distribution given a covariate X , with constant (unit) scale parameter and shape parameter specified by $1/X$. Let X be uniformly distributed, which in practice can (and should) be attained by transforming the covariates X through the probability integral transform (PIT), that is by applying the empirical distribution function $\mathbb{F}_X^{(n)}(\cdot)$ of the data (X_n) to the data itself. We simulate $n = 5000$ joint observations. By applying the naïve plug-in strategy to find $\hat{h}$ delineated above, we get a satisfactory performance as shown in Figure 2.2, where we have also provided two sets of confidence bounds constructed by: a) not considering the pathwise properties (separately for each t); and b) considering the pathwise properties and using the result of Theorem 2.8 and the fact that the distribution of

$$\text{KS} = \sup_t |\mathbb{B}(t)|$$

is the well-known Kolmogorov–Smirnov distribution, which can explicitly be computed through

$$\mathbb{P}(\text{KS} \leq x) = 1 - 2 \sum_{k=1}^{\infty} (-1)^{k-1} e^{-2k^2 x^2} = \frac{\sqrt{2\pi}}{x} \sum_{k=1}^{\infty} e^{-(2k-1)^2 \pi^2 / (8x^2)}.$$

The difference between confidence bands at their thickest is not negligible. Notice also the simplicity of the approach, which automatically captures the nonlinear dependence structure of the vectors (Y, X) , which can serve as a particularly useful tool to verify the adequacy of other models. $\diamond$

2.4 k -nearest neighbours distributional regression

We now parallel the kernel-based distributional-regression analysis with a k -nearest-neighbour localisation. The strategy is the same as before, where we aim to establish pathwise weak convergence for the conditional cdf estimator, then obtain the conditional quantile process by the functional δ -method.

Definition 2.6 (k -nearest-neighbour conditional empirical distribution function) For $t \in [0, \infty)$ and fixed $x \in \mathbb{R}$, define

$$\mathbb{F}_{\kappa}^{(n)}(t | x) = \frac{1}{\kappa} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} 1_{(Y_i \leq t)},$$

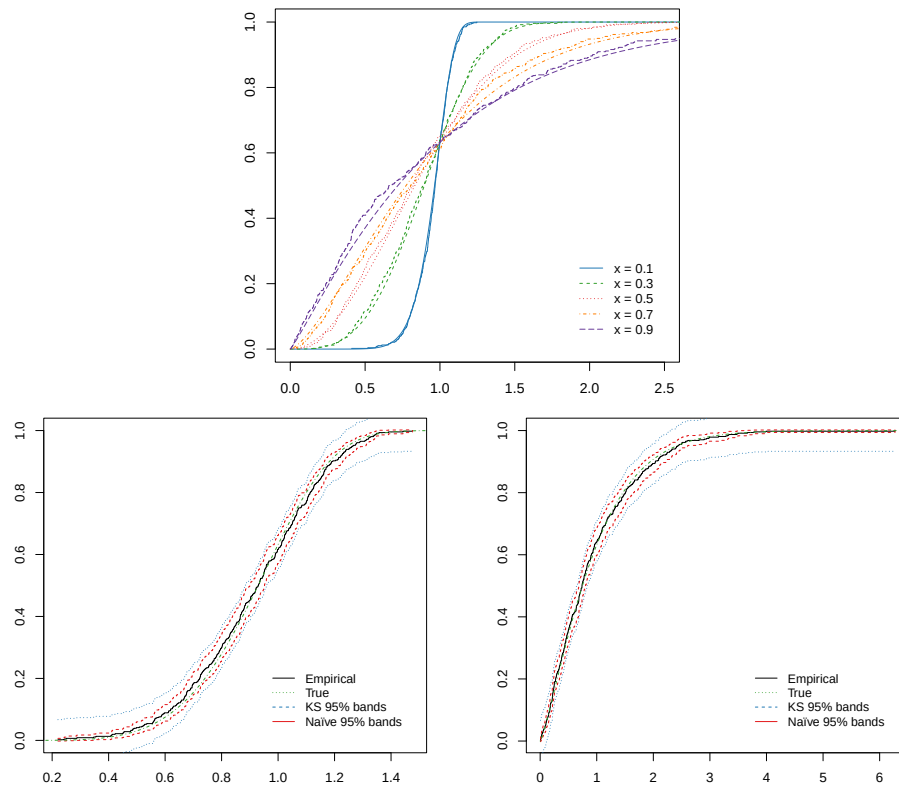


Figure 2.2: Nonparametric distributional regression: conditional distribution functions $y \mapsto F(y|x)$ for various x -values (top), and Kolmogorov–Smirnov and naïve confidence bands for $x = 0.2$ and $x = 0.8$ (bottom).

where $\mathcal{K}_{n,\kappa}(x)$ denotes the indices of the κ nearest covariates X_i to x .

The next result mirrors Theorem 2.8 and is obtained by verifying the conditions of Theorem 1.21 and Corollary 1.2 for the indicator class.

Theorem 2.9 (Pathwise weak convergence of the k -NN cdf) *Let x be such that $f(x) > 0$, with f continuous at x . Let $\kappa = \kappa_n$ satisfy $\kappa \rightarrow \infty$ and $\kappa^{3/2}/n \rightarrow 0$. Assume:*

(i) *For every $t \in \mathbb{R}$, the map $z \mapsto F(t | z)$ is continuous at x .*

(ii) *There exist constants $\tau_x, L_x > 0$ such that*

$$\sup_{t \in \mathbb{R}} |F(t | z) - F(t | x)| \leq L_x \|z - x\|, \quad z \in B(x, \tau_x).$$

Then

$$\sqrt{\kappa}(\mathbb{F}_{\kappa}^{(n)}(\cdot | x) - F(\cdot | x)) \xrightarrow{d} \mathbb{B} \circ F(\cdot | x)$$

in $\ell^\infty([0, \infty))$, where $\mathbb{B}$ is a standard Brownian bridge.

Proof. Set

$$\mathcal{F} = \{f_t : t \in \mathbb{R}, f_t(y) = 1_{(-\infty, t]}(y)\}.$$

Let $F \equiv 1$ be the envelope of $\mathcal{F}$. We verify (1.10)–(1.13) in Theorem 1.21. First, for fixed Q on $\mathbb{R}$ and $\varepsilon \in (0, 1)$, the same bracketing construction as in the proof of Theorem 1.22 (class of half-line indicators) gives

$$N_{[]}(\varepsilon \|F\|_{L^2(Q)}, \mathcal{F}, L^2(Q)) \leq 2/\varepsilon^2 + 1.$$

The bound is distribution-free, hence valid for all $Q = \mu_{x,u}$ with $|u| \leq \delta$. Hence, for every $\delta_n \rightarrow 0$,

$$\sup_{|u| \leq \delta} J(\mathcal{F}, L^2(\mu_{x,u}), \delta_n) \leq \int_0^{\delta_n} \sqrt{\log(2/\varepsilon^2 + 1)} \, d\varepsilon \rightarrow 0,$$

which is exactly (1.10). Second, the local Lindeberg condition is immediate, since

$$\sup_{|u| \leq \delta} \mu_{x,u}(F^2 1_{(F > M)}) = 0, \quad M > 1.$$

Thus (1.11) holds. Third, we verify (1.12). Fix $\eta > 0$. For each $u \in [0, \delta]$, by the same bracketing bound there exists a finite partition $-\infty = t_0^{(u)} < t_1^{(u)} < \dots < t_{m_u}^{(u)} = \infty$ with

$$\mu_{x,u}((t_{j-1}^{(u)}, t_j^{(u)}]) \leq \eta^2/4, \quad j = 1, \dots, m_u.$$

For fixed $(a, b]$, the map $v \mapsto \mu_{x,v}((a, b])$ is continuous on $[0, \delta]$; therefore each partition above remains bounded by $\eta^2/2$ on some neighbourhood U_u of u . By

compactness, choose $u_1, \dots, u_M$ with $[0, \delta] \subset \bigcup_{m=1}^M U_{u_m}$ and merge all corresponding partition endpoints into one grid $-\infty = s_0 < s_1 < \dots < s_K = \infty$. Then for every $v \in [0, \delta]$ and every $r = 1, \dots, K$,

$$\mu_{x,v}((s_{r-1}, s_r]) \leq \eta^2/2.$$

Hence, if $t \in (s_{r-1}, s_r]$,

$$\rho_\delta(f_t, f_{s_{r-1}})^2 = \sup_{|v| \leq \delta} \mu_{x,v}((s_{r-1}, t]) \leq \sup_{|v| \leq \delta} \mu_{x,v}((s_{r-1}, s_r]) \leq \eta^2/2.$$

Therefore

$$\sup_{f \in \mathcal{F}} \min_{0 \leq r \leq K} \rho_\delta(f, f_{s_r}) \leq \eta,$$

which is (1.12). Fourth, continuity of the covariance map is immediate, as for $s, t \in \mathbb{R}$,

$$z \mapsto (\mu_z(f_s f_t), \mu_z(f_t), \mu_z(F^2)) = (F(s \wedge t | z), F(t | z), 1),$$

is continuous at x by (i), so (1.13) holds. Finally, we check (1.14)–(1.15) from Corollary 1.2. For $\mathcal{F}$,

$$\sup_{f \in \mathcal{F}} |(\mu_z - \mu_x)(f)| = \sup_{t \in \mathbb{R}} |F(t | z) - F(t | x)| \leq L_x \|z - x\|$$

by (ii), which is (1.14). Also, here $d = 1$, so (1.15) reads $\kappa^{(d+2)/2}/n = \kappa^{3/2}/n \rightarrow 0$, which is assumed. Therefore Corollary 1.2 applies directly and yields the claim. $\square$

Remark 2.2 Condition (ii) is often verifiable directly. For instance, if $f_{Y|X}(\cdot | z)$ exists for z near x and

$$\sup_{|z-x| \leq \tau} \int_{\mathbb{R}} \left| \frac{\partial}{\partial z} f_{Y|X}(y | z) \right| dy < \infty,$$

then $\sup_t |F(t | z) - F(t | x)| \leq L_x |z - x|$ for some $L_x < \infty$ by the mean value theorem. $\circ$

Remark 2.3 Compared with Theorem 2.8, the normalisation changes from $\sqrt{n\hbar}$ to $\sqrt{\kappa}$ and the variance factor $\int K^2/f(x)$ disappears. This reflects the density-adaptive localisation of k -NN weighting.

The proof of the k -NN distributional regression weak convergence, Theorem 2.9, might seem more straightforward and less layered than its kernel smoothing counterpart, Theorem 2.8, but this is only because much of the k -NN tailoring is done behind the scenes in Theorem 1.21 and Corollary 1.2. $\circ$

2.5 Notes and Comments

The kernel estimators used in this chapter trace back to early ideas independently developed by Rosenblatt [1956] and Parzen [1962]. Their setup then became the template for modern treatments of kernel bias, variance, and asymptotic normality, with classic expositions including Silverman [2018], Wand and Jones [1994], Scott [2015]. Regression smoothing via kernels emerged soon after, with the conditional mean estimator independently introduced by Nadaraya [1964] and Watson [1964], and the closely related product of the regression function and marginal density considered by Priestley and Chao [1972]. As seen in this chapter, under mild conditions, these estimators are weakly consistent and, after centring and appropriate scaling, converge to Gaussian limits. Now, while their joint asymptotics follow from standard theorems such as the Cramér–Wold device and the Lindeberg–Feller CLT, the empirical process approach given here follows the techniques of Shorack and Wellner [2009] and van der Vaart and Wellner [1996]. From a general perspective, the minimax benchmarks that motivate kernel choices and bandwidth rates were explored in Stone [1980] and, in the nonparametric regression setting, in Stone [1982]. For regression smoothing beyond simple kernels, local polynomial methods provide bias reduction and better boundary behaviour; see Fan and Gijbels [2018]. Broader theoretical surveys include Györfi et al. [2002], Tsybakov [2009], while Wasserman [2006] offers a compact and applied overview. Classical results on kernel-shape efficiency include Epanechnikov [1969]. Practical issues, such as boundary bias on one-sided supports and the effects of kernel shape, are discussed throughout the above-cited works and in focused papers, for instance Hall et al. [1999].

After these core estimators, the chapter turns to distributional regression and bandwidth choice. Reliable plug-in rules and related data-driven criteria for density estimation are given by Sheather and Jones [1991] and for local least-squares regression by Ruppert et al. [1995]; a general treatment is given in Scott [2015], including the multivariate case. Foundational predictive cross-validation papers include Stone [1974], Allen [1974], Geisser [1975], and density-oriented cross-validation smoothing is classically treated in Bowman [1984]. For mixed continuous and discrete covariates, from an econometrics perspective, see Li and Racine [2007]. In distributional regression built from a conditional empirical distribution, the pathwise weak-convergence arguments use bracketing and empirical-process tools of van der Vaart and Wellner [1996], Shorack and Wellner [2009], while almost sure convergence results for conditional empirical distribution functions were established by Stute [1986]. Bandwidth choice is more delicate for conditional distribution estimators. It was shown by Altman and Léger [1995] that naïve leave-one-out cross-validation can behave unsatisfactorily. Empirical processes for k -nearest neighbours were considered in Portier [2025], extending classical nearest-neighbour roots in classification, density and regression such as Cover and Hart [1967], Loftsgaarden and Quesenberry [1965], Stone [1977]; see also references therein. Although our emphasis here is theoretical, these smoothing ideas are central for practical work in insurance data exploration, ratemaking and claim size modelling, where flexible nonparametric structure is a valuable applied tool, see Silverman [2018], Scott [2015].

2.6 Exercises

1. The Epanechnikov kernel is given by

$$K(x) = \frac{3}{4}(1 - x^2),$$

for $x \in [-1, 1]$ and 0 elsewhere. Show that it satisfies the kernel requirements.

2. Recall the Priestly–Chao estimator $\mathfrak{m}^{(n)}(x) = (nh)^{-1} \sum_{i=1}^n Y_i K(h^{-1}(x - X_i))$. The mean squared error of $\mathfrak{m}^{(n)}(x)$ can be decomposed into a sum of the variance and the squared bias as follows:

$$\mathbb{E}[(\mathfrak{m}^{(n)}(x) - m(x))^2] = \mathbb{E}[(\mathfrak{m}^{(n)}(x) - \mathbb{E}[\mathfrak{m}^{(n)}(x)])^2] + (\mathbb{E}[\mathfrak{m}^{(n)}(x)] - m(x))^2,$$

where we examined the latter two quantities in Theorem 2.3. Plugging these quantities into the mean squared error, we obtain that

$$\begin{aligned} \mathbb{E}[(\mathfrak{m}^{(n)}(x) - m(x))^2] &= \left(\frac{h^2}{2} m''(x) \int u^2 K(u) \, du \right)^2 + o(h^4) \\ &\quad + \frac{\mathbb{E}[Y^2 | X = x] f(x)}{nh} \int K^2(u) \, du + O(1/n). \end{aligned}$$

For large n , define the asymptotic mean squared error as

$$\begin{aligned} \text{AMSE}(\mathfrak{m}^{(n)}(x)) &= \left(\frac{h^2}{2} m''(x) \int u^2 K(u) \, du \right)^2 \\ &\quad + \frac{\mathbb{E}[Y^2 | X = x] f(x)}{nh} \int K^2(u) \, du. \end{aligned}$$

- (a) For fixed x , show that the optimal bandwidth that minimises the $\text{AMSE}(\mathfrak{m}^{(n)}(x))$ is given by

$$\hat{h}(x) = \left(\frac{\mathbb{E}[Y^2 | X = x] f(x) \int K^2(u) \, du}{(m''(x))^2 (\int u^2 K(u) \, du)^2} \right)^{\frac{1}{5}} n^{-\frac{1}{5}}.$$

- (b) Show that the optimal bandwidth that minimises $\int \text{AMSE}(\mathfrak{m}^{(n)}(x)) \, dx$, is given by

$$\hat{h} = \left(\frac{\mathbb{E}[Y^2] \int K^2(u) \, du}{\int (m''(x))^2 \, dx (\int u^2 K(u) \, du)^2} \right)^{\frac{1}{5}} n^{-\frac{1}{5}}.$$

- (c) Using the above optimal bandwidths, determine the decay rate, as a function of n , of $\text{AMSE}(\mathfrak{m}^{(n)}(x))$ and $\int \text{AMSE}(\mathfrak{m}^{(n)}(x)) \, dx$, and compare it with maximum likelihood estimation in a parametric setting.

3. Recall the definition of the cross-validation risk estimator:

$$\hat{J}(h) = \int \mathfrak{f}^{(n)}(x)^2 \, dx - \frac{2}{n} \sum_{i=1}^n \mathfrak{f}_{-i}^{(n)}(X_i).$$

Let $K_{(2)}$ denote the convolution of K with itself:

$$K_{(2)}(v) = \int_{-\infty}^{\infty} K(u) K(v - u) \, du.$$

- (a) Show that $\int \mathbb{F}^{(n)}(x)^2 dx = 1/(n^2 h) \sum_{i=1}^n \sum_{j=1}^n K_{(2)}(h^{-1}(X_j - X_i))$.
- (b) If we use the Gaussian kernel $K(u) = e^{-u^2/2}/\sqrt{2\pi}$, then $K_{(2)}(u) = 1/(2\sqrt{\pi})e^{-u^2/4}$. Show that in this case

$$\begin{aligned} \hat{J}(h) &= \frac{1}{n^2 h} \sum_{i=1}^n \sum_{j=1}^n 1/(2\sqrt{\pi}) e^{-\left(\frac{X_i - X_j}{h}\right)^2/4} \\ &\quad + \frac{2}{\sqrt{2\pi} h(n-1)} - \frac{2}{n(n-1)h} \sum_{i=1}^n \sum_{j=1}^n 1/\sqrt{2\pi} e^{-\left(\frac{X_i - X_j}{h}\right)^2/2}. \end{aligned}$$

4. Go through the details of the proof of Theorem 2.8.
5. Let $F(\cdot | x)$ be the conditional distribution function of Y given $X = x$, and let

$$q_\alpha(x) = F^{\leftarrow}(\alpha | x), \quad \alpha \in (0, 1),$$

denote the conditional quantile. Define the conditional empirical quantile

$$\hat{q}_{n,\alpha}(x) = \mathbb{F}^{(n)\leftarrow}(\alpha | x).$$

Assume the conditions of Theorem 2.8, and assume that $F(\cdot | x)$ has a density $f(\cdot | x)$ that is continuous and strictly positive at $q_\alpha(x)$. Show that the inverse map is Hadamard differentiable at $F(\cdot | x)$ and use the functional δ -method to show that, for fixed $\alpha \in (0, 1)$,

$$\sqrt{nh}(\hat{q}_{n,\alpha}(x) - q_\alpha(x)) \xrightarrow{d} N\left(0, \frac{\int K^2(u) du}{f(x)} \frac{\alpha(1-\alpha)}{f(q_\alpha(x) | x)^2}\right).$$

6. Let $\sigma^2(x) = \text{Var}(Y_1 | X_1 = x)$ and define

$$\mathfrak{m}_2^{(n)}(x) = \frac{1}{nh} \sum_{i=1}^n Y_i^2 K(h^{-1}(x - X_i)), \quad \hat{\sigma}^2(x) = \frac{\mathfrak{m}_2^{(n)}(x)}{\mathbb{F}^{(n)}(x)} - (\mathfrak{e}^{(n)}(x))^2.$$

Assume the conditions of Theorem 2.4 and that $\mathbb{E}[Y_1^4 | X_1 = x] < \infty$. Derive the leading bias of $\hat{\sigma}^2(x)$ and show that

$$\sqrt{nh}(\hat{\sigma}^2(x) - \sigma^2(x)) \xrightarrow{d} N(0, \tau^2(x)),$$

for a variance $\tau^2(x)$ that you should express in terms of conditional moments of Y given $X = x$.

7. Assume X has support $[0, 1]$ and the kernel K has support on $[-1, 1]$. Consider the conditional empirical distribution function $\mathbb{F}^{(n)}(t | x)$ at the boundary point $x = 0$. Proceed in two stages, first for the naive estimator and then for the reflection correction.
- (a) Show that $\mathbb{E}[\mathbb{F}^{(n)}(t | 0)]$ has a non-vanishing boundary bias, and identify its leading term.
- (b) Define a reflection-based estimator $\mathbb{F}_R^{(n)}(t | 0)$ by adding reflected covariates $-X_i$ with the same responses. Show that the leading boundary bias vanishes and give the resulting order of the bias.

- (c) Discuss a condition on the joint distribution of (X, Y) at $x = 0$ under which the reflection estimator achieves an $O(h^2)$ bias rate.
- 8. State and prove a functional central limit theorem for the conditional mean excess function.
- 9. State and prove a functional central limit theorem for the conditional cumulative hazard $\Lambda^{(n)}(\cdot|x)$, without product integration, suitably adapted from the first chapter.

Practical exercises

1. Consider the following simulation study:
 - (a) Simulate 1000 observations of a standard normal random variable. Implement the Parzen–Rosenblatt density estimator using the Epanechnikov kernel. Use the following bandwidths: 0.1, 0.3, 0.5, 0.7, 1.0.
 - (b) Let $X_1 \sim \text{Unif}(0, 1)$, $X_2 \sim \text{Unif}(1, 2)$ and $p \sim \text{Bernoulli}(0.8)$. Define

$$Y = pX_1 + (1 - p)X_2.$$

Use the same bandwidths.

- (c) Implement κ -nearest neighbours in both cases, for $\kappa = 10, 20, 50, 100, 200$.
2. We wish to investigate when it is appropriate to use kernel estimation.
 - (a) Simulate 1000 observations from a standard normal random variable. Estimate the density via kernel estimation and with the MLE. Plot the densities together with the theoretical density, and assess the fit overall and in the tail.
 - (b) Do the same experiment, but simulate now from a Pareto distribution with shape parameter 2 and threshold 1.
 - (c) Let $X_1 \sim \text{Unif}(0, 1)$, $X_2 \sim \text{Unif}(1, 2)$ and $p \sim \text{Bernoulli}(0.8)$. Define

$$Y = pX_1 + (1 - p)X_2$$

Estimate the density with kernels and assess the fit.

- (d) Repeat the three experiments with a different kernel using the R function `density()`. Assess the influence of the kernel choice and indicate when it matters.
- (e) Repeat the three experiments with a κ -nearest neighbours estimator with κ of your choice. Comment on the differences.
3. Consider the dataset `frecomfire`, from the R package `CASdatasets` [Dutang and Charpentier \[2024\]](#). Compute $\hat{h} = \text{argmin}_h \hat{J}(h)$ from Exercise 3 for the covariate `ClaimCost2007` using the first

$$n = 100, 500, 2000, 3000, 4500, 7000, 9613$$

observations, and plot $\hat{h}$ as a function of n .

4. Consider the dataset `brvehins1a` from the R package `CASdatasets` [Dutang and Charpentier \[2024\]](#), which is a Brazilian vehicle insurance dataset (by policy) with risk features. In this exercise, the response variable is `ClaimAmountTotColl` (denoted Y), which is the claim sizes caused by total collisions, given the covariate, sum insured `SumInsAvg` (denoted X).
 - (a) Load the data, select the variables `SumInsAvg` and `ClaimAmountTotColl`, and filter the data such that `ClaimAmountTotColl` $\in (0, M)$, with $M = 200,000$.
 - (b) Transform `SumInsAvg` = X to uniform variables by applying the empirical distribution function $\mathbb{F}_X^{(n)}(\cdot)$ on X .
 - (c) Let n be the number of observations in the dataset after filtering, and define a bandwidth $h_n = 3n^{-1/5}/\log \log n$. Let $x = 0.7$. Consider the grid of indices from 0 to M with step size 100. Compute the empirical distribution function $\mathbb{F}_Y^{(n)}(\cdot)$ and the conditional empirical distribution function, as per Definition 2.3, $\mathbb{F}^{(n)}(\cdot | 0.7)$ using the Gaussian kernel.
 - (d) We now compare the 95% naive confidence bands with the pathwise confidence bands. Plot $\mathbb{F}^{(n)}(\cdot | 0.7)$ with the 95% pointwise confidence bands arising from Theorem 2.4, and add the pathwise 95% confidence bands that arise from the limiting distribution of the quantity $\sqrt{nh_n} \sup_{t \in (0, M)} |\mathbb{F}^{(n)}(t | 0.7) - F(t | 0.7)|$, using Theorem 2.8.
Hint: The 95% quantile of the Kolmogorov distribution is $\text{KS}^{-1}(1 - 0.05) \approx 1.36$.

Chapter 3

Heavy-tailed distributions

Large claims and catastrophic losses are very common in non-life insurance, and they are, by definition, driven by tail behaviour. In this chapter, we first clarify the main heavy-tail classes and their relationships. We then introduce extreme value theory, develop estimation tools for regularly varying tails, and close with conditional extremes and their limit theory.

3.1 Definitions of heavy-tailed distributions

There are several definitions of heavy-tailed risk. We start by setting terminology and the main tail classes, since the statistical tools later in the chapter depend on these distinctions. In probability theory, a random variable Z is loosely said to have a heavy-tailed distribution if it lacks a moment generating function in some neighbourhood of the origin. More precisely, define the moment generating function of Z as $M_Z(t) = \mathbb{E}[e^{tZ}]$, $t \in \mathbb{R}$. A distribution is *heavy-tailed* if there exists no $\varepsilon > 0$ such that $M_Z(t) < \infty$ for every $t \in (-\varepsilon, \varepsilon)$. Intuitively, this definition indicates that the tails of the distribution decay more slowly than those of an exponential distribution, and can be equivalently stated, for positive random variables, in terms of the survival function $\bar{F} = 1 - F$ of Z as

$$\lim_{z \rightarrow \infty} \exp(tz) \bar{F}(z) = \infty, \quad \text{for all } t > 0. \quad (3.1)$$

A more specific class of heavy-tailed distributions is the class of long-tailed distributions. A distribution is called *long-tailed* if for all $x \geq 0$, the following limit holds:

$$\lim_{z \rightarrow \infty} \frac{\mathbb{P}(Z > z + x)}{\mathbb{P}(Z > z)} = 1.$$

This property means that the tail of the distribution is asymptotically invariant under translations, implying that a large value is not significantly affected by adding any fixed amount x . Perhaps surprisingly, long-tailed distributions are always heavy-tailed. The converse is generally not true.

Regularly varying distributions are another important and widely used subclass of heavy-tailed distributions. A distribution F or random variable Z is *regularly varying* if its survival function is a regularly varying function at infinity, i.e. there exists a function ℓ , slowly varying at infinity, and an index $1/\gamma > 0$ such that:

$$\mathbb{P}(Z > z) = z^{-1/\gamma} \ell(z).$$

Here, ℓ is said to be slowly varying if for every $t > 0$, $\lim_{z \rightarrow \infty} \ell(tz)/\ell(z) = 1$. Regularly varying distributions have heavy tails, as can be directly seen from (3.1) if ℓ is increasing. The general case can be shown using the fact that ℓ cannot asymptotically decrease faster than any power; this is made precise below. Many important distributions such as Pareto and stable distributions are regularly varying.

Finally, subexponential distributions are characterised by the property that the sum of two independent random variables with the same distribution has a tail that is asymptotically equivalent to the tail of the maximum of the two. Formally, a distribution is *subexponential* if:

$$\lim_{z \rightarrow \infty} \frac{\mathbb{P}(Z_1 + Z_2 > z)}{2\mathbb{P}(Z > z)} = 1, \quad (3.2)$$

where Z_1 and Z_2 are independent copies of Z . Subexponential distributions are a superset of regularly varying distributions and capture the idea that large sums are governed by their largest term. Indeed if $M_2 = Z_1 \vee Z_2$ then $\mathbb{P}(M_2 > z) = 1 - F(z)^2 = 1 - (1 - \bar{F}(z))^2 \approx 2\bar{F}(z)$ for large z , and the concept is equivalent to the analogous property for n copies of Z , and replacing 2 with n in (3.2). This is also known as the *single big event principle*.

The relationships between these classes can be summarised as follows: regularly varying distributions are subexponential, subexponential distributions are long-tailed, and long-tailed distributions are heavy-tailed. In this chapter, we focus on regularly varying distributions, which are arguably the most of the above definitions in statistical terms.

3.2 Introduction to extreme value theory

We now connect tail classes to asymptotic behaviour of maxima, which motivates much of the estimation theory that follows. A standard introductory reference for extreme value theory is [Embrechts et al. \[1997\]](#).

3.2.1 General max-domains of attraction

Before delving into regularly varying distributions, we first present the Fisher–Tippett theorem, which describes the limiting behaviour of the maxima of iid random variables. This theorem introduces the Generalised Extreme Value (GEV) distribution.

Theorem 3.1 (Fisher–Tippett) *Let (Z_n) be a sequence of iid random variables with common distribution function F . If there exist sequences of constants $a_n >$*

0 and $b_n \in \mathbb{R}$ such that the distribution of the normalised maximum converges to a non-degenerate distribution function H , that is,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{\max(Z_1, Z_2, \dots, Z_n) - b_n}{a_n} \leq z \right) = H(z), \quad (3.3)$$

then H belongs to one of the three types of distributions, jointly parametrised by the Generalised Extreme Value (GEV) distribution:

$$H(z) = \exp \left(- \left(1 + \gamma \frac{z - \mu}{\sigma} \right)^{-1/\gamma} \right),$$

defined on the set $\{z : 1 + \gamma \frac{z - \mu}{\sigma} > 0\}$, where $\mu \in \mathbb{R}$, $\sigma > 0$, and $\gamma \in \mathbb{R}$. The $\gamma = 0$ expression can be understood as a limit.

The GEV distribution encompasses three families: the Gumbel, Fréchet, and Weibull distributions, which correspond to $\gamma = 0$, $\gamma > 0$, and $\gamma < 0$, respectively.

A distribution F is said to belong to the Gumbel, Fréchet, and Weibull *max-domain of attraction* if (3.3) holds with $\gamma = 0$, $\gamma > 0$, and $\gamma < 0$, respectively. We have the following:

1. All distributions in the Weibull max-domain of attraction have finite right-endpoints, and thus are not heavy-tailed at all.
2. Distributions in the Gumbel max-domain of attraction include light-tailed ones such as the exponential or normal distributions, but also heavy-tailed ones such as the Weibull or lognormal distributions.
3. For a distribution in the Fréchet max-domain of attraction, its tails are heavier than in the other two max-domains of attraction. In fact, we may characterise them in terms of regularly varying functions as follows.

The next theorem makes this characterisation precise.

Theorem 3.2 (Fréchet max-domain and regular variation) *A distribution function F belongs to the Fréchet max-domain of attraction if and only if its survival function is regularly varying with index $1/\gamma > 0$.*

This result shows that regularly varying distributions form a significantly large class within the heavy-tailed distributions and that polynomial decay of tails is not an arbitrary choice, but rather one of three ways that maxima behave asymptotically. A similar result is as follows, in terms of the Generalised Pareto Distribution (GPD).

Definition 3.1 (Generalised Pareto Distribution (GPD)) The Generalised Pareto Distribution is given by the cumulative distribution function:

$$G(y) = \begin{cases} 1 - (1 + \gamma \frac{y - \mu}{\sigma})^{-1/\gamma} & \text{if } \gamma \neq 0, \\ 1 - \exp(-\frac{y - \mu}{\sigma}) & \text{if } \gamma = 0, \end{cases} \quad (3.4)$$

for $y \geq \mu$ when $\gamma \geq 0$, and $\mu \leq y \leq \mu - \sigma/\gamma$ when $\gamma < 0$, where $\mu \in \mathbb{R}$ is the location parameter, $\sigma > 0$ is the scale parameter, and $\gamma \in \mathbb{R}$ is the shape parameter.

Theorem 3.3 (Pickands–Balkema–de Haan) *Let Z be a random variable with distribution function F . Then F is in the max-domain of attraction of some extreme value distribution if and only if the exceedances over high thresholds converge to a Generalised Pareto Distribution (GPD). Formally, if and only if there exists a positive measurable function $a(u)$ such that*

$$\lim_{u \rightarrow x_F} \mathbb{P}((Z - u)/a(u) \leq z \mid Z > u) = G(z),$$

where G is the GPD with $\mu = 0$ and $\sigma = 1$, and where $x_F = \sup\{z : F(z) < 1\}$ is the right endpoint of F .

Thus, if the maxima of blocks of some data follow approximately a GEV distribution, then the exceedances over a high threshold of the same data are approximately GPD distributed, with the same γ . This relationship highlights that both the GPD and GEV describe different aspects of the same underlying extremal behaviour. The unification of all max-domains of attraction under the GEV/GPD framework is convenient for statistical estimation when data are bundled into *block-maxima* or treated as *peaks-over-thresholds*. In that case, maximum likelihood estimation of H and G is a viable option.

3.2.2 Results for regularly varying distributions

We survey some of the main properties of regularly varying distributions. These results are also helpful before moving to nonparametric estimation of $\gamma > 0$, the key parameter for this class of heavy-tailed distributions, by Theorem 3.2. A standard reference is ?. The order is deliberate. Uniform convergence gives the baseline tail approximation, Potter bounds provide robust inequalities, and the representation and Karamata results provide structural and integral tools used later in the asymptotic theory.

Theorem 3.4 (Uniform convergence property of regularly varying functions) *Let F be a regularly varying distribution with index $1/\gamma$, i.e. $\bar{F}(z) = z^{-1/\gamma}\ell(z)$ where ℓ is slowly varying. Then for any fixed $a > 0$, the following convergence holds:*

$$\lim_{z \rightarrow \infty} \sup_{t \in [a, \infty)} \left| \frac{\bar{F}(tz)}{\bar{F}(z)} - t^{-1/\gamma} \right| = 0. \quad (3.5)$$

The first version of Potter's bounds below deals with the survival function directly, and shows, in particular, that the oscillation of the slowly varying component can be bounded by any power.

Theorem 3.5 (Potter's bounds, first version) *Let F be a regularly varying distribution with index $1/\gamma$. For any $\varepsilon \in (0, 1/\gamma)$ there exist positive finite constants such that*

$$\begin{aligned} \bar{F}(z) &\leq cst \, z^{-1/\gamma+\varepsilon}, & z > 0, \\ \bar{F}(z) &\geq cst \, z^{-1/\gamma-\varepsilon}, & z > \varepsilon. \end{aligned}$$

Proof. We rewrite the tail in terms of the slowly varying function ℓ and bound it uniformly above and below. Set $\ell(z) = z^{1/\gamma}\bar{F}(z)$, which is a slowly varying

function. Let $\varepsilon \in (0, 1/\gamma)$, and note that $z \mapsto z^{-\varepsilon}\ell(z)$ is a regularly varying function with tail index $-\varepsilon$. We use that $\lim_{z \rightarrow \infty} z^{-\varepsilon}\ell(z) = 0$. In particular, there exists a b' such that for all $b \geq b'$ it holds that $\sup_{z \geq b} z^{-\varepsilon}\ell(z) < \infty$. In addition we note that $z^{-\varepsilon}\ell(z) = z^{-\varepsilon+1/\gamma}\bar{F}(z)$ is bounded on the compact interval $[1, b']$ i.e

$$\sup_{z \in [1, b']} z^{-\varepsilon}\ell(z) < \infty.$$

For $z \in (0, 1)$ we see that $z^{-\varepsilon}\ell(z) = z^{1/\gamma-\varepsilon}\bar{F}(z) \leq 1$. Consequently $z \mapsto z^{-\varepsilon}\ell(z)$ is uniformly bounded on $(0, \infty)$ and thus for all $z > 0$,

$$\bar{F}(z) = z^{-1/\gamma+\varepsilon} z^{-\varepsilon}\ell(z) \leq z^{-1/\gamma+\varepsilon} \sup_{x>0} x^{-\varepsilon}\ell(x) = \text{cst } z^{-1/\gamma+\varepsilon}.$$

This proves the first statement.

For the second statement, note that $z \mapsto z^{\varepsilon}\ell(z)$ is regularly varying with tail index ε . We use that $\lim_{z \rightarrow \infty} z^{\varepsilon}\ell(z) = \infty$. Thus for any $z > \varepsilon$, it holds that $z \mapsto z^{\varepsilon}\ell(z) = z^{\varepsilon+1/\gamma}\bar{F}(z)$ is strictly positive and diverging to infinity. Consequently $\inf_{x \geq \varepsilon} x^{\varepsilon}\ell(x) > 0$ and hence

$$\bar{F}(z) = z^{-1/\gamma-\varepsilon} z^{\varepsilon}\ell(z) \geq z^{-1/\gamma-\varepsilon} \inf_{x \geq \varepsilon} x^{\varepsilon}\ell(x) = \text{cst } z^{-1/\gamma-\varepsilon}.$$

□

The second version of Potter's bounds provides a similar control, but for the ratio of tails, which can be used to quantify the quality of the regular variation property.

Theorem 3.6 (Potter's bounds, second version) *Let F be a regularly varying distribution with index $1/\gamma$. For any $\varepsilon > 0$, there exists a finite positive constant such that for all $x \geq 1$ and $y > 0$,*

$$\begin{aligned} \frac{\bar{F}(x/y)}{\bar{F}(x)} &\leq \text{cst } (y^{1/\gamma-\varepsilon} \vee y^{1/\gamma+\varepsilon}), \\ &= \text{cst } y^{1/\gamma-\varepsilon} 1_{(y<1)} + \text{cst } y^{1/\gamma+\varepsilon} 1_{(y \geq 1)}. \end{aligned}$$

It turns out that all regularly varying distribution functions can be represented in a convenient exponential form, which can in particular help identify the regular variation, and prove limits of specific parametric classes.

Theorem 3.7 (Representation for regularly varying distributions) *A distribution function F is regularly varying if and only if it can be represented in the form*

$$\bar{F}(z) = c(z) \exp \left(\int_{z_0}^z \frac{\delta(u)}{u} du \right), \quad z \geq z_0$$

for some $z_0 > 0$, where $c(z) \rightarrow c_0 \in (0, \infty)$ and $\delta(z) \rightarrow -1/\gamma$ as $z \rightarrow \infty$.

The following result, as a special case, helps with the transfer of the regular variation property from densities to tails.

Theorem 3.8 (Karamata's Theorem) *Let $f(z) = z^{-1/\gamma-1}\ell(z)/\gamma$ be a regularly varying function at infinity with index $-1/\gamma-1$, and let ℓ be locally bounded in $[z_0, \infty)$. Then for $\gamma > 0$ we have*

$$\int_z^\infty f(t) dt \sim z^{-1/\gamma}\ell(z) \quad \text{as } z \rightarrow \infty.$$

In a nutshell, the first three results help with the taming of regular variation for limit theorems. From the second it follows that any regularly varying distribution is heavy-tailed. The last two results, apart from being theoretically significant, concretely help identify regularly varying distributions.

Example 3.1 (Some regularly varying distributions) Let $\gamma > 0$.

1. *Pareto distribution.* Define

$$F_P(x) = 1 - x^{-1/\gamma}, \quad x \geq 1, \quad F_P(x) = 0, \quad x < 1.$$

Then $\bar{F}_P(x) = x^{-1/\gamma}$ for $x \geq 1$, so F_P is regularly varying with index $1/\gamma$ and slowly varying function $\ell(x) \equiv 1$.

2. *Shifted Pareto (US-Pareto).* Define

$$F_{SP}(x) = 1 - (x+1)^{-1/\gamma}, \quad x \geq 0, \quad F_{SP}(x) = 0, \quad x < 0.$$

Then $\bar{F}_{SP}(x) = (x+1)^{-1/\gamma} = x^{-1/\gamma}\ell(x)$ with

$$\ell(x) = (1+x^{-1})^{-1/\gamma}, \quad x > 0.$$

Since $\ell(x) \rightarrow 1$ as $x \rightarrow \infty$, the function ℓ is slowly varying and hence F_{SP} is regularly varying with index $1/\gamma$.

3. *Student t distribution.* Define for the parameter $\nu > 0$ (degrees of freedom) the density

$$f_\nu(x) = c_\nu(1+x^2/\nu)^{-(\nu+1)/2}, \quad x \in \mathbb{R}, \quad c_\nu = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)}.$$

Then, as $x \rightarrow \infty$,

$$f_\nu(x) = x^{-(\nu+1)}\ell(x), \quad \ell(x) = c_\nu x^{\nu+1}(1+x^2/\nu)^{-(\nu+1)/2}.$$

Since $\ell(x) \rightarrow c_\nu \nu^{(\nu+1)/2}$, the function ℓ is slowly varying. By Karamata's Theorem 3.8, the tail satisfies

$$\bar{F}_\nu(x) = \int_x^\infty f_\nu(t) dt \sim \frac{1}{\nu} x^{-\nu}\ell(x),$$

and therefore F_ν is regularly varying with index $1/\gamma = \nu$.

4. *Fréchet distribution.* Define

$$F_{Fr}(x) = \exp(-x^{-1/\gamma}), \quad x > 0.$$

Then $\bar{F}_{Fr}(x) = 1 - \exp(-x^{-1/\gamma}) = x^{-1/\gamma}\ell(x)$ with

$$\ell(x) = x^{1/\gamma}(1 - \exp(-x^{-1/\gamma})).$$

Since $\exp(-y) = 1 - y + o(y)$ as $y \downarrow 0$, we have $\ell(x) \rightarrow 1$ as $x \rightarrow \infty$, and hence ℓ is slowly varying. Thus F_{Fr} is regularly varying with index $1/\gamma$.



A common heavy-tailed distribution in insurance is the lognormal distribution, which in fact is not regularly varying.

Example 3.2 (Heavy tailed distribution which is not regularly varying) Let $Z = \exp(\sigma W)$ with $W \sim N(0, 1)$ and $\sigma > 0$, so that Z follows a *lognormal distribution*. For any $t > 0$,

$$\mathbb{E}[e^{tZ}] = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp(te^{\sigma x} - x^2/2) dx = \infty,$$

since for x large enough one has $te^{\sigma x} \geq x^2$, and thus the integrand is bounded below by $\exp(x^2/2)$ on a tail interval, making the integral diverge. Hence Z is heavy-tailed. Writing Φ for the standard normal distribution function, we have by Mill's ratio that

$$\bar{F}_Z(x) = 1 - \Phi(\log x / \sigma) \sim \frac{\sigma}{\log x} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(\log x)^2}{2\sigma^2}\right),$$

so for any $t > 1$,

$$\frac{\bar{F}_Z(tx)}{\bar{F}_Z(x)} \sim \frac{\log x}{\log(tx)} \exp\left(-\frac{\log t}{\sigma^2} \log x - \frac{(\log t)^2}{2\sigma^2}\right) \rightarrow 0,$$

which is incompatible with regular variation. Therefore the lognormal distribution is heavy-tailed but not regularly varying. ◇

3.3 Statistics for regularly varying distributions

We now pivot from the general probabilistic theory to estimation, focusing on tools tailored to regularly varying tails. For exploratory analysis, QQ plots and the mean excess function are two valuable tools for investigating heavy-tailed behaviour in data. The former allows for a visual assessment of how well the tail of a distribution matches a theoretical heavy-tailed model, with deviations suggesting heavy tails, which may be mathematically investigated with empirical quantile processes. The mean excess function does not need a candidate distribution and instead reveals heavy-tailed patterns, though it can be quite volatile for large thresholds (as per the variance in Theorem 1.11). Despite these useful tools, the Hill estimator, defined below, is considered the gold standard for visualising and quantifying heavy-tailed behaviour.

In this section, we let $u_n = F^{\leftarrow}(1 - k/n)$ be a sequence of thresholds where $k = k_n$ is an intermediate sequence satisfying $k_n \rightarrow \infty$ and $k_n/n \rightarrow 0$. Note that in this case $n(1 - F(u_n)) = k$. Large k is necessary for having sufficient data to perform estimation, while small k/n indicates the need to focus solely on the tail.

The Hill estimator is motivated by the following three facts: a) regularly varying distributions are Pareto in the tail; b) the logarithm of a Pareto is an exponential with rate $1/\gamma$; c) the mean can be estimated by a sample average.

Definition 3.2 (Hill estimator) Let (Z_n) be a sequence of positive random variables with order statistics $(Z_{n,j})$, the latter non-decreasing in j for fixed n . The Hill estimator is given by

$$\gamma_{n,k} = \frac{1}{k} \sum_{i=1}^k \log(Z_{n,n-i+1}/Z_{n,n-k}).$$

3.3.1 Tail processes

Tail processes are stochastic processes that are defined in the far tail of a sample. Their analysis, and in particular establishing their convergence to a Pareto distribution, is a fundamental task for distributions in the Fréchet max-domain of attraction. In essence, this convergence allows us to dissect the $z^{-1/\gamma}$ part from the $\ell(z)$ part for regularly varying survival functions, which in turn makes estimating γ easier.

Definition 3.3 (Tail empirical distribution function) Define the tail empirical function and its expectation as

$$\tilde{T}_n(s) = \frac{1}{k} \sum_{i=1}^n 1_{(Z_i > u_n s)}, \quad T_n(s) = \mathbb{E}[\tilde{T}_n(s)] = \frac{1 - F(u_n s)}{1 - F(u_n)}, \quad s > 0.$$

Define also $T(s) = s^{-1/\gamma}$.

Note that, for $s \geq 1$, $\tilde{T}_n(s)$ estimates the conditional tail probability $\mathbb{P}(Z > u_n s \mid Z > u_n)$, so the Pareto limit corresponds to asymptotic scale-invariance of exceedances.

Theorem 3.9 (Consistency of tail empirical function) *We have $\tilde{T}_n(s) \xrightarrow{\mathbb{P}} T(s)$ uniformly on intervals bounded away from zero. Moreover $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$.*

Proof. We decompose into a stochastic term and a bias term, then use Dini's theorem for uniformity. Consider the decomposition $\tilde{T}_n(s) - T(s) = (\tilde{T}_n(s) - T_n(s)) + (T_n(s) - T(s)) = A_n + B_n$. By definition of regular variation, $B_n \rightarrow 0$. Then note that A_n has mean zero and

$$\text{Var}(A_n) = \frac{1}{n^2 \bar{F}(u_n)^2} n F(su_n) \bar{F}(su_n) = \frac{F(su_n) \bar{F}(su_n)}{k \bar{F}(u_n)} \rightarrow 0,$$

which implies that $A_n \xrightarrow{\mathbb{P}} 0$, and thus $\tilde{T}_n(s) \xrightarrow{\mathbb{P}} T(s)$ for fixed s . By Dini's Theorem 1.15, the convergence is uniform on bounded intervals. Finally, observe that for any $\varepsilon > 0$, if $Z_{n,n-k}/u_n > (1 + \varepsilon)$, then necessarily at least k terms are non-zero in the sum of $\tilde{T}_n$, or in other words $\tilde{T}_n(1 + \varepsilon) > 1$. Then by the previous result,

$$\limsup_{n \rightarrow \infty} \mathbb{P}(Z_{n,n-k}/u_n > 1 + \varepsilon) \leq \lim_{n \rightarrow \infty} \mathbb{P}(\tilde{T}_n(1 + \varepsilon) > 1) = 0,$$

and similarly $\limsup_{n \rightarrow \infty} \mathbb{P}(Z_{n,n-k}/u_n < 1 - \varepsilon) = 0$. $\square$

This requirement is essential; a fixed threshold breaks the limit, as the next example shows.

Example 3.3 (Failure without intermediate thresholds) Let the survival function be

$$\bar{F}(z) = z^{-1/\gamma} \ell(z), \quad z \geq e, \quad \ell(z) = \frac{1}{1 + \log z}.$$

Suppose the threshold is fixed, e.g. $u_n \equiv e$, so that $k/n = \bar{F}(e) > 0$ and u_n does not diverge. Then

$$T_n(s) = \frac{\bar{F}(u_n s)}{\bar{F}(u_n)} = s^{-1/\gamma} \frac{1 + \log u_n}{1 + \log(u_n s)} = s^{-1/\gamma} \frac{1 + \log e}{1 + \log(es)} \neq s^{-1/\gamma},$$

so we still obtain $\tilde{T}_n(s) \xrightarrow{\mathbb{P}} T_n(s)$ but no longer $T_n(s) \not\xrightarrow{\mathbb{P}} T(s)$. This highlights the importance of the assumption $k/n \not\xrightarrow{\mathbb{P}} 0$. $\diamond$

Definition 3.4 (Tail empirical process) Define for a sequence u_n of thresholds the tail empirical process as

$$\tilde{\mathbb{T}}_n(s) = \sqrt{k}(\tilde{T}_n(s) - T_n(s)), \quad \mathbb{T}_n(s) = \sqrt{k}(\tilde{T}_n(s) - T(s)), \quad s > 0.$$

Theorem 3.10 (Weak convergence of the tail empirical process) *On compacts bounded away from zero, $\tilde{\mathbb{T}}_n \xrightarrow{d} \mathbb{T}$ where $\mathbb{T} = \mathbb{W} \circ T$ and $\mathbb{W}$ is a standard Brownian motion.*

Proof. Let us find the limit covariance structure. Define

$$Z_{ni}(s) = k^{-1/2} \{1_{(Z_i > u_n s)} - \mathbb{P}(Z_i > u_n s)\},$$

so that $\tilde{\mathbb{T}}_n = \sum_{i=1}^n Z_{ni}$. We have:

$$\begin{aligned} n \operatorname{Cov}(Z_{ni}(s), Z_{ni}(t)) &= \frac{\mathbb{P}(Z_1 > u_n(s \vee t)) - \mathbb{P}(Z_1 > u_n s) \mathbb{P}(Z_1 > u_n t)}{1 - F(u_n)} \\ &\rightarrow (s \vee t)^{-1/\gamma} - 0. \end{aligned}$$

To prove functional convergence, we make use of the Bracketing CLT for changing functions, Theorem 1.20. Consider the bounded interval $I = [a, b]$ with $a > 0$, and define

$$\mathcal{F}_n = \{f_{n,t} : [0, \infty) \rightarrow [0, \infty) \text{ s.t. } f_{n,t}(z) = \bar{F}(u_n)^{-1/2} 1_{[u_n t, \infty)}(z) \text{ for } t \in [a, b]\}.$$

Clearly the envelope is given by $F_n(Z_1) = \bar{F}(u_n)^{-1/2} 1_{(Z_1 > u_n a)}$. It satisfies by regular variation

$$\mathbb{E}[F_n(Z_1)^2] = \frac{\bar{F}(u_n a)}{\bar{F}(u_n)} = O(1),$$

showing condition (1.6). We also have by regular variation and $k \rightarrow \infty$ that

$$\begin{aligned} \mathbb{E}[F_n(Z_1)^2 1_{(F_n(Z_1) > c\sqrt{n})}] &\leq \frac{1}{c\sqrt{n}} \mathbb{E}[F_n(Z_1)^3] = \frac{1}{c(n\bar{F}(u_n))^{1/2}} \frac{\bar{F}(u_n a)}{\bar{F}(u_n)} \\ &\rightarrow 0 \cdot a^{-1/\gamma} / c = 0. \end{aligned}$$

This shows (1.7). The bracketing number is calculated as follows. Consider for $j = 1, \dots, m$ brackets on the form $l_j(y) = \bar{F}(u_n)^{-1/2} 1_{[u_n t_j, \infty)}(y)$, and similarly $u_j(y) = \bar{F}(u_n)^{-1/2} 1_{[u_n t_{j-1}, \infty)}(y)$ for sequences on the form $a = t_0 < t_1 < \dots < t_m = b$. Let $\mathcal{F}_j^n = \{f \in \mathcal{F}_n : l_j \leq f \leq u_j\}$. Then clearly we have the partition $\mathcal{F}_n = \bigcup_{j=1}^m \mathcal{F}_j^n$. Let $\varepsilon > 0$ be given, and pick a sequence for $m \leq \lceil 2/(\varepsilon^2/2) \rceil$ such that

$$T(t_{j-1}) - T(t_j) \leq \varepsilon^2 a^{-1/\gamma} / 2 \quad \text{for } 0 \leq j \leq m,$$

which is possible because T is a continuous and bounded function on $[a, b]$. Then this sequence covers all of $\mathcal{F}_n$ with brackets satisfying

$$\begin{aligned} \|u_j(Z_1) - l_j(Z_1)\|_{L^2} &= \frac{\mathbb{P}(u_n t_j \geq Z_1 > u_n t_{j-1})^{1/2}}{(\bar{F}(u_n))^{1/2}} \\ &= (T(t_{j-1}) - T(t_j))^{1/2} (1 + o(1)) \\ &= (T(t_{j-1}) - T(t_j))^{1/2} a^{1/(2\gamma)} \|F_n(Z_1)\|_{L^2} (1 + o(1)) \\ &\leq \varepsilon \|F_n(Z_1)\|_{L^2}, \end{aligned}$$

where we have used regular variation twice. Consequently for some constant C ,

$$N_{[]}(\varepsilon \|F_n(Z_1)\|_{L^2}, \mathcal{F}_n, L^2(\mathbb{P})) \leq m \leq \lceil C/\varepsilon^2 \rceil.$$

Integration now shows (1.9). Note that the above $o(1)$ term is independent of the partition by (3.5). In particular, the partition is independent of n and (1.8) does not need to be verified. $\square$

We now define another key process which plays an important role when finding a suitable estimator for the deterministic and unobservable threshold u_n .

Definition 3.5 (Empirical quantile process) The empirical quantile process is defined by

$$\tilde{Q}_n(t) = \tilde{T}_n^{\leftarrow}(\lfloor kt \rfloor / k) = \frac{Z_{n, n - \lfloor kt \rfloor}}{u_n}, \quad t \in (0, n/k).$$

Setting $Q(t) = t^{-\gamma}$ we define the associated empirical process as

$$\mathbb{Q}_n(t) = \sqrt{k}(\tilde{Q}_n(t) - Q(t)), \quad t \in (0, n/k).$$

Define $B_n(s_0) = \sup_{t \geq s_0} |T_n(t) - T(t)|$, which converges uniformly to 0 for each $s_0 > 0$.

Theorem 3.11 (Joint convergence of tail empirical and quantile processes) *Let $k = k_n$ grow slowly enough such that for some $s_0 \in (0, 1)$, $\sqrt{k}B_n(s_0) \rightarrow 0$. Let $b > 1$. Then*

$$(\mathbb{T}_n, \mathbb{Q}_n) \xrightarrow{d} (\mathbb{T}, -Q' \cdot \mathbb{T} \circ Q) = (\mathbb{W} \circ T, -Q' \mathbb{W})$$

on $\ell^\infty([s_0, \infty)) \times \ell^\infty((T(b), T(s_0)])$. In particular,

$$(\mathbb{T}_n, \sqrt{k}(Z_{n, n-k}/u_n - 1)) \xrightarrow{d} (\mathbb{T}, \gamma \cdot \mathbb{T}(1)) = (\mathbb{W} \circ T, \gamma \cdot \mathbb{W}(1)).$$

Proof. We combine the tail empirical CLT with the functional δ -method for the inverse map. We may write, on $[s_0, \infty)$, by assumption, $\mathbb{T}_n = \tilde{\mathbb{T}}_n + o(1)$, and the latter converges to $\mathbb{T}$ in $\ell^\infty([s_0, \infty))$. Then the desired joint convergence follows from an application of the functional δ -method, Theorem 1.17, applied with the inverse mapping (also called Vervaat's Lemma in the literature), as per Lemma 1.10, by recalling that for Q a differentiable function, $Q' = 1/T' \circ Q$. $\square$

We may now define a tail empirical process which is observable from data, and which also has tractable asymptotic properties.

Definition 3.6 (Tail empirical distribution function with random levels) Define the tail empirical distribution function with random levels and its empirical process as

$$\hat{T}_n(s) = \frac{1}{k} \sum_{i=1}^n 1_{(Z_i > Z_{n,n-k} s)}, \quad \hat{\mathbb{T}}_n(s) = \sqrt{k}(\hat{T}_n(s) - T(s)), \quad s > 0.$$

Theorem 3.12 (Weak convergence of tail empirical process with random levels) *Let $k = k_n$ grow slowly enough such that for some $s_0 \in (0, 1)$, $\sqrt{k}B_n(s_0) \rightarrow 0$. Then $\hat{\mathbb{T}}_n \xrightarrow{d} \mathbb{T} - T \cdot \mathbb{T}(1) = \mathbb{B} \circ T$, on $\ell^\infty([s_0, \infty))$, where $\mathbb{B}$ is a standard Brownian bridge.*

Proof. We expand around the random level $Z_{n,n-k}$ and show each term is controlled by earlier limits. We simply write

$$\begin{aligned} \hat{\mathbb{T}}_n(s) &= \sqrt{k}(\tilde{T}_n(sZ_{n,n-k}/u_n) - T_n(sZ_{n,n-k}/u_n)) \\ &\quad + \sqrt{k}(T_n(sZ_{n,n-k}/u_n) - T(sZ_{n,n-k}/u_n)) \\ &\quad + \sqrt{k}(T(sZ_{n,n-k}/u_n) - T(s)). \end{aligned}$$

Since by Theorem 3.9 we have $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$, the first term has pathwise convergence to $\mathbb{T}$ by Theorem 3.10, Slutsky's lemma, Lemma 1.9, and the continuous mapping theorem, Theorem 1.14. Again by $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$ we get

$$\mathbb{P}(|T_n(sZ_{n,n-k}/u_n) - T(sZ_{n,n-k}/u_n)| \leq B_n(s_0)) \rightarrow 1,$$

for any $s \geq s_0$, so by assumption the second term becomes pathwise $o_{\mathbb{P}}(1)$.

Finally, we may pull out $T(s)$ from the last term, and then apply the δ -method, Theorem 1.10, to the convergence of Theorem 3.11 and the transform $y \mapsto T(y)$ with derivative $T'(y) = -y^{-1/\gamma-1}/\gamma$ to obtain pathwise convergence of the last term (jointly with the first), to $s^{-1/\gamma}(-1/\gamma) \cdot \gamma \mathbb{T}(1) = -T(s)\mathbb{T}(1)$. $\square$

3.3.2 The Hill estimator

We now introduce the Hill estimator and establish its basic asymptotic properties. It is the canonical tail index estimator and admits a simple integral representation in terms of the tail empirical function. Indeed, notice that we may re-express

$$\gamma_{n,k} = \int_1^\infty \hat{T}_n(t)/t \, dt.$$

In practice, for a fixed value of n , the value of k must be chosen with care. Taking k too small yields high variance, while taking k too large introduces bias from non-Pareto tail behaviour. A common diagnostic is the Hill plot, which plots $\gamma_{n,k}$ as a function of the smallest k -values. Then one should look for a stable region in k where the estimator is relatively flat, and take the largest k in that region.

Theorem 3.13 (Weak consistency of the Hill estimator) *We have that $\gamma_{n,k} \xrightarrow{\mathbb{P}} \gamma$.*

Proof. We write

$$\begin{aligned} \gamma_{n,k} &= \int_1^\infty \frac{\widehat{T}_n(t)}{t} dt = \int_1^\infty \frac{\widetilde{T}_n(tZ_{n,n-k}/u_n)}{t} dt \\ &= \int_{Z_{n,n-k}/u_n}^\infty \frac{\widetilde{T}_n(t)}{t} dt = \int_{Z_{n,n-k}/u_n}^1 \frac{\widetilde{T}_n(t)}{t} dt + \int_1^\infty \frac{\widetilde{T}_n(t)}{t} dt \end{aligned} \quad (3.6)$$

Since $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$, the first integral is $o_{\mathbb{P}}(1)$. By consistency of the tail empirical function on bounded intervals, Theorem 3.9, for any $m > 0$,

$$\gamma_{nm} := \int_1^m \frac{\widetilde{T}_n(t)}{t} dt \xrightarrow{\mathbb{P}} \int_1^m \frac{T(t)}{t} dt =: \gamma_m$$

and $\lim_{m \rightarrow \infty} \gamma_m = \gamma$. Thus, by the approximation theorem for random elements, the proof is complete by Theorem 1.5 if we can show that $\gamma_{n,k} - \gamma_{nm}$ vanishes as $n \rightarrow \infty$ first and then $m \rightarrow \infty$, i.e. for $\eta > 0$

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_m^\infty \frac{\widetilde{T}_n(t)}{t} dt > \eta \right) = 0.$$

Note that by Potter's bounds, Proposition 3.6, we have for any $\varepsilon \in (0, 1/\gamma)$

$$\mathbb{E} \int_m^\infty \frac{\widetilde{T}_n(t)}{t} dt = \int_m^\infty \frac{1 - F(u_n t)}{t(1 - F(u_n))} dt \leq \text{cst} \int_m^\infty t^{-1/\gamma-1+\varepsilon} dt = O(m^{-1/\gamma+\varepsilon}).$$

The above expression is vanishing, so an application of Markov's inequality (equation (1.1) with φ the identity function) establishes the desired result. $\square$

Theorem 3.14 (Weak convergence of the Hill estimator) *Let $k = k_n$ grow slowly enough such that for some $s_0 \in (0, 1)$, $\sqrt{k}B_n(s_0) \rightarrow 0$ and*

$$\lim_{n \rightarrow \infty} \sqrt{k} \int_1^\infty \frac{T_n(t) - T(t)}{t} dt = 0. \quad (3.7)$$

Then $\sqrt{k}(\gamma_{n,k} - \gamma) \xrightarrow{d} N(0, \gamma^2)$.

Proof. By the decomposition (3.6) we get

$$\begin{aligned} &\sqrt{k}(\gamma_{n,k} - \gamma) \\ &= \int_1^\infty \frac{\widetilde{T}_n(t)}{t} dt + \sqrt{k} \int_1^\infty \frac{T_n(t) - T(t)}{t} dt + \sqrt{k} \int_{Z_{n,n-k}/u_n}^1 \frac{\widetilde{T}_n(t)}{t} dt. \end{aligned} \quad (3.8)$$

The middle term vanishes by assumption. Consider the last term. Since $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$ we may assume w.l.o.g. that for a given $\varepsilon \in (0, 1)$ we have $Z_{n,n-k}/u_n \in (1 - \varepsilon, 1 + \varepsilon)$ for n large enough. Then by consistency of the tail empirical function and the last limit result of Theorem 3.11,

$$\begin{aligned} & \sqrt{k} \left| \int_{Z_{n,n-k}/u_n}^1 \frac{\tilde{T}_n(t)}{t} dt - \int_{Z_{n,n-k}/u_n}^1 \frac{T(t)}{t} dt \right| \\ & \leq \frac{\sqrt{k}}{1 - \varepsilon} \sup_{t \in (1 - \varepsilon, 1 + \varepsilon)} |\tilde{T}_n(t) - T(t)| |Z_{n,n-k}/u_n - 1| = o_{\mathbb{P}}(1). \end{aligned}$$

Thus the asymptotic of the last term is equivalent to that of

$$\sqrt{k} \int_{Z_{n,n-k}/u_n}^1 T(t)/t dt,$$

which by the δ -method, Theorem 1.10, applied to the function $x \mapsto \int_x^1 \frac{T(t)}{t} dt$, has the same limit as

$$\sqrt{k}(Z_{n,n-k}/u_n - 1)\{-T(\zeta_n)/\zeta_n\}$$

for some ζ_n between $Z_{n,n-k}/u_n$ and 1. It remains to note that the term in curly brackets converges in probability to -1 . Next consider the first term in (3.8), where we use a similar argument as in the proof of consistency. Namely, we know from Theorem 3.10 that $\int_1^m \tilde{\mathbb{T}}_n(t)/t dt \xrightarrow{d} \int_1^m \mathbb{T}(t)/t dt$ and that as $m \rightarrow \infty$, $\int_m^\infty \mathbb{T}(t)/t dt \xrightarrow{\mathbb{P}} 0$ (it is centred and has variance going to zero). Thus the proof is complete by Theorem 1.5 if we can show that for every $\eta > 0$

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_m^\infty \frac{\tilde{\mathbb{T}}_n(t)}{t} dt > \eta \right) = 0.$$

It suffices by Markov's inequality to show the corresponding limit property for the variance, which is given as follows:

$$\begin{aligned} \text{Var} \left(\int_m^\infty \frac{\tilde{\mathbb{T}}_n(t)}{t} dt \right) &= \frac{1}{k} \text{Var} \left(\int_m^\infty \frac{\sum_{i=1}^n \{1_{(Z_i > u_n t)} - \mathbb{P}(Z_i > u_n t)\}}{t} dt \right) \\ &= \frac{n}{k} \text{Var} \left(\int_m^{(Z_1/u_n) \vee m} \frac{1}{t} dt \right) = \frac{1}{\bar{F}(u_n)} \text{Var} \left(\log_+(Z_1/(m \cdot u_n)) \right) \\ &\leq \frac{\mathbb{E}[\log_+(Z_1/(m \cdot u_n))^2]}{\bar{F}(u_n)} = \frac{2 \int_{mu_n}^\infty \log(t/(mu_n)) \bar{F}(t) \frac{1}{t} dt}{\bar{F}(u_n)} \\ &= \frac{2 \int_1^\infty \log(t) \bar{F}(mu_n t) \frac{1}{t} dt}{\bar{F}(u_n)} \leq m^{-1/\gamma+\varepsilon} \cdot 2 \int_1^\infty \log(t) t^{-1/\gamma+\varepsilon-1} dt \end{aligned}$$

where we have used Potter's bounds, Theorem 3.6, in the last step with $\varepsilon \in (0, 1/\gamma)$. The final quantity does not depend on n and furthermore vanishes as $m \rightarrow \infty$, as desired. Finally, we get that the weak limit of $\sqrt{k}(\gamma_{n,k} - \gamma)$ has the following distributional representation

$$\int_1^\infty \frac{\mathbb{W}(T(t))}{t} dt - \gamma \mathbb{W}(1) = \int_1^\infty \frac{\mathbb{B}(T(t))}{t} dt = \gamma \int_0^1 \frac{\mathbb{B}(t)}{t} dt,$$

which has variance given by $\gamma^2 \int_0^1 \int_0^1 (s^{-1} \wedge t^{-1} - 1) ds dt = \gamma^2$. $\square$

To verify the bias condition in practice, it is useful to have simple tail-expansion criteria.

Proposition 3.1 (Class of distributions with reasonable bias) *Let F be a continuous distribution function on $\mathbb{R}$ and assume that there exist $c > 0$ and $\beta > 1/\gamma > 0$ such that*

$$\sup_{t \geq 1} t^\beta |\bar{F}(t) - ct^{-1/\gamma}| < \infty.$$

Let $u_n = F^{\leftarrow}(1 - k/n)$ so that $n\bar{F}(u_n) = k$, and define $T_n(t) = k^{-1}n\bar{F}(u_nt)$. If $k = o(n^\rho)$ with $\rho = 2(\beta\gamma - 1)/(2\beta\gamma - 1)$, then (3.7) holds.

Proof. For $t \geq 1$,

$$\begin{aligned} |T_n(t) - t^{-1/\gamma}| &\leq k^{-1}n|\bar{F}(u_nt) - c(u_nt)^{-1/\gamma}| + k^{-1}n|cu_n^{-1/\gamma} - \bar{F}(u_n)| t^{-1/\gamma} \\ &\leq \text{cst } k^{-1}nu_n^{-\beta}(t^{-\beta} + t^{-1/\gamma}) \leq \text{cst } k^{-1}nu_n^{-\beta}t^{-1/\gamma}, \end{aligned}$$

since $\beta > 1/\gamma$ and $t \geq 1$. The tail expansion implies $u_n = O((n/k)^\gamma)$. Therefore

$$\begin{aligned} \sqrt{k} \int_1^\infty \frac{|T_n(t) - t^{-1/\gamma}|}{t} dt &\leq \text{cst } \sqrt{k} k^{-1}nu_n^{-\beta} \int_1^\infty t^{-1/\gamma-1} dt \\ &\leq \text{cst } k^{-1/2+\beta\gamma} n^{1-\beta\gamma} = \text{cst } \frac{k^{(2\beta\gamma-1)/2}}{n^{\beta\gamma-1}}. \end{aligned}$$

If $k = o(n^\rho)$ with $\rho = 2(\beta\gamma - 1)/(2\beta\gamma - 1)$, the last term tends to 0, which yields (3.7). $\square$

The next example shows that the bias condition can be quite restrictive even for regularly varying tails with mild slowly varying components.

Example 3.4 (Big bias example) Consider the Pareto-type distribution with a heavy slowly varying component:

$$\bar{F}(z) = z^{-1/\gamma} \log z, \quad z \geq e.$$

Here, the slowly varying function is $\ell(z) = \log z$. Let u_n be the threshold associated with the sample fraction k , such that $k \sim n\bar{F}(u_n)$. We analyse the behaviour of the tail process:

$$T_n(t) = t^{-1/\gamma} \frac{\ell(u_nt)}{\ell(u_n)} = t^{-1/\gamma} \frac{\log(u_nt)}{\log u_n}.$$

Consequently

$$\int_1^\infty \frac{T_n(t) - T(t)}{t} dt = \int_1^\infty t^{-1/\gamma-1} \frac{\log t}{\log u_n} dt = \frac{1}{\log u_n} \int_1^\infty t^{-1/\gamma-1} \log t dt,$$

The integral $\int_1^\infty t^{-1/\gamma-1} \log t dt$ evaluates to γ^2 . Thus, the deterministic bias is exactly $\gamma^2/\log u_n$. The bias condition (3.7) then requires the vanishing of

$$\sqrt{k} \frac{\gamma^2}{\log u_n} \sim \frac{\gamma^2 \sqrt{k}}{\gamma \log(n/k)} \sim \gamma \frac{\sqrt{k}}{\log n}.$$

This means that k cannot be taken as a power of n , and the estimator has a strong systematic bias; for instance for a sample size of $n = 100,000$ we require $k \ll \log(n)^2 = 25$.

The pathology is even more severe if $\ell(z) = \exp((\log z)^\eta)$ for $0 < \eta < 1$. A first-order Taylor expansion yields:

$$\sqrt{k} \int_1^\infty \frac{T_n(t) - T(t)}{t} dt \sim \gamma \eta \frac{\sqrt{k}}{(\log n)^{1-\eta}}.$$

For asymptotic normality to hold, we require $k \ll (\log n)^{2(1-\eta)}$. If $\eta = 0.95$, then $k \ll (\log n)^{0.1}$. Even with $n = 10^{80}$ (number of atoms in the universe) we require $k \ll 1.684731$. So while $k \rightarrow \infty$ technically exists, one cannot obtain this regime in practice. $\diamond$

3.3.3 Probability and quantile estimation

Small tail probability and high quantile estimators for regularly varying distributions are motivated by a two-step procedure. First, a high quantile is estimated using the available data, providing a baseline from which further extrapolation can be made. In the second step, the tail index is used as the leading coefficient for extrapolation beyond the observed data. The Hill estimator is employed to estimate the latter index. This combination of direct high quantile estimation and tail index extrapolation sometimes goes under the name of *Weissman estimation*.

Focusing first on high quantiles, and writing in this subsection $Q(p) = F^\leftarrow(p)$ for the distribution quantile, inverting the Pareto tail approximation gives the *quantile estimator* of $Q(p)$

$$\hat{Q}(p) = ((1-p)n/k)^{-\gamma_{n,k}} Z_{n,n-k}, \quad p \in [1-k/n, 1).$$

The consistency of the ingredients already follows from Theorems 3.9 and 3.13. For statistical inference one needs a finer statement, and the extra work is to control the deterministic extrapolation bias at the level where the probability $1-p_n$ is evaluated. The next theorem gives a convenient condition directly in terms of the tail ratio $T_n(t) = \bar{F}(u_n t)/\bar{F}(u_n)$.

Theorem 3.15 (Asymptotic normality of the Weissman quantile estimator) *Let $k = k_n$ grow slowly enough such that for some $s_0 \in (0, 1)$, $\sqrt{k}B_n(s_0) \rightarrow 0$ and (3.7) holds. Let (p_n) be a sequence with $c_n := n(1-p_n)/k \rightarrow 0$. Assume also that F is continuous, that $\log(1/c_n) = o(\sqrt{k})$, and that*

$$\lim_{n \rightarrow \infty} \frac{\sqrt{k}}{\log(1/c_n)} \sup_{t \geq 1} \left| t^{1/\gamma} \frac{\bar{F}(u_n t)}{\bar{F}(u_n)} - 1 \right| = 0. \quad (3.9)$$

Then

$$\frac{\sqrt{k}}{\log(1/c_n)} \left(\frac{\hat{Q}(p_n)}{Q(p_n)} - 1 \right) \xrightarrow{d} N(0, \gamma^2).$$

Proof. Let $t_n = Q(p_n)/u_n$. Since $n(1-p_n)/k = c_n \rightarrow 0$, we have $1-p_n < k/n$ for n large enough and therefore $Q(p_n) \geq u_n$, so $t_n \geq 1$ eventually. By continuity

of F and the definition of u_n ,

$$\frac{\bar{F}(u_n t_n)}{\bar{F}(u_n)} = \frac{\bar{F}(Q(p_n))}{\bar{F}(u_n)} = c_n.$$

Now write

$$\frac{\hat{Q}(p_n)}{Q(p_n)} - 1 = A_n B_n C_n - 1,$$

where

$$A_n = \frac{Z_{n,n-k}}{u_n}, \quad B_n = c_n^{-(\gamma_{n,k}-\gamma)}, \quad C_n = \frac{c_n^{-\gamma} u_n}{Q(p_n)} = \frac{c_n^{-\gamma}}{t_n}.$$

Accordingly,

$$A_n B_n C_n - 1 = (A_n - 1) B_n C_n + (B_n - 1) C_n + (C_n - 1).$$

We show that the first and third terms are negligible after scaling by $\sqrt{k}/\log(1/c_n)$, while the second term carries the asymptotic normality.

By Theorem 3.9, $A_n \xrightarrow{\mathbb{P}} 1$, and by Theorem 3.14, $\gamma_{n,k} \xrightarrow{\mathbb{P}} \gamma$. Moreover, since $\sqrt{k}(\gamma_{n,k} - \gamma) = O_{\mathbb{P}}(1)$ and $\log(1/c_n) = o(\sqrt{k})$, we get

$$(\gamma_{n,k} - \gamma) \log(1/c_n) \xrightarrow{\mathbb{P}} 0,$$

so $B_n \xrightarrow{\mathbb{P}} 1$. To analyse C_n , define

$$r_n(t) = t^{1/\gamma} \frac{\bar{F}(u_n t)}{\bar{F}(u_n)} - 1, \quad t \geq 1.$$

Then $T_n(t) = t^{-1/\gamma}(1 + r_n(t))$, and at $t = t_n$ the identity above gives

$$c_n = t_n^{-1/\gamma}(1 + r_n(t_n)).$$

Therefore

$$C_n = \frac{c_n^{-\gamma}}{t_n} = (1 + r_n(t_n))^{-\gamma}.$$

By (3.9) and $\log(1/c_n) = o(\sqrt{k})$ we have $\sup_{t \geq 1} |r_n(t)| \rightarrow 0$, hence $C_n \rightarrow 1$. In addition, for n large enough, $\sup_{t \geq 1} |r_n(t)| \leq 1/2$, and then the mean value theorem gives

$$|C_n - 1| = |(1 + r_n(t_n))^{-\gamma} - 1| \leq \text{cst } |r_n(t_n)| \leq \text{cst } \sup_{t \geq 1} |r_n(t)|.$$

Consequently, we even get

$$\frac{\sqrt{k}}{\log(1/c_n)}(C_n - 1) \xrightarrow{\mathbb{P}} 0, \tag{3.10}$$

which deals with the third term. The first term is negligible because Theorem 3.11 yields $\sqrt{k}(A_n - 1) = O_{\mathbb{P}}(1)$, while $\log(1/c_n) \rightarrow \infty$ and $B_n C_n \xrightarrow{\mathbb{P}} 1$. Finally, for the middle term, by Taylor expansion we obtain

$$\frac{\sqrt{k}}{\log(1/c_n)}(B_n - 1)C_n = \sqrt{k}(\gamma_{n,k} - \gamma)(1 + o_{\mathbb{P}}(1)) \xrightarrow{d} N(0, \gamma^2)$$

by Theorem 3.14. The result now follows from Slutsky's theorem. $\square$

The rate condition (3.9) is abstract on purpose: it isolates exactly the extrapolation error that matters for the Weissman step. It is always satisfied if one imposes second-order conditions found in standard literature. The next example shows that it can be verified explicitly for oscillating slowly varying functions (which do not obey the classic second order conditions), and that one can choose concrete sequences (k_n) and (p_n) satisfying all requirements.

Example 3.5 (Oscillating slowly varying tail and admissible sequences) Fix $\gamma > 0$ and $A_0 \in \mathbb{R}$. For sufficiently large z , let

$$\bar{F}(z) = z^{-1/\gamma} \ell(z), \quad \ell(z) = 1 + \frac{A_0 \cos(\log z)}{\log z}.$$

Since $\ell(z) \rightarrow 1$ and $\ell'(z) = O((z \log z)^{-1})$, the derivative of $z^{-1/\gamma} \ell(z)$ is negative for z large enough. After modifying the distribution on a bounded interval (and, if needed, starting the tail representation at a larger point), this defines a distribution with regularly varying tail index γ . The factor ℓ is slowly varying, but the oscillation prevents the usual monotone second-order parametrisations from holding. Nevertheless, the condition in Theorem 3.15 is easy to verify.

Indeed, for $t \geq 1$,

$$t^{1/\gamma} \frac{\bar{F}(u_n t)}{\bar{F}(u_n)} - 1 = \frac{\ell(u_n t)}{\ell(u_n)} - 1.$$

Since $\log(u_n t) \geq \log u_n$ and $\ell(u_n) \rightarrow 1$, we have uniformly in $t \geq 1$ that

$$|\ell(u_n t) - \ell(u_n)| \leq \frac{2|A_0|}{\log u_n},$$

and therefore

$$\sup_{t \geq 1} \left| t^{1/\gamma} \frac{\bar{F}(u_n t)}{\bar{F}(u_n)} - 1 \right| = O(1/\log u_n).$$

Because $k/n = \bar{F}(u_n) = u_n^{-1/\gamma} \ell(u_n)$ and $\ell(u_n) \rightarrow 1$, we also have that $\log u_n$ is asymptotically equivalent to $\log(n/k)$. Hence the rate condition follows if we can prescribe

$$\frac{\sqrt{k}}{\log(n/k) \log(1/c_n)} \rightarrow 0.$$

As a concrete choice, take

$$k_n = \lfloor (\log n)^{3/2} \rfloor, \quad c_n = \exp\{-(\log n)^{1/4}\}, \quad p_n = 1 - \frac{k_n}{n} c_n.$$

Then $k_n \rightarrow \infty$, $k_n/n \rightarrow 0$, and $n(1 - p_n)/k_n = c_n \rightarrow 0$. Moreover,

$$\log(1/c_n) = (\log n)^{1/4} = o(\sqrt{k_n}).$$

Finally,

$$\frac{\sqrt{k_n}}{\log(n/k_n) \log(1/c_n)} \sim \frac{(\log n)^{3/4}}{(\log n)(\log n)^{1/4}} = (\log n)^{-1/2} \rightarrow 0,$$

so the speed conditions of Theorem 3.15 are satisfied. It remains to check the Hill assumptions. Fix $s_0 \in (0, 1)$. For $t \geq s_0$ and n large enough, the same argument as above gives

$$\left| \frac{\ell(u_n t)}{\ell(u_n)} - 1 \right| \leq \frac{\text{cst}}{\log u_n},$$

and therefore

$$B_n(s_0) = \sup_{t \geq s_0} |T_n(t) - T(t)| \leq \text{cst } s_0^{-1/\gamma} \frac{1}{\log u_n} = O(1/\log u_n).$$

Hence $\sqrt{k_n} B_n(s_0) \rightarrow 0$. Likewise,

$$\left| \int_1^\infty \frac{T_n(t) - T(t)}{t} dt \right| \leq \frac{\text{cst}}{\log u_n} \int_1^\infty t^{-1/\gamma-1} dt = O(1/\log u_n),$$

so $\sqrt{k_n} \int_1^\infty (T_n(t) - T(t)) t^{-1} dt \rightarrow 0$, i.e. condition (3.7) holds. Thus the assumptions of Theorem 3.15 are satisfied. $\diamond$

The same extrapolation principle can be written in probability form. The corresponding Weissman tail probability estimator is

$$\hat{\bar{F}}(z) = \frac{k}{n} \left(\frac{z}{Z_{n,n-k}} \right)^{-1/\gamma_{n,k}}, \quad z > Z_{n,n-k}.$$

When the extrapolation point is chosen as $z_n = Q(p_n)$, the same rate $c_n = n(1-p_n)/k$ appears as in Theorem 3.15, and the proof may be carried out in a similar fashion.

Theorem 3.16 (Asymptotic normality of the Weissman probability estimator) *Let $k = k_n$ grow slowly enough such that for some $s_0 \in (0, 1)$, $\sqrt{k} B_n(s_0) \rightarrow 0$ and (3.7) holds. Let (p_n) be a sequence with $c_n := n(1-p_n)/k \rightarrow 0$. Assume also that F is continuous, that $\log(1/c_n) = o(\sqrt{k})$, and that (3.9) holds. Then*

$$\frac{\gamma \sqrt{k}}{\log(1/c_n)} \left(\frac{\hat{\bar{F}}(Q(p_n))}{1-p_n} - 1 \right) \xrightarrow{d} N(0, \gamma^2).$$

Proof. Let $t_n = Q(p_n)/u_n$. Since $c_n = n(1-p_n)/k \rightarrow 0$, we have $1-p_n < k/n$ for n large enough and therefore $Q(p_n) \geq u_n$, so $t_n \geq 1$ eventually. By continuity of F and the definition of u_n ,

$$\frac{\bar{F}(u_n t_n)}{\bar{F}(u_n)} = \frac{\bar{F}(Q(p_n))}{\bar{F}(u_n)} = c_n.$$

Now write

$$\frac{\hat{\bar{F}}(Q(p_n))}{1-p_n} = A_n B_n C_n,$$

where

$$A_n = \left(\frac{Z_{n,n-k}}{u_n} \right)^{1/\gamma}, \quad B_n = \left(\frac{Z_{n,n-k}}{Q(p_n)} \right)^{1/\gamma_{n,k}-1/\gamma},$$

$$C_n = \frac{(Q(p_n)/u_n)^{-1/\gamma}}{c_n} = \frac{t_n^{-1/\gamma}}{c_n}.$$

Accordingly,

$$A_n B_n C_n - 1 = (A_n - 1) B_n C_n + (B_n - 1) C_n + (C_n - 1).$$

We now use the same decomposition strategy as in Theorem 3.15: the first and third terms are negligible after scaling by $\gamma\sqrt{k}/\log(1/c_n)$, while the second term carries the asymptotic normality.

By Theorem 3.9, $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$, hence $A_n \xrightarrow{\mathbb{P}} 1$. By Theorem 3.14, $\gamma_{n,k} \xrightarrow{\mathbb{P}} \gamma$ and $\sqrt{k}(\gamma_{n,k} - \gamma) = O_{\mathbb{P}}(1)$. To analyse C_n , define

$$r_n(t) = t^{1/\gamma} \frac{\bar{F}(u_n t)}{\bar{F}(u_n)} - 1, \quad t \geq 1.$$

Then $T_n(t) = t^{-1/\gamma}(1 + r_n(t))$, and at $t = t_n$ the identity above gives

$$C_n = \frac{t_n^{-1/\gamma}}{c_n} = (1 + r_n(t_n))^{-1}.$$

Exactly as in the proof of Theorem 3.15, condition (3.9) implies $\sup_{t \geq 1} |r_n(t)| \rightarrow 0$, hence $C_n \rightarrow 1$, and the mean value theorem yields $|C_n - 1| \leq \text{cst} \sup_{t \geq 1} |r_n(t)|$. Thus

$$\frac{\gamma\sqrt{k}}{\log(1/c_n)}(C_n - 1) \xrightarrow{\mathbb{P}} 0, \quad (3.11)$$

which deals with the third term. Since $C_n = t_n^{-1/\gamma}/c_n \rightarrow 1$, we also get

$$\frac{\log t_n}{\gamma \log(1/c_n)} = 1 + \frac{\log C_n}{\log(1/c_n)} \xrightarrow{\mathbb{P}} 1.$$

As $\log(Z_{n,n-k}/u_n) = o_{\mathbb{P}}(1)$ and $\log(1/c_n) \rightarrow \infty$, it follows that

$$\frac{\log(Q(p_n)/Z_{n,n-k})}{\gamma \log(1/c_n)} = \frac{\log t_n - \log(Z_{n,n-k}/u_n)}{\gamma \log(1/c_n)} \xrightarrow{\mathbb{P}} 1. \quad (3.12)$$

In particular, $\log(Q(p_n)/Z_{n,n-k}) = O_{\mathbb{P}}(\log(1/c_n))$, so $B_n \xrightarrow{\mathbb{P}} 1$ because $\gamma_{n,k} \xrightarrow{\mathbb{P}} \gamma$ and $(\gamma_{n,k} - \gamma) \log(1/c_n) \xrightarrow{\mathbb{P}} 0$. The first-term estimate is also identical in structure to the quantile case: Theorem 3.11 and the δ -method yield $\sqrt{k}(A_n - 1) = O_{\mathbb{P}}(1)$, and since $\log(1/c_n) \rightarrow \infty$ with $B_n C_n \xrightarrow{\mathbb{P}} 1$, we get

$$\frac{\gamma\sqrt{k}}{\log(1/c_n)}(A_n - 1) B_n C_n \xrightarrow{\mathbb{P}} 0,$$

which deals with the first term. Finally, for the middle term, by Taylor expansion, and since $C_n \xrightarrow{\mathbb{P}} 1$, we obtain

$$\begin{aligned} \frac{\gamma\sqrt{k}}{\log(1/c_n)}(B_n - 1)C_n &= \gamma\sqrt{k} \left(\frac{1}{\gamma} - \frac{1}{\gamma_{n,k}} \right) \frac{\log(Q(p_n)/Z_{n,n-k})}{\log(1/c_n)} (1 + o_{\mathbb{P}}(1)) \\ &= \sqrt{k}(\gamma_{n,k} - \gamma) \frac{\gamma}{\gamma_{n,k}} \frac{\log(Q(p_n)/Z_{n,n-k})}{\gamma \log(1/c_n)} (1 + o_{\mathbb{P}}(1)) \\ &\xrightarrow{d} N(0, \gamma^2), \end{aligned}$$

by Theorem 3.14, (3.12), and Slutsky's theorem. Combining the three terms completes the proof. $\square$

Numeric example 3.1 (Weissman versus empirical tail-probability estimation) To compare extrapolation with a non-extrapolating benchmark at a fixed sample size, consider the Burr distribution with distribution function

$$F(z) = 1 - (1 + (z/\theta)^\eta)^{-\beta}, \quad z \geq 0, \quad (3.13)$$

where $\theta, \eta, \beta > 0$. In this example we take $\theta = 1$ and $\eta = \beta = \sqrt{2}$, so the extreme-value index is $\gamma = (\beta\eta)^{-1} = 1/2$.

We fix $n = 5000$ and simulate iid samples from (3.13). For each sample, each intermediate value k , and each tail level z , we compare the Weissman estimator

$$\hat{\bar{F}}(z) = \frac{k}{n} \left(\frac{z}{Z_{n,n-k}} \right)^{-1/\gamma_{n,k}}, \quad z > Z_{n,n-k},$$

with the empirical tail estimator

$$1 - \mathbb{F}^{(n)}(z) = \frac{1}{n} \sum_{i=1}^n 1_{(Z_i > z)}.$$

To study the effect of extrapolation, we evaluate both estimators against the true tail probability $\bar{F}(z)$ and report the relative root mean squared error

$$\text{RRMSE}(\hat{p}(z)) = (\mathbb{E}[\{\hat{p}(z)/\bar{F}(z) - 1\}^2])^{1/2},$$

approximated by Monte Carlo averages. Figure 3.1 is based on 600 repetitions. The left panel shows the RRMSE as a function of z for three representative values of k , together with the empirical benchmark. The right panel shows the RRMSE as a function of k for three fixed tail levels z ; for each colour, the dashed horizontal line is the empirical estimator and the solid curve is the Weissman estimator. The comparison illustrates the usual bias-variance trade-off in the choice of k , and also the gain from extrapolation once z is far into the tail. $\diamond$

3.4 Other estimators in extremes

There exist other famous estimators for γ when there is no restriction on the max-domain of attraction. We briefly survey three of them below, and provide, mostly without proofs, their asymptotic distributions. This gives a broader context for comparing the Hill estimator in practice.

We begin by introducing the Pickands estimator, which is simply defined, but has the drawback of being based on only three distinct points, as opposed to using an entire set of top order statistics.

Definition 3.7 (Pickands estimator) The Pickands estimator for the index $\gamma \in \mathbb{R}$ is given by:

$$\gamma_{n,k}^P = \frac{1}{\log 2} \log \left(\frac{Z_{n,n-k+1} - Z_{n,n-2k+1}}{Z_{n,n-2k+1} - Z_{n,n-4k+1}} \right).$$

Theorem 3.17 (Consistency and asymptotic normality of Pickands estimator) Let (Z_n) be an iid sequence satisfying (3.3) for some $\gamma \in \mathbb{R}$. We have that

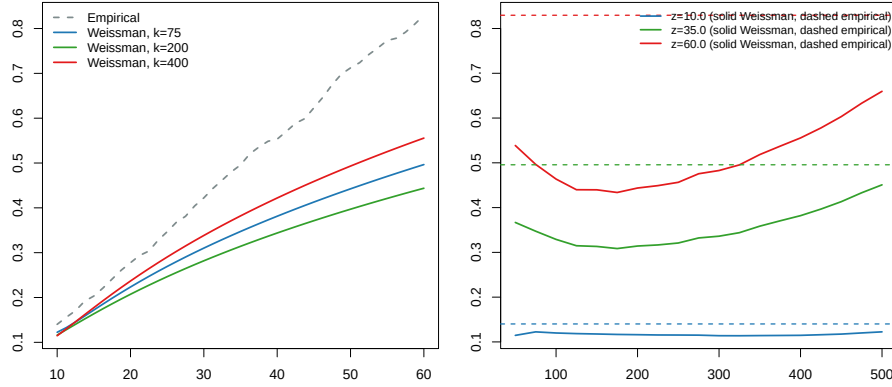


Figure 3.1: Finite-sample comparison of the Weissman tail-probability estimator and the empirical tail estimator under the Burr model (3.13) with $(\theta, \eta, \beta) = (1, \sqrt{2}, \sqrt{2})$, based on $n = 5000$ and 600 Monte Carlo repetitions. Left panel: relative root mean squared error as a function of the tail level z for $k \in \{75, 200, 400\}$. Right panel: relative root mean squared error as a function of k for three fixed tail levels z . In both panels, solid curves denote Weissman estimates and dashed curves denote empirical tail estimates.

$\gamma_{n,k}^P \xrightarrow{\mathbb{P}} \gamma$. Furthermore, if $k/\log \log n \rightarrow \infty$ then $\gamma_{n,k}^P \xrightarrow{a.s.} \gamma$. Finally, under suitable conditions on F and k provided in ?,

$$\sqrt{k}(\gamma_{n,k}^P - \gamma) \xrightarrow{d} N\left(0, \gamma^2 \frac{2^{2\gamma+1} + 1}{(2(2^\gamma - 1) \log 2)^2}\right).$$

Next, we define the maximum likelihood estimators applied to peaks over thresholds, which is a parametric estimator, relying on the GPD.

Definition 3.8 (Peaks over thresholds (POT) maximum likelihood estimator (MLE)) The MLE estimator for excesses (Y_n) over the thresholds u for the index $\gamma \in \mathbb{R}$ is given by the argument that minimises the likelihood constructed using densities from the GPD (3.4), with $\mu = 0$ and (γ, σ) as parameters. We denote the resulting estimators $(\gamma_{n,u}, \sigma_{n,u})$.

Theorem 3.18 (Asymptotic normality of the GPD MLE estimator) Let (Z_n) be an iid sequence satisfying (3.3) for some $\gamma > -1/2$. Let k be the number of exceedances over the threshold u , and denote by $Y_1, \dots, Y_k$ the sequence of excesses over u . Then, under suitable conditions on F and the speed at which $k \rightarrow \infty$ and $k/n \rightarrow 0$ provided in ?, we get

$$k^{1/2}(\gamma_{k,u} - \gamma, \sigma_{k,u}/\sigma - 1) \xrightarrow{d} N(0, \Sigma(\gamma)),$$

where

$$\Sigma(\gamma) = (1 + \gamma) \begin{pmatrix} 1 + \gamma & 1 \\ 1 & 2 \end{pmatrix}.$$

Finally, we present an estimator which is very similar to the Hill estimator in that it uses the top k (scaled) order statistics. It is worth noting that it is known in advance that $\gamma > 0$, then the Hill estimator has a lower variance and consequently is preferable.

Definition 3.9 (Moment Estimator) Define

$$M_n^{(r)} = \frac{1}{k} \sum_{i=1}^k \log(Z_{n,n-i+1}/Z_{n,n-k})^r, \quad r = 1, 2.$$

The Moment estimator for the index $\gamma \in \mathbb{R}$ is given by:

$$\gamma_{n,k}^M := M_n^{(1)} + 1 - \frac{1}{2} \left\{ 1 - \frac{(M_n^{(1)})^2}{M_n^{(2)}} \right\}^{-1}.$$

The following result, for regularly varying tails, shows that the moment estimator is indeed a consistent estimator.

Proposition 3.2 (Weak consistency of the moment estimator) *We have that*
 $\gamma_{n,k}^M \xrightarrow{\mathbb{P}} \gamma$.

Proof. We only show the $\gamma > 0$ case. Since the first moment is the Hill estimator, we focus on the second moment. First note

$$M_n^{(2)} = - \int_1^\infty \log(t)^2 \hat{T}_n(dt) = 2 \int_1^\infty \frac{\log(t)}{t} \hat{T}_n(t) dt$$

By following the same technique as in the consistency of the Hill estimator we get

$$\begin{aligned} M_n^{(2)} &= 2 \int_{Z_{n,n-k}/u_n}^1 \frac{\log(t)}{t} \tilde{T}_n(t) dt + 2 \int_1^\infty \frac{\log(t)}{t} \tilde{T}_n(t) dt \\ &\quad + 2 \log(Z_{n,n-k}/u_n) \int_{Z_{n,n-k}/u_n}^\infty \frac{1}{t} \tilde{T}_n(t) dt. \end{aligned}$$

By consistency of the Hill estimator and $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$, i.e. Theorem 3.13 and Theorem 3.9, we have that the last term vanishes in probability. Using the same argument as in Theorem 3.9, the first term is bounded in probability. By uniform consistency of the tail empirical function on bounded intervals, Theorem 3.9, then for any $m > 0$,

$$\gamma_{nm} := \int_1^m \frac{\tilde{T}_n(t) \log(t)}{t} dt \xrightarrow{\mathbb{P}} \int_1^m \frac{T(t) \log(t)}{t} dt =: \gamma_m$$

and $\lim_{m \rightarrow \infty} \gamma_m = \gamma^2$. Thus, we have shown the result if for $\eta > 0$

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_m^\infty \frac{\tilde{T}_n(t) \log(t)}{t} dt > \eta \right) = 0.$$

For large enough m , we can pick any $1/\gamma > \delta > 0$ such that $\log(x) < x^\delta$ for $x > m$. Furthermore, we can pick a $\varepsilon \in (0, 1/\gamma - \delta)$.

$$\begin{aligned} \mathbb{E} \int_m^\infty \frac{\log(t) \tilde{T}_n(t)}{t} dt &= \int_m^\infty \log(t) \frac{1 - F(u_n t)}{t(1 - F(u_n))} dt \\ &\leq \text{cst} \int_m^\infty t^{-1/\gamma-1+\delta+\varepsilon} dt = O(m^{-1/\gamma+\delta+\varepsilon}) \end{aligned}$$

We get $M_n^{(2)} \xrightarrow{\mathbb{P}} 2\gamma^2$ by Markov's inequality. The result follows. $\square$

The general result, for all max-domains of attraction, including asymptotic normality, is given below.

Theorem 3.19 (Consistency and asymptotic normality of the moment estimator) *Let (Z_n) be an iid sequence satisfying (3.3) for some $\gamma \in \mathbb{R}$ and having positive right-endpoint. Under suitable conditions on F and k provided in ?, then $\gamma_{n,k}^M \xrightarrow{a.s.} \gamma$, and*

$$\sqrt{k}(\gamma_{n,k}^M - \gamma) \xrightarrow{d} N(0, \sigma^2(\gamma)),$$

where

$$\sigma^2(\gamma) = \begin{cases} 1 + \gamma^2, & \gamma \geq 0, \\ (1 - \gamma)^2(1 - 2\gamma) \left\{ 4 - 8\frac{1-2\gamma}{1-3\gamma} + \frac{(5-11\gamma)(1-2\gamma)}{(1-3\gamma)(1-4\gamma)} \right\}, & \gamma < 0. \end{cases}$$

The standard way of plotting all the above estimators is as a function of k , the number of top-order statistics in question. We look for stability, ideally a prolonged region of small k -values where the estimator is constant. Automatic techniques for selecting such a region exist, but we instead propose to always plot the estimator and select the optimal k from visual inspection.

Numeric example 3.2 (Tail-index estimators across models) In Figure 3.2 we illustrate how plots from Definitions 3.2, 3.8, 3.9 and 3.7 look in practice. There, we have simulated two regularly varying distributions with $\gamma = 1/2$: a standard Pareto distribution and a Burr distribution as in (3.13), with $\theta = 1$ and $\eta = \beta = \sqrt{2}$. We also provide a $\gamma = 0$ case corresponding to a one-sided Normal distribution (constructed by taking the absolute value of a Normal random variable). Apart from the Pickands estimator's poor behaviour, which follows from the variance in Theorem 3.17, we observe that there is no overall best estimator, highlighting the difficulty of accurate inference for γ in finite samples.

The especially flat behaviour of the exact Pareto is not a surprise, since in that case $B_n(s_0) = \sup_{t \geq s_0} |T_n(t) - T(t)|$ is exactly zero, which is a bias term appearing in both Theorems 3.11 and 3.14. $\diamond$

3.5 Conditional estimation of regularly varying tails

Covariates are an important part of non-life insurance modelling, and it is sometimes desirable to have covariate-dependent tail behaviour. In this section we

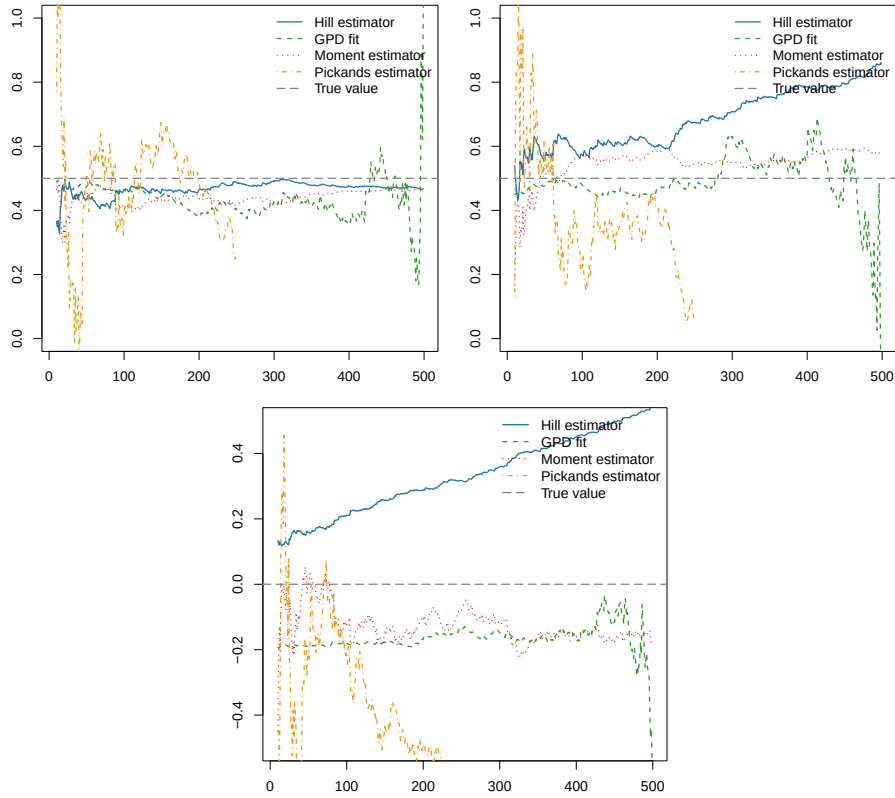


Figure 3.2: Estimation of γ through Definitions 3.2, 3.8, 3.9 and 3.7. Curves show $\hat{\gamma}$ against the number k of upper order statistics, for $n = 1000$ observations from the Pareto (top left), Burr (top right) and one-sided Normal (bottom) distributions.

introduce estimators for the conditional tail function and for the conditional tail index, for a sample where the covariate influences the shape of the tail. The mathematical goal is to let the tail index vary with covariates, while still preserving the tractable local regular variation structure at a given conditioning value x .

3.5.1 Conditional regular variation

Let (X_n, Y_n) be a bivariate iid sequence where (Y_n) is a positive process of interest and (X_n) is a process of covariates. Let the distribution function of Y_1 be denoted by F , the density of X_1 be denoted by g , which is assumed twice continuously differentiable, and let the distribution function of Y_1 given $X_1 = x$ be denoted by F^x .

Let $x \in \mathbb{R}$ with $g(x) > 0$ be given. The key assumption is that the conditional distribution of Y_1 , given $X_1 = x$, is regularly varying with tail index $\gamma(x) > 0$, and that this holds in a neighbourhood around x , that is

$$\bar{F}^u(y) = \mathbb{P}(Y_1 > y | X_1 = u) = L^u(y) y^{-\frac{1}{\gamma(u)}}, \quad y > 0, \quad (3.14)$$

for all u in an open set $U \subseteq \mathbb{R}$ containing x , and where L^u is slowly varying at infinity, that is $L^u(tv)/L^u(v) \rightarrow 1$ as $v \rightarrow \infty$ for any $t > 0$.

Let K be a kernel, supported on $[-1, 1]$, let (h_n) be the corresponding bandwidth sequence, and let (u_n^x) be a scaling sequence diverging to infinity.

The conditional survival function $\bar{F}^x$ is estimated with the Nadaraya–Watson pathwise estimator as

$$\bar{F}_n^x(y) = \sum_{j=1}^n \frac{1_{(Y_j > y)} K\left(\frac{x-X_j}{h_n}\right)}{\sum_{j=1}^n K\left(\frac{x-X_j}{h_n}\right)}, \quad y > 0. \quad (3.15)$$

Definition 3.10 (Conditional tail empirical distribution function) Let the corresponding tail empirical distribution function, for a deterministic threshold, be given by

$$\tilde{T}_n^x(s) = \frac{\bar{F}_n^x(su_n^x)}{\bar{F}_n^x(u_n^x)} = \frac{\frac{1}{nh_n \bar{F}_n^x(u_n^x)} \sum_{j=1}^n K\left(\frac{x-X_j}{h_n}\right) 1_{(Y_j > su_n^x)}}{\frac{1}{nh_n} \sum_{j=1}^n K\left(\frac{x-X_j}{h_n}\right)} =: \frac{\tilde{V}_n^x(s)}{g_n(x)}, \quad (3.16)$$

for $s > 0$. The target of $\tilde{T}_n^x$ is the conditional tail function $s \mapsto T^x(s) = s^{-1/\gamma(x)}$.

The above convergence, both in probability and weakly, is shown under suitable conditions below. However, as in the unconditional case, the deterministic thresholds u_n^x are not observable. Below we provide asymptotic results also for when these thresholds are estimated from data.

3.5.2 Conditional tail function and conditional Hill estimators

To transfer the tail function behaviour to the conditional tail index, the quantity of interest, the following key identity is analogous to the unconditional case:

$$\gamma(x) = \int_1^\infty \frac{T^x(s)}{s} ds.$$

We assume again that an *intermediate sequence* is given, that is a sequence $k_n \leq n$ such that $k_n \rightarrow \infty$, $k_n/n \rightarrow 0$. Let $u_n^x = F^{\leftarrow, x}(1 - k_n/n)$ be an unobserved sequence of deterministic thresholds, diverging to infinity, where $F^{\leftarrow, x}$ denotes the generalised inverse of F^x . We may and shall estimate this sequence by its empirical counterpart $\{q_n^x, n \in \mathbb{N}\}$ given by

$$q_n^x = F_n^{\leftarrow, x}(1 - k_n/n).$$

The conditional tail function is thus estimated by

$$\hat{T}_n^x(s) = \frac{\bar{F}_n^x(sq_n^x)}{\bar{F}^x(u_n^x)} = \tilde{T}_n^x(sq_n^x/u_n^x),$$

which no longer depends on the sequence u_n^x , as can be seen from the definition of $\tilde{T}_n^x$ and from the fact that $\bar{F}^x(u_n^x) = k_n/n$.

Definition 3.11 (Conditional Hill estimator) The conditional Hill estimator is given by

$$\gamma_{n,k}(x) = \int_1^\infty \frac{\hat{T}_n^x(s)}{s} ds = \frac{n}{k_n} \frac{\sum_{j=1}^n \log_+(\frac{Y_j}{q_n^x}) K(\frac{x-X_j}{h_n})}{\sum_{j=1}^n K(\frac{x-X_j}{h_n})}.$$

It is worth noticing that the effective sample size used in this estimator is proportional to $h_n k_n$ instead of k_n , which is also the rate for weak convergence. Consequently, the parameter k_n has a different interpretation than in the unconditional case. In particular, $\gamma_{n,k}(x)$ can be plotted for the entire range $k_n \in \{2, 3, \dots, n\}$.

Numeric example 3.3 Before delving into the mathematical treatment of our estimators, we provide an example of a conditional Hill plot. In Figure 3.3 we have taken (Y_j) to be the (absolute value of) the negative log-returns of the West Texas Intermediate (WTI) grade crude oil spot prices from 2007–2024. As a covariate (X_j) , we take the Crude Oil Volatility Index (CBOE OVX), transformed to have uniform marginals. For large x , we clearly observe a different, heavier tail regime. In contrast, for medium and small x , the tail regime is similar and lighter. The assumption that (Y_n, X_n) is an iid sequence is not fulfilled for this data. However, the conditional Hill estimator for stationary time-series also possesses large sample asymptotics, which happen to agree with the iid case under suitably lax assumptions; see ?.

3.6 Limit theorems for conditional extremes

In this section we provide consistency and weak convergence results for the estimators of the conditional tail function and conditional tail index under suitable

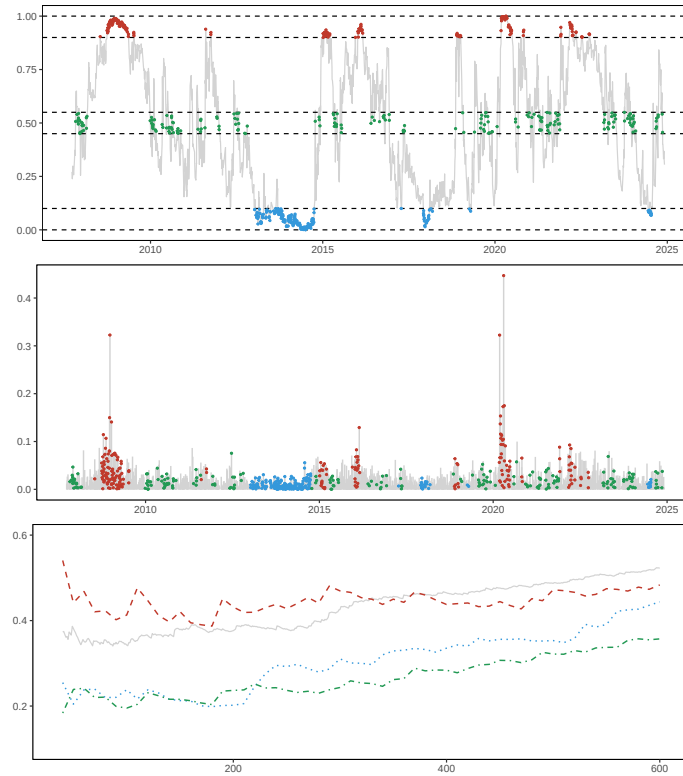


Figure 3.3: Top panel: covariate process (X_j) given by the Crude Oil Volatility Index (CBOE OVX), as a function of calendar year. The effective sample is calculated using a uniform kernel centred at $x = 0.05, 0.50, 0.95$ (highlighted in red, green and blue, respectively) and with $h_n = 0.05$. Middle panel: target process (Y_j) given by the absolute value of the negative log-returns of the West Texas Intermediate (WTI) grade crude oil spot prices with highlighted subsamples, as a function of calendar year. Bottom panel: conditional Hill estimates $\gamma_{n,k}(x)$ in dotted blue, dashdotted green and dashed red, respectively, as a function of k ; in solid grey we have the unconditional Hill estimator of the entire sample.

conditions. Furthermore, second order conditions are formulated in a very general and abstract form. It can be shown that broad classes of regularly varying tails satisfy them.

3.6.1 Consistency

A fundamental underlying result is the uniform consistency of the conditional tail empirical function, for deterministic thresholds. This provides the backbone for all subsequent asymptotic statements in the conditional setting.

Condition 3.1 The conditions for consistency are as follows.

3.1.1 The map $z \mapsto 1/\gamma(z)$ is c_γ -Lipschitz continuous,

3.1.2 It holds that $\lim_{n \rightarrow \infty} h_n \log(u_n^x) = 0$,

3.1.3 There exists $y_0 \in \mathbb{R}$ and $c_L > 0$ such that

$$\sup_{y \geq y_0} \left| \frac{\log(L^x(y)) - \log(L^u(y))}{\log(y)} \right| \leq c_L |x - u|,$$

uniformly in $x, u \in U$,

3.1.4 It holds that $\lim_{n \rightarrow \infty} nh_n \bar{F}^x(u_n^x) = \infty$,

We begin by proving some preliminary results related to the process $\tilde{V}_n^x$.

Lemma 3.1 (Mean of the tail process) *Assume conditions 3.1.1-3.1.3. Then*

$$\mathbb{E} \tilde{V}_n^x(s) \rightarrow s^{-1/\gamma(x)} g(x), \quad s > 0.$$

Proof. If $g(x) > 0$ it holds that

$$\begin{aligned} \mathbb{E} \tilde{V}_n^x(s) &= \\ g(x) \frac{\bar{F}^x(su_n^x)}{\bar{F}^x(u_n^x)} \int (su_n^x)^{-(1/\gamma(x-uh_n)-1/\gamma(x))} \frac{L^{x-uh_n}(su_n^x)}{L^x(su_n^x)} \frac{g(x-h_nu)}{g(x)} K(u) \, du. \end{aligned}$$

The expression before the integral converges to the desired limit, and hence it remains to show that the integral converges to 1. We consider each of the three n -dependent factors in the integral

$$\begin{aligned} (su_n^x)^{-(1/\gamma(x-uh_n)-1/\gamma(x))} &= \exp(-(1/\gamma(x-uh_n)-1/\gamma(x)) \log(su_n^x)) \\ &\leq \exp(c_\gamma h_n |u| \log(su_n^x)) = \exp(c_\gamma |u| h_n \log(u_n^x)) \exp(c_\gamma h_n |u| \log(s)) \rightarrow 1, \end{aligned}$$

and similarly

$$(su_n^x)^{-(1/\gamma(x-uh_n)-1/\gamma(x))} \geq \exp(-c_\gamma |u| h_n \log(u_n^x)) \exp(-c_\gamma h_n |u| \log(s)) \rightarrow 1,$$

uniformly in $u \in [-1, 1]$, using condition 3.1.1 and $h_n \log(u_n^x) \rightarrow 0$. Now, condition 3.1.3 implies

$$\exp(-\log(su_n^x) h_n c_L |u|) \leq \frac{L^{x-uh_n}(su_n^x)}{L^x(su_n^x)} \leq \exp(\log(su_n^x) h_n c_L |u|)$$

and both sides converge to 1, uniformly in $u \in [-1, 1]$. The density g is assumed twice continuously differentiable which implies Lipschitz continuity, so that

$$1 - \frac{c_g |u| h_n}{g(x)} \leq \frac{g(x-h_nu)}{g(x)} \leq 1 + \frac{c_g |u| h_n}{g(x)}$$

and both sides converge to 1, uniformly in $u \in [-1, 1]$. By the dominated convergence theorem we get $\mathbb{E} \tilde{V}_n^x(s) \rightarrow g(x) s^{-1/\gamma(x)}$. If $g(x) = 0$, similar arguments, using Lipschitz continuity of g , yield the result. $\square$

The following lemma underpins several of the asymptotic arguments to follow.

Lemma 3.2 (Kernel tail expectation) *Assume conditions 3.1.1-3.1.3. For any $t, q > 0$ it holds that*

$$\frac{\mathbb{E}[1_{(Y_1 > u_n^x t)} K^q(\frac{x-X_1}{h_n})]}{h_n \bar{F}^x(u_n^x)} \rightarrow t^{-1/\gamma(x)} g(x) \int K^q(u) du.$$

Proof. Calculations similar to those of Lemma 3.1 yield the result. $\square$

Theorem 3.20 (Consistency of the tail empirical process) *Assume that Condition 3.1 holds. Then*

$$\tilde{T}_n^x(s) \xrightarrow{\mathbb{P}} T^x(s) = s^{-1/\gamma(x)}, \quad s > 0, \quad (3.17)$$

the convergence being uniform on compacts bounded away from zero.

Proof. We have $g_n(x) \xrightarrow{\mathbb{P}} g(x)$, and hence, using the relation $\tilde{T}_n^x(s) = \tilde{V}_n^x(s)/g_n(x)$, it remains to show that $\tilde{V}_n^x(s) \xrightarrow{\mathbb{P}} g(x)s^{-1/\gamma(x)}$. Independence and Lemma 3.2 yield that

$$\mathbb{E}\tilde{V}_n^x(s) = \frac{\mathbb{E}[1_{(Y > u_n^x s)} K(\frac{x-X}{h_n})]}{h_n \bar{F}^x(u_n^x)} \rightarrow s^{-1/\gamma(x)} g(x),$$

and

$$\text{Var}(\tilde{V}_n^x(s)) = \frac{1}{nh_n \bar{F}^x(u_n^x)} \left[\frac{\mathbb{E}[1_{(Y_1 > u_n^x s)} K^2(\frac{x-X_1}{h_n})]}{h_n \bar{F}^x(u_n^x)} - \frac{\mathbb{E}[1_{(Y_1 > u_n^x s)} K(\frac{x-X_1}{h_n})]^2}{h_n \bar{F}^x(u_n^x)} \right],$$

which vanishes by Lemma 3.2. This finishes the proof, by Dini's Theorem 1.15. $\square$

As a first application we get that consistency of the conditional tail empirical function implies that the intermediate empirical quantile q_n^x is equivalent to the deterministic u_n^x . This is of practical importance, since the former is estimable from data. Consistency of the conditional Hill estimator then directly follows.

Proposition 3.3 (Consistency of conditional quantiles) *If (3.17) holds, then $q_n^x/u_n^x \xrightarrow{\mathbb{P}} 1$.*

Proof. Let $\varepsilon > 0$ be given and assume that $q_n^x/u_n^x > 1 + \varepsilon$ for all $n \in \mathbb{N}$. Since $\bar{F}_n^x$ is decreasing we have

$$\tilde{T}_n^x(1 + \varepsilon) = \frac{\bar{F}_n^x((1 + \varepsilon)u_n^x)}{\bar{F}^x(u_n^x)} \geq \frac{\bar{F}_n^x(q_n^x)}{\bar{F}^x(u_n^x)} = \frac{\bar{F}_n^x(F_n^{\leftarrow, x}(1 - k_n/n))}{\bar{F}^x(F^{\leftarrow, x}(1 - k_n/n))} = 1.$$

In particular it holds that

$$\mathbb{P}(q_n^x/u_n^x > 1 + \varepsilon) \leq \mathbb{P}(\tilde{T}_n^x(1 + \varepsilon) \geq 1).$$

Since (3.17) holds, then $\tilde{T}_n^x(1 + \varepsilon) \xrightarrow{\mathbb{P}} (1 + \varepsilon)^{-1/\gamma(x)}$ and thus

$$\limsup_{n \rightarrow \infty} \mathbb{P}(q_n^x/u_n^x > 1 + \varepsilon) \leq \lim_{n \rightarrow \infty} \mathbb{P}(\tilde{T}_n^x(1 + \varepsilon) \geq 1) = 0.$$

Using similar arguments we obtain that $\limsup_{n \rightarrow \infty} \mathbb{P}(q_n^x/u_n^x < 1 - \varepsilon) = 0$. $\square$

Theorem 3.21 (Consistency of the conditional Hill estimator) *Assume conditions 3.1.1-3.1.3, and (3.17). Then $\gamma_{n,k}(x) \xrightarrow{\mathbb{P}} \gamma(x)$.*

Proof. We consider the following decomposition

$$\gamma_{n,k}(x) = \int_{q_n^x/u_n^x}^1 \frac{\tilde{T}_n^x(s)}{s} ds + \int_1^\infty \frac{\tilde{T}_n^x(s)}{s} ds.$$

Proposition 3.3 yields that $q_n^x/u_n^x \xrightarrow{\mathbb{P}} 1$ and thus the first integral is $o_{\mathbb{P}}(1)$. The convergence (3.17) is uniform on compacts of $[1, \infty)$ and hence for all $m > 1$ we have that

$$\int_1^m \frac{\tilde{T}_n^x(s)}{s} ds \xrightarrow{\mathbb{P}} \int_1^m \frac{T^x(s)}{s} ds, \quad \text{as } n \rightarrow \infty.$$

Since $\lim_{m \rightarrow \infty} \int_1^m (T^x(s)/s) ds = \gamma(x)$, consistency of $\gamma_{n,k}$ follows by Theorem 1.5 if for every $\eta > 0$

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_m^\infty \frac{\tilde{T}_n^x(s)}{s} ds > \eta \right) = 0,$$

Now, since

$$\lim_{m \rightarrow \infty} \int_m^\infty \frac{\tilde{T}_n^x(s)}{s} ds = \frac{g(x)}{g_n(x)} \lim_{m \rightarrow \infty} \int_m^\infty \frac{\tilde{V}_n^x(s)}{g(x)s} ds,$$

and $g(x)/g_n(x) \xrightarrow{\mathbb{P}} 1$, it suffices to show that for every $\eta > 0$,

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_m^\infty \frac{\tilde{V}_n^x(s)}{g(x)s} ds > \eta \right) = 0.$$

Fix $\eta > 0$. We get

$$\begin{aligned} \mathbb{P} \left(\int_m^\infty \frac{\tilde{V}_n^x(s)}{g(x)s} ds > \eta \right) &\leq \frac{1}{\eta} \int_m^\infty \frac{\mathbb{E} \tilde{V}_n^x(s)}{g(x)s} ds \\ &= \frac{1}{\eta} \int_m^\infty \frac{\bar{F}^x(su_n^x)}{s\bar{F}^x(u_n^x)} \int (su_n^x)^{-(1/\gamma(x)-uh_n)-1/\gamma(x)} \frac{L^{x-uh_n}(su_n^x)}{L^x(su_n^x)} \frac{g(x-h_nu)}{g(x)} K(u) du ds \\ &\leq \frac{\text{cst}}{\eta} \int_m^\infty \frac{\bar{F}^x(su_n^x)}{s\bar{F}^x(u_n^x)} \sup_{u \in [-1,1]} \left((su_n^x)^{-(1/\gamma(x)-uh_n)-1/\gamma(x)} \frac{L^{x-uh_n}(su_n^x)}{L^x(su_n^x)} \frac{g(x-h_nu)}{g(x)} \right) ds \\ &\leq \frac{\text{cst}}{\eta} \int_m^\infty \frac{\bar{F}^x(su_n^x)}{s\bar{F}^x(u_n^x)} \sup_{u \in [-1,1]} \left(\frac{g(x-h_nu)}{g(x)} \right) s^{(c_\gamma+c_L)h_n} (u_n^x)^{(c_\gamma+c_L)h_n} ds \\ &\leq \frac{\text{cst}}{\eta} \sup_{u \in [-1,1]} \left(\frac{g(x-h_nu)}{g(x)} \right) (u_n^x)^{(c_\gamma+c_L)h_n} \int_m^\infty s^{-1/\gamma(x)+\varepsilon-1+(c_\gamma+c_L)h_n} ds, \end{aligned}$$

for $\varepsilon \in (0, 1/\gamma(x))$, using Potter's bound. Let $\varepsilon' \in (0, 1/\gamma(x) - \varepsilon)$ be given. It is possible to find n' such that $(c_\gamma + c_L)h_n \leq \varepsilon'$ for all $n \geq n'$. Hence for $n \geq n'$ we have that the integrand $s \mapsto s^{-1/\gamma(x)+\varepsilon-1+(c_\gamma+c_L)h_n}$ is integrable and consequently

$$\limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_m^\infty \frac{\tilde{V}_n^x(s)}{g(x)s} ds > \eta \right) = O(m^{-1/\gamma(x)+\varepsilon+\varepsilon'}),$$

which vanishes as $m \rightarrow \infty$. $\square$

3.6.2 Central limit theorems

We now consider the univariate conditional tail empirical process for deterministic thresholds

$$\tilde{T}_n^x(s) = \sqrt{k_n h_n}(\tilde{T}_n^x(s) - T^x(s)), \quad s > 0,$$

for which a functional central limit is established. Notice that the speed of convergence is determined by the effective sample size $nh_n \bar{F}^x(u_n^x) = h_n k_n$, which is slower than in the unconditional case, which is simply k_n .

For $s_0 \in (0, 1)$, define

$$B_n^x(s_0) = \sup_{t \geq s_0} |\mathbb{E} \tilde{V}_n^x(t) - t^{-1/\gamma(x)} g(x)|.$$

Condition 3.2 The conditions for normality are as follows.

3.2.1 There exists $s_0 \in (0, 1)$, such that $\sqrt{k_n h_n} B_n^x(s_0) \rightarrow 0$,

3.2.2 It holds that

$$\lim_{n \rightarrow \infty} \sqrt{k_n h_n} \int_1^\infty \frac{\mathbb{E} \tilde{V}_n^x(t) - t^{-1/\gamma(x)} g(x)}{t} dt = 0.$$

Define the auxiliary empirical process

$$\tilde{V}_n^x(s) = \sqrt{k_n h_n}(\tilde{V}_n^x(s) - \mathbb{E}[\tilde{V}_n^x(s)]), \quad s > 0,$$

Lemma 3.3 (Covariance of the tail process) *Assume Condition 3.1. Then for all $s, t > 0$*

$$\lim_{n \rightarrow \infty} \text{Cov}(\tilde{V}_n^x(s), \tilde{V}_n^x(t)) = (s \vee t)^{-1/\gamma(x)} g(x) \int K^2(u) du. \quad (3.18)$$

Proof. Independence and identical distribution yield

$$\begin{aligned} \text{Cov}(\tilde{V}_n^x(s), \tilde{V}_n^x(t)) &= \frac{1}{h_n \bar{F}^x(u_n^x)} \mathbb{E} \left[1_{(Y_1 > (s \vee t) u_n^x)} K^2 \left(\frac{x - X_1}{h_n} \right) \right] \\ &\quad - h_n \bar{F}^x(u_n^x) \frac{\mathbb{E} \left[K \left(\frac{x - X_1}{h_n} \right) 1_{(Y_1 > s u_n^x)} \right]}{h_n \bar{F}^x(u_n^x)} \frac{\mathbb{E} \left[K \left(\frac{x - X_1}{h_n} \right) 1_{(Y_1 > t u_n^x)} \right]}{h_n \bar{F}^x(u_n^x)} \end{aligned}$$

The limit of the first term follows from Lemma 3.2:

$$\lim_{n \rightarrow \infty} \frac{1}{h_n \bar{F}^x(u_n^x)} \mathbb{E} \left[1_{(Y_1 > (s \vee t) u_n^x)} K^2 \left(\frac{x - X_1}{h_n} \right) \right] = (s \vee t)^{-1/\gamma(x)} g(x) \int K^2(u) du,$$

which is the asymptotic covariance. The second term disappears asymptotically due to Lemma 3.2. This finishes the proof. $\square$

With the covariance limit in place, we next establish finite-dimensional Gaussian limits and then lift this to functional weak convergence.

Theorem 3.22 (Finite-dimensional convergence) *Assume Conditions 3.1. Then the finite-dimensional distributions of $\tilde{\mathbb{V}}_n^x$ converge to those of a Gaussian process $\mathbb{V}^x$ with covariance function defined in (3.18).*

Proof. Fix $s > 0$. We only show the result for one marginal, but the general case follows by the Cramér–Wold device. Note first that $\mathbb{V}_n^x$ can be written as

$$\tilde{\mathbb{V}}_n^x(s) = \sum_{j=1}^n \frac{1_{(Y_j > su_n^x)} K\left(\frac{x-X_j}{h_n}\right) - \mathbb{E}\left[1_{(Y_1 > su_n^x)} K\left(\frac{x-X_1}{h_n}\right)\right]}{\sqrt{k_n h_n}} =: \sum_{j=1}^n Z_{j,n}(s)$$

where the summands are mean zero iid random variables. In the calculations below we show that the conditions of the Lindeberg CLT are fulfilled. By Lemma 3.3 it only remains to show Lindeberg’s condition. Let $c > 0$ be given. Then we have that

$$\begin{aligned} \sum_{j=1}^n \int_{(|Z_{j,n}(s)| > c)} Z_{j,n}^2(s) \, d\mathbb{P} &= n \mathbb{E}\left[Z_{1,n}^2(s) 1_{(|Z_{1,n}(s)| > c)}\right] = \\ &= \frac{n}{k_n h_n} \mathbb{E}\left[\left(1_{(Y_1 > su_n^x)} K\left(\frac{x-X_1}{h_n}\right) + \mathbb{E}\left[1_{(Y_1 > su_n^x)} K\left(\frac{x-X_1}{h_n}\right)\right]\right)^2 1_{(|Z_{1,n}(s)| > c)}\right] \\ &\quad - \frac{2n}{k_n h_n} \mathbb{E}\left[K\left(\frac{x-X_1}{h_n}\right) 1_{(Y_1 > su_n^x)} \mathbb{E}\left[1_{(Y_1 > su_n^x)} K\left(\frac{x-X_1}{h_n}\right)\right] 1_{(|Z_{1,n}(s)| > c)}\right]. \end{aligned}$$

By two applications of the Cauchy–Schwarz inequality, and invoking Lemma 3.2 and noticing that $1_{(|Z_{1,n}(s)| > c)}$ converges almost surely to zero, the above term vanishes. $\square$

Theorem 3.23 (Functional CLT for the tail process) *Assume Condition 3.1. Then $\tilde{\mathbb{V}}_n^x$ converges weakly to the centred Gaussian process $\mathbb{V}^x$ in $\ell^\infty((0, \infty))$, and the process $\mathbb{V}^x$ admits an almost surely continuous version.*

Proof. Lemma 3.3 yields that the covariance structure of $\tilde{\mathbb{V}}_n^x$ converges to that of the proposed Gaussian process. Define

$$Z_{ni}(s) = (k_n h_n)^{-1/2} \{1_{(Y_i > u_n^x s)} K(h_n^{-1}(X_i - x)) - \mathbb{E}[1_{(Y_i > u_n^x s)} K(h_n^{-1}(X_i - x))]\},$$

so that $\tilde{\mathbb{V}}_n^x = \sum_{i=1}^n Z_{ni}$. To prove functional convergence, we make use of the Bracketing CLT for changing functions, Theorem 1.20. Consider the bounded interval $I = [a, b]$ with $a > 0$, and define

$$\begin{aligned} \mathcal{F}_n &= \{f_{n,t} : [0, \infty) \times \mathbb{R} \rightarrow [0, \infty) \\ \text{s.t. } f_{n,t}(y, z) &= (h_n \bar{F}(u_n^x))^{-1/2} 1_{[u_n^x t, \infty)}(y) K(h_n^{-1}(z - x)) \text{ for } t \in [a, b]\}. \end{aligned}$$

Clearly the envelope is given by $F_n(Y_1, X_1) = (h_n \bar{F}(u_n^x))^{-1/2} 1_{(Y_1 > u_n^x a)} K(h_n^{-1}(X_1 - x))$. It satisfies by Lemma 3.2 that $\mathbb{E}[F_n(Y_1, X_1)^2] = O(1)$, showing condition (1.6). We also have that by the same argument as in the Lindeberg CLT, Theorem 3.22, when proving finite-dimensional distributions that

$$\mathbb{E}[F_n(Y_1, X_1)^2 1_{(F_n(Y_1, X_1) > c\sqrt{n})}] \rightarrow 0.$$

This shows (1.7). The bracketing number is calculated as follows. Consider for $j = 1, \dots, m$ lower brackets on the form

$$l_j(y, z) = (h_n \bar{F}(u_n^x))^{-1/2} 1_{[u_n^x t_j, \infty)}(y) K(h_n^{-1}(z - x)),$$

and similarly for the upper brackets

$$u_j(y, z) = (h_n \bar{F}(u_n^x))^{-1/2} 1_{[u_n^x t_{j-1}, \infty)}(y) K(h_n^{-1}(z - x)),$$

for sequences on the form $a = t_0 < t_1 < \dots < t_m = b$. Let $\mathcal{F}_j^n = \{f \in \mathcal{F}_n : l_j \leq f \leq u_j\}$. Then clearly we have the partition $\mathcal{F}_n = \bigcup_{j=1}^m \mathcal{F}_j^n$. Let $\varepsilon > 0$ be given, and pick a sequence for $m \leq \lceil 2/(\varepsilon^2/2) \rceil$ such that

$$T^x(t_{j-1}) - T^x(t_j) \leq \varepsilon^2 a^{-1/\gamma(x)} / (2g(x)) \quad \text{for } 0 \leq j \leq m,$$

which is possible because T is a continuous and bounded function on $[a, b]$. Then this sequence covers all of $\mathcal{F}_n$ with brackets satisfying

$$\begin{aligned} & \|u_j(Y_1, X_1) - l_j(Y_1, X_1)\|_{L^2} \\ &= \frac{\mathbb{E} \left[1_{(u_n^x t_j \geq Y_1 > u_n^x t_{j-1})} K^2 \left(\frac{x - X_1}{h_n} \right) \right]^{1/2}}{(h_n \bar{F}(u_n^x))^{1/2}} \\ &= (g(x) \int K^2(u) \, du)^{1/2} (t_{j-1}^{-1/\gamma(x)} - t_j^{-1/\gamma(x)})^{1/2} (1 + o(1)) \\ &\leq \varepsilon \|F_n(Y_1, X_1)\|_{L^2}, \end{aligned}$$

Consequently for some constant C ,

$$N_{[\cdot]}(\varepsilon \|F_n(Y_1, X_1)\|_{L^2}, \mathcal{F}_n, L^2(\mathbb{P})) \leq m \leq \lceil C/\varepsilon^2 \rceil.$$

Integration now shows (1.9). Note that the above $o(1)$ term is independent of the partition by Theorem 3.20. In particular, the partition is independent of n and (1.8) does not need to be verified. $\square$

Proposition 3.4 (Weak convergence of the conditional tail empirical process)
Assume that conditions 3.1 and 3.2.1 hold. Then it holds that $\tilde{\mathbb{T}}_n^x$ converges weakly to a centred Gaussian process $\mathbb{T}^x$ in $\ell^\infty([s_0, \infty))$, with covariance function

$$\Gamma(s, t) = \frac{(s \vee t)^{-1/\gamma(x)}}{g(x)} \int K^2(u) \, du, \quad (3.19)$$

and the process $\mathbb{T}^x$ has an almost surely continuous version.

Proof. Consider the following decomposition

$$\begin{aligned} \tilde{\mathbb{T}}_n^x(s) &= \sqrt{k_n h_n} (g(x) - g_n(x)) \frac{\tilde{T}_n^x(s)}{g(x)} + \frac{1}{g(x)} \tilde{\mathbb{V}}_n^x(s) \\ &\quad + \sqrt{k_n h_n} \frac{1}{g(x)} (\mathbb{E} \tilde{\mathbb{V}}_n^x(s) - g(x) s^{-1/\gamma(x)}). \end{aligned}$$

The absolute value of the last term vanishes, since it is bounded by $\sqrt{k_n h_n} B_n^x(s_0) \rightarrow 0$, so normality of the Parzen–Rosenblatt kernel density estimator, and Theorem 3.23 imply that

$$\tilde{T}_n^x(s) = \sqrt{k_n h_n} (g(x) - g_n(x)) \frac{\tilde{T}_n^x(s)}{g(x)} + \frac{1}{g(x)} \tilde{V}_n^x(s) + o(1) \xrightarrow{d} \frac{1}{g(x)} \mathbb{V}^x(s).$$

□

Recall the definitions $u_n^x = F_n^{\leftarrow, x}(1 - k_n/n)$ and $q_n^x = F_n^{\leftarrow, x}(1 - k_n/n)$. Then $n\bar{F}^x(u_n^x) = k_n$. Define the conditional tail empirical quantile function $\tilde{Q}_n^x$ on $(0, k_n/n)$ by

$$\tilde{Q}_n^x(t) = \tilde{T}_n^{\leftarrow, x}(\lfloor k_n t \rfloor / k_n) = \frac{q_{\lfloor k_n t \rfloor}}{u_n^x},$$

and let $Q^x(t) = t^{-\gamma(x)}$, $t > 0$. From here the argument is map-based. We first establish weak convergence of the tail process, then invert to the quantile process, and finally evaluate at random levels for the Hill estimator.

Theorem 3.24 (Joint CLT for tail and quantile processes) *Assume that conditions 3.1 and 3.2.1 hold. Then*

$$\sqrt{h_n k_n} (\tilde{T}_n^x - T^x, \tilde{Q}_n^x - Q^x) \xrightarrow{d} (\mathbb{T}^x, -(Q^x)' \cdot \mathbb{T}^x \circ Q^x),$$

in $\ell^\infty([s_0, \infty)) \times \ell^\infty((0, s_0^{-\gamma(x)}])$. Consequently

$$\sqrt{h_n k_n} (q_n^x / u_n^x - 1) \xrightarrow{d} \gamma(x) \mathbb{T}^x(1) \stackrel{d}{=} N(0, \frac{\gamma(x)^2}{g(x)} \int K^2(u) \, du),$$

jointly with the convergence of $\sqrt{h_n k_n} (\tilde{T}_n^x - T^x)$ to $\mathbb{T}^x$.

Proof. Note that $[s_0, \infty) \ni s \mapsto T^x(s)$ is surjective on $(0, s_0^{-\gamma(x)}]$, continuous and decreasing with negative derivative. As the processes $\tilde{T}_n^x$ have non-increasing sample paths for each $n \geq 1$, an application of the functional δ -method, Theorem 1.17, applied with the inverse mapping, as per Lemma 1.10, yields that

$$\sqrt{h_n k_n} (\tilde{T}_n^x - T^x, \tilde{T}_n^{\leftarrow, x} - T^{\leftarrow, x}) \xrightarrow{d} (\mathbb{T}^x, -(T^{\leftarrow, x})' \cdot \mathbb{T}^x \circ T^{\leftarrow, x}).$$

This concludes the proof. □

The sequence (u_n^x) is unobserved and we may estimate it with (q_n^x) . Let

$$\hat{\mathbb{T}}_n^x(s) = \sqrt{k_n h_n} (\hat{T}_n^x(s) - T^x(s)),$$

which we refer to as the tail empirical process with random levels.

Theorem 3.25 (Weak convergence of the tail empirical process with random levels) *Assume that conditions 3.1 and 3.2.1 hold. Then*

$$\hat{\mathbb{T}}_n^x \xrightarrow{d} \mathbb{T}^x - T^x \times \mathbb{T}^x(1),$$

in $\ell^\infty([s_0, \infty))$.

Proof. Decompose $\widehat{\mathbb{T}}_n^x$ into the following components

$$\widehat{\mathbb{T}}_n^x = \widetilde{\mathbb{T}}_n^x(sq_n^x/u_n^x) + \sqrt{k_n h_n}(T^x(sq_n^x/u_n^x) - T^x(s)).$$

By Theorem 3.20, the tail consistency (3.17) holds, while Proposition 3.3 yields that $q_n^x/u_n^x \xrightarrow{\mathbb{P}} 1$, and the continuous mapping theorem thus yields that the first term above converges to $\mathbb{T}^x$. As for the second term, Theorem 3.24 and the delta-method yields that

$$\begin{aligned} \sqrt{k_n h_n}(T^x(sq_n^x/u_n^x) - T^x(s)) &= s^{-1/\gamma(x)} \sqrt{k_n h_n}(T^x(q_n^x/u_n^x) - T^x(1)) \\ &\xrightarrow{d} s^{-1/\gamma(x)}(T^x)'(1) \cdot \gamma(x)\mathbb{T}^x(1) = -T^x(s)\mathbb{T}^x(1), \end{aligned}$$

jointly with $\widetilde{\mathbb{T}}_n^x(sq_n^x/u_n^x)$. The argument can be transferred unchanged to path-wise weak convergence. $\square$

It turns out that Theorem 3.25 is not required to establish convergence of the conditional Hill estimator. However, it is an interesting result in its own right, since the empirical tail process can be used to measure how closely the top order statistics behave like a strict Pareto tail.

Theorem 3.26 (Asymptotic normality of the conditional Hill estimator) *Assume that Condition 3.1 and Condition 3.2 hold. Then*

$$\sqrt{k_n h_n}(\gamma_{n,k}(x) - \gamma(x)) \xrightarrow{d} N(0, \frac{\gamma(x)^2}{g(x)} \int K^2(u) du).$$

Proof. We consider the decomposition

$$\sqrt{k_n h_n}(\gamma_{n,k}(x) - \gamma(x)) = \sqrt{k_n h_n} \int_{q_n^x/u_n^x}^1 \frac{\widetilde{T}_n^x(s)}{s} ds + \int_1^\infty \frac{\widetilde{\mathbb{T}}_n^x(s)}{s} ds.$$

We examine the first term. Since $q_n^x/u_n^x \xrightarrow{\mathbb{P}} 1$, we may assume without loss of generality that for given $\varepsilon \in (0, 1)$, there exists $N \in \mathbb{N}$ such that for $n \geq N$ it holds that $q_n^x/u_n^x \in (1 - \varepsilon, 1 + \varepsilon)$. Thus for $n \geq N$ we have that

$$\begin{aligned} &\sqrt{k_n h_n} \int_{q_n^x/u_n^x}^1 \frac{\widetilde{T}_n^x(s) - T^x(s)}{s} ds \\ &\leq \frac{\sqrt{k_n h_n}}{1 - \varepsilon} |q_n^x/u_n^x - 1| \sup_{t \in (1 - \varepsilon, 1 + \varepsilon)} |\widetilde{T}_n^x(t) - T^x(t)|. \end{aligned}$$

Moreover, Theorem 3.24 and the continuous mapping theorem yield that

$$\sqrt{k_n h_n} |q_n^x/u_n^x - 1| \xrightarrow{d} \gamma(x) |\mathbb{T}^x(1)|,$$

while Theorem 3.20 yields $\sup_{t \in (1 - \varepsilon, 1 + \varepsilon)} |\widetilde{T}_n^x(t) - T^x(t)| \xrightarrow{\mathbb{P}} 0$ and thus

$$\sqrt{k_n h_n} \int_{q_n^x/u_n^x}^1 \frac{\widetilde{T}_n^x(s)}{s} ds = \sqrt{k_n h_n} \int_{q_n^x/u_n^x}^1 \frac{T^x(s)}{s} ds + o_{\mathbb{P}}(1).$$

Applying the delta-method with $x \mapsto \int_x^1 \frac{T^x(s)}{s} ds$, on the sequence (q_n^x/u_n^x) we obtain

$$\sqrt{k_n h_n} \int_{q_n^x/u_n^x}^1 \frac{T^x(s)}{s} ds \xrightarrow{d} -\gamma(x) \mathbb{T}^x(1).$$

This provides the asymptotics of the first term. We proceed to the second term

$$\begin{aligned} \int_1^\infty \frac{\tilde{T}_n^x(s)}{s} ds &= \sqrt{k_n h_n} \frac{1}{g(x)} (g(x) - g_n(x)) \int_1^\infty \frac{\tilde{T}_n^x(s)}{s} ds + \frac{1}{g(x)} \int_1^\infty \frac{\tilde{V}_n^x(s)}{s} ds \\ &\quad + \sqrt{k_n h_n} \frac{1}{g(x)} \int_1^\infty \frac{\mathbb{E} \tilde{V}_n^x(s) - g(x) s^{-1/\gamma(x)}}{s} ds. \end{aligned}$$

The last term is negligible due to condition 3.2.2. In the proof of Theorem 3.21 we showed $\int_1^\infty \tilde{T}_n^x(s)/s ds \xrightarrow{\mathbb{P}} \gamma(x)$. Combined with the asymptotic normality of the Parzen–Rosenblatt kernel density estimator we have that the first term also vanishes. Finally we consider $g(x)^{-1} \int_1^\infty \tilde{V}_n^x(s)/s ds$. We have that

$$\begin{aligned} \text{Var} \left(\int_m^\infty \tilde{V}_n^x(s)/s ds \right) &= \frac{1}{h_n \bar{F}^x(u_n^x)} \text{Var} \left(K \left(\frac{x - X_0}{h_n} \right) \int_m^\infty 1_{(Y_0 > s u_n)} \frac{1}{s} ds \right) \\ &= \frac{1}{h_n \bar{F}^x(u_n^x)} \text{Var} \left(K \left(\frac{x - X_0}{h_n} \right) \log \left(\frac{(Y_1/u_n^x) \vee m}{m} \right) \right) \\ &\leq \frac{1}{h_n \bar{F}^x(u_n^x)} \mathbb{E} \left[K^2 \left(\frac{x - X_0}{h_n} \right) \log_+^2 \left(\frac{Y_1}{u_n^x m} \right) \right]. \end{aligned}$$

Since the kernel K has support $[-1, 1]$, we have that $1_{((z-x)/h_n \leq 1)} = 1_{(z \leq x+h_n)}$ and hence we have the further bound

$$\begin{aligned} &\frac{1}{h_n \bar{F}^x(u_n^x)} \int K^2 \left(\frac{x-z}{h_n} \right) \mathbb{E} \left[\log_+^2 \left(\frac{Y_1}{u_n^x m} \right) \middle| X_0 = z \right] g(z) dz \\ &\leq \sup_{u \in [-h_n+x, x+h_n]} \left(\frac{1}{\bar{F}^u(u_n^x)} \mathbb{E} \left[\log_+^2 \left(\frac{Y_1}{u_n^x m} \right) \middle| X_0 = u \right] \right) \\ &\quad \times \frac{1}{h_n} \int K^2 \left(\frac{x-z}{h_n} \right) g(z) \frac{\bar{F}^z(u_n^x)}{\bar{F}^x(u_n^x)} dz, \end{aligned}$$

where it holds that

$$\begin{aligned} &\frac{1}{h_n} \int K^2 \left(\frac{x-z}{h_n} \right) g(z) \frac{\bar{F}^z(u_n^x)}{\bar{F}^x(u_n^x)} dz \\ &\leq \frac{1}{h_n} \int K^2 \left(\frac{x-z}{h_n} \right) g(z) (u_n^x)^{-(c_\gamma+c_L)|x-z|} dz \\ &\leq (u_n^x)^{(c_\gamma+c_L)h_n} \int K^2(w) g(x-wh_n) dw = O(1), \end{aligned}$$

and

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \sup_{u \in [-h_n + x, x + h_n]} \left(\frac{1}{\bar{F}^u(u_n^x)} \mathbb{E} \left[\log_+ \left(\frac{Y_1}{u_n^x m} \right)^2 \middle| X_0 = u \right] \right) \\
& \leq \lim_{n \rightarrow \infty} \left(\frac{1}{\bar{F}^x(u_n^x)} \mathbb{E} \left[\log_+ \left(\frac{Y_1}{u_n^x m} \right)^2 \middle| X_0 = x \right] \right) \\
& \quad + \lim_{n \rightarrow \infty} \sup_{u \in [-h_n + x, x + h_n]} \left| \frac{1}{\bar{F}^u(u_n^x)} \mathbb{E} \left[\log_+ \left(\frac{Y_1}{u_n^x m} \right)^2 \middle| X_0 = u \right] \right. \\
& \quad \quad \left. - \frac{1}{\bar{F}^x(u_n^x)} \mathbb{E} \left[\log_+ \left(\frac{Y_1}{u_n^x m} \right)^2 \middle| X_0 = x \right] \right| \\
& = \int_m^\infty \log^2(t) 1/\gamma(x) t^{-1/\gamma(x)-1} dt + 0,
\end{aligned}$$

where we have used Potter's bound to invoke the dominated convergence theorem for the first limit, and the term in the difference can be bounded using Potter's bound and applying twice the Lipschitz conditions on L^x and $1/\gamma(x)$ as was done in Theorem 3.21.

Thus, taking limit as $m \rightarrow \infty$ establishes the required asymptotic normality by Theorem 1.5. By construction, $\mathbb{T}^x(\cdot) = \mathbb{V}^x(\cdot)/g(x)$, and thus we have the representation

$$\sqrt{k_n h_n} (\gamma_{n,k}(x) - \gamma(x)) \xrightarrow{d} \frac{1}{g(x)} \int_1^\infty \frac{\mathbb{V}^x(s)}{s} ds - \frac{\gamma(x)}{g(x)} \mathbb{V}^x(1),$$

which has variance $\gamma(x)^2 \frac{1}{g(x)} \int K^2(u) du$. $\square$

Remark 3.1 The bias of the conditional Hill estimator vanishes due to condition 3.2.2. If this condition is relaxed to

$$\lim_{n \rightarrow \infty} \sqrt{k_n h_n} \int_1^\infty \frac{\mathbb{E} \tilde{V}_n^x(t) - t^{-1/\gamma(x)} g(x)}{t} dt = B \in \mathbb{R},$$

the asymptotic variance in Theorem 3.26 remains similar, while the bias becomes $B/g(x)$.

A theoretical bandwidth choice can be made as a function of k_n and n . For instance, $h_n = \sqrt{\log(k_n)/n}$ falls into the framework of conditions 3.1.2 and 3.1.4. It can also be seen to satisfy equation (3.21), related to second order behaviour, for a suitable (k_n) sequence. $\circ$

3.7 Notes and Comments

The terminology heavy-tailed, long-tailed, subexponential, and regularly varying goes back to classical asymptotic analysis and renewal theory, with modern probabilistic treatments given in monographs on regular variation and in general texts on regular variation, extremes and heavy tails such as [Resnick \[1987\]](#), [de Haan and Ferreira \[2006\]](#), [de Haan and Ferreira \[2006\]](#), [de Haan and Ferreira \[2006\]](#). The detailed relationships between different definitions of heavy tails are standard and can be found with careful proofs in these references. For statistical practice, the insurance and finance perspective is well treated in [Embrechts et al. \[1997\]](#), which also treats modelling choices, diagnostics, a practical statistical section, and risk implications. The latter reference complements the more measure-theoretic treatments in [Resnick \[1987\]](#), [de Haan and Ferreira \[2006\]](#).

The classical probabilistic treatment of extreme value theory goes from the extremal types limit theorem in seminal works of Fisher–Tippett and Gnedenko [\[1930\]](#) to the Generalised Extreme Value family, and from there to characterisations of max-domains of attraction; see [de Haan and Ferreira \[2006\]](#) for a stationary-sequence viewpoint and [de Haan and Ferreira \[2006\]](#) for systematic treatments. The equivalence between membership in the Fréchet max-domain and regular variation of the tail is an important result connecting distributional limits of maxima to tail-regularity assumptions. The threshold-exceedance analogue, namely the Pickands–Balkema–de Haan limit, justifies modelling large exceedances by the Generalised Pareto family and leads to peaks-over-threshold methods, see [de Haan and Ferreira \[2006\]](#), [Embrechts et al. \[1997\]](#).

Regarding the manipulation of regularly varying tails, three tools recur throughout the chapter: uniform convergence of tail ratios on ranges bounded away from the origin; Potter-type bounds; and integral or representation theorems (Karamata-type results and de Haan’s representation), for which we refer to [Resnick \[1987\]](#). The former reference also provides a very detailed account on properties of regularly varying functions, beyond the special case of distribution functions or quantile functions.

The statistical part of univariate extreme value estimation centres on tail-index and extreme-quantile estimation. In this simple context, the Hill estimator remains a benchmark for estimating the tail index, with consistency and asymptotic normality under standard intermediate-sequence and second-order conditions given in [de Haan and Ferreira \[2006\]](#). The tail empirical and quantile process limits, specifically weak convergence to Brownian motion under a tail-driven time change, and the induced Gaussian limits for functionals such as Hill, are standard in empirical process approaches to extremes and are well shown in [Resnick \[1987\]](#), with the present treatment mirroring the slightly adapted approach and notation of [de Haan and Ferreira \[2006\]](#), but specialising to the independent case. The block-maxima and threshold-based statistical counterparts of the Hill estimator include maximum likelihood for the Generalised Pareto model and alternative tail-index estimators such as the Pickands and the moment estimators; asymptotics and variance expressions for these estimators can be found in [de Haan and Ferreira \[2006\]](#). Extrapolation of extreme quantiles via Weissman-type estimators is classical and investigated in various methodological papers and also in the general monographs cited above, see [de Haan and Ferreira \[2006\]](#). Finally, conditional tail estimation has been considered in various settings, where we notably mention [de Haan and Ferreira \[2006\]](#) and the time-series

version ? upon which the present exposition is based, again specialised to the independent case. In particular, the final consistency and central-limit-theorem sections for conditional extremes make explicit how these conditional estimators involve the same empirical-process structure as their simpler unconditional counterparts.

3.8 Exercises

1. Show that regularly varying distributions are subexponential, that subexponential distributions are long-tailed, and that long-tailed distributions are heavy-tailed.
2. Provide a counterexample for each reversed inclusion in the previous exercise.
3. Prove that $\gamma = \int_1^\infty \frac{T(s)}{s} ds$. Relate the sum version and integral version of the Hill estimator.
4. Let Z have Weibull distribution with survival function $\bar{F}(x) = \exp(-x^a)$, $x \geq 0$, for some $a > 0$.
 - (a) Show that the moment generating function is finite for some $t > 0$ if and only if $a \geq 1$, and conclude that the Weibull distribution is heavy-tailed if and only if $a < 1$.
 - (b) Show that $\bar{F}(tx)/\bar{F}(x) \rightarrow 0$ for any $t > 1$, and hence that the Weibull distribution is not regularly varying for any $a > 0$.
5. The distribution function of a Burr-distributed random variable is given by

$$F(x) = 1 - \left(1 + \left(\frac{x}{\theta}\right)^\eta\right)^{-\beta}, \quad x \geq 0, \theta > 0, \eta > 0, \beta > 0.$$

Show that the tail is regularly varying and find the corresponding index.

6. Discuss in detail the proof of Theorem 3.5 (Potter's bounds, first version). Prove Theorem 3.6 (Potter's bounds, second version).
7. Prove the following result, known as Breiman's lemma.

Lemma 3.4 (Breiman's Lemma) *Let Z and W be independent non-negative random variables, such that Z is regularly varying with tail index $1/\gamma > 0$ and there exists $\varepsilon > 0$ such that $\mathbb{E}[W^{1/\gamma+\varepsilon}] < \infty$. Then ZW is also regularly varying with index $1/\gamma$ and*

$$\lim_{x \rightarrow \infty} \frac{\mathbb{P}(ZW > x)}{\mathbb{P}(Z > x)} = \mathbb{E}[W^{1/\gamma}].$$

8. Recall that the weak limit of the Hill estimator has the following distributional representation:

$$\int_1^\infty \frac{\mathbb{W}(T(t))}{t} dt - \gamma \mathbb{W}(1) = \int_1^\infty \frac{\mathbb{B}(T(t))}{t} dt = \gamma \int_0^1 \frac{\mathbb{B}(t)}{t} dt.$$

Show that it has variance given by $\gamma^2 \int_0^1 \int_0^1 (s^{-1} \wedge t^{-1} - 1) ds dt = \gamma^2$.

9. Go through the details of the proof of consistency of the moment estimator in the $\gamma > 0$ case, Proposition 3.2.

10. Let F be a continuous distribution function on $\mathbb{R}$ and assume that there exist $c > 0$, $\beta > 0$ and $\gamma > 0$ such that

$$\sup_{t \geq 1} t^\beta |\overline{F}(t) - c \log(t) t^{-1/\gamma}| < \infty. \quad (9.6.3)$$

Deduce that $k = o(\log^2(n))$ is a suitable choice for (3.7) to hold.

11. Prove Lemma 3.2.
12. Recall that the weak limit of the conditional Hill estimator has the following distributional representation:

$$\sqrt{k_n h_n} (\hat{\gamma}_n(x) - \gamma(x)) \xrightarrow{d} \frac{1}{g(x)} \int_1^\infty \frac{\mathbb{V}^x(s)}{s} ds - \frac{\gamma(x)}{g(x)} \mathbb{V}^x(1),$$

Show that it has variance given by $\gamma(x)^2 \frac{1}{g(x)} \int K^2(u) du$.

13. Provide the details for the following limit, used in the proof of Theorem 3.26:

$$\begin{aligned} \lim_{n \rightarrow \infty} \sup_{u \in [-h_n + x, x + h_n]} \left| \frac{1}{\overline{F}^u(u_n^x)} \mathbb{E} \left[\log_+ \left(\frac{Y_1}{u_n^x m} \right)^2 \middle| X_0 = u \right] \right. \\ \left. - \frac{1}{\overline{F}^x(u_n^x)} \mathbb{E} \left[\log_+ \left(\frac{Y_1}{u_n^x m} \right)^2 \middle| X_0 = x \right] \right| \rightarrow 0. \end{aligned}$$

14. Assume Condition 3.1, that F^x is a continuous distribution and that there exist $c_x, 1/\gamma(x) > 0$, and $\beta_x > 1/\gamma(x)$ such that

$$\sup_{t \geq 1} t^{\beta_x} |\overline{F}^x(t) - c_x t^{-1/\gamma(x)}| < \infty. \quad (3.20)$$

In this exercise we wish to show that if there exist sequences (k_n) and (h_n) such that

$$\lim_{n \rightarrow \infty} \sqrt{k_n h_n^3} \log(n/k_n) = \lim_{n \rightarrow \infty} \sqrt{h_n k_n}^{\gamma(x)\beta_x - \frac{1}{2}} n^{1-\gamma(x)\beta_x} = 0, \quad (3.21)$$

then Condition 3.2 holds.

- (a) Argue that (3.20) implies that $u_n^x = O((n/k)^{\gamma(x)})$, and show that the following decomposition holds

$$\begin{aligned} \mathbb{E}[\tilde{V}_n^x(t)] - t^{-1/\gamma(x)} g(x) \\ = \int K(u) t^{-1/\gamma(x)} (g(x - h_n u) - g(x)) du \\ + \int K(u) \left(\frac{\overline{F}^{x-uh_n}(tu_n^x)}{\overline{F}^x(u_n^x)} - \frac{\overline{F}^x(tu_n^x)}{\overline{F}^x(u_n^x)} \right) g(x - h_n u) du \\ + \int K(u) \left(\frac{\overline{F}^x(tu_n^x)}{\overline{F}^x(u_n^x)} - t^{-1/\gamma(x)} \right) g(x - h_n u) du. \end{aligned}$$

- (b) Show that

$$\int K(u) t^{-1/\gamma(x)} (g(x - h_n u) - g(x)) du \leq O(t^{-1/\gamma(x)} h_n).$$

Hint: Use that g is assumed twice differentiable.

(c) Show that

$$\begin{aligned} & \left(\frac{\overline{F}^{x-uh_n}(tu_n^x)}{\overline{F}^x(u_n^x)} - \frac{\overline{F}^x(tu_n^x)}{\overline{F}^x(u_n^x)} \right) \\ & \leq \text{cst } t^{-1/\gamma(x)+\varepsilon} \log(tu_n^x) |u| h_n \left(1 + \log(tu_n^x) |u| h_n (1 + (tu_n^x)^{(c_\gamma+c_L)|u|h_n})/2 \right), \end{aligned}$$

and argue that the leading term is of order

$$O \left(t^{-1/\gamma(x)+\varepsilon} \log(tn/k_n) h_n \right),$$

for $\varepsilon \in (0, 1/\gamma(x))$.

Hint: Use that $|e^z - 1| \leq |z| + (1 + e^{|z|}) z^2/2$ along with the calculations in Lemma 3.1. You may use that Potter's bound still holds in the conditional setting.

(d) Show that

$$\left| \frac{\overline{F}^x(tu_n^x)}{\overline{F}^x(u_n^x)} - t^{-1/\gamma(x)} \right| \leq O(k_n^{\beta_x \gamma(x)-1} n^{1-\beta_x \gamma(x)} t^{-1/\gamma(x)}).$$

Hint: Use the assumption in (3.20) and follow the same strategy as in the unconditional case.

(e) Argue that all three O -terms are uniformly bounded for $t \geq s_0$ for some $s_0 \in (0, 1)$ and conclude that condition 3.2.1 holds.

(f) By using the calculations from the previous subexercises, show that

$$\begin{aligned} & \int_1^\infty \frac{\mathbb{E}[\tilde{V}_n^x(t)] - t^{-1/\gamma(x)} g(x)}{t} dt \\ & \leq O(h_n) \int_1^\infty t^{-1/\gamma(x)-1} dt + O(k_n^{\beta_x \gamma(x)-1} n^{1-\beta_x \gamma(x)}) \int_1^\infty t^{-1/\gamma(x)-1} dt \\ & + O(h_n) \int_1^\infty t^{-1/\gamma(x)+\varepsilon-1} \log(tu_n^x) \left(1 + \log(tu_n^x) h_n (1 + (tu_n^x)^{(c_\gamma+c_L)h_n|u|})/2 \right) dt. \end{aligned}$$

(g) Prove that

$$O(h_n) \int_1^\infty t^{-1/\gamma(x)+\varepsilon-1} \log(tu_n^x) \left(1 + \log(tu_n^x) h_n (1 + (tu_n^x)^{(c_\gamma+c_L)h_n|u|})/2 \right) dt$$

is bounded for all n , and conclude that condition 3.2.2 also holds.

Hint: First show that $(u_n^x)^{h_n} = O(1)$ and that there exists n' such that for all $n \geq n'$ then $(c_\gamma + c_L)h_n \leq \varepsilon' \in (0, 1/\gamma(x) - \varepsilon)$, then analyse each sub-integral separately.

15. Assume Condition 3.1, that F^x is a continuous distribution and that there exist $c_x, 1/\gamma(x) > 0$, and $\beta_x > 1/\gamma(x)$ such that

$$\sup_{t \geq 1} t^{\beta_x} \left| \overline{F}^x(t) - c_x \log(t) t^{-1/\gamma(x)} \right| < \infty. \quad (3.22)$$

Show that if there exist sequences (k_n) and (h_n) such that

$$\lim_{n \rightarrow \infty} \sqrt{k_n h_n^3} \log(n/k_n) = \lim_{n \rightarrow \infty} \frac{\sqrt{k_n h_n}}{\log(n/k_n)} = 0, \quad (3.23)$$

then Condition 3.2 holds.

16. Assume that F^x is continuous and that conditions 3.1.1, 3.1.3, and condition (3.20) hold. Show that there exist sequences (k_n) , (h_n) such that all the conditions for consistency and normality are satisfied.
17. Assume that F^x is continuous and that conditions 3.1.1, 3.1.3, and condition (3.22) hold. Show that there exist sequences (k_n) , (h_n) such that all the conditions for consistency and normality are satisfied.

Practical exercises

1. Simulate 1000 observations of a Burr-distributed random variable with $\beta = 1$, $\eta = 2$ and $\theta = 1$ and of a standard normal random variable.
 - (a) Implement the Hill estimator as a function of k .
 - (b) Make a Hill plot of the Burr data. Also make one for a normally distributed sample. Estimate the tail index by selecting a suitable k .
 - (c) Plot the empirical mean excess functions. Qualify their behaviour based on their theoretical counterparts.
 - (d) Estimate the 90%, 99%, 99.99% quantile of the Burr-distributed data by using Section 3.3.3. Compare with the true quantiles.
 - (e) Estimate the 90%, 99%, 99.99% quantile of the Burr-distributed data based on the empirical distribution function. Compare the result with the previous subexercise.
2. Consider now the dataset `beaonre` from the R package `CASdatasets` [Dutang and Charpentier \[2024\]](#), which contains claim sizes caused by fire losses for which a reinsurance company Aon is liable. Consider the column `ClaimCost`.
 - (a) Construct the Hill estimator $\gamma_{n,k}$ as a function of the threshold k and use it to construct the Hill plot. Based on the plot, provide an estimated value for the tail index γ .
 - (b) Randomly sample 300 observations from `ClaimCost` and construct the Hill estimator, call it $\hat{\gamma}_{n,k}$. Repeat this experiment 100 times (sampling with replacement), and plot a histogram of the 100 samples of $\hat{\gamma}_{n,k}$ for $k = n/5$. Compare with the theoretical limit distribution.
3. Consider the datasets `freMTPL2sev` and `freMTPL2freq` from the R package `CASdatasets` [Dutang and Charpentier \[2024\]](#). The first dataset contains policy ID and claim size, the second contains policy ID along with covariates and claim number. We are interested in the conditional tail behaviour of the claim sizes (`ClaimAmount`) given driver age (`DrvAge`). The first task is to join the two datasets. Transform the covariate into

uniform random variables and consider the uniform kernel on $[-1, 1]$. We now implement the conditional Hill estimator of the claim severities (Y_n) given the chosen covariate (X_n) .

- (a) Recall that $q_n^x = (\mathbb{F}^{(n)})^{\leftarrow}(1 - k_n/n|x)$. In **R**, implement a function q_n^x that, for given $x \in \mathbb{R}$, returns a vector corresponding to each k_n , where (k_n) is chosen to live on an equidistant grid **K_grid** of 100 points on $[0.01n, n/2]$.
- (b) In **R**, implement the conditional Hill estimator: Construct an equidistant grid **X_grid** of 101 points on $[0, 1]$. Your function should return the value of $\hat{\gamma}_n(x)$ for each pair of (x, k_n) on $(\mathbf{X_grid}) \times (\mathbf{K_grid})$.
- (c) Consider $(x_1, \dots, x_5, x_{97}, \dots, x_{101}) \in (\mathbf{X_grid})$, which are the five smallest and five largest values on the x grid. For each of these values, plot the conditional Hill estimator as a function of k_n . Add corresponding 95% confidence bands arising from (3.26).
- (d) Consider $(k_1, \dots, k_5, k_{97}, \dots, k_{101}) \in (\mathbf{K_grid})$, which are the five smallest and five largest values on the k grid. For each of these values, plot the conditional Hill estimator as a function of x . Assess whether it makes sense to plot the corresponding 95% confidence bands that arise from (3.26), and justify your conclusion.
- (e) Construct a 3D plot of $\hat{\gamma}_n(x)$ on $(\mathbf{X_grid}) \times (\mathbf{K_grid})$.

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